

# Reasoning Tests Workbook for EPSO Exams

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*400+ Practice Questions: Verbal, Numerical and Abstract Reasoning, Situational Judgement and Assistant-Level Skills, 3 Timed Mock Exams and Worked Solutions*

CONCOURS PREP

### **Reasoning Tests Workbook for EPSO Exams**

400+ Practice Questions: Verbal, Numerical and Abstract Reasoning, Situational Judgement and Assistant-Level Skills, 3 Timed Mock Exams and Worked Solutions

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# How to Use This Book

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This workbook is built around one idea: you improve fastest when you practise under realistic conditions and then study every mistake. Each of the 413 questions in this book is original and has a full worked solution, so a wrong answer becomes a lesson rather than a mystery.

## What is inside

- **Part I — Understanding the tests.** What each test measures, how the questions are built, and proven methods for each question type.
- **Part II — Practice sets.** 65 verbal, 75 numerical and 75 abstract reasoning questions, graded from foundation to advanced, plus 24 situational judgement scenarios, 30 accuracy & precision questions and 24 prioritising & organising questions.
- **Part III — Three timed mock exams.** Each mock combines 20 verbal, 10 numerical and 10 abstract reasoning questions.
- **Part IV — Answers and worked solutions.** A quick answer key for every set, then a full explanation of every question, including why each wrong option is wrong.
- **Back matter.** Answer sheets, a score tracker and an error log.

## A simple routine that works

1. **Learn the method first.** Read the Part I chapter for a test type before you attempt its practice set.
2. **Practise untimed, then timed.** Work through the foundation questions without a clock and concentrate on method. Start timing yourself from the intermediate questions onwards.
3. **Mark immediately, review deeply.** Check your answers straight after each block. For every error, read the full solution and write one line in the error log: *what did I do, what should I have done, and what will I check next time?*
4. **Re-attempt.** A week later, redo the questions you got wrong without looking at your notes.
5. **Simulate the real thing.** Sit the mock exams in one go, at a desk, with the suggested time limit and no interruptions.

**Tip.** Accuracy comes before speed. Candidates who rush early practice build bad habits that are hard to unlearn. Get the method right, and speed follows.

## Study plans

### Four-week plan (about 1 hour a day)

- **Week 1:** Part I, Chapters 1–3. Verbal questions 1–30 and numerical questions 1–30. Start your error log.
- **Week 2:** Chapter 4. Abstract questions 1–40. Verbal 31–65. Numerical 31–55.
- **Week 3:** Chapters 5–7. Abstract 41–75. Numerical 56–75. Situational judgement, accuracy and prioritising sets if your procedure includes them. Mock Exam 1.
- **Week 4:** Review the error log. Mock Exam 2 early in the week and Mock Exam 3 two days before your test. Light revision only on the last day.

### Eight-week plan (about 30 minutes a day)

- **Weeks 1–2:** Part I. Foundation questions in every test type, untimed.
- **Weeks 3–5:** Intermediate questions, timed. Weekly review of the error log.
- **Week 6:** Advanced questions. Mock Exam 1.
- **Week 7:** The other test types your procedure requires. Mock Exam 2. Re-attempt every error.
- **Week 8:** Mock Exam 3. Revise your strategy notes, then rest.

PART I

# Understanding the Tests

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*What each test measures, how the questions are built, and proven methods for answering them quickly and accurately.*





# Chapter 1 — EU Selection Tests at a Glance

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The European Personnel Selection Office (EPSO) organises selection procedures (often called competitions) for the institutions and agencies of the European Union. Most procedures begin with computer-based multiple-choice tests. Candidates who do well enough in these tests move on to later stages, such as an assessment of their qualifications or an assessment centre.

## Which tests does this book cover?

The core reasoning tests appear across EPSO's generalist administrator (AD) and assistant (AST) procedures, and in many specialist ones:

- **Verbal reasoning:** understanding a text and identifying the statement it supports.
- **Numerical reasoning:** extracting and processing information from tables and charts.
- **Abstract reasoning:** identifying the rule that governs a sequence of figures.

Some procedures, mainly assistant-level (secretaries/clerks, AST-SC), contract-agent (CAST) and certain specialist competitions, have also used other test types:

- **Situational judgement:** choosing the most and least effective response to a workplace scenario.
- **Accuracy & precision:** checking data quickly and without errors.
- **Prioritising & organising:** planning time, tasks and resources under constraints.

**Not covered:** this book does not cover EU-knowledge tests, digital-skills tests, language-comprehension tests or field-specific tests. Use the official sources for those.

**Always check the Notice of Competition.** Test types, numbers of questions, time limits, pass marks and whether tests are eliminatory differ between procedures and change over time. The Notice of Competition (or call for expressions of interest) for *your* procedure is the authoritative source. This book teaches the skills that transfer across formats.

## Typical format

- **On screen, multiple choice, one answer per question.** Verbal and numerical questions typically offer four options and abstract questions five.
- **Strict time limits.** The test is designed so that time pressure matters. Most candidates do not finish comfortably.
- **No negative marking in typical formats.** A blank answer and a wrong answer usually score the same, so you should never leave a question blank. Confirm this in your procedure's instructions.
- **Your chosen language.** Reasoning tests are generally taken in a language you select from those allowed. This book is in English. The reasoning skills are the same in every language.

- **Test centre or remote.** Tests may be taken at an accredited test centre or remotely under online supervision, depending on the procedure. Both use similar on-screen tools.

## How results are used

Reasoning tests are often combined into a single score or marked against pass thresholds. In some procedures, a high mark in one test can compensate for a lower mark in another; in others, each test has its own minimum. Either way, aim to be solid in **all three** reasoning tests rather than brilliant in one.

## Chapter 2 — Verbal Reasoning

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In a verbal reasoning question you read a short passage, typically 120–220 words, and choose the one statement that the passage supports. The skill being measured is disciplined reading: separating what the text actually says from what you know, assume or expect.

### The golden rule: only the text counts

Treat the passage as the only source of truth. A statement can be accurate in the real world and still be the wrong answer, because the passage does not say it. Equally, the correct answer may sound surprising, as long as the passage supports it.

**The test question to ask of every option:** “Where exactly in the passage is this stated or necessarily implied?” If you cannot point to the words, the option is wrong.

### The six classic traps

Most wrong options use one of a small number of traps. Learn to name them and you will spot them quickly.

1. **Extreme wording.** The option says *always*, *never*, *all*, *only*, *entirely* or *guarantees*, while the passage says *often*, *some*, *may* or *in most cases*.
2. **Scope shift.** The passage makes a claim about one group, place or period; the option widens or narrows it (for example, “in coastal regions” becomes “across the country”).
3. **Cause and effect.** The passage reports that two things occur together; the option claims one causes the other, or reverses the direction of cause.
4. **Outside knowledge.** The option is plausible or even true in reality, but the passage never says it.
5. **Partial truth.** Most of the option matches the passage, but one detail — a number, an actor, an order of events, a condition — is wrong.
6. **Contradiction.** The option states the opposite of the passage, often in different words.

Watch for two more advanced variants. *Unsupported comparison* turns a statement about A alone into a comparison between A and B. *Unsupported inference* draws a conclusion that might be likely but is not required by the text. A **proposal is not a decision**, and a **plan is not a result**: be precise about the status of what the passage describes.

## A reliable method

1. **Read the question first** so that you know whether you are looking for the statement that *is* supported or the one that is *not*.
2. **Read the passage once, actively.** Note the main claim, and circle qualifiers (*some, most, may, unless*), numbers and conditions.
3. **Test each option against the text.** For each one, find the relevant sentence and compare the wording closely. Cross out any option you can name a trap for.
4. **Choose the option that survives.** If two seem to survive, re-read the qualifiers: one of them almost always overstates.

**Speed tip.** Do not try to memorise the passage. Read it for structure, then go back to the exact sentence each option refers to. Most errors come from relying on memory instead of re-checking.

## Common mistakes

- Choosing the option that *sounds* most sensible or most like the main idea, rather than the one the text supports.
- Accepting a paraphrase that quietly changes “may” to “will” or “some” to “most”.
- Spending too long on one passage. Make your best choice, flag it and move on.

## Chapter 3 — Numerical Reasoning

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Numerical reasoning questions give you one or more tables or charts and ask a question that requires one to three calculation steps. The maths is rarely beyond lower-secondary level. The difficulty lies in choosing the right numbers, using the right method and working accurately under time pressure. An on-screen calculator is normally available, but check what your procedure allows.

### The essential toolkit

#### Percentage of a total.

Share =  $\text{part} \div \text{total} \times 100$ . Make sure the “total” is the one the question refers to: a row total, a column total or the grand total.

#### Percentage change.

Change =  $(\text{new} - \text{old}) \div \text{old} \times 100$ . The base is always the *earlier* or *reference* value. A rise from 80 to 100 is +25%; a fall from 100 to 80 is –20%.

#### Percentage points versus per cent.

If unemployment falls from 10% to 8%, it has fallen by 2 **percentage points**, which is a fall of 20 **per cent**. Read the question carefully: it tells you which one it wants.

#### Reverse percentages.

If a value *after* a 25% increase is 500, the original was  $500 \div 1.25 = 400$ , not  $500 \times 0.75 = 375$ .

#### Compound growth.

Growing at 10% a year for 3 years multiplies a value by  $1.1 \times 1.1 \times 1.1 = 1.331$ , not by 1.30.

#### Weighted averages.

To average across groups of different sizes, multiply each group's average by its size, add the results and divide by the total size. A simple average of the group averages is a classic trap.

#### Ratios.

Divide both parts by their highest common factor:  $45:30 = 3:2$ . Keep the order the question asks for.

#### Index numbers.

With a base year of 100, the change from an index of 110 to 121 is  $(121 - 110) \div 110 = 10\%$ , not 11%.

#### Currency conversion.

If €1 = 1.50 units of a currency, a price of 300 units is  $300 \div 1.50 = \text{€}200$ . Divide when converting *into* euros and multiply when converting *out of* euros.

**Rates and time.**

Time = distance  $\div$  speed. A 20% rise in speed does not cut journey time by 20%.

**A reliable method**

1. **Read the question before the data.** Identify exactly what is being asked, including units (thousands? millions? per cent?) and the time period.
2. **Locate the numbers.** Find the precise row, column and year. Many errors are reading errors, not maths errors.
3. **Plan the calculation in words** (“Year 3 minus Year 1, divided by Year 1”), then compute.
4. **Estimate before you calculate.** A rough figure (for example, “a bit over 20%”) lets you eliminate options and catch keying errors.
5. **Check units and rounding** against the options. If two options are close, recompute more carefully.

**Distractors are built from typical mistakes:** the wrong base, percentage points instead of per cent, a simple average instead of a weighted one, simple growth instead of compound growth. If your answer matches an option, ask yourself whether you made one of these mistakes before you commit.

**Reading charts**

- Read the axis labels, units and legend before anything else.
- Where data labels are printed, use them; do not estimate from bar heights.
- With line charts, identify the series by marker shape and line style, not position alone.
- With pie charts, every slice is a share of the stated total; amounts = share  $\times$  total.

# Chapter 4 — Abstract Reasoning

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Abstract reasoning questions show a series of five figures and ask which of five options comes next. Each figure contains one or more elements that change according to a rule. The test measures your ability to spot patterns, form a hypothesis and check it systematically, with no words or numbers to help you.

## What can change

The figures in this book (and in typical tests) are built from a small set of variables:

- **Position:** an element moves around the frame, clockwise or anticlockwise, by a fixed or changing number of positions. The frame has eight positions: the four corners and the four edge midpoints.
- **Rotation:** an element turns by a fixed angle, such as  $45^\circ$ ,  $90^\circ$  or  $135^\circ$ , in a fixed direction.
- **Shading:** an element cycles through white, grey and black, or alternates between two of them.
- **Number:** the count of elements rises or falls steadily, or alternates (for example +2, -1, +2, -1).
- **Shape:** a shape gains a side each time, or cycles through a fixed sequence (triangle, square, pentagon, ...).

## The rule-hunting checklist

Work through the variables one at a time, for one element at a time.

1. **Isolate one element.** Follow it through all five figures and ignore everything else.
2. **Ask the five questions:** Where is it? Which way does it point? What shading does it have? How many are there? What shape is it?
3. **Describe the change between each pair of figures** (for example, “+2 positions clockwise”) and look for a constant, alternating or steadily growing step.
4. **Predict the sixth figure** for that element only, then eliminate options that disagree.
5. **Repeat for the next element** until one option remains.

**Elimination is faster than construction.** You rarely need to picture the whole answer. Each rule you confirm usually eliminates one or two options.

## Common patterns and traps

- **Alternating steps:** movements of 1, 2, 1, 2 ... or rotations of  $45^\circ$ ,  $90^\circ$ ,  $45^\circ$ ,  $90^\circ$  ... Check the step after the last figure carefully.
- **Growing steps:** movements of 1, 2, 3, 4 ... positions.

- **Independent rules:** two elements following different rules in the same frame. Treat them separately.
- **Option traps:** the right element in the wrong position, the right position with the wrong shading, or a figure that simply repeats the fifth one.



# Chapter 5 — Situational Judgement

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Situational judgement tests (SJTs) present realistic workplace scenarios and four possible actions. You choose the **most** effective and the **least** effective. In procedures that use them, SJTs assess behavioural competencies such as working with others, communicating and delivering quality.

## Typical competency themes

EU selection procedures have long assessed a stable set of competencies. Their exact names vary between procedures, but the themes are typically:

- **Analysing & problem solving:** identifying the facts and finding practical solutions.
- **Communicating:** being clear and precise, and adapting the message to the audience.
- **Delivering quality & results:** taking responsibility and meeting standards and deadlines.
- **Learning & development:** improving your skills and learning from feedback.
- **Prioritising & organising:** focusing on what matters most and planning realistically.
- **Resilience:** staying effective under pressure, change and setbacks.
- **Working with others:** co-operating, sharing information and resolving conflict constructively.
- **Leadership:** guiding and motivating others, and taking ownership. This mainly applies to higher grades.

## How to rate the options

Effective actions in an EU institutional setting tend to be:

- **Proactive and proportionate:** dealing with the problem at the right level, neither ignoring it nor over-escalating.
- **Constructive and respectful:** talking directly to the person concerned before involving others, and protecting working relationships.
- **Service-oriented and rule-aware:** following procedures, meeting obligations to colleagues and stakeholders, and respecting confidentiality.
- **Transparent:** keeping the right people informed early, especially about risks to deadlines or quality.

Ineffective actions tend to avoid the problem, shift blame, break rules or confidentiality, damage relationships, or escalate before trying a reasonable direct step.

**Answer as an effective colleague would, not as you might under stress.** Think about what the institution would consider best practice. Bear in mind that the most effective option is not always the longest or the most cautious.

# Chapter 6 — Accuracy & Precision and Prioritising & Organising

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These tests have been used mainly in assistant-level and contract-agent procedures. Both reward careful, systematic work under time pressure.

## Accuracy & precision

You compare information, such as reference numbers, names, amounts and codes, against a source table and spot discrepancies.

- **Compare field by field, character by character.** Do not read a code as a word; read it in chunks (for example “TR | 48 | 21 | KX”).
- **Know the usual errors:** transposed digits (4821 → 4812), look-alike characters (0/O, 1/I, 5/S), missing or added accents (Novák → Novak), similar names (Janssen/Jansen), and similar country codes (SI/SK, LT/LV).
- **Use a pointer.** Following the line with a finger or the cursor reduces skipped characters.
- **“No error” is a real option.** Do not invent a discrepancy because you expect one.

## Prioritising & organising

You plan meetings, sequence tasks or allocate work under constraints.

- **Write the constraints down as a list** before looking at the options.
- **For scheduling,** check every required person for each candidate slot, and remember any excluded days and time limits. If the question asks for the *earliest* slot, check slots in time order.
- **For task sequencing,** work out the earliest finish time of each task: its own duration plus the latest finish among its prerequisites. The project length is the longest chain (the critical path), not the sum of all durations.
- **For allocation,** start with the task or person that has the fewest possible options; one forced choice usually unlocks the rest.

# Chapter 7 — Test-Day Strategy

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## Pacing

- **Divide the time by the number of questions** so you know your average budget per question, and check your progress at the quarter, half and three-quarter marks.
- **Don't get stuck.** If a question takes more than about one and a half times your budget, make your best choice, flag it if the system allows, and move on.
- **No blanks.** Where there is no negative marking, answer every question, even if you have to guess in the final seconds.

## Guessing intelligently

Eliminate what you can, then choose from the options that remain. In numerical questions, a quick estimate often eliminates two options. In verbal questions, an extreme or out-of-scope option can usually be crossed out quickly.

## The day before and on the day

- Check the technical and identity requirements in your invitation, especially for remote tests, which may require a system check, a clear desk and a quiet room.
- Know the rules on the calculator, scrap paper and breaks for your sitting.
- Sleep, eat and arrive (or log in) early. A calm start is worth several marks.

## After each mock exam

Score it, enter the result in the tracker and spend at least as long reviewing it as you spent sitting it. That review is where the improvement happens.



PART II

# Practice Sets

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*293 questions across six test types, graded from foundation to advanced. Mark each block straight away and study every solution in Part IV.*



# Practice Set — Verbal Reasoning

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Read each passage and choose the ONE statement that is supported by the passage alone. Do not use outside knowledge. Suggested pace: about 1 minute 45 seconds per question once you move on to timed practice.

*65 questions · answers and worked solutions in Part IV.*

## Foundation

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### Question 1

In a dual vocational training system, apprentices divide their time between a company, where they learn on the job, and a vocational school, where they receive theoretical and general instruction. The apprentice usually signs a contract with the employer and receives a wage, which is lower than that of a qualified worker but rises during the training period. Curricula are typically drawn up jointly by employers' organisations, trade unions and public authorities, so that qualifications reflect the needs of the labour market while being recognised across a whole sector rather than by a single firm. Countries with well-established dual systems tend to record relatively low youth unemployment, although analysts point out that this may also reflect other features of their economies. Attempts to transplant the model elsewhere have met with varying success, often because it depends on employers being willing to invest in training young people whom they may not ultimately retain.

Which of the following statements is NOT supported by the passage?

- A Apprentices in a dual system are generally paid during their training.
- B Qualifications obtained in a dual system are intended to be valid beyond the company that provided the training.
- C Dual training is the main reason for low youth unemployment in the countries that use it.
- D The success of a dual system depends partly on employers' willingness to invest in training.

**Question 2**

When analysts summarise a set of figures, they usually choose between two measures of the “typical” value. The arithmetic mean is obtained by adding all the values and dividing the total by their number; the median is the value that lies in the middle once the figures have been arranged in order. In many everyday datasets the two measures are close, but they can diverge sharply when a distribution is skewed. Household income is the textbook case: a small number of very high earners pull the mean upwards, while the median is barely affected by them. For this reason, statistical offices frequently prefer the median when describing the income of a typical household. The mean nonetheless remains valuable. Because it takes every observation into account, it is the appropriate measure when the total matters – for instance, when a regional authority wishes to estimate the total income of its residents from their average income. Neither measure is inherently superior; the choice depends on the question being asked, and careful reports often publish both.

**Which of the following statements is correct?**

- A** The median is always a more accurate measure than the mean and should be preferred in official statistics.
- B** With a few very high values, the mean can overstate what is typical; the median barely moves.
- C** Statistical offices have abandoned the mean in favour of the median when reporting on household income.
- D** The mean and the median of a dataset will always be identical unless some of the figures contain errors.

**Question 3**

When groups of people who share no common language need to communicate regularly – for example, in the course of trade – they sometimes develop a simplified form of speech known as a pidgin. A pidgin typically draws most of its vocabulary from one language, often that of the economically dominant group, while its grammar is reduced to essentials. Crucially, a pidgin has no native speakers: everyone who uses it has learned another language first. If, however, children grow up in a community where the pidgin is the main shared language, they may acquire it as their first language. In this process the language tends to expand rapidly, developing a fuller grammar and a larger vocabulary so that it can serve every purpose of daily life. Linguists call the resulting language a creole. Creoles are complete languages in their own right, and several are now used in education, broadcasting and government in the places where they are spoken. Not every pidgin becomes a creole, however; many disappear once the circumstances that produced them come to an end.

**Which of the following statements is correct?**

- A** A pidgin is usually the first language of the children who speak it.
- B** Unlike a pidgin, a creole has speakers who acquired it as their mother tongue.
- C** Every pidgin eventually develops into a creole.
- D** A pidgin usually takes its grammar, rather than its vocabulary, from the language of the economically dominant group.



#### Question 4

The principle of comparative advantage, developed by classical economists in the early nineteenth century, explains why countries can gain from trade even when one of them is more efficient at producing everything. What matters is not absolute efficiency but opportunity cost: what must be given up to produce one more unit of a good. Suppose that Country X can produce both cloth and wine using fewer workers than Country Y, but that its advantage is much greater in cloth than in wine. If X concentrates on cloth and Y on wine, and the two countries trade, total output of both goods can be higher than if each tried to produce everything for itself. The principle does not imply that every individual or industry within a country benefits from trade. Workers in industries that face new foreign competition may lose their jobs, and the overall gains depend partly on how easily resources can move from one sector to another. For this reason, many economists who support open trade also favour policies to help affected workers retrain.

Which of the following statements is correct?

- A** A country that is less efficient at producing every good can still gain from specialising and trading.
- B** According to the principle, every worker in a trading country benefits from trade.
- C** What determines specialisation under the principle is which country can produce a good with the fewest workers.
- D** Economists who favour open trade generally oppose schemes to retrain workers.

#### Question 5

A regional authority replaced its paper-based system for building-permit applications with a single online portal. Under the previous arrangement, applicants had to submit separate forms to three departments – planning, environmental health and fire safety – each of which kept its own file and applied its own deadline. The new portal allows an applicant to upload documents once, after which they are shared automatically with all three departments. An internal review, conducted after the first full year of operation, found that the average time taken to reach a decision had fallen by roughly a third. However, the review noted that the improvement was concentrated among straightforward residential applications; complex commercial projects, which often require site visits and consultation with neighbours, showed little change in processing time. Staff reported that the main benefit was the elimination of duplicate data entry rather than any change in the way applications were assessed. The authority has kept a paper option available for applicants without internet access, although relatively few people have chosen to use it.

Which of the following statements is correct?

- A** The portal reduced the average decision time by about a third for every category of application, including complex commercial projects.
- B** Staff saw the main gain as ending repeated entry of the same data, not changing how applications are assessed.
- C** Paper applications are no longer accepted now that the online portal is in operation.
- D** The faster decisions resulted mainly from departments changing the way in which they assess applications.

**Question 6**

A common misconception holds that the seasons are caused by changes in the Earth's distance from the Sun. In fact, the Earth's orbit is nearly circular, and the planet is actually slightly closer to the Sun during the northern winter than during the northern summer. The seasons arise instead from the tilt of the Earth's axis, which is inclined at roughly twenty-three and a half degrees relative to the plane of its orbit. As the Earth travels around the Sun, this tilt causes each hemisphere to lean towards the Sun for part of the year and away from it for another part. When a hemisphere is tilted towards the Sun, sunlight strikes it at a steeper angle, concentrating energy on a smaller area, and the days are longer. Both effects contribute to warmer conditions. This also explains why the seasons are reversed between the two hemispheres: when it is summer in Europe, it is winter in southern Africa and Australia. Near the Equator, where the angle of sunlight changes less over the year, seasonal differences in temperature are generally small.

**Which of the following statements is correct?**

- A** Summer in the northern hemisphere occurs at the time of year when the Earth is closest to the Sun.
- B** Longer days are the only reason why summers are warmer than winters.
- C** The Earth's axial inclination is why the hemispheres have opposite seasons simultaneously.
- D** Regions near the Equator experience the largest seasonal differences in temperature.

**Question 7**

Satellite navigation systems determine a receiver's position by measuring how long signals take to arrive from several satellites whose positions are precisely known. Because the signals travel at the speed of light, even tiny timing errors translate into large positional errors: an error of one millionth of a second corresponds to roughly 300 metres. Each satellite therefore carries highly accurate atomic clocks. Remarkably, the system must also allow for effects predicted by Einstein's theories of relativity. Owing to their high speed, the satellites' clocks run slightly slow compared with clocks on the ground; owing to the weaker gravity at their altitude, they run slightly fast. The gravitational effect is the larger of the two, so the net result is that the satellite clocks gain a small amount of time each day. Engineers compensate by adjusting the clocks' frequency before launch and by applying corrections in the system's calculations. Without these adjustments, positioning errors would accumulate rapidly. A receiver needs signals from at least four satellites, since it must solve for its own clock error as well as its three-dimensional position.

**Which of the following statements is correct?**

- A** Without correction, satellite clocks would run increasingly ahead of clocks on the ground as each day passed.
- B** The high speed of the satellites makes their clocks run faster than clocks on Earth.
- C** Signals from two satellites are sufficient for a receiver to calculate its position.
- D** Relativistic corrections are needed only for military users of satellite navigation.

**Question 8**

In many destinations, tourism is highly seasonal: coastal resorts may receive most of their annual visitors in a few summer months, and mountain areas during the skiing season. Seasonality creates distinctive economic problems. Hotels, restaurants and transport services must be large enough to cope with peak demand, yet much of this capacity stands idle for the rest of the year, which raises the average cost of each visitor served. Employment is often temporary, making it harder for businesses to retain trained staff and for workers to secure a stable income. Local infrastructure, such as water supply and waste collection, must be sized for the peak, while being paid for partly by a much smaller permanent population. Authorities therefore frequently try to spread demand by promoting events, conferences or cultural attractions outside the main season. Such efforts can lengthen the season, but they rarely eliminate the peak, since its timing is largely determined by climate and by the dates of school holidays.

Which of the following statements is correct?

- A** Promoting events outside the main season usually removes the seasonal peak altogether.
- B** Facilities left idle out of season make each visitor more expensive to serve on average.
- C** Temporary tourism jobs mainly suit workers who prefer flexible working patterns.
- D** The timing of the peak season is set mainly by the marketing decisions of hotels and tour operators.

**Question 9**

The road network of the Roman Empire was built primarily to allow the rapid movement of troops and officials, but over time it came to serve a much wider range of purposes. Major roads were typically constructed in layers, with a foundation of larger stones beneath finer material, and many were given a cambered surface so that rainwater would drain to the sides. Where possible, engineers favoured direct routes, although they adapted alignments to the terrain in hilly or marshy areas. Along the principal roads, the state maintained a system of stations where official messengers could change horses and travellers on government business could obtain lodging. This courier service was generally reserved for those carrying official authorisation rather than open to the public at large. Merchants and private travellers nonetheless used the roads extensively, and many towns grew up at junctions and river crossings. Sections of some Roman roads remained in use for centuries after the empire in the west had ceased to exist, and modern roads in several European countries still follow their courses.

Which of the following statements is correct?

- A** Roman roads were built chiefly to encourage private trade between towns.
- B** Any traveller could use the state stations to change horses and obtain lodging.
- C** Roman engineers built every road in a straight line, regardless of hills, marshes or other features of the terrain.
- D** Merchants and private travellers made wide use of roads designed mainly for military and official purposes.

**Question 10**

Before the late eighteenth century, units of length and weight in Europe varied from one country, and often from one town, to the next. A “foot” or a “pound” in one market might differ noticeably from its namesake a day’s journey away, which complicated trade and invited fraud. The metric system, devised in France at the end of that century, aimed to replace this patchwork with units based on nature and related to one another by multiples of ten. The metre was originally defined as one ten-millionth of the distance from the North Pole to the Equator, measured along a meridian, and the unit of mass was linked to the mass of a given volume of water. Adoption was neither immediate nor universal. Even in France, traditional units persisted in everyday use for decades, and some countries never fully abandoned their customary measures. Over time, the definitions themselves were revised: physical reference objects were introduced, and later the units were redefined in terms of fundamental constants of nature, which can be reproduced in any properly equipped laboratory.

**Which of the following statements is correct?**

- A** Once introduced, the metric system was immediately adopted in everyday use throughout France.
- B** Every country eventually gave up its traditional units in favour of the metric system.
- C** The metre was originally defined by reference to the speed of light.
- D** One aim of the metric system was to simplify the relationships between different units.

**Question 11**

In the mid-twentieth century, many planners favoured separating urban functions into distinct zones: housing in one area, offices in another, industry and shopping elsewhere. Separation promised to protect residents from noise and pollution, and it suited a growing reliance on the private car. The American writer and activist Jane Jacobs became one of the most influential critics of this approach. Drawing on her observations of city neighbourhoods, she argued that streets are safest and liveliest when they are used by different people for different purposes at different times of day. A street with homes, shops, workshops and cafés, she suggested, has “eyes on the street”: residents and shopkeepers who notice what is happening and informally discourage crime. She also defended short blocks, older buildings that offer low rents to new businesses, and a dense concentration of people. Her ideas were at first dismissed by many professionals, but they later shaped the revival of mixed-use development in numerous cities. Some critics have since argued that her prescriptions, when applied to fashionable districts, can contribute to rising property prices that displace the very residents and small businesses she valued.

**Which of the following statements is correct?**

- A** Jacobs argued that separating housing from commerce made streets safer.
- B** Jacobs believed that informal watchfulness by people using a street could help to deter crime.
- C** Jacobs’s ideas were immediately accepted by most urban planners.
- D** Jacobs opposed keeping older buildings, arguing that they held back the new businesses on which lively streets depend.

### Question 12

The “span of control” refers to the number of employees who report directly to a single manager. Organisations with wide spans have fewer management layers and are often described as “flat”; those with narrow spans have more layers and are described as “tall”. Flat structures can speed up decision-making, because information passes through fewer levels, and they reduce the cost of management. They may also give employees greater autonomy. Yet a manager with too many direct reports may struggle to provide adequate supervision, feedback and support. The appropriate span depends on the nature of the work. Where tasks are routine and staff are experienced, a manager can oversee a large team effectively; where work is complex, varied or carried out by new recruits, closer supervision is usually needed. Many organisations have reduced their management layers in pursuit of efficiency, but some have later reintroduced intermediate posts after finding that the workloads of the remaining managers had become unsustainable.

Which of the following statements is correct?

- A Flat organisational structures are always more efficient than tall ones.
- B In a tall structure, each manager typically supervises a large number of employees.
- C Routine work by experienced staff suits a wide span of control.
- D Most organisations that removed management layers have since reinstated them.

### Question 13

Albedo is the proportion of incoming sunlight that a surface reflects. Fresh snow and sea ice are highly reflective, returning most of the solar radiation that reaches them to space, whereas open ocean water is dark and absorbs the great majority of it. This contrast underlies one of the best-understood amplifying mechanisms in the climate system, the ice–albedo feedback. When warming causes sea ice to retreat, more dark water is exposed; the water absorbs more solar energy and warms further, which in turn makes it harder for new ice to form and easier for the remaining ice to melt. Because a change in one direction triggers further change in the same direction, scientists describe this as a positive feedback, a term that refers to reinforcement rather than to any beneficial effect. The mechanism helps to explain why the Arctic has been observed to warm faster than the global average. Its strength varies with the seasons: it operates mainly in spring and summer, since during the polar night there is little or no sunlight to absorb. Other processes, including changes in cloud cover and in the transport of heat by the atmosphere and the oceans, also contribute to Arctic warming.

Which of the following statements is NOT supported by the passage?

- A Open water absorbs a larger share of incoming sunlight than sea ice does.
- B The ice–albedo feedback alone explains the Arctic’s rapid warming.
- C Here, the word “positive” does not imply that the effect is desirable.
- D The feedback achieves little during the months of polar darkness.

**Question 14**

For centuries, farmers in many parts of Europe have alternated the crops grown on a given field rather than planting the same crop year after year. This practice, known as crop rotation, serves several purposes. Pests and diseases that specialise in one crop tend to build up in the soil when that crop is grown repeatedly; changing the crop interrupts their life cycle and reduces the pressure they exert. Rotation can also improve soil fertility. Leguminous plants such as peas, beans and clover host bacteria in their root nodules that convert nitrogen from the air into a form plants can use, so a cereal sown after a legume often requires less added fertiliser. In the traditional four-field system that spread across parts of northern Europe, a fodder crop such as turnips or clover replaced the fallow year, in which land had previously been left unplanted to recover. Livestock fed on these fodder crops produced manure that was returned to the fields. Rotation is not without drawbacks: it demands more careful planning, and farmers may be unable to specialise in the single crop that would fetch the highest price in a given season.

**Which of the following statements is correct?**

- A** Crop rotation removes the need to apply fertiliser to cereal fields.
- B** Crop rotation was first developed in northern Europe as part of the four-field system.
- C** The four-field system used formerly fallow land to grow animal feed.
- D** Rotation allows farmers to concentrate on whichever crop is most profitable in a given season.

**Question 15**

A mid-sized city introduced a charge for private cars entering its historic centre on weekdays between seven in the morning and seven in the evening. Residents of the zone receive a substantial discount, and vehicles used by people with disabilities are exempt. Revenue from the charge is reserved by law for public transport and cycling infrastructure. In the first year, traffic entering the zone during charging hours fell by about a fifth, and bus journey times through the centre became more reliable. Some retailers in the zone reported fewer customers, although the city's own analysis found that overall retail spending in the centre was broadly unchanged, with a shift from shoppers arriving by car towards pedestrians and cyclists. Traffic on the ring road around the zone increased slightly, as some drivers chose to travel around the centre rather than through it. The city intends to review the level of the charge after three years of operation.

**Which of the following statements is correct?**

- A** Cars are charged for entering the centre at any time of the day or week.
- B** Every retailer in the zone lost customers as a result of the charge.
- C** Money raised by the charge must be spent on public transport and cycling infrastructure.
- D** Traffic fell on all roads in and around the city centre.

### Question 16

The builders of ancient Rome made extensive use of a form of concrete, a material that allowed them to create structures that would have been difficult or impossible to build in cut stone alone. Its key ingredients were lime, water, rubble and, in many cases, a volcanic ash found near the Bay of Naples. The ash reacted with the lime to produce a mortar that could set even under water, which made it especially valuable for harbour construction. Concrete also gave architects freedom of form: it could be poured into wooden moulds or between facings of brick to produce vaults and domes of great span. The best-known example is the dome of the Pantheon in Rome, which remains one of the largest unreinforced concrete domes in the world. Its builders reduced the weight of the structure by using progressively lighter materials towards the top. Roman concrete was not identical to the modern material, which normally relies on a different kind of cement and on steel reinforcement, but it shared the same basic principle of a binder holding together an aggregate.

Which of the following statements is correct?

- A** The volcanic ash used in some Roman concrete allowed it to harden in water.
- B** Roman concrete was made with the same type of cement and steel reinforcement as modern concrete.
- C** The dome of the Pantheon is the largest concrete dome ever built.
- D** The Pantheon's builders used heavier materials towards the top of the dome to strengthen it.

### Question 17

Pasteurisation, named after the French scientist Louis Pasteur, is a heat treatment that reduces the number of harmful micro-organisms in food and drink. Pasteur developed the technique in the 1860s, initially to prevent the spoilage of wine, before it was applied to milk. Unlike sterilisation, pasteurisation does not aim to kill all micro-organisms. It uses temperatures that destroy the most common disease-causing bacteria while limiting changes to flavour and nutritional value. Milk, for example, is commonly heated to around 72 degrees Celsius for about fifteen seconds and then rapidly cooled. Because some heat-resistant bacteria and spores survive, pasteurised milk must still be refrigerated and has a limited shelf life. Ultra-high-temperature treatment, which heats milk to around 135 degrees for a few seconds, produces a product that can be stored unopened at room temperature for months, though many consumers notice a difference in taste. The widespread adoption of milk pasteurisation in the late nineteenth and early twentieth centuries contributed to a marked decline in diseases transmitted through milk, such as tuberculosis.

Which of the following statements is correct?

- A** Pasteurised milk contains no living bacteria.
- B** Pasteur first used the method to stop wine spoiling; milk came later.
- C** Many consumers find that ultra-high-temperature milk tastes the same as pasteurised milk.
- D** Pasteurisation was the only reason for the decline in milk-borne tuberculosis.

**Question 18**

For most of history, books were costly, and collections of any size belonged to monasteries, universities, rulers or wealthy individuals. Access was usually restricted to members or to those with personal connections. In the eighteenth century, subscription libraries spread in many European towns: members paid an annual fee and could borrow from a shared stock, which made reading cheaper for the middle classes but still excluded most working people. The idea of a library funded from public revenue and open to all residents free of charge gained ground during the nineteenth century. Its advocates argued that free access to books would encourage self-improvement, serve the growing number of literate workers and, some claimed, reduce drunkenness and crime. Opponents objected to the cost to ratepayers and questioned whether public money should be spent on novels, which were then often regarded as frivolous. Early public libraries frequently kept books behind counters, so that readers had to request each title from a librarian; open shelves, allowing readers to browse freely, became common only later.

**Which of the following statements is correct?**

- A** Subscription libraries made books freely available to most working people.
- B** Advocates of public libraries demonstrated that free access to books reduced drunkenness and crime.
- C** Open shelves on which readers could browse were a feature of public libraries from the outset.
- D** Some critics doubted that taxpayers' money should pay for fiction.

**Question 19**

The Hanseatic League was a loose association of merchant communities and towns that dominated much of the trade of the North Sea and the Baltic during the late Middle Ages. Rather than a state with a single government, it functioned as a network: member towns sent representatives to occasional general assemblies, but decisions were not always applied uniformly, and individual towns frequently pursued their own interests. The League's merchants maintained major trading posts, known as kontors, in several foreign cities, including Bergen and Novgorod, where they enjoyed privileges such as reduced tolls and the right to be governed by their own rules. Goods moving through the network included grain, timber, furs, wax and salted fish. The League's influence declined gradually from the fifteenth century onwards, as stronger territorial monarchies asserted control over trade and as Dutch and English merchants became increasingly competitive. It was never formally dissolved in a single act; its last general assembly took place in the seventeenth century.

**Which of the following statements is correct?**

- A** The League was run by a central government whose decisions were binding on every member town.
- B** The League came to a sudden end when it was formally dissolved in the fifteenth century.
- C** Hanseatic merchants operating abroad could benefit from special rights, including lower charges on trade.
- D** The League declined chiefly because demand for grain and timber fell.



**Question 20**

Under a deposit-return scheme, consumers pay a small additional sum, or deposit, when they buy a drink in a can or bottle, and the deposit is refunded when the empty container is returned to a shop or collection point. The aim is to give consumers a financial incentive to return containers rather than discard them. Schemes of this kind have operated in a number of countries for many years, and where they are well established, return rates for the containers covered are typically very high. Because returned containers are collected separately from other waste, the material tends to be cleaner and therefore easier to recycle into new containers. Deposit-return schemes do, however, involve costs. Retailers may need to provide storage space for returned containers or install machines to accept them, and a central body is usually needed to manage the flow of deposits and materials. The design of a scheme – including which types of container are covered and the size of the deposit – has a strong influence on how well it performs.

Which of the following statements is correct?

- A** Returned containers are generally easier to recycle because they are kept apart from other waste.
- B** Deposit-return schemes impose no additional costs or obligations on the retailers that take part.
- C** Consumers can obtain a refund for returning any kind of packaging.
- D** Very high return rates are achieved as soon as a deposit-return scheme is first introduced in a country.

**Intermediate****Question 21**

A national tax administration reviewed the standard letters it sends to citizens after finding that a large share of telephone enquiries concerned the meaning of its own correspondence. A working group rewrote a sample of letters according to plain-language principles: the main message and any required action were placed in the opening paragraph, sentences were shortened, legal terms were either replaced or explained, and deadlines were shown in bold. Legal advisers reviewed every revised letter to confirm that its legal effect was unchanged. In a controlled test, recipients of a payment reminder were randomly allocated either the original version or the revised one. Among those who received the revised letter, a higher proportion paid by the deadline and fewer telephoned for clarification. The working group cautions that the test covered only one type of letter and that results for more complex notices, such as assessments that can be appealed, may differ. The administration has decided to revise all of its payment reminders, and will consider extending the approach to other correspondence once further tests have been carried out.

Which of the following statements is NOT supported by the passage?

- A** Rewriting the letters was not intended to alter their legal consequences.
- B** Recipients of the revised reminder were less likely to telephone for an explanation.
- C** Plain-language versions of all the administration's letters are now in use.
- D** The test results may not hold for more complex notices.

**Question 22**

Linguists generally consider a language to be endangered not when its number of speakers is small, but when children are no longer acquiring it at home. A language spoken by only a few thousand people may be perfectly secure if every new generation learns it, whereas one with hundreds of thousands of speakers can be in serious danger if those speakers are mostly elderly.

Intergenerational transmission typically breaks down when parents come to believe that another language offers their children better economic or social prospects. Once this happens, decline can be rapid: within two or three generations, a community may retain only a handful of fluent speakers. Revitalisation efforts have had mixed results. Programmes that immerse young children in the language – such as so-called language nests, in which pre-school care is conducted entirely in the endangered language – have in several cases produced new fluent speakers. By contrast, initiatives limited to documentation, such as recording elderly speakers and compiling dictionaries, preserve valuable knowledge but do not, by themselves, create new speakers.

**Which of the following statements is correct?**

- A** A language with a large number of speakers may nonetheless be endangered if few children are learning it.
- B** Language nests have succeeded in producing new fluent speakers wherever they have been introduced.
- C** Documentation projects are of little value to communities whose language is endangered.
- D** Parents abandon a heritage language mainly because schools forbid its use.

**Question 23**

When museums abolish admission charges, visitor numbers usually rise, sometimes sharply. Studies of such changes suggest, however, that the composition of the audience shifts less than advocates of free entry often hope. Much of the increase tends to come from people who already visited museums – and now come more often – rather than from groups that previously stayed away. Researchers attribute this to the fact that price is only one barrier among several: lack of time, distance and a feeling that museums are “not for people like me” can matter as much as cost. Free admission also has financial consequences. A museum that loses ticket revenue must replace it through public subsidy, donations or income from shops and cafés, which may themselves benefit from higher footfall. Some institutions adopt intermediate approaches, such as free entry on particular evenings or reduced prices for young people, while charging for temporary exhibitions. Evaluations stress that outreach work with schools and community organisations is often more effective at widening participation than price changes alone.

**Which of the following statements is NOT supported by the passage?**

- A** Abolishing admission charges tends to increase the number of museum visits.
- B** Free admission mainly draws in people who had not previously visited museums.
- C** Cost is not the only factor that deters people from visiting museums.
- D** A museum that stops charging for entry may need other sources of income to make up for the revenue it has lost.

**Question 24**

Open-plan offices, in which most staff work in a shared space without walls or individual rooms, have been widely adopted on the grounds that they reduce accommodation costs and encourage communication. The first claim is generally accepted: more employees can be accommodated in a given floor area. The second has proved less straightforward. Several studies that tracked interactions before and after moves to open-plan layouts found that face-to-face conversation fell rather than rose, while electronic messaging increased. Researchers suggest that, lacking privacy, employees withdraw – wearing headphones, avoiding conversations that might be overheard or seeking quiet spaces elsewhere. Noise and frequent interruptions are also the most common complaints in surveys of open-plan workers, particularly among those whose tasks require sustained concentration. Some organisations have responded by providing a mixture of spaces: open areas for collaborative work, small rooms for confidential or focused tasks, and quiet zones. The effects of any layout appear to depend on how well it matches the nature of the work carried out in it.

**Which of the following statements is NOT supported by the passage?**

- A** Moves to open-plan layouts have consistently increased face-to-face contact between colleagues.
- B** Open-plan layouts allow more employees to be housed in the same floor area.
- C** Staff in open-plan offices may adopt habits that limit their interaction with colleagues.
- D** Employees whose work requires long periods of concentration appear particularly likely to complain about noise and interruptions in open-plan offices.

**Question 25**

For centuries, navigators on the open sea could determine their latitude – their position north or south of the Equator – with reasonable accuracy by measuring the height of the Sun or certain stars above the horizon. Longitude, their position east or west, posed a far harder problem. Because the Earth rotates through 360 degrees in twenty-four hours, each hour of difference between local time and the time at a reference port corresponds to fifteen degrees of longitude. Local time could be found from the Sun; the difficulty lay in knowing, far from land, what time it was at the reference port. Pendulum clocks, accurate on land, were unreliable on a moving ship. Two solutions emerged in the eighteenth century. One was the marine chronometer, a clock robust and precise enough to keep reference time throughout a long voyage. The other relied on astronomical tables that allowed navigators to deduce reference time from the position of the Moon relative to particular stars. Early chronometers were costly, and for some decades many ships continued to use the astronomical method, which required only instruments and tables but involved lengthy calculations.

**Which of the following statements is correct?**

- A** If a navigator knew the time both on board and at the reference port, longitude could be worked out.
- B** Until the eighteenth century, latitude was as difficult to establish at sea as longitude.
- C** The marine chronometer immediately replaced the astronomical method on all ships.
- D** A difference of two hours between local time and reference time corresponds to fifteen degrees of longitude.

**Question 26**

A European research consortium has been digitising the handwritten registers of several regional archives, some of them centuries old. The project produces high-resolution images of every page and, for a selection of registers, a machine-generated transcription that users can search by keyword. The consortium stresses that digitisation is not a substitute for preservation. Digital files must be migrated to new formats and storage systems as technology changes, which requires continuous funding, whereas a well-kept paper register can survive for centuries with relatively little intervention. The originals will therefore remain in the archives, although access to them will be restricted to cases where the digital image is insufficient – for example, when a researcher needs to examine the paper or the ink. Early feedback from users has been positive, but the automatic transcriptions contain errors, particularly in passages written in older scripts. The consortium has accordingly advised users to check search results against the page images, and it has invited volunteers to correct transcriptions, a step that it hopes will improve accuracy over time.

**Which of the following statements is correct?**

- A** A searchable transcription is being produced for every register that the consortium digitises.
- B** Original registers may still be consulted when a digital image does not meet a researcher's needs.
- C** The consortium regards digital copies as a cheaper long-term alternative to keeping the original registers.
- D** Corrections made by volunteers have already removed the errors from the automatic transcriptions.

**Question 27**

Reef-building corals live in partnership with microscopic algae that inhabit their tissues. The algae photosynthesise and supply the coral with much of the energy it needs, while the coral provides shelter and compounds that the algae use. The algae also give many corals their colour. When water temperatures remain unusually high for a prolonged period, this partnership breaks down: the coral expels the algae, and its white calcium carbonate skeleton becomes visible through its now translucent tissue. This is known as bleaching. A bleached coral is not necessarily dead. If temperatures return to normal reasonably quickly, the algae can recolonise the coral and it may recover, although repeated or prolonged episodes leave less time for recovery and increase the likelihood of death. Heat stress is the main cause of mass bleaching events, but other stresses, such as pollution or unusually low tides that expose corals to the air, can also trigger bleaching locally. Because reefs shelter a great variety of fish and invertebrates, extensive coral mortality can have consequences well beyond the corals themselves.

**Which of the following statements is NOT supported by the passage?**

- A** Corals obtain much of their energy from the algae living in their tissues.
- B** Bleaching can be triggered by factors other than high water temperatures.
- C** A coral that has bleached is unable to survive.
- D** The loss of corals can affect other species that depend on reefs.

**Question 28**

In several European ports of the late seventeenth and eighteenth centuries, coffee houses became informal centres of commercial life. Merchants, shipowners and captains gathered there to exchange news of arrivals, departures and losses at sea, and some proprietors began to post shipping information for their customers. These gatherings proved well suited to marine insurance. A merchant seeking cover for a cargo would circulate a document describing the voyage and the vessel, and individuals willing to bear part of the risk would write their names beneath it, each stating the share he accepted. Because no single backer usually took on the whole risk, a loss could be absorbed by many people rather than ruining one. The premium depended on the information available: a sound ship on a familiar route in peacetime attracted a lower rate than an ageing vessel sailing in wartime. Access to reliable news was therefore a competitive advantage, and coffee houses that offered it attracted more business. Over time, some of these informal arrangements developed into formal institutions with membership rules and written procedures.

**Which of the following statements is correct?**

- A** Coffee-house proprietors were responsible for paying merchants' claims whenever an insured ship was lost.
- B** Each individual backer normally accepted the whole risk of a voyage in return for a single premium payment.
- C** Dividing a risk among several backers made it less likely that one loss would ruin any of them.
- D** All of the informal insurance arrangements made in coffee houses eventually turned into formal institutions.

**Question 29**

A national civil service carried out a pilot in which the names, ages and addresses of candidates were removed from application forms for selected junior posts before the forms were assessed by selection panels. The aim was to reduce the risk that panels might, consciously or unconsciously, favour candidates from particular backgrounds. The pilot covered six ministries over eighteen months. Its evaluation found that the share of candidates from under-represented groups who were invited to interview rose in four of the six ministries but remained unchanged in the other two. The evaluators noted that the two ministries where no change occurred had already used structured scoring grids at the sifting stage, which may have limited the scope for bias. They also found that anonymisation had no measurable effect on who was eventually appointed, since interviews could not be conducted anonymously. The evaluation recommended extending anonymised sifting to all ministries, combined with training for interview panels. The civil service management board accepted the first part of the recommendation but asked for further analysis of the cost of the proposed training.

**Which of the following statements is correct?**

- A** The board has accepted that every ministry should sift applications anonymously, but not yet the panel training.
- B** The pilot increased the proportion of appointments going to candidates from under-represented groups.
- C** Invitations to interview for under-represented candidates rose in every ministry taking part in the pilot.
- D** The evaluators concluded that structured scoring grids had introduced bias in the two ministries where no change occurred.

### Question 30

Beneath the familiar wind-driven currents at the ocean surface lies a slower, global pattern of circulation driven by differences in water density. Density depends chiefly on temperature and salinity: colder and saltier water is denser and tends to sink. In certain high-latitude regions, notably parts of the North Atlantic, surface water cools as it releases heat to the atmosphere, and the formation of sea ice leaves salt behind in the remaining water. This dense water sinks to great depths and spreads slowly southwards, while warmer surface water flows northwards to replace it. Because of the roles of temperature and salt, the system is known as the thermohaline circulation. Oceanographers estimate that an individual parcel of water may take on the order of a thousand years to complete the full circuit. The northward flow of warm surface water in the Atlantic carries substantial amounts of heat and is one of several factors contributing to the comparatively mild climate of north-western Europe. Scientists have long studied whether large inflows of fresh water, for example from melting ice, could weaken the sinking process by making surface water less dense.

Which of the following statements is correct?

- A The thermohaline circulation is driven mainly by winds at the ocean surface.
- B Sea-ice formation can aid sinking by making the remaining water saltier.
- C The mild climate of north-western Europe is due entirely to the northward flow of warm Atlantic water.
- D Melting ice is known to have weakened the circulation.

### Question 31

Cities are often noticeably warmer than the surrounding countryside, a phenomenon known as the urban heat island effect. Several factors contribute to it. Dark surfaces such as asphalt and many roofing materials absorb a large proportion of incoming sunlight and release the stored heat slowly, particularly after sunset. Buildings reduce wind speeds and trap heat between them, while the replacement of vegetation and soil by impermeable surfaces reduces evaporation, a process that otherwise has a cooling effect. Heat produced directly by vehicles, air-conditioning units and industry adds to the total. The difference in temperature between a city and its surroundings is usually most pronounced at night and in calm, clear weather. Measures to reduce the effect include planting trees, creating green roofs and using lighter-coloured paving and roofing materials that reflect more sunlight. However, the effectiveness of such measures depends on local conditions: trees, for example, provide shade and cooling through evaporation, but they need an adequate supply of water to do so during dry periods.

Which of the following statements is correct?

- A The urban heat island effect is caused mainly by heat released from vehicles and air-conditioning units.
- B The city–countryside temperature gap is usually greatest on calm, clear nights.
- C Planting trees will lower urban temperatures whatever the amount of water available.
- D Light-coloured paving and roofing materials absorb more sunlight than dark surfaces such as asphalt.

**Question 32**

For most of history, encrypted communication required both parties to share a secret key in advance, which made secure communication between strangers difficult. Public-key cryptography, developed in the 1970s, solved this problem by using a pair of mathematically related keys. The public key can be distributed openly, while the private key is kept secret by its owner. A message encrypted with someone's public key can be decrypted only with the corresponding private key, so anyone can send that person a confidential message without first agreeing a secret. The same pair of keys can be used in reverse to create a digital signature: the owner applies the private key to a message, and anyone holding the public key can verify that the message came from the owner and has not been altered. The security of widely used schemes rests on mathematical problems, such as factoring very large numbers, that are believed to be extremely hard for conventional computers to solve. Because public-key operations are relatively slow, practical systems usually use them only to exchange a shorter secret key, which then encrypts the bulk of the data by faster methods.

Which of the following statements is correct?

- A Public-key cryptography requires the two parties to agree on a secret key before they communicate.
- B A digital signature is created by applying the recipient's public key to a message.
- C It has been proven that conventional computers cannot factor very large numbers.
- D Public-key methods are usually paired with faster ones for encrypting bulk data.

**Question 33**

Archivists organise records according to a principle known as provenance, or respect des fonds. Under this principle, the documents created or accumulated by a single body – a ministry, a family, a trading company – are kept together as a distinct group and are not merged with records from other sources, even when their subjects overlap. A related rule, known as original order, holds that the arrangement given to the records by their creator should be preserved wherever it can still be identified. The reasoning is that the meaning of a document depends partly on its context: a letter filed among a department's complaints says something different from the same letter filed among its commendations. Rearranging material by topic, as a library might arrange its books, can therefore destroy evidence of how an organisation actually worked. These principles do not prevent researchers from searching by subject. Archives produce catalogues, indexes and, increasingly, searchable databases that allow users to locate material on a given theme across many collections, while the arrangement of each collection remains intact.

Which of the following statements is correct?

- A The principle of respect des fonds was first formulated in France, which explains its French name.
- B Under the principle of provenance, records from different creators are combined whenever they concern the same subject.
- C Original order must be preserved in every case, even where the creator's arrangement can no longer be identified.
- D Preserving a creator's arrangement can reveal how the creating body actually worked.



**Question 34**

Educational psychologists have repeatedly found that the act of retrieving information from memory strengthens learning more than simply reviewing it. In typical experiments, students who study a text and then attempt to recall its content perform better on a later test than students who spend the same time rereading it, even when the recall attempt is incomplete and receives no feedback. This “testing effect” is strongest when the final assessment takes place days or weeks later; on a test taken within minutes, rereading can produce equal or even superior results. Spacing study sessions over time, rather than concentrating them in a single block, has a similarly durable benefit. Intriguingly, students themselves tend to prefer rereading and massed study, which create a sense of fluency and therefore feel more effective. Researchers suggest that this mismatch between perceived and actual effectiveness helps to explain why the most efficient strategies are underused, and recommend that teachers build low-stakes quizzes into ordinary lessons.

**Which of the following statements is correct?**

- A** Retrieval practice produces better results than rereading, whenever learning is assessed.
- B** Retrieval practice is beneficial only when students receive feedback on their answers.
- C** Students’ own impressions of which study methods work best are not a reliable guide to their effectiveness.
- D** Researchers recommend replacing end-of-term examinations with frequent low-stakes quizzes.

**Question 35**

A team of education researchers compared library membership with school examination results across twenty districts of a region. The pattern was striking: the five districts with the highest proportion of residents holding a library card also recorded the highest average results, while the five with the lowest membership recorded the lowest. Some commentators cited the findings as proof that library use improves educational performance and called for new branches in districts with weaker results. The researchers themselves urged caution. They pointed out that the districts with high membership were also, on average, the most affluent, and that household income is itself associated with both library use and examination results. The data, they argued, were consistent with a causal effect of libraries but could not establish one. To test the hypothesis more directly, they proposed following the results of pupils in two districts where new branches are due to open, and comparing them with pupils in similar districts where no branch is planned. Even this design, they acknowledged, would not be perfect, since the districts receiving branches had not been selected at random.

**Which of the following statements is NOT supported by the passage?**

- A** The researchers concluded that libraries have no effect on examination results.
- B** Household income may help to explain why high library membership and high examination results occur in the same districts.
- C** The proposed follow-up would compare pupils in districts gaining a branch with pupils in comparable districts that are not.
- D** The researchers accepted that even their proposed follow-up study would have limitations.

**Question 36**

The principle of subsidiarity holds that decisions should be taken at the lowest level of government capable of handling them effectively, and that a higher level should act only where a lower one cannot adequately achieve the objective in question. The idea has roots in social and political philosophy, where it was originally applied to the relationship between the state and smaller associations such as families and voluntary bodies. In multi-level systems of government, it serves both as a guide for allocating responsibilities and as a check on the expansion of central power. Its application is rarely straightforward. Whether a lower level can “adequately” achieve an objective is often a matter of judgement, and the answer may change as circumstances do: problems that cross boundaries, such as river pollution, may be handled more effectively by a higher authority. Critics argue that the principle is too vague to be enforced by courts; supporters reply that its value lies chiefly in obliging decision-makers to justify why action at a higher level is needed.

**Which of the following statements is correct?**

- A** Subsidiarity requires that every decision be taken by the lowest level of government.
- B** The principle was originally developed to govern relations between national and international bodies.
- C** Courts have generally found subsidiarity too vague to enforce.
- D** The appropriate level for dealing with an issue may change if the issue comes to extend across several jurisdictions.

**Question 37**

Population screening programmes aim to detect disease in people who have no symptoms, so that treatment can begin earlier. Their value, however, depends on more than the accuracy of the test. A test with high sensitivity will identify most people who have the condition, but if it also produces many false positives, large numbers of healthy people may experience anxiety and undergo further investigations, some of which carry their own risks. A further concern is overdiagnosis: the detection of genuine conditions that would never have caused symptoms or shortened life. Treating such conditions brings side-effects without benefit. For these reasons, public health bodies typically recommend screening only where the condition is an important health problem, where an effective treatment exists, where earlier treatment improves outcomes and where the benefits of the programme as a whole are judged to outweigh its harms. Recommendations are often restricted to particular age groups in which the balance of benefit and harm is most favourable, and they are revised as new evidence emerges.

**Which of the following statements is correct?**

- A** A screening test that detects most cases of a condition is necessarily worth introducing.
- B** Overdiagnosis occurs when a test wrongly indicates that a healthy person has a condition.
- C** Public health bodies usually recommend that screening be offered to the whole population regardless of age.
- D** Even when a condition has been correctly detected, treating it may cause harm without bringing any benefit.

**Question 38**

Unexpected inflation redistributes wealth between different groups in society. Borrowers whose debts are fixed in money terms benefit, because the real value of what they owe falls as prices rise. Lenders who receive fixed repayments lose for the same reason. Households living on incomes that are not adjusted for rising prices, such as some pensions or interest on fixed-rate savings, also see their purchasing power decline. Where inflation is anticipated, many of these effects can be reduced, since lenders can demand higher interest rates and wages and benefits can be linked to prices. Even anticipated inflation, however, imposes costs. Firms must update their prices more frequently, and people may spend time and effort managing their money to avoid holding cash. High and variable inflation also makes it harder for households and firms to distinguish changes in relative prices from changes in the general price level, which can lead to poor decisions about spending and investment. For these reasons, many central banks aim to keep inflation low and stable.

**Which of the following statements is NOT supported by the passage?**

- A** Unexpected inflation tends to reduce the real burden of debts that are fixed in money terms.
- B** Once inflation is fully anticipated, it no longer imposes any costs on the economy.
- C** Linking incomes to prices can reduce the losses that inflation would otherwise cause.
- D** Volatile inflation can make it more difficult to interpret changes in prices correctly.

**Question 39**

A national agency has issued internal guidance for staff who suspect wrongdoing at work. Staff are encouraged to raise concerns first with their line manager. If that is inappropriate – for example, because the manager is involved – they may report directly to an independent integrity office within the agency. Reports may be made anonymously, although the guidance notes that anonymous reports can be harder to investigate, since the office cannot ask follow-up questions. The guidance protects staff who report in good faith from disciplinary action or other detriment, even if the concern proves unfounded. It does not protect those who knowingly make false allegations. Where a member of staff has reported internally and the integrity office has not responded within three months, he or she may take the matter to an external oversight body. The guidance emphasises that it does not replace the general duty of all staff to report suspected criminal offences to the relevant authorities, a duty that applies regardless of internal procedures.

**Which of the following statements is correct?**

- A** Staff must in all cases raise their concern with their line manager first, before they may contact the integrity office.
- B** Staff who report a concern in good faith lose their protection if the concern turns out to be unfounded.
- C** If the integrity office leaves an internal report unanswered for three months, staff may go to an external body.
- D** The integrity office does not accept anonymous reports, because they cannot be investigated.

**Question 40**

Before the electric telegraph, information could travel overland no faster than a rider on horseback, except where chains of visual signalling stations had been built. The commercial telegraph, which spread rapidly from the 1840s, separated the movement of information from the movement of people and goods on a large scale for the first time. Its effects were felt quickly in several fields. Railway companies used it to coordinate train movements on single-track lines, improving safety and allowing more trains to run. Commodity traders could compare prices in distant markets almost immediately, which narrowed price differences between cities. Newspapers began to carry reports from far-off places within a day rather than weeks. The first lasting transatlantic cable, completed in 1866, reduced the time needed to send a message between Europe and North America from around ten days by ship to a matter of minutes. Sending telegrams remained expensive, however, so for decades most private correspondence continued to go by post, and the telegraph was used mainly by businesses, governments and the press.

**Which of the following statements is correct?**

- A** The telegraph was the first means of sending information overland faster than a rider on horseback.
- B** The telegraph quickly replaced the post as the main channel for private correspondence.
- C** The telegraph helped railways to run more trains safely on lines with a single track.
- D** Messages could be sent across the Atlantic within minutes as soon as the commercial telegraph began to spread.

**Question 41**

Herd immunity describes the indirect protection that unvaccinated individuals receive when a sufficiently large proportion of a population is immune to an infectious disease. If most people cannot catch and transmit a pathogen, chains of transmission are broken before they reach those who remain susceptible, such as infants too young to be vaccinated or people whose immune systems are weakened. The proportion of immune individuals needed to achieve this effect is not fixed. It depends chiefly on how contagious the disease is: the more people a single infected person would infect in a fully susceptible population, the higher the threshold. For a highly contagious disease such as measles, the threshold is commonly estimated at around 95 per cent, whereas for less transmissible diseases it is considerably lower. Herd immunity can be undermined locally even when national coverage appears high, because unvaccinated people often cluster in particular communities or schools. Moreover, the concept applies only to diseases transmitted from person to person; it offers no protection against an infection such as tetanus, which is acquired from spores in the environment.

**Which of the following statements is correct?**

- A** Once 95 per cent of a population is immune, herd immunity is achieved for any infectious disease.
- B** High national vaccination coverage ensures that every local community is protected.
- C** Vaccinating a high proportion of a population against tetanus indirectly protects those who remain unvaccinated.
- D** Less contagious diseases have lower herd-immunity thresholds.

**Question 42**

The oceans absorb a substantial share of the carbon dioxide released into the atmosphere by human activities. This uptake slows the rise of atmospheric concentrations, but it alters seawater chemistry. Dissolved carbon dioxide forms carbonic acid, lowering the water's pH, and, through a series of reactions, reduces the concentration of carbonate ions. Many marine organisms, including corals, oysters, mussels and some types of plankton, use carbonate to build shells or skeletons, and laboratory experiments show that some of them grow more slowly or form weaker structures when carbonate is scarce. Responses vary considerably between species, however, and a few organisms appear unaffected or even benefit, such as certain seagrasses that use the additional carbon dioxide for photosynthesis. The term "acidification" can mislead: seawater remains slightly alkaline, and the word refers to a shift towards the acidic end of the scale rather than to the oceans actually becoming acidic. Because colder water absorbs carbon dioxide more readily, polar seas are expected to experience some of the earliest and largest effects.

**Which of the following statements is correct?**

- A** The oceans have become acidic as a result of absorbing carbon dioxide.
- B** Ocean uptake of carbon dioxide slows its rise in the air but can harm some shell-builders.
- C** All organisms that build shells are harmed by lower carbonate concentrations.
- D** Warm tropical seas absorb carbon dioxide more readily than the colder waters of polar regions.

**Question 43**

Before the spread of mass literacy, much of a community's knowledge – its genealogies, laws, songs and stories – was transmitted orally. Scholars once assumed that such transmission was inherently unreliable, producing ever-greater distortion with each retelling. Fieldwork among oral poets has complicated this picture. Performers of long epic songs were found not to memorise a fixed text word for word; instead, they recomposed the poem at each performance, drawing on a large stock of formulaic phrases and recurring scenes. Two performances of the "same" song by the same singer could differ considerably in length and wording, yet the singers themselves insisted that they were repeating it exactly. Where precise wording mattered, as in ritual formulae or certain legal pronouncements, communities often developed specific techniques – metre, rhyme and collective recitation – that helped to fix the text. Oral transmission is thus best understood not as uniformly reliable or unreliable, but as varying in stability according to the purpose for which a text is preserved.

**Which of the following statements is correct?**

- A** How fixed an orally transmitted text remains can depend on the function it serves.
- B** Epic singers memorised their songs word for word and reproduced them exactly.
- C** Oral transmission is generally less reliable than written transmission.
- D** Singers acknowledged that their performances of a song varied from one occasion to the next.

## Advanced

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### Question 44

How can scientific theories be distinguished from claims that merely look scientific? In the early seventeenth century, Francis Bacon championed induction: the careful accumulation of observations from which general laws could gradually be inferred. Later philosophers pointed out a difficulty. However many confirming instances are observed, they cannot logically guarantee that a universal statement is true; the next observation might contradict it. In the twentieth century, Karl Popper proposed a different criterion. A theory, he argued, is scientific only if it is falsifiable – that is, if it forbids certain observable outcomes, so that it could in principle be shown to be false by experience. On this view, a theory that can accommodate any conceivable result is not thereby strengthened but rendered unscientific. Popper did not claim that falsifiable theories are true, nor that unfalsifiable claims are meaningless; he sought only to mark the boundary of empirical science. His critics noted that, in practice, scientists confronted with an anomalous result often adjust auxiliary assumptions rather than abandon a theory, which makes the decisive refutation Popper described harder to achieve than his account suggests.

Which of the following statements is correct?

- A** For Popper, a theory compatible with every possible result is not scientific.
- B** Popper held that a falsifiable theory which has withstood testing has thereby been proved true.
- C** Popper regarded claims that cannot be falsified as meaningless.
- D** Bacon maintained that a sufficient number of observations can logically guarantee the truth of a general law.

**Question 45**

Economists distinguish between the legal incidence of a tax – who is formally required to pay it to the authorities – and its economic incidence, meaning who ultimately bears its cost. The two frequently differ. Suppose a regional authority imposes a levy on the sellers of a product. Sellers may respond by raising prices, in which case part of the burden passes to buyers; how large a part depends on how strongly each side reacts to changes in price. If buyers have few alternatives and continue to purchase almost the same quantity whatever the price, sellers can pass on most of the levy. If, on the other hand, buyers can easily switch to substitutes, sellers will find it harder to raise prices and will absorb a larger share themselves through lower margins. Over the longer term, part of the burden may also shift to workers or suppliers. One consequence is that a decision about whom to tax formally does not, by itself, determine who pays. Analysts assessing the fairness of a tax are therefore advised to examine market conditions rather than simply reading the text of the law.

**Which of the following statements is correct?**

- A** A levy imposed on sellers is always borne entirely by those sellers.
- B** When buyers can readily switch to substitutes, sellers are able to pass most of a levy on to them.
- C** Two levies imposed on the same group could place their real burden on different groups, depending on the markets concerned.
- D** An authority can ensure that buyers bear the full cost of a tax simply by making them legally liable for it.

**Question 46**

A European cultural foundation offers grants to publishers to cover part of the cost of translating literary works written in less widely spoken languages. An external evaluation found that supported titles were frequently reviewed in the press of the countries where the translations appeared, but it could not determine whether the grants had increased sales, since many publishers did not report sales data. The evaluators also noted that most grants went to translations into a small number of widely spoken languages, partly because publishers in those markets submitted more applications. They recommended that the foundation make sales reporting a condition of future grants, and that it consider reserving a share of its budget for translations between less widely spoken languages. The foundation has agreed to require sales reporting from the next funding round. On the second recommendation, it has stated that it will consult publishers' associations before reaching a decision.

**Which of the following statements is correct?**

- A** The evaluation showed that the grant scheme had increased sales of translated works.
- B** The foundation has decided to reserve part of its budget for translations between less widely spoken languages.
- C** Publishers receiving grants from the next funding round onwards will be expected to report sales data.
- D** The concentration of grants on widely spoken target languages results solely from the foundation's selection criteria.

**Question 47**

Some public administrations and large companies move staff periodically between posts, a practice known as job rotation. Its supporters point to several benefits. Employees who have worked in different units develop a broader understanding of the organisation, which is considered valuable in those who later take on management responsibilities. Rotation can also reduce the risk that one person becomes indispensable, and in posts exposed to pressure from outside interests – such as those involving inspections or the award of contracts – it can limit the formation of relationships that might compromise impartiality. The practice nevertheless carries costs. Each move involves a period during which the newcomer is less productive, and specialist knowledge built up over years may be lost if the previous post-holder is moved too soon. Rotation is therefore generally considered most suitable where the knowledge required can be acquired relatively quickly or documented and handed over. Where a post demands deep technical expertise, organisations that rotate staff tend to do so at longer intervals, or rely on documentation and overlapping hand-over periods to limit the loss.

**Which of the following statements is correct?**

- A** Rotation can support impartiality in some posts but may also cost specialist knowledge.
- B** Job rotation is unsuitable for posts that require deep technical expertise.
- C** Employees who have been rotated through several units go on to perform better as managers than colleagues who have remained in a single post.
- D** The main purpose of job rotation in public administrations is to prepare staff for management roles.

**Question 48**

The idea that shared resources are bound to be overused has long influenced policy. According to this view, when access to a pasture, fishery or forest is open to all, each user has an incentive to take as much as possible, since the benefits of doing so accrue to the individual while the costs of depletion are spread across everyone. The remedies usually proposed were either to place the resource under state control or to divide it into private property. Later research, based on detailed studies of communities around the world, challenged the assumption that these were the only options. It found that many user groups had sustained shared resources over long periods through rules they had developed themselves. Such arrangements tended to succeed where the boundaries of the resource and the membership of the group were clearly defined, where users could take part in changing the rules, where compliance was monitored, and where sanctions for breaking the rules were graduated rather than immediately severe. The research did not claim that community management always works; it identified conditions under which success is more likely.

**Which of the following statements is correct?**

- A** The later research established that community management will succeed provided that the group's membership is clearly defined.
- B** According to the earlier view, overuse occurs because each user bears the full cost of depleting the resource.
- C** Arrangements that endured tended to punish any breach of the rules severely from the outset.
- D** Rules were likelier to endure when the people bound by them could help to revise them.



**Question 49**

Standard-form contracts, drafted in advance by a business and offered to consumers on a take-it-or-leave-it basis, are a familiar feature of modern commerce. Because consumers rarely read such terms in full and have no opportunity to negotiate them, many legal systems allow courts to set aside terms that are found to be unfair. Under a typical regime of this kind, a term is regarded as unfair if, contrary to the requirement of good faith, it causes a significant imbalance in the parties' rights and obligations to the detriment of the consumer. An unfair term is not binding on the consumer, but the rest of the contract continues to apply if it can operate without that term. Terms that have been individually negotiated generally fall outside this protection. So, in many such regimes, do the terms defining the main subject matter of the contract and the adequacy of the price, provided that they are written in plain and intelligible language. The reasoning is that consumers can be expected to pay attention to these core elements, whereas they cannot be expected to scrutinise the small print governing, for example, cancellation fees or the seller's liability.

**Which of the following statements is correct?**

- A** Under a regime of the kind described, a contract containing an unfair term is void in its entirety.
- B** Under such a regime, consumers can challenge a price as unfair whenever it exceeds the usual market rate.
- C** An obscurely drafted price term may be open to review for unfairness.
- D** Under such a regime, a clearly worded term defining the main subject matter of the contract will always be open to review for unfairness.

**Question 50**

In the early twentieth century, the German meteorologist Alfred Wegener proposed that the continents had once been joined in a single landmass and had since drifted apart. He pointed to the way the coastlines of South America and Africa appear to fit together, to matching rock formations on opposite sides of the Atlantic, and to fossils of the same land-dwelling species found on continents now separated by ocean. Many geologists rejected the idea, chiefly because Wegener could not offer a convincing mechanism capable of moving entire continents. The theory gained acceptance only decades later, after the Second World War, when surveys of the ocean floor revealed that new crust forms at mid-ocean ridges and that the rocks on either side record reversals of the Earth's magnetic field in symmetrical bands. These findings indicated that the sea floor spreads outwards from the ridges, providing the kind of mechanism that Wegener had lacked. The modern theory of plate tectonics holds that the Earth's outer shell is divided into rigid plates whose movement explains not only the distribution of the continents but also the location of most earthquakes and volcanoes.

**Which of the following statements is correct?**

- A** Wegener's critics rejected his theory mainly because they disputed that the coastlines of South America and Africa fit together.
- B** Wegener identified sea-floor spreading as the force that moves the continents.
- C** Plate tectonics accounts for the location of all earthquakes and volcanoes.
- D** The chief objection to continental drift was eventually answered by evidence obtained from the ocean floor.

**Question 51**

A regional health authority piloted video consultations for follow-up appointments in three outpatient specialties: dermatology, diabetes care and mental health. Patients were offered the choice between a video consultation and a traditional visit, and about two-fifths chose video. Those who did so were, on average, younger and lived further from the hospital than those who attended in person. Video consultations saved patients an average of more than an hour of travel and waiting time, and missed appointments were less frequent in the video group. Clinicians in mental health and diabetes care rated video consultations as generally adequate, but dermatologists reported that image quality was often insufficient to assess skin lesions, and a notable share of their video patients had to be called in for a further in-person visit. The authority's evaluation team cautions that, because patients selected their own type of appointment, differences between the groups cannot be attributed with confidence to the consultation format itself. The authority intends to continue offering video follow-ups in two of the three specialties.

**Which of the following statements is correct?**

- A** Video consultations caused patients to miss fewer appointments.
- B** Video follow-ups seemed less suitable for dermatology than for the other specialties.
- C** Most patients preferred a video consultation to a visit to the hospital.
- D** Patients living far from the hospital were less likely than others to choose video consultations.

### Question 52

A statistics office that wishes to know what proportion of residents use public transport cannot ask everyone, so it surveys a sample. Provided the sample is selected at random from the whole population, the laws of probability allow the office to estimate how far its result is likely to deviate from the true figure – a range usually reported as the margin of error. That margin shrinks as the sample grows, but more slowly than one might expect: to halve it, the sample must be roughly quadrupled. Crucially, the margin of error measures only the uncertainty that arises from chance in random selection. It says nothing about systematic bias. If a survey is conducted by telephone during working hours, for example, people in full-time employment will be under-represented, and if they use public transport at a different rate from others, the result will be distorted in a consistent direction. Enlarging such a sample does not remove the bias; it merely produces a more precise estimate of the wrong quantity. For this reason, survey designers often devote more effort to how respondents are selected than to how many are selected.

Which of the following statements is correct?

- A** Doubling the size of a random sample will halve its margin of error.
- B** A small margin of error shows that a survey's result is free from systematic bias.
- C** Telephone surveys conducted during working hours will always overestimate the use of public transport.
- D** Adding respondents to a survey with a biased selection method can make its estimate more precise without making it more accurate.

### Question 53

A European research consortium of universities and public laboratories has been debating how the results of its jointly funded projects should be shared. At present, the consortium agreement requires peer-reviewed articles arising from joint projects to be made freely available to readers within twelve months of publication, but it leaves the treatment of underlying data to each project team. A working group has now recommended two changes. First, the embargo on articles would be removed, so that they became freely available immediately on publication. Second, research data would have to be deposited in an open repository unless a project team could justify an exception, for instance on grounds of personal-data protection or of commercial confidentiality agreed with an industrial partner. The working group argues that earlier access accelerates follow-on research; several member institutions, however, are concerned about the publication fees that some journals charge for immediate open access. The consortium's governing board will vote on the recommendations at its next meeting. If they are adopted, they will apply only to projects that begin after that date.

Which of the following statements is correct?

- A** Current rules let an article stay behind a paywall for some months.
- B** Project teams are currently obliged to deposit their research data in an open repository.
- C** If adopted, the new rules would require ongoing projects to make their articles freely available immediately.
- D** The working group's proposals would oblige every project to share its data openly without exception.

**Question 54**

A national government has introduced a requirement that all new regulations affecting businesses include a “sunset clause”, under which the regulation automatically lapses after a fixed period – normally seven years – unless it is reviewed and expressly renewed. The policy applies only to regulations adopted after the requirement came into force; existing rules are unaffected unless they are substantially amended. Exemptions may be granted where a regulation implements an international obligation or where a minister certifies that automatic expiry would create unacceptable risks to public safety. Supporters argue that sunset clauses oblige officials to check whether rules are still fit for purpose. Critics respond that reviews can become a box-ticking exercise, with regulations routinely renewed without serious scrutiny, and that the prospect of expiry may create uncertainty for firms planning long-term investments. A parliamentary committee has proposed extending the requirement to regulations that affect voluntary organisations, but the government has said only that it will consider the proposal once the first round of reviews has been completed.

**Which of the following statements is correct?**

- A** Every regulation affecting businesses, whenever it was adopted, will now lapse after seven years unless it is renewed.
- B** The sunset requirement has been extended to regulations affecting voluntary organisations.
- C** Critics are concerned that sunset clauses will cause too many regulations to be abolished.
- D** A new business regulation that implements an international obligation may be exempted from automatic expiry.

### Question 55

Linguists group languages into families on the basis of systematic correspondences rather than isolated resemblances. Two languages may share a word by chance, or because one has borrowed it from the other; neither case demonstrates common descent. What the comparative method looks for are regular sound correspondences recurring across large parts of the basic vocabulary – words for body parts, close kin, low numbers and the like, which are less often borrowed than terms for trade goods or technology. Where such correspondences are found, linguists infer that the languages descend from a common ancestor, and they can partly reconstruct that ancestor's words, even if no written record of it survives. The Indo-European family, which includes most languages of Europe as well as several of Iran and northern India, was established in this way. The method has limits, however. Because sound and meaning change continuously, resemblances fade over time, and beyond a certain depth – usually estimated at several thousand years – the evidence becomes too thin to distinguish genuine relationship from coincidence. Claims about very ancient super-families therefore remain contested.

Which of the following statements is correct?

- A** Linguists may reconstruct elements of an ancestral language for which no written evidence exists.
- B** Words for trade goods and technology provide the strongest evidence that two languages are related.
- C** If two languages share a word, this demonstrates that they descend from a common ancestor.
- D** The comparative method has shown that every language of Europe belongs to the Indo-European family.

### Question 56

Conservation managers on a small offshore island have been trying to protect a colony of ground-nesting seabirds whose numbers have fallen sharply. Monitoring indicates that the main cause of nest failure is predation of eggs and chicks by rats, which were introduced accidentally by ships many decades ago. A programme to eradicate the rats was approved by the island's management board after a public consultation, and baiting is scheduled to begin outside the breeding season, when the birds are at sea and less likely to be affected. Eradication on islands has a good record elsewhere, but success depends on removing every last individual: if even a few rats survive, the population can recover within a few years. For this reason, the plan includes two years of follow-up monitoring and permanent biosecurity measures, such as inspections of boats landing supplies. Some residents have argued that a fence around the colony would be cheaper. The managers' view is that fencing would protect only part of the island and would require continual maintenance, but they have not ruled out using it as a temporary measure.

Which of the following statements is NOT supported by the passage?

- A** Baiting has been timed to reduce the risk to the seabirds themselves.
- B** The managers regard fencing as unsuitable even as a short-term measure.
- C** The eradication programme was approved following a consultation of the public.
- D** If some rats escape eradication, their numbers could rise again within a few years.

**Question 57**

In some education systems, pupils who fail to reach a required standard repeat the school year; in others, almost all pupils progress automatically with their age group, receiving additional support where needed. Advocates of repetition argue that it gives struggling pupils time to master material before moving on. Research has generally been sceptical. Studies comparing pupils who repeated a year with similar pupils who were promoted often find that any initial improvement fades within a few years, and some find that repeaters are more likely to leave school early. Such comparisons are difficult, however, because teachers select pupils for repetition on the basis of characteristics that researchers may not observe. Repetition is also costly, since each repeated year adds a year of schooling to be funded. Several researchers conclude that the resources involved might achieve more if spent on targeted support during the year in which difficulties first appear. They acknowledge, even so, that the effects of repetition may vary with the age at which it occurs and the support that accompanies it.

**Which of the following statements is correct?**

- A** Repeating a year causes pupils to leave school early.
- B** Studies show that repetition brings no improvement in pupils' attainment at any stage.
- C** Comparisons between repeaters and promoted pupils may be distorted by differences that researchers cannot see.
- D** Researchers have established that targeted support in the year in which difficulties appear is more effective than repetition.

**Question 58**

A regional transport authority must replace an ageing fleet of diesel trains on several rural lines that have no overhead electrification. Its options appraisal compared three alternatives. Full electrification would allow conventional electric trains to be used but would require substantial civil works, including the reconstruction of some bridges, and would take the longest to deliver. Battery-electric trains could recharge from short electrified sections at terminal stations; they performed well in the appraisal on the shorter lines but, with current battery capacities, would be unable to complete the longest line without an additional charging point midway. Hydrogen fuel-cell trains could serve all the lines without new electrification, but would require a refuelling depot and a reliable supply of hydrogen, and their overall energy efficiency is lower than that of battery trains. The appraisal recommends battery-electric trains for the shorter lines and further study of the longest line, where either a midway charging point or hydrogen traction may prove preferable. The authority's board has endorsed the recommendation for the shorter lines and will decide on the longest line once the further study is complete.

**Which of the following statements is correct?**

- A** The authority has decided to use hydrogen trains on the longest line.
- B** The appraisal found battery-electric trains unsuitable for all of the authority's lines.
- C** Battery trains could currently serve the longest line only with an added midway charging point.
- D** The appraisal rejected hydrogen trains because of their lower energy efficiency.

### Question 59

Public debates often blur two distinct concepts: the budget deficit and public debt. The deficit is a flow – the amount by which a government's expenditure exceeds its revenue within a single year. The debt is a stock – the total amount the government owes at a given moment, accumulated through past borrowing. A deficit adds to the debt; a surplus, if used to repay creditors, reduces it. It follows that a government that cuts its deficit has not thereby reduced its debt: as long as any deficit remains, the debt continues to grow in absolute terms. Analysts usually express both figures as a share of national output, because a given sum is more manageable for a large economy than for a small one. This ratio can fall even while the debt rises in absolute terms, provided that the economy grows faster than the debt. Conversely, a recession can push the ratio up without any additional borrowing, simply by shrinking output. For this reason, budget rules that target the ratio rather than the absolute figure are sensitive to economic conditions that lie partly outside a government's control.

Which of the following statements is correct?

- A** Reducing the annual deficit is sufficient to bring down the total amount of public debt.
- B** A fall in the ratio of debt to national output shows that the government has repaid part of its debt.
- C** A recession raises the debt ratio because governments always borrow more during economic downturns.
- D** A government in deficit every year can still see its debt shrink relative to national output.

### Question 60

Desertification is often imagined as the advance of sand dunes across fertile land, but the term is generally used for something broader: the degradation of land in dry regions, resulting in a lasting loss of biological and economic productivity. It can occur far from any existing desert. Both climatic variation and human activity contribute. Prolonged drought reduces plant cover, but land already under pressure from overgrazing, the clearing of vegetation or poorly managed irrigation is far less able to recover when the rains return. Irrigation can itself cause damage: where drainage is inadequate, water evaporates and leaves salts behind, gradually making the soil unfit for crops. Once vegetation is lost, exposed soil is more easily removed by wind and water, which further reduces the land's capacity to support plants – a self-reinforcing process. Degradation is nonetheless not always irreversible. Controlled grazing, the planting of drought-resistant species and the revival of traditional water-harvesting techniques have, in some areas, allowed productivity to recover, although results depend heavily on local conditions.

Which of the following statements is correct?

- A** Because desertification is self-reinforcing, degraded land cannot be restored once vegetation has been lost.
- B** Drought tends to cause more lasting harm where land is already under pressure from human use.
- C** Desertification is driven chiefly by climatic variation, with human activity playing only a minor role.
- D** Restoration measures such as controlled grazing have reversed land degradation wherever they have been applied, whatever the local conditions.

**Question 61**

Groundwater – water held in the pores and fractures of rocks and sediments known as aquifers – supplies drinking water to a large share of the world’s population and irrigates a substantial part of its farmland. Aquifers are replenished, or recharged, mainly by rainfall and snowmelt seeping down from the surface, but recharge can be extremely slow; some deep aquifers contain water that accumulated thousands of years ago and is effectively not being replaced. When abstraction persistently exceeds recharge, the water table falls. Wells must then be deepened, and pumping costs rise because water has to be lifted further. In coastal areas, a falling water table can allow seawater to move inland into the aquifer, making the water too salty for drinking or irrigation; such contamination can take decades to reverse. Where aquifers consist of compressible clays, prolonged pumping can also cause the ground to subside, and this compaction is largely permanent: even if water levels later recover, the aquifer’s storage capacity may not. Hydrologists stress that the sustainable rate of abstraction is not simply equal to the recharge rate, since rivers, wetlands and springs fed by groundwater also depend on part of that inflow.

**Which of the following statements is correct?**

- A** Abstraction that matches an aquifer’s recharge rate is, by definition, sustainable.
- B** Once water levels recover, land that subsided over clay aquifers will return to its former level.
- C** Wherever a water table falls, seawater can be expected to move into the aquifer.
- D** Sustained pumping from an aquifer with negligible recharge will lower its water table and raise the cost of pumping.

**Question 62**

Global mean sea level has been rising since at least the late nineteenth century, and two processes account for most of the increase. The first is thermal expansion: as seawater warms, it occupies a greater volume. The second is the addition of water from land-based ice, namely mountain glaciers and the large ice sheets of Greenland and Antarctica. The melting of floating sea ice, by contrast, makes almost no direct contribution, because floating ice already displaces a mass of water roughly equal to its own. Sea level does not change uniformly around the world. Ocean currents, prevailing winds and variations in the Earth’s gravitational field all cause regional departures from the global average. In some places the land itself is moving: regions that were weighed down by thick ice sheets during the last glacial period are still slowly rising, while other coasts are subsiding, sometimes because of the compaction of sediments. For coastal communities, therefore, the relevant measure is relative sea-level change – the combined effect of changes in the water level and vertical movement of the land.

**Which of the following statements is correct?**

- A** The melting of Arctic sea ice is one of the two main drivers of global sea-level rise.
- B** Every coastal area must be seeing a rise in relative sea level.
- C** A coastal town on land that is rising could experience a smaller relative sea-level rise than the global average would suggest.
- D** Sediment compaction is the main reason why some coasts are subsiding.



**Question 63**

In the 1840s, Ignaz Semmelweis, a physician at a large maternity hospital in Vienna, investigated why one of the hospital's two maternity clinics had a far higher death rate from puerperal fever than the other. The clinic with the higher rate was staffed by doctors and medical students, who often came directly from performing autopsies; the other was staffed by midwives, who did not perform them. After a colleague died from an infection contracted through a cut received during an autopsy, Semmelweis concluded that "cadaverous particles" carried on the hands were responsible. He introduced compulsory handwashing with a chlorinated lime solution, and the death rate in the doctors' clinic fell sharply. Yet his ideas were widely resisted. He published a full account only years later, his explanation did not fit prevailing theories of disease, and some doctors were offended by the suggestion that they themselves transmitted illness. Wider acceptance of hand hygiene came only in the following decades, once the work of later scientists had established the germ theory of disease.

Which of the following statements is NOT supported by the passage?

- A Semmelweis's explanation conflicted with the dominant medical thinking of his time.
- B The difference in death rates between the two clinics was linked to the work of the staff in each.
- C Semmelweis drew on the germ theory of disease when reaching his conclusion.
- D Some of the resistance to Semmelweis's ideas reflected doctors' reluctance to accept that they spread illness.

**Question 64**

The board of a regional museum of decorative arts has been reviewing how it handles objects whose ownership history contains gaps. At present, the museum's acquisition policy requires curators to document an object's provenance before purchase, but it imposes no obligation to research items already in the collection. Following a review, an advisory panel recommended three changes: that the museum systematically examine the history of all objects acquired before its current policy took effect; that it publish online any object whose provenance cannot be established; and that it establish a procedure for considering claims from individuals or communities who believe an object was taken from them unlawfully. The board has accepted the first recommendation and allocated funding for two research posts. It has deferred a decision on online publication pending legal advice, and has asked the panel to draft the claims procedure in more detail before it is considered. The panel noted that research of this kind rarely produces conclusive results quickly.

Which of the following statements is correct?

- A The museum now publishes online the details of objects with incomplete ownership histories.
- B The board has rejected the panel's recommendation concerning claims from individuals or communities who believe that objects were taken from them unlawfully.
- C The museum's current policy already requires research into objects acquired in the past.
- D Of the panel's three recommendations, only the one concerning research into earlier acquisitions has so far been adopted.

**Question 65**

The great cathedrals built in western Europe from the twelfth century onwards are commonly described as Gothic, a label applied much later and originally intended as disparaging. Their builders sought height and light, and three structural devices made these aims attainable. The pointed arch directs the weight of the vault more steeply downwards than a semicircular arch does, reducing the outward thrust on the walls. The ribbed vault concentrates loads along a network of stone ribs, so that they are carried to particular points rather than spread evenly along the wall. At those points, flying buttresses – external arches springing from detached piers – receive the remaining thrust and transfer it to the ground. Because the walls no longer had to carry the whole load, large sections of them could be opened up and filled with stained glass. The techniques were not entirely new; pointed arches and rib vaults appear in earlier buildings. What distinguished the Gothic cathedral was the systematic combination of these elements, which allowed masons to build higher, and with thinner walls, than their predecessors had generally managed.

**Which of the following statements is NOT supported by the passage?**

- A** The builders of these cathedrals did not themselves describe their work as “Gothic”.
- B** The pointed arch was invented by Gothic builders as a means of reducing the outward thrust on walls.
- C** Flying buttresses carry part of the thrust of the vault away from the walls and down to the ground.
- D** Concentrating loads at particular points made it possible to replace much of the wall surface with glass.

# Practice Set — Numerical Reasoning

Use the data provided to answer each question. A basic calculator is allowed. All data are fictitious. Suggested pace: about 2 minutes per question.

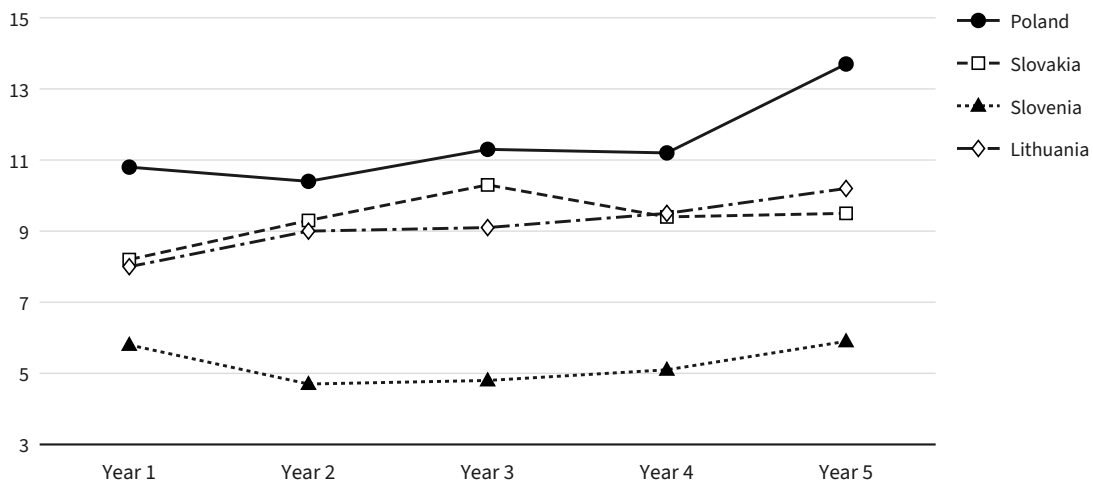
75 questions · answers and worked solutions in Part IV.

## Foundation

### Question 1

Unemployment rate (% of labour force)

Unit: %



| Series    | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|-----------|--------|--------|--------|--------|--------|
| Poland    | 10.8   | 10.4   | 11.3   | 11.2   | 13.7   |
| Slovakia  | 8.2    | 9.3    | 10.3   | 9.4    | 9.5    |
| Slovenia  | 5.8    | 4.7    | 4.8    | 5.1    | 5.9    |
| Lithuania | 8.0    | 9.0    | 9.1    | 9.5    | 10.2   |

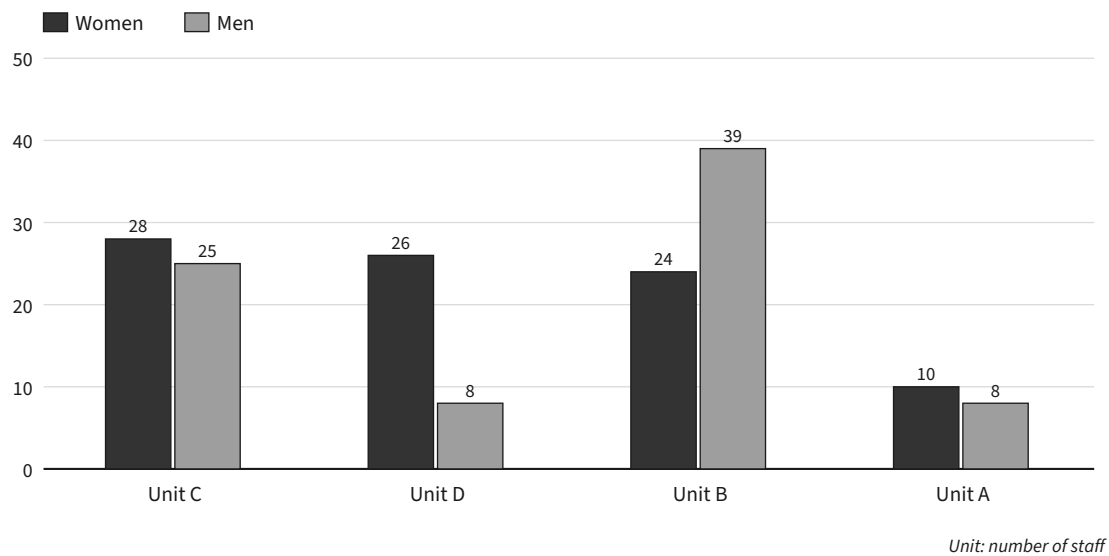
All data are fictitious.

By how many percentage points did the unemployment rate in Poland rise between Year 3 and Year 5?

- A 0.9 percentage points
- B 0.5 percentage points
- C 4.3 percentage points
- D 2.4 percentage points

Question 2

Staff by gender in four units



All data are fictitious.

If the total staff of Unit A increases by 50% and the proportion of women stays the same, how many women will Unit A employ?

- A 19
- B 15
- C 12
- D 14

Question 3

Visitors to national science museums (thousands)

| Country  | Year 1 | Year 2 | Year 3 |
|----------|--------|--------|--------|
| Italy    | 930    | 1,056  | 973    |
| Hungary  | 363    | 356    | 414    |
| Sweden   | 753    | 669    | 675    |
| Greece   | 663    | 615    | 583    |
| Slovakia | 233    | 261    | 260    |

All data are fictitious.

By approximately what percentage did the number of visitors to science museums in Hungary increase between Year 1 and Year 3?

- A 11.8%
- B 16.3%
- C 12.3%
- D 14.0%

**Question 4**

Translation requests received by a language unit in one month, by source language

| Source language | Requests |
|-----------------|----------|
| Italian         | 32       |
| English         | 40       |
| Swedish         | 60       |
| Polish          | 146      |

*All data are fictitious.*

What is the ratio of requests from English to requests from Italian, in its simplest form?

- A** 5:3
- B** 6:5
- C** 5:4
- D** 4:5

**Question 5**

Regional development fund allocations by sector (€ million)

| Region | Transport | Health | Education | Agriculture | Total |
|--------|-----------|--------|-----------|-------------|-------|
| West   | 210       | 930    | 650       | 580         | 2,370 |
| East   | 720       | 790    | 870       | 850         | 3,230 |
| North  | 490       | 890    | 440       | 260         | 2,080 |
| South  | 210       | 860    | 660       | 910         | 2,640 |

*All data are fictitious.*

What percentage of the South region's total allocation went to Education?

- A** 33.3%
- B** 25.0%
- C** 16.7%
- D** 40.5%

### Question 6

Overnight stays in campsites (thousands)

| Country   | Year 1 | Year 2 | Year 3 |
|-----------|--------|--------|--------|
| Romania   | 192    | 217    | 212    |
| Germany   | 666    | 631    | 619    |
| Greece    | 419    | 373    | 369    |
| Lithuania | 399    | 463    | 491    |
| Austria   | 484    | 447    | 508    |

All data are fictitious.

By approximately what percentage did the number of overnight stays in campsites in Lithuania increase between Year 1 and Year 3?

- A 22.1%
- B 16.0%
- C 23.1%
- D 18.7%

### Question 7

Regional development fund allocations by sector (€ million)

| Region | Digital | Culture | Agriculture | Research | Total |
|--------|---------|---------|-------------|----------|-------|
| North  | 280     | 250     | 410         | 360      | 1,300 |
| East   | 160     | 860     | 380         | 300      | 1,700 |
| South  | 950     | 800     | 810         | 910      | 3,470 |
| West   | 420     | 450     | 600         | 270      | 1,740 |

All data are fictitious.

What percentage of the North region's total allocation went to Culture?

- A 10.6%
- B 19.2%
- C 23.8%
- D 11.8%

**Question 8**

Translation requests received by a language unit in one month, by source language

| Source language | Requests |
|-----------------|----------|
| Italian         | 138      |
| French          | 75       |
| German          | 45       |
| Swedish         | 74       |

*All data are fictitious.*

What is the ratio of requests from German to requests from French, in its simplest form?

- A 3:4
- B 3:5
- C 5:6
- D 5:3

**Question 9**

Grant applications received and approval rate

| Country     | Applications | Approval rate |
|-------------|--------------|---------------|
| Netherlands | 1,100        | 12%           |
| Slovakia    | 3,000        | 35%           |
| Italy       | 2,200        | 40%           |
| Ireland     | 2,100        | 15%           |

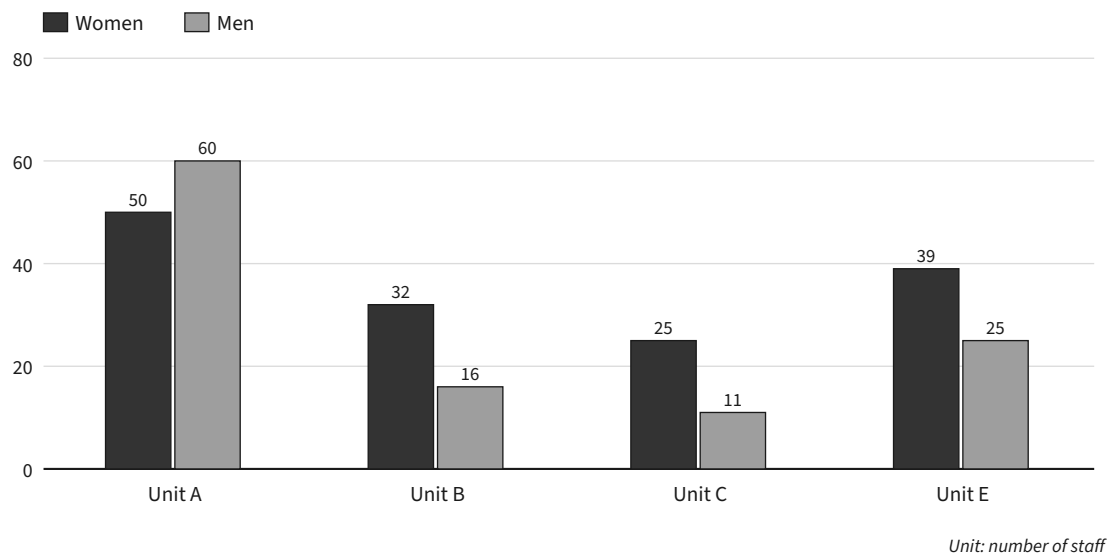
*All data are fictitious.*

How many applications from Italy were rejected?

- A 880
- B 1,558
- C 1,320
- D 1,785

Question 10

Staff by gender in four units



All data are fictitious.

If the total staff of Unit A increases by 10% and the proportion of women stays the same, how many women will Unit A employ?

- A 55
- B 66
- C 61
- D 49

Question 11

Translation requests received by a language unit in one month, by source language

| Source language | Requests |
|-----------------|----------|
| English         | 140      |
| German          | 85       |
| Spanish         | 16       |
| French          | 40       |

All data are fictitious.

What is the ratio of requests from French to requests from Spanish, in its simplest form?

- A 7:3
- B 5:1
- C 5:2
- D 2:5



**Question 12**

Grant applications received and approval rate

| Country  | Applications | Approval rate |
|----------|--------------|---------------|
| Portugal | 800          | 40%           |
| Spain    | 500          | 40%           |
| Germany  | 2,900        | 30%           |
| Croatia  | 600          | 40%           |

*All data are fictitious.*

How many applications from Croatia were rejected?

- A** 425
- B** 240
- C** 360
- D** 480

**Question 13**

Overnight stays in campsites (thousands)

| Country | Year 1 | Year 2 | Year 3 |
|---------|--------|--------|--------|
| France  | 390    | 356    | 464    |
| Finland | 333    | 392    | 421    |
| Italy   | 346    | 357    | 428    |
| Estonia | 614    | 615    | 579    |
| Ireland | 770    | 725    | 705    |

*All data are fictitious.*

By approximately what percentage did the number of overnight stays in campsites in France increase between Year 1 and Year 3?

- A** 15.9%
- B** 21.6%
- C** 30.3%
- D** 19.0%

**Question 14**

Regional development fund allocations by sector (€ million)

| Region  | Transport | Digital | Culture | Education | Total |
|---------|-----------|---------|---------|-----------|-------|
| North   | 220       | 930     | 680     | 280       | 2,110 |
| South   | 560       | 450     | 170     | 210       | 1,390 |
| East    | 380       | 950     | 450     | 360       | 2,140 |
| Central | 690       | 480     | 190     | 140       | 1,500 |

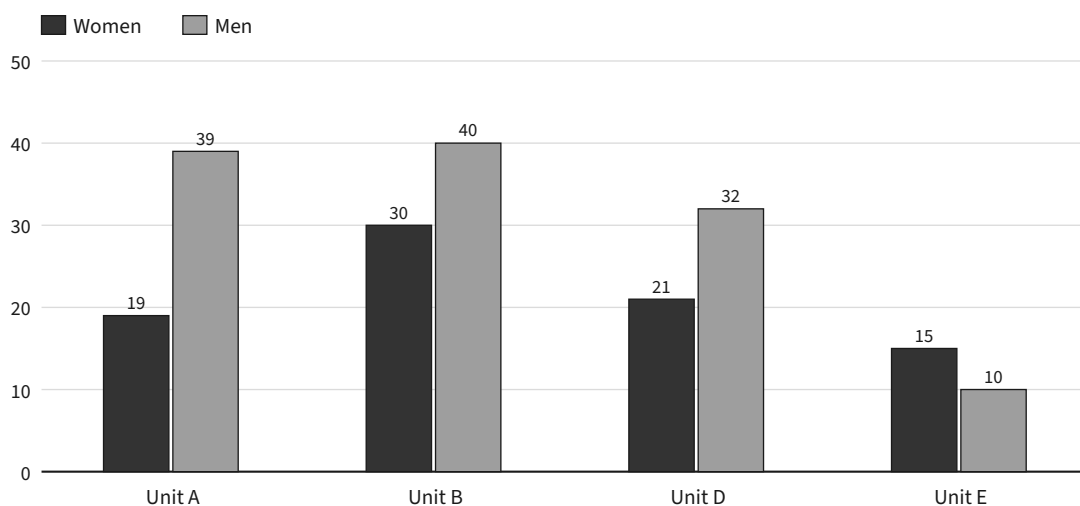
*All data are fictitious.*

What percentage of the North region's total allocation went to Transport?

- A 10.4%
- B 9.0%
- C 13.5%
- D 11.9%

**Question 15**

Staff by gender in four units

*Unit: number of staff**All data are fictitious.*

If the total staff of Unit B increases by 20% and the proportion of women stays the same, how many women will Unit B employ?

- A 44
- B 50
- C 36
- D 48

**Question 16**

Regional development fund allocations by sector (€ million)

| Region | Energy | Culture | Digital | Education | Total |
|--------|--------|---------|---------|-----------|-------|
| West   | 280    | 240     | 410     | 230       | 1,160 |
| South  | 640    | 380     | 940     | 780       | 2,740 |
| North  | 950    | 250     | 770     | 450       | 2,420 |
| East   | 550    | 260     | 510     | 630       | 1,950 |

*All data are fictitious.*

What percentage of the South region's total allocation went to Digital?

- A** 37.2%
- B** 55.6%
- C** 52.2%
- D** 34.3%

**Question 17**

Visitors to national science museums (thousands)

| Country  | Year 1 | Year 2 | Year 3 |
|----------|--------|--------|--------|
| Germany  | 465    | 414    | 416    |
| Ireland  | 630    | 570    | 548    |
| Spain    | 743    | 713    | 756    |
| Czechia  | 629    | 584    | 749    |
| Portugal | 845    | 773    | 888    |

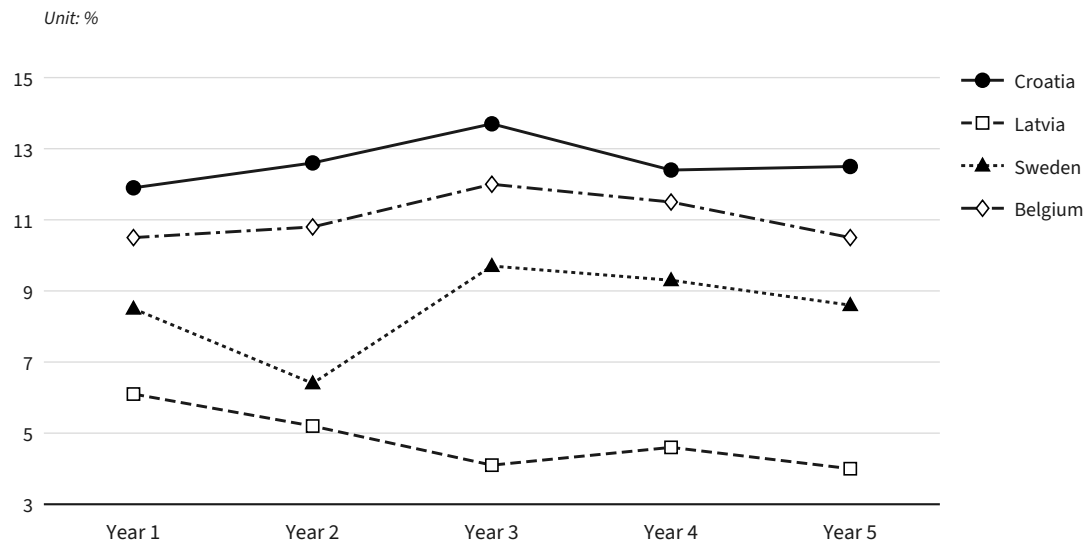
*All data are fictitious.*

By approximately what percentage did the number of visitors to science museums in Czechia increase between Year 1 and Year 3?

- A** 21.1%
- B** 28.3%
- C** 19.1%
- D** 16.0%

Question 18

Unemployment rate (% of labour force)



| Series  | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|---------|--------|--------|--------|--------|--------|
| Croatia | 11.9   | 12.6   | 13.7   | 12.4   | 12.5   |
| Latvia  | 6.1    | 5.2    | 4.1    | 4.6    | 4.0    |
| Sweden  | 8.5    | 6.4    | 9.7    | 9.3    | 8.6    |
| Belgium | 10.5   | 10.8   | 12.0   | 11.5   | 10.5   |

All data are fictitious.

By how many percentage points did the unemployment rate in Sweden fall between Year 1 and Year 2?

- A 1.2 percentage points
- B 3.0 percentage points
- C 2.1 percentage points
- D 0.8 percentage points

**Question 19**

Translation requests received by a language unit in one month, by source language

| Source language | Requests |
|-----------------|----------|
| Polish          | 26       |
| Italian         | 36       |
| French          | 139      |
| English         | 65       |

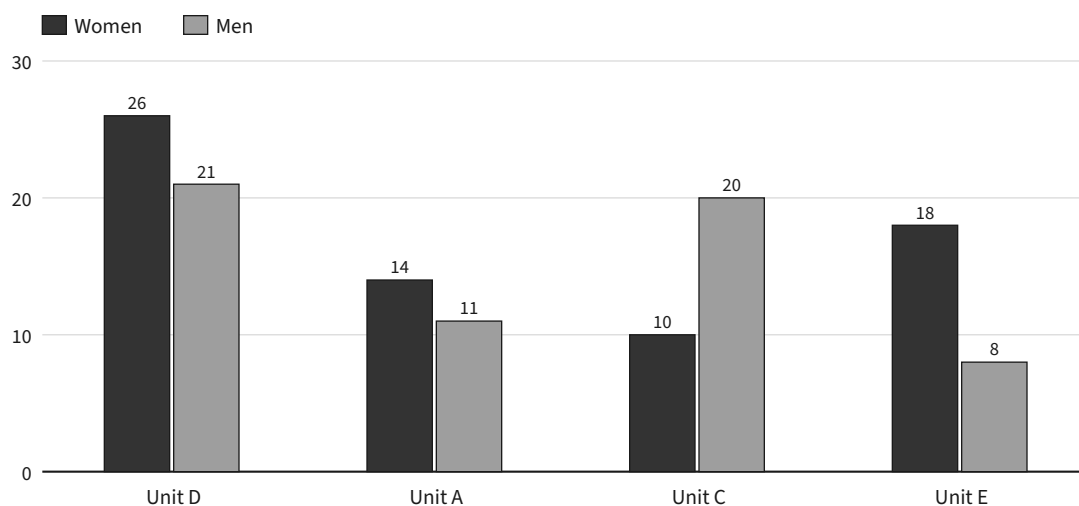
*All data are fictitious.*

What is the ratio of requests from Polish to requests from English, in its simplest form?

- A 5:2
- B 2:5
- C 3:5
- D 1:5

**Question 20**

Staff by gender in four units



*Unit: number of staff*

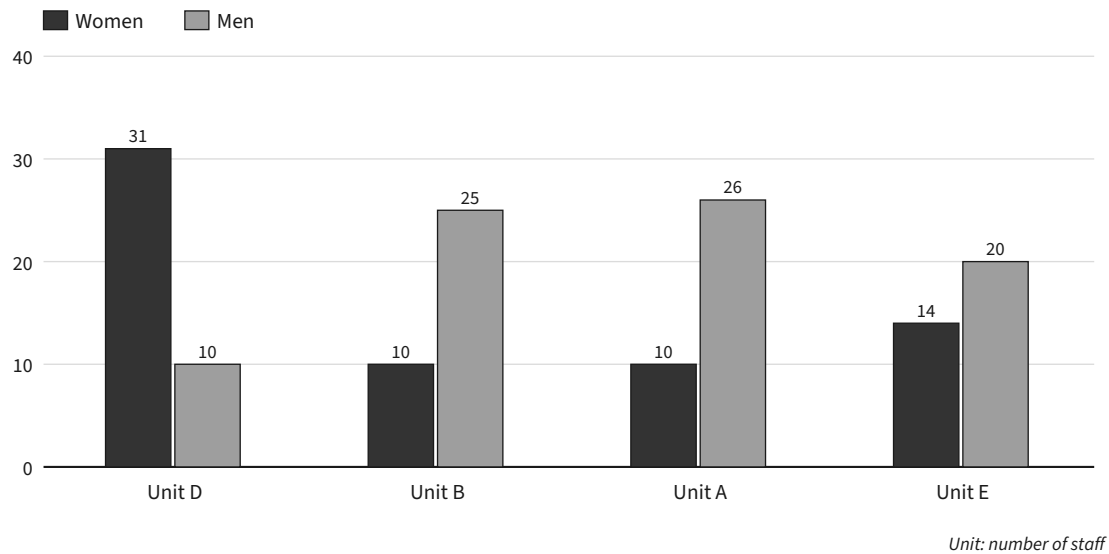
*All data are fictitious.*

If the total staff of Unit C increases by 40% and the proportion of women stays the same, how many women will Unit C employ?

- A 14
- B 22
- C 21
- D 28

Question 21

Staff by gender in four units



All data are fictitious.

If the total staff of Unit B increases by 20% and the proportion of women stays the same, how many women will Unit B employ?

- A 30
- B 21
- C 12
- D 17

Question 22

Regional development fund allocations by sector (€ million)

| Region  | Education | Culture | Digital | Agriculture | Total |
|---------|-----------|---------|---------|-------------|-------|
| Central | 150       | 200     | 880     | 680         | 1,910 |
| North   | 620       | 150     | 460     | 530         | 1,760 |
| West    | 430       | 830     | 280     | 850         | 2,390 |
| South   | 230       | 660     | 930     | 850         | 2,670 |

All data are fictitious.

What percentage of the Central region’s total allocation went to Digital?

- A 34.5%
- B 46.1%
- C 85.4%
- D 18.3%

## Intermediate

### Question 23

Population and public spending on culture in five regions of one country

| Region          | Population (million) | Spending (€ million) |
|-----------------|----------------------|----------------------|
| Northern region | 4.38                 | 780.9                |
| Central region  | 5.48                 | 1,127.9              |
| Eastern region  | 2.53                 | 459.3                |
| Island region   | 1.04                 | 171.6                |
| Coastal region  | 3.86                 | 906.2                |

All data are fictitious.

Which of the following regions had the highest spending on culture per inhabitant?

- A Coastal region
- B Central region
- C Island region
- D Eastern region

### Question 24

Index of average house prices (Year 1 = 100)

| Country  | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|----------|--------|--------|--------|--------|--------|
| Portugal | 100.0  | 105.4  | 110.8  | 120.0  | 124.9  |
| Austria  | 100.0  | 106.4  | 106.6  | 113.5  | 118.2  |
| Poland   | 100.0  | 104.9  | 108.2  | 116.7  | 117.0  |
| Latvia   | 100.0  | 102.2  | 109.8  | 111.8  | 121.9  |

All data are fictitious.

By approximately what percentage did average house prices in Latvia rise between Year 3 and Year 5?

- A 12.1%
- B 11.0%
- C 21.9%
- D 9.9%

**Question 25****Table 1 — Participants per training course**

| Course               | Q1 | Q2 | Q3 | Q4 |
|----------------------|----|----|----|----|
| Project management   | 6  | 8  | 16 | 12 |
| Drafting legislation | 4  | 20 | 16 | 19 |
| Data analysis        | 26 | 15 | 10 | 18 |
| Financial rules      | 26 | 22 | 19 | 25 |

**Table 2 — Cost per participant (€)**

| Course               | Cost (€) |
|----------------------|----------|
| Project management   | 140      |
| Drafting legislation | 450      |
| Data analysis        | 520      |
| Financial rules      | 440      |

*All data are fictitious.***What was the total cost of the “Project management” course for Q1 to Q3 inclusive?**

- A** €3,080
- B** €5,880
- C** €4,200
- D** €5,320

**Question 26****Price of an ergonomic office chair in four markets (fictitious prices and rates)**

| Currency        | Code | Units per €1 | Local price |
|-----------------|------|--------------|-------------|
| Japanese yen    | JPY  | 153.9        | 69,950      |
| Swiss franc     | CHF  | 0.97         | 437         |
| Icelandic króna | ISK  | 146.6        | 52,120      |
| US dollar       | USD  | 1.08         | 406         |

*All data are fictitious.***In euros, approximately how much more does an ergonomic office chair cost in the most expensive of the four markets than in the cheapest?**

- A** €79
- B** €20
- C** €148
- D** €99



**Question 27**

Cases closed per month by three teams

| Team       | Jan | Feb | Mar | Apr | May | Jun |
|------------|-----|-----|-----|-----|-----|-----|
| Team North | 46  | 38  | 55  | 21  | 50  | 46  |
| Team East  | 52  | 34  | 48  | 26  | 60  | —   |
| Team West  | 62  | 21  | 61  | 49  | 58  | 27  |

— = figure not yet available

*All data are fictitious.*

How many cases would Team East need to close in June for its average over January to June to be exactly 50 cases per month?

- A** 80
- B** 30
- C** 44
- D** 50

**Question 28**

Staff numbers and average monthly salary by unit

| Unit   | Staff | Average monthly salary (€) |
|--------|-------|----------------------------|
| Unit B | 44    | 9,100                      |
| Unit E | 29    | 8,700                      |
| Unit A | 16    | 5,800                      |
| Unit C | 18    | 7,800                      |

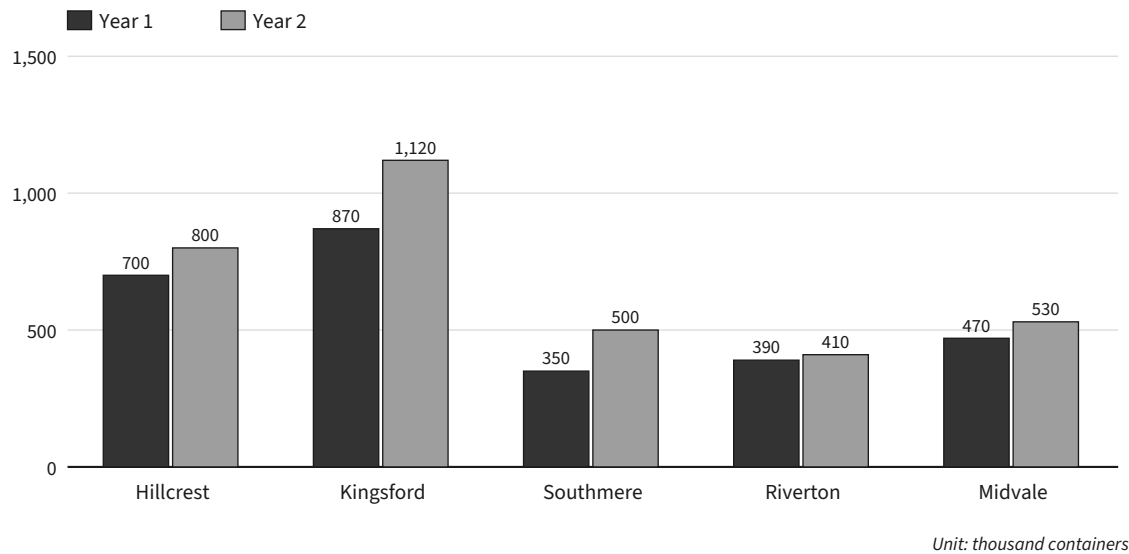
*All data are fictitious.*

What is the average monthly salary across all staff in the four units combined (to the nearest euro)?

- A** €7,450
- B** €8,941
- C** €7,850
- D** €8,279

Question 29

Containers handled at five ports (thousand units)



All data are fictitious.

Which of the following ports recorded the largest percentage increase in containers handled between Year 1 and Year 2?

- A Southmere
- B Riverton
- C Kingsford
- D Hillcrest

Question 30

Grant applications received and approval rate

| Country  | Applications | Approval rate |
|----------|--------------|---------------|
| Slovenia | 600          | 20%           |
| Estonia  | 1,900        | 18%           |
| France   | 900          | 15%           |
| Slovakia | 1,100        | 12%           |

All data are fictitious.

If the number of applications stayed the same, how many additional applications from Estonia would need to be approved for its approval rate to reach 45%?

- A 641
- B 270
- C 513
- D 855

**Question 31**

Rail routes: distance and average train speed

| Route                  | Distance (km) | Average speed (km/h) |
|------------------------|---------------|----------------------|
| Hillcrest–Fairhaven    | 114           | 140                  |
| Brookhaven–Stonebridge | 316           | 140                  |
| Southmere–Westfield    | 153           | 110                  |
| Lakeside–Kingsford     | 273           | 110                  |

All data are fictitious.

If the average speed on the Southmere–Westfield route rose to 160 km/h, approximately how many minutes would the journey save?

- A 13 minutes
- B 57 minutes
- C 38 minutes
- D 26 minutes

**Question 32**

Table 1 — Participants per training course

| Course               | Q1 | Q2 | Q3 | Q4 |
|----------------------|----|----|----|----|
| Project management   | 21 | 24 | 28 | 7  |
| Leadership basics    | 27 | 10 | 28 | 7  |
| Public speaking      | 18 | 11 | 7  | 14 |
| Drafting legislation | 15 | 23 | 14 | 25 |

Table 2 — Cost per participant (€)

| Course               | Cost (€) |
|----------------------|----------|
| Project management   | 570      |
| Leadership basics    | 450      |
| Public speaking      | 130      |
| Drafting legislation | 250      |

All data are fictitious.

What was the total cost of the “Leadership basics” course for Q1 to Q2 inclusive?

- A €13,050
- B €16,650
- C €32,400
- D €29,250

### Question 33

Rail routes: distance and average train speed

| Route                 | Distance (km) | Average speed (km/h) |
|-----------------------|---------------|----------------------|
| Fairhaven–Riverton    | 196           | 100                  |
| Midvale–Southmere     | 299           | 90                   |
| Stonebridge–Hillcrest | 371           | 100                  |
| Westfield–Lakeside    | 197           | 100                  |

All data are fictitious.

If the average speed on the Midvale–Southmere route rose to 110 km/h, approximately how many minutes would the journey save?

- A 36 minutes
- B 54 minutes
- C 18 minutes
- D 44 minutes

### Question 34

Index of average house prices (Year 1 = 100)

| Country   | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|-----------|--------|--------|--------|--------|--------|
| Belgium   | 100.0  | 105.2  | 111.2  | 113.9  | 123.7  |
| Lithuania | 100.0  | 102.2  | 103.4  | 103.8  | 107.5  |
| Estonia   | 100.0  | 108.3  | 117.5  | 126.3  | 133.6  |
| Austria   | 100.0  | 107.7  | 112.8  | 115.9  | 116.0  |

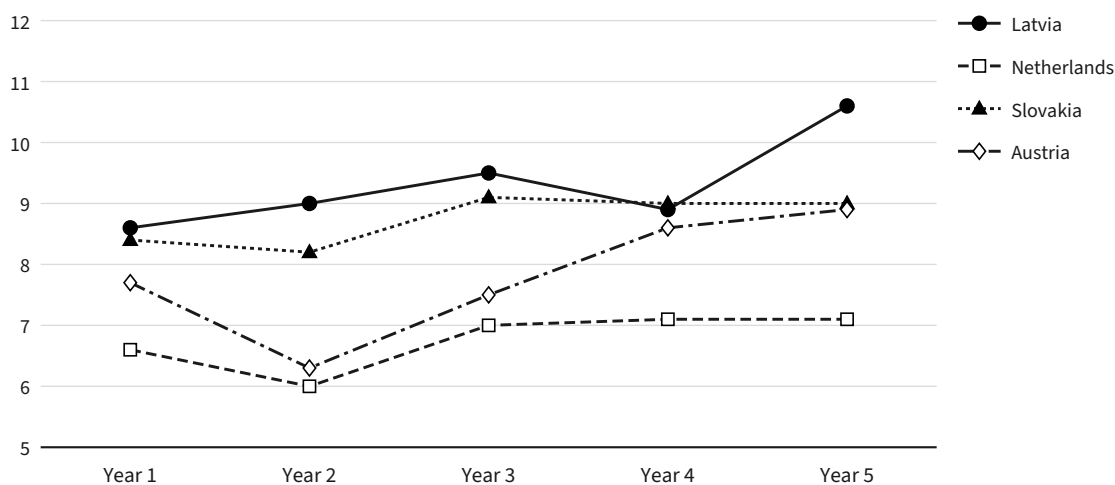
All data are fictitious.

By approximately what percentage did average house prices in Belgium rise between Year 3 and Year 5?

- A 11.2%
- B 12.5%
- C 10.1%
- D 23.7%

**Question 35****Unemployment rate (% of labour force)**

Unit: %



All data are fictitious.

**By approximately what percentage did the unemployment rate in Latvia rise between Year 4 and Year 5?**

- A** 6.7%
- B** 3.4%
- C** 16.0%
- D** 19.1%

**Question 36**

Cases closed per month by three teams

| Team       | Jan | Feb | Mar | Apr | May | Jun |
|------------|-----|-----|-----|-----|-----|-----|
| Team East  | 44  | 52  | 51  | 45  | 28  | 31  |
| Team North | 40  | 45  | 21  | 46  | 32  | —   |
| Team West  | 49  | 53  | 64  | 28  | 28  | 53  |

— = figure not yet available

*All data are fictitious.*

How many cases would Team North need to close in June for its average over January to June to be exactly 40 cases per month?

- A 40
- B 37
- C 16
- D 56

**Question 37**

Population and public spending on culture in five regions of one country

| Region          | Population (million) | Spending (€ million) |
|-----------------|----------------------|----------------------|
| Central region  | 4.24                 | 679.6                |
| Northern region | 6.50                 | 1,637.2              |
| Southern region | 3.08                 | 679.0                |
| Western region  | 3.49                 | 723.0                |
| Alpine region   | 3.52                 | 868.1                |

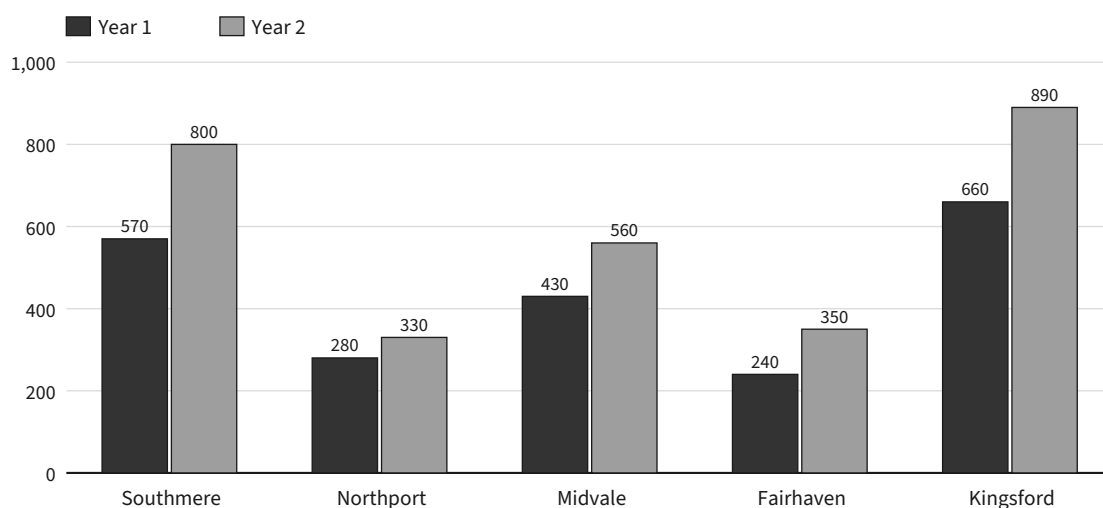
*All data are fictitious.*

Which of the following regions had the lowest spending on culture per inhabitant?

- A Southern region
- B Northern region
- C Central region
- D Western region

**Question 38**

Containers handled at five ports (thousand units)



Unit: thousand containers

All data are fictitious.

Which of the following ports recorded the largest percentage increase in containers handled between Year 1 and Year 2?

- A Fairhaven
- B Northport
- C Southmere
- D Kingsford

**Question 39**

Staff numbers and average monthly salary by unit

| Unit   | Staff | Average monthly salary (€) |
|--------|-------|----------------------------|
| Unit C | 27    | 7,200                      |
| Unit E | 47    | 8,800                      |
| Unit D | 24    | 4,900                      |
| Unit B | 9     | 7,100                      |

All data are fictitious.

What is the average monthly salary across all staff in the four units combined (to the nearest euro)?

- A €6,850
- B €7,000
- C €6,265
- D €7,379

### Question 40

Price of a standard office laptop in four markets (fictitious prices and rates)

| Currency        | Code | Units per €1 | Local price |
|-----------------|------|--------------|-------------|
| Canadian dollar | CAD  | 1.42         | 709         |
| Pound sterling  | GBP  | 0.83         | 403         |
| US dollar       | USD  | 1.08         | 680         |
| Icelandic króna | ISK  | 153.1        | 100,810     |

All data are fictitious.

In euros, approximately how much more does a standard office laptop cost in the most expensive of the four markets than in the cheapest?

- A €202
- B €144
- C €173
- D €259

### Question 41

Index of average house prices (Year 1 = 100)

| Country | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|---------|--------|--------|--------|--------|--------|
| Latvia  | 100.0  | 105.6  | 106.0  | 111.5  | 118.3  |
| France  | 100.0  | 108.4  | 111.7  | 115.3  | 119.4  |
| Belgium | 100.0  | 106.8  | 110.4  | 115.1  | 121.2  |
| Denmark | 100.0  | 101.3  | 104.8  | 111.0  | 115.0  |

All data are fictitious.

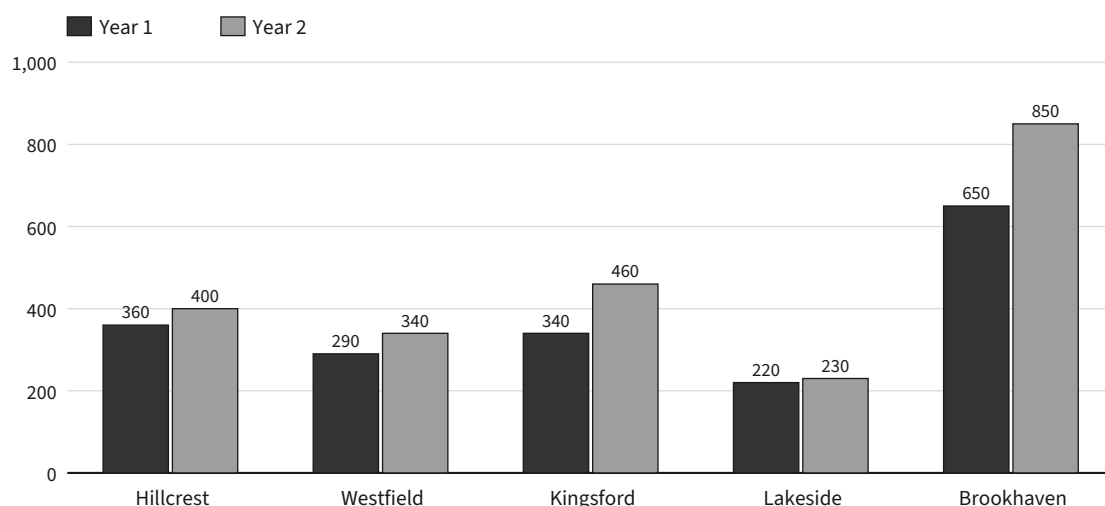
By approximately what percentage did average house prices in Latvia rise between Year 3 and Year 5?

- A 18.3%
- B 10.4%
- C 6.0%
- D 11.6%



**Question 42**

Containers handled at five ports (thousand units)



Unit: thousand containers

All data are fictitious.

Which of the following ports recorded the largest percentage increase in containers handled between Year 1 and Year 2?

- A Brookhaven
- B Kingsford
- C Hillcrest
- D Westfield

**Question 43**

Price of an ergonomic office chair in four markets (fictitious prices and rates)

| Currency       | Code | Units per €1 | Local price |
|----------------|------|--------------|-------------|
| Japanese yen   | JPY  | 164.7        | 90,720      |
| Swiss franc    | CHF  | 0.92         | 646         |
| US dollar      | USD  | 1.10         | 576         |
| Pound sterling | GBP  | 0.86         | 545         |

All data are fictitious.

In euros, approximately how much more does an ergonomic office chair cost in the most expensive of the four markets than in the cheapest?

- A €68
- B €151
- C €110
- D €179

**Question 44**

Cases closed per month by three teams

| Team       | Jan | Feb | Mar | Apr | May | Jun |
|------------|-----|-----|-----|-----|-----|-----|
| Team West  | 52  | 32  | 62  | 31  | 43  | 60  |
| Team East  | 33  | 50  | 24  | 27  | 44  | —   |
| Team North | 55  | 23  | 38  | 23  | 49  | 60  |

— = figure not yet available

*All data are fictitious.*

How many cases would Team East need to close in June for its average over January to June to be exactly 40 cases per month?

- A 62
- B 40
- C 22
- D 36

**Question 45**

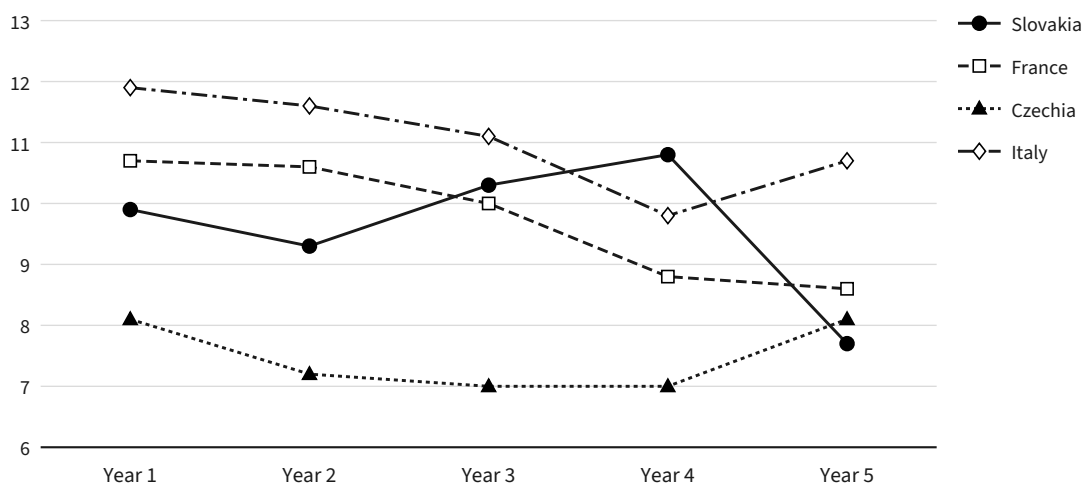
Population and public spending on culture in five regions of one country

| Region          | Population (million) | Spending (€ million) |
|-----------------|----------------------|----------------------|
| Alpine region   | 5.16                 | 707.3                |
| Coastal region  | 0.83                 | 89.2                 |
| Western region  | 3.15                 | 287.7                |
| Central region  | 2.60                 | 638.1                |
| Northern region | 1.40                 | 315.3                |

*All data are fictitious.*

Which of the following regions had the highest spending on culture per inhabitant?

- A Alpine region
- B Central region
- C Western region
- D Coastal region

**Question 46****Unemployment rate (% of labour force)***Unit: %**All data are fictitious.***By approximately what percentage did the unemployment rate in Slovakia fall between Year 2 and Year 5?**

- A** 20.8%
- B** 6.5%
- C** 10.8%
- D** 17.2%

**Question 47****Table 1 — Participants per training course**

| Course               | Q1 | Q2 | Q3 | Q4 |
|----------------------|----|----|----|----|
| Public speaking      | 9  | 6  | 15 | 24 |
| Data analysis        | 12 | 30 | 7  | 12 |
| Financial rules      | 18 | 10 | 13 | 30 |
| Drafting legislation | 23 | 26 | 16 | 24 |

**Table 2 — Cost per participant (€)**

| Course               | Cost (€) |
|----------------------|----------|
| Public speaking      | 290      |
| Data analysis        | 160      |
| Financial rules      | 550      |
| Drafting legislation | 250      |

*All data are fictitious.*

**What was the total cost of the “Data analysis” course for Q2 to Q3 inclusive?**

- A** €5,920
- B** €3,680
- C** €9,760
- D** €7,840

**Question 48****Rail routes: distance and average train speed**

| Route                 | Distance (km) | Average speed (km/h) |
|-----------------------|---------------|----------------------|
| Stonebridge–Fairhaven | 317           | 120                  |
| Westfield–Kingsford   | 258           | 110                  |
| Riverton–Midvale      | 328           | 80                   |
| Brookhaven–Northport  | 193           | 140                  |

*All data are fictitious.*

**If the average speed on the Stonebridge–Fairhaven route rose to 150 km/h, approximately how many minutes would the journey save?**

- A** 127 minutes
- B** 48 minutes
- C** 40 minutes
- D** 32 minutes

**Question 49**

Staff numbers and average monthly salary by unit

| Unit   | Staff | Average monthly salary (€) |
|--------|-------|----------------------------|
| Unit D | 38    | 4,000                      |
| Unit E | 26    | 5,500                      |
| Unit C | 11    | 7,600                      |
| Unit A | 20    | 8,900                      |

*All data are fictitious.*

What is the average monthly salary across all staff in the four units combined (to the nearest euro)?

- A** €7,098
- B** €5,859
- C** €6,500
- D** €4,609

**Question 50**

Grant applications received and approval rate

| Country  | Applications | Approval rate |
|----------|--------------|---------------|
| Romania  | 2,400        | 24%           |
| Czechia  | 900          | 24%           |
| Slovakia | 2,300        | 20%           |
| Hungary  | 1,000        | 20%           |

*All data are fictitious.*

If the number of applications stayed the same, how many additional applications from Slovakia would need to be approved for its approval rate to reach 45%?

- A** 250
- B** 719
- C** 575
- D** 1,035

**Question 51****Rail routes: distance and average train speed**

| Route                | Distance (km) | Average speed (km/h) |
|----------------------|---------------|----------------------|
| Midvale–Eastbay      | 118           | 90                   |
| Kingsford–Lakeside   | 292           | 120                  |
| Brookhaven–Southmere | 68            | 160                  |
| Westfield–Fairhaven  | 124           | 140                  |

*All data are fictitious.*

If the average speed on the Kingsford–Lakeside route rose to 140 km/h, approximately how many minutes would the journey save?

- A 21 minutes
- B 10 minutes
- C 24 minutes
- D 31 minutes

**Question 52****Price of an annual museum pass in four markets (fictitious prices and rates)**

| Currency        | Code | Units per €1 | Local price |
|-----------------|------|--------------|-------------|
| Canadian dollar | CAD  | 1.44         | 676         |
| Japanese yen    | JPY  | 158.4        | 96,340      |
| Swiss franc     | CHF  | 0.95         | 476         |
| Pound sterling  | GBP  | 0.82         | 420         |

*All data are fictitious.*

In euros, approximately how much more does an annual museum pass cost in the most expensive of the four markets than in the cheapest?

- A €107
- B €43
- C €139
- D €32

**Question 53**

Population and public spending on culture in five regions of one country

| Region          | Population (million) | Spending (€ million) |
|-----------------|----------------------|----------------------|
| Island region   | 1.26                 | 189.9                |
| Alpine region   | 1.15                 | 293.2                |
| Northern region | 5.68                 | 1,244.0              |
| Western region  | 6.11                 | 974.3                |
| Southern region | 5.42                 | 548.6                |

*All data are fictitious.*

Which of the following regions had the lowest spending on culture per inhabitant?

- A** Island region
- B** Southern region
- C** Alpine region
- D** Northern region

**Question 54**

Staff numbers and average monthly salary by unit

| Unit   | Staff | Average monthly salary (€) |
|--------|-------|----------------------------|
| Unit E | 13    | 8,200                      |
| Unit A | 16    | 8,500                      |
| Unit B | 44    | 4,700                      |
| Unit D | 17    | 7,300                      |

*All data are fictitious.*

What is the average monthly salary across all staff in the four units combined (to the nearest euro)?

- A** €6,372
- B** €6,600
- C** €7,972
- D** €7,175

### Question 55

Index of average house prices (Year 1 = 100)

| Country  | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|----------|--------|--------|--------|--------|--------|
| Denmark  | 100.0  | 101.3  | 102.9  | 108.0  | 116.9  |
| Czechia  | 100.0  | 107.4  | 110.5  | 116.8  | 125.7  |
| Portugal | 100.0  | 107.9  | 112.3  | 113.6  | 114.0  |
| Slovakia | 100.0  | 105.8  | 112.2  | 119.4  | 129.9  |

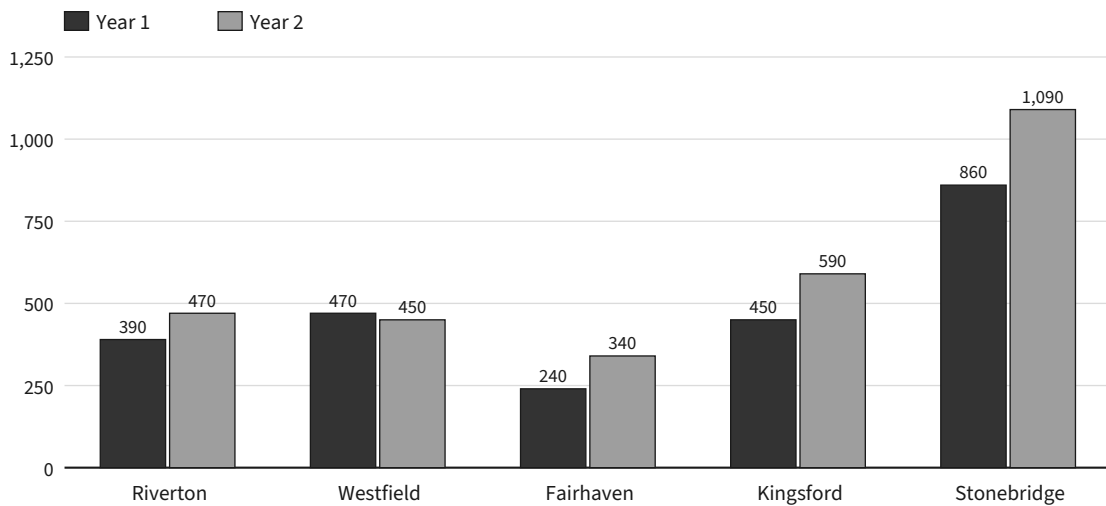
All data are fictitious.

By approximately what percentage did average house prices in Denmark rise between Year 3 and Year 5?

- A 12.0%
- B 13.6%
- C 16.9%
- D 15.4%

### Question 56

Containers handled at five ports (thousand units)



Unit: thousand containers

All data are fictitious.

Which of the following ports recorded the largest percentage increase in containers handled between Year 1 and Year 2?

- A Westfield
- B Kingsford
- C Stonebridge
- D Fairhaven



**Question 57**

Cases closed per month by three teams

| Team       | Jan | Feb | Mar | Apr | May | Jun |
|------------|-----|-----|-----|-----|-----|-----|
| Team East  | 50  | 20  | 41  | 24  | 38  | —   |
| Team South | 63  | 36  | 37  | 19  | 28  | 33  |
| Team North | 21  | 35  | 38  | 29  | 25  | 36  |

— = figure not yet available

*All data are fictitious.*

How many cases would Team East need to close in June for its average over January to June to be exactly 45 cases per month?

- A 35
- B 45
- C 52
- D 97

**Question 58**

Table 1 — Participants per training course

| Course               | Q1 | Q2 | Q3 | Q4 |
|----------------------|----|----|----|----|
| Negotiation skills   | 27 | 15 | 10 | 12 |
| Drafting legislation | 19 | 30 | 24 | 30 |
| Financial rules      | 28 | 23 | 24 | 6  |
| Public speaking      | 6  | 30 | 26 | 23 |

Table 2 — Cost per participant (€)

| Course               | Cost (€) |
|----------------------|----------|
| Negotiation skills   | 500      |
| Drafting legislation | 140      |
| Financial rules      | 410      |
| Public speaking      | 150      |

*All data are fictitious.*

What was the total cost of the “Negotiation skills” course for Q2 to Q4 inclusive?

- A €42,000
- B €32,000
- C €18,500
- D €13,500

**Question 59****Table 1 — Participants per training course**

| Course             | Q1 | Q2 | Q3 | Q4 |
|--------------------|----|----|----|----|
| Negotiation skills | 25 | 11 | 9  | 25 |
| Financial rules    | 12 | 10 | 29 | 17 |
| Data analysis      | 25 | 16 | 15 | 5  |
| Leadership basics  | 24 | 20 | 29 | 28 |

**Table 2 — Cost per participant (€)**

| Course             | Cost (€) |
|--------------------|----------|
| Negotiation skills | 250      |
| Financial rules    | 580      |
| Data analysis      | 260      |
| Leadership basics  | 250      |

*All data are fictitious.*

**What was the total cost of the “Financial rules” course for Q2 to Q4 inclusive?**

- A** €14,560
- B** €39,440
- C** €15,660
- D** €32,480

**Question 60****Index of average house prices (Year 1 = 100)**

| Country  | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|----------|--------|--------|--------|--------|--------|
| Latvia   | 100.0  | 101.8  | 107.9  | 115.1  | 116.1  |
| Portugal | 100.0  | 102.9  | 109.7  | 111.3  | 113.5  |
| Slovakia | 100.0  | 107.1  | 109.1  | 111.4  | 112.5  |
| France   | 100.0  | 105.1  | 108.0  | 117.0  | 127.3  |

*All data are fictitious.*

**By approximately what percentage did average house prices in France rise between Year 3 and Year 5?**

- A** 15.2%
- B** 17.9%
- C** 19.3%
- D** 27.3%

**Question 61**

Price of a conference registration in four markets (fictitious prices and rates)

| Currency        | Code | Units per €1 | Local price |
|-----------------|------|--------------|-------------|
| Swiss franc     | CHF  | 0.96         | 690         |
| Icelandic króna | ISK  | 148.6        | 107,260     |
| Pound sterling  | GBP  | 0.84         | 609         |
| Japanese yen    | JPY  | 158.2        | 95,660      |

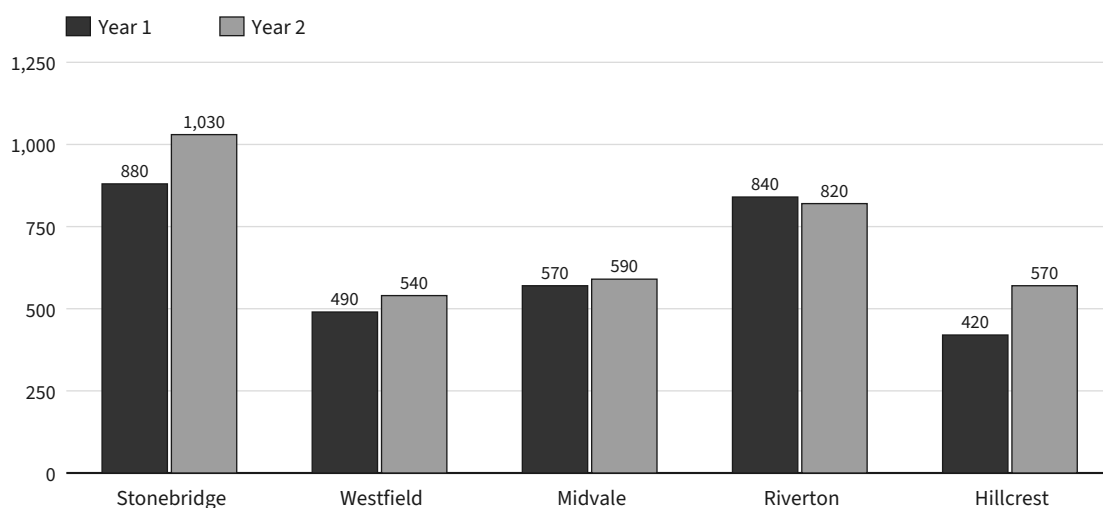
*All data are fictitious.*

In euros, approximately how much more does a conference registration cost in the most expensive of the four markets than in the cheapest?

- A €120
- B €60
- C €180
- D €142

**Question 62**

Containers handled at five ports (thousand units)



*Unit: thousand containers*

*All data are fictitious.*

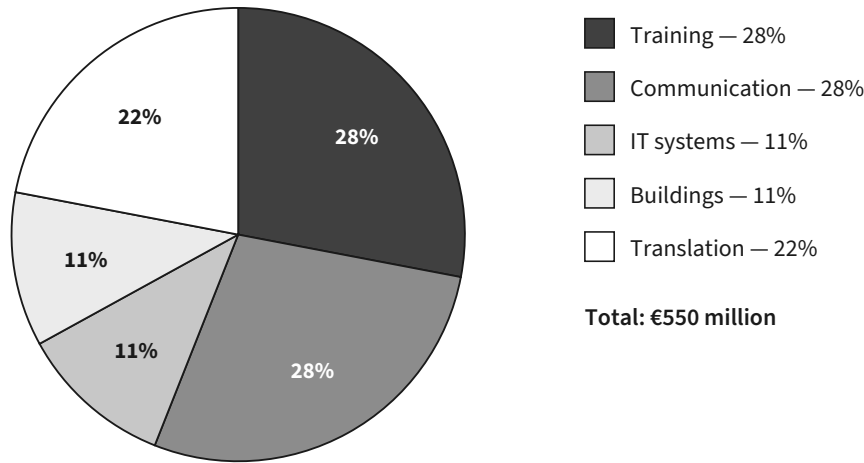
Which of the following ports recorded the largest percentage increase in containers handled between Year 1 and Year 2?

- A Midvale
- B Riverton
- C Hillcrest
- D Stonebridge

Advanced

Question 63

Breakdown of an agency’s annual budget (total €550 million)



All data are fictitious.

Next year the Communication budget is to rise by 10% while all other items stay unchanged. By how much will the Communication budget then exceed the Buildings budget?

- A €108.9 million
- B €93.5 million
- C €148.5 million
- D €102.9 million

Question 64

Plastic collected for recycling by four municipal schemes this year, and expected annual growth

| Municipality | Collected this year (tonnes) | Expected annual growth |
|--------------|------------------------------|------------------------|
| Midvale      | 6,370                        | 12%                    |
| Lakeside     | 4,390                        | 8%                     |
| Fairhaven    | 7,300                        | 8%                     |
| Westfield    | 2,070                        | 8%                     |

All data are fictitious.

If the amount collected in Westfield grows at its expected annual rate, compounded, approximately how many tonnes will be collected three years from now?

- A 2,414 t
- B 2,816 t
- C 2,608 t
- D 2,567 t

**Question 65**

Budget for this year and change compared with last year

| Programme                     | This year (€ thousand) | Change on last year |
|-------------------------------|------------------------|---------------------|
| Cultural heritage restoration | 2,064.0                | -20%                |
| Student mobility grants       | 4,233.6                | +8%                 |
| Youth employment schemes      | 2,600.0                | +30%                |
| Coastal flood defences        | 2,210.0                | -15%                |

*All data are fictitious.*

What was last year's budget for the "Coastal flood defences" programme?

- A €1,921.7k
- B €2,600.0k
- C €2,541.5k
- D €2,360.0k

**Question 66**

Budget for this year and change compared with last year

| Programme                | This year (€ thousand) | Change on last year |
|--------------------------|------------------------|---------------------|
| Coastal flood defences   | 2,976.0                | +20%                |
| Clean-bus procurement    | 1,955.0                | +15%                |
| Student mobility grants  | 2,499.2                | -12%                |
| Youth employment schemes | 528.0                  | -12%                |

*All data are fictitious.*

What was last year's budget for the "Student mobility grants" programme?

- A €2,619.2k
- B €2,231.4k
- C €2,840.0k
- D €2,499.2k

Question 67

Plastic collected for recycling by four municipal schemes this year, and expected annual growth

| Municipality | Collected this year (tonnes) | Expected annual growth |
|--------------|------------------------------|------------------------|
| Brookhaven   | 2,160                        | 8%                     |
| Eastbay      | 2,060                        | 15%                    |
| Fairhaven    | 7,870                        | 10%                    |
| Hillcrest    | 5,790                        | 12%                    |

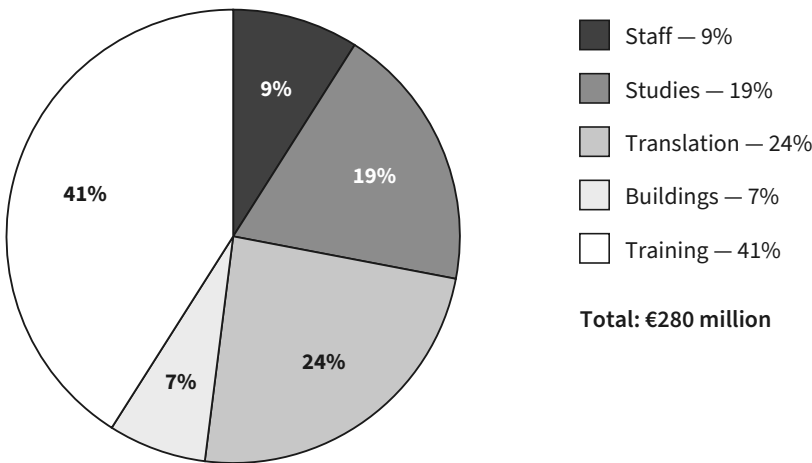
All data are fictitious.

If the amount collected in Eastbay grows at its expected annual rate, compounded, approximately how many tonnes will be collected three years from now?

- A 3,603 t
- B 3,133 t
- C 2,987 t
- D 2,724 t

Question 68

Breakdown of an agency’s annual budget (total €280 million)



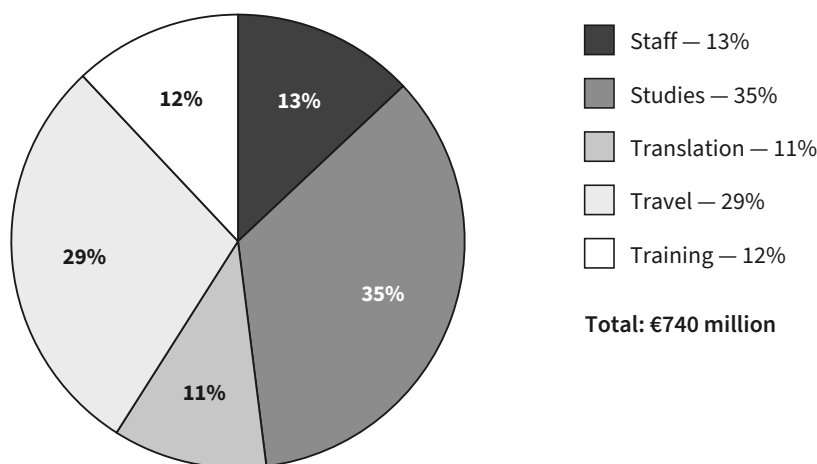
All data are fictitious.

Next year the Training budget is to rise by 20% while all other items stay unchanged. By how much will the Training budget then exceed the Staff budget?

- A €112.6 million
- B €89.6 million
- C €145.6 million
- D €107.5 million

**Question 69**

Breakdown of an agency's annual budget (total €740 million)



All data are fictitious.

Next year the Staff budget is to rise by 20% while all other items stay unchanged. By how much will the Staff budget then exceed the Translation budget?

- A €14.8 million
- B €34.0 million
- C €17.8 million
- D €19.2 million

**Question 70**

Plastic collected for recycling by four municipal schemes this year, and expected annual growth

| Municipality | Collected this year (tonnes) | Expected annual growth |
|--------------|------------------------------|------------------------|
| Lakeside     | 8,800                        | 8%                     |
| Brookhaven   | 1,410                        | 10%                    |
| Stonebridge  | 1,610                        | 10%                    |
| Eastbay      | 8,670                        | 20%                    |

All data are fictitious.

If the amount collected in Stonebridge grows at its expected annual rate, compounded, approximately how many tonnes will be collected three years from now?

- A 2,143 t
- B 2,093 t
- C 2,357 t
- D 1,948 t

**Question 71**

Budget for this year and change compared with last year

| Programme                     | This year (€ thousand) | Change on last year |
|-------------------------------|------------------------|---------------------|
| Clean-bus procurement         | 2,136.0                | +20%                |
| Student mobility grants       | 2,112.0                | -20%                |
| Coastal flood defences        | 3,606.4                | -8%                 |
| Cultural heritage restoration | 3,288.0                | +20%                |

*All data are fictitious.*

What was last year's budget for the "Coastal flood defences" programme?

- A €3,339.3k
- B €3,920.0k
- C €3,686.4k
- D €3,606.4k

**Question 72**

Plastic collected for recycling by four municipal schemes this year, and expected annual growth

| Municipality | Collected this year (tonnes) | Expected annual growth |
|--------------|------------------------------|------------------------|
| Hillcrest    | 2,950                        | 12%                    |
| Fairhaven    | 8,730                        | 20%                    |
| Eastbay      | 8,370                        | 10%                    |
| Brookhaven   | 8,020                        | 15%                    |

*All data are fictitious.*

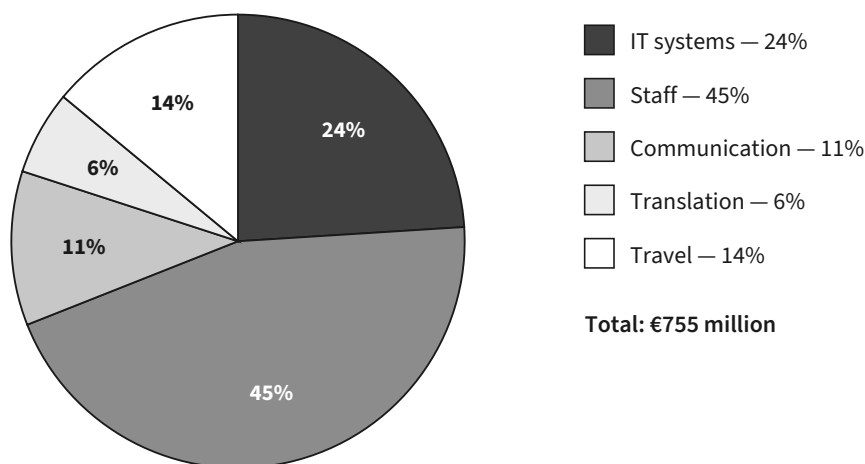
If the amount collected in Brookhaven grows at its expected annual rate, compounded, approximately how many tonnes will be collected three years from now?

- A 10,606 t
- B 12,197 t
- C 11,629 t
- D 14,027 t



**Question 73**

Breakdown of an agency's annual budget (total €755 million)



All data are fictitious.

Next year the IT systems budget is to rise by 10% while all other items stay unchanged. By how much will the IT systems budget then exceed the Communication budget?

- A €108.0 million
- B €173.7 million
- C €98.1 million
- D €116.3 million

**Question 74**

Budget for this year and change compared with last year

| Programme                     | This year (€ thousand) | Change on last year |
|-------------------------------|------------------------|---------------------|
| Coastal flood defences        | 5,076.0                | +8%                 |
| Youth employment schemes      | 2,570.4                | +8%                 |
| Cultural heritage restoration | 3,408.0                | +20%                |
| Rural broadband support       | 1,742.0                | +30%                |

All data are fictitious.

What was last year's budget for the "Coastal flood defences" programme?

- A €4,700.0k
- B €4,996.0k
- C €5,076.0k
- D €5,517.4k

**Question 75****Budget for this year and change compared with last year**

| Programme                     | This year (€ thousand) | Change on last year |
|-------------------------------|------------------------|---------------------|
| Cultural heritage restoration | 4,816.8                | +8%                 |
| Rural broadband support       | 3,977.6                | -12%                |
| Clean-bus procurement         | 5,219.2                | +12%                |
| Coastal flood defences        | 4,009.6                | +12%                |

*All data are fictitious.***What was last year's budget for the "Rural broadband support" programme?**

- A** €3,977.6k
- B** €4,097.6k
- C** €3,551.4k
- D** €4,520.0k

# Practice Set — Abstract Reasoning

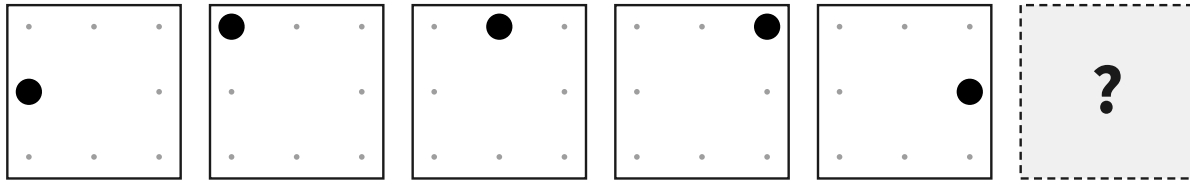
Each series of five figures follows one or more rules. Choose the figure (A–E) that comes next.  
Suggested pace: about 1 minute per question.

75 questions · answers and worked solutions in Part IV.

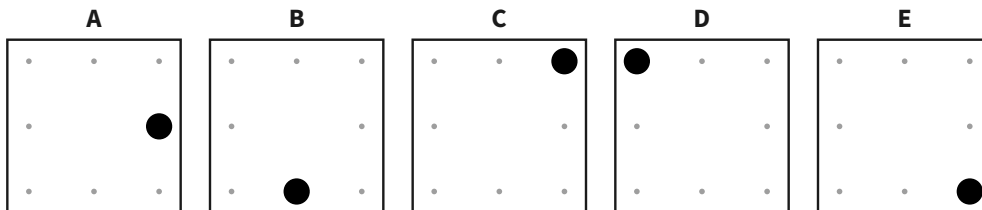
## Foundation

### Question 1

Which figure comes next in the series?

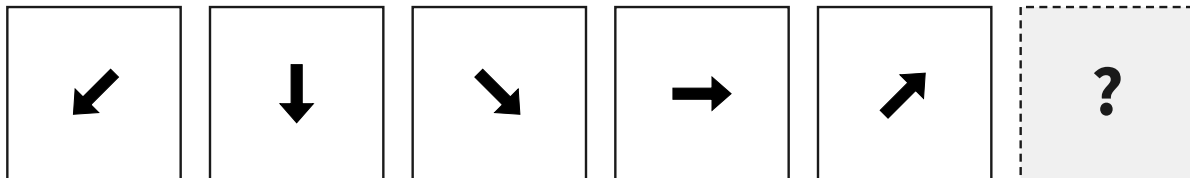


Answer options

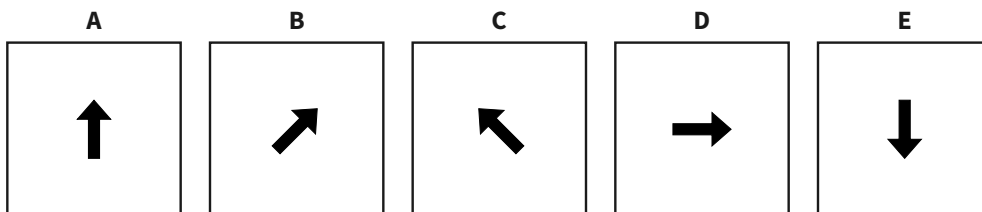


### Question 2

Which figure comes next in the series?

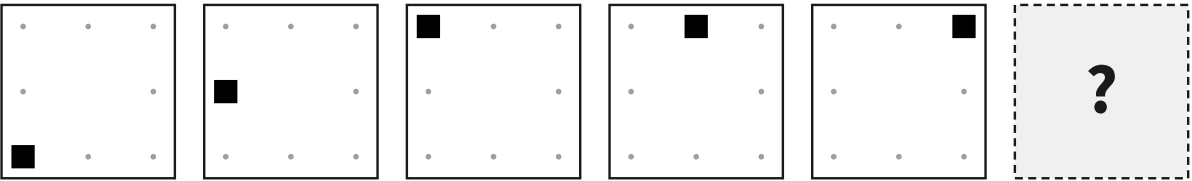


Answer options

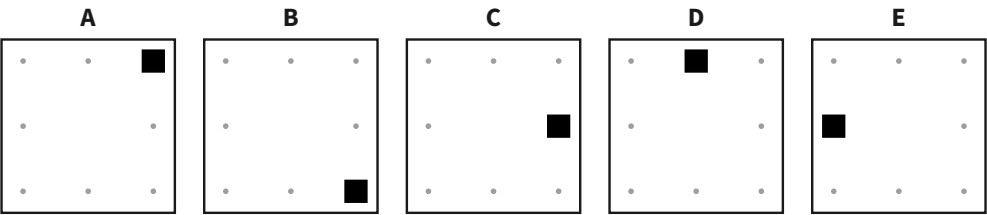


Question 3

Which figure comes next in the series?

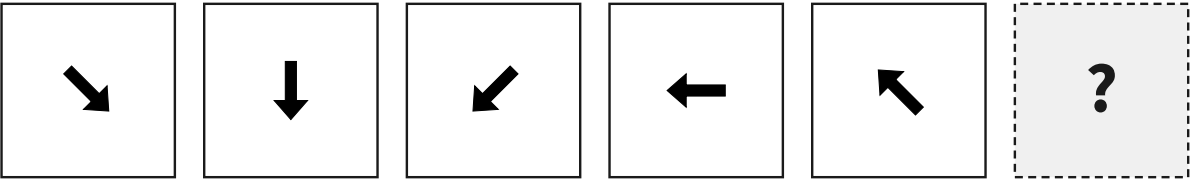


Answer options

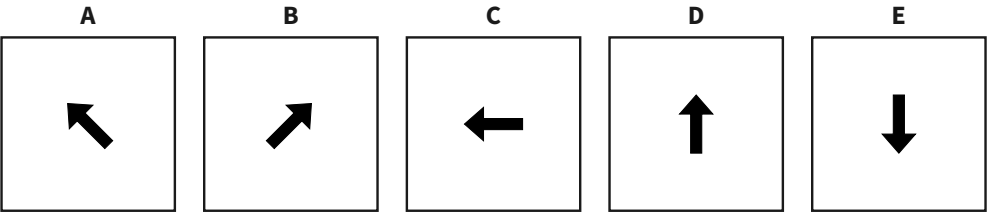


Question 4

Which figure comes next in the series?

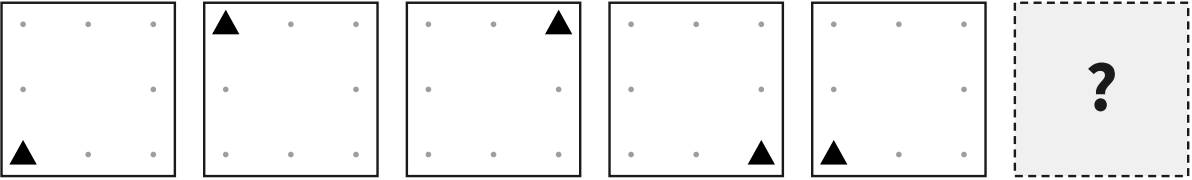


Answer options

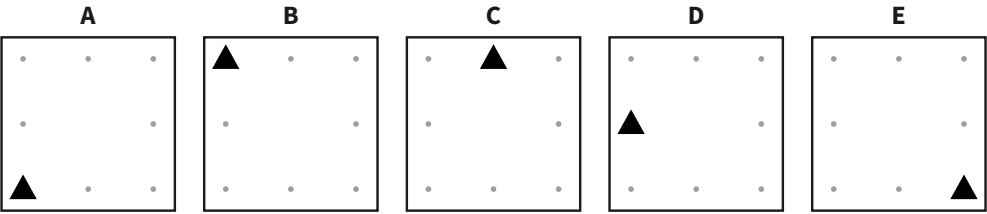


Question 5

Which figure comes next in the series?

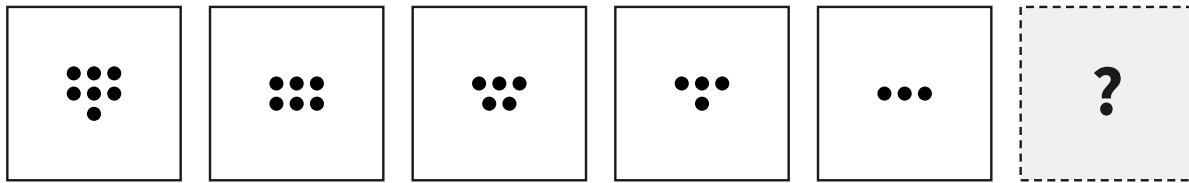


Answer options

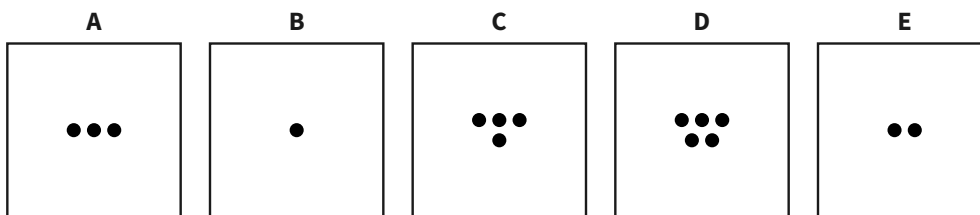


### Question 6

Which figure comes next in the series?

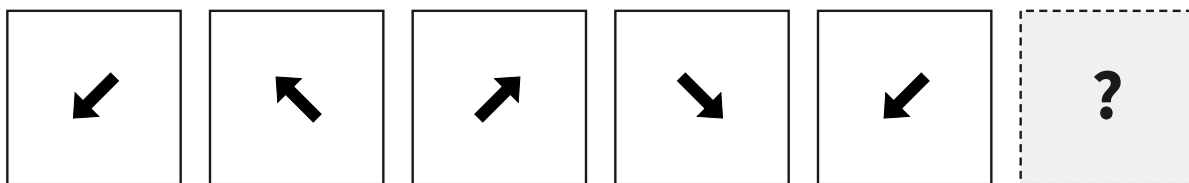


Answer options

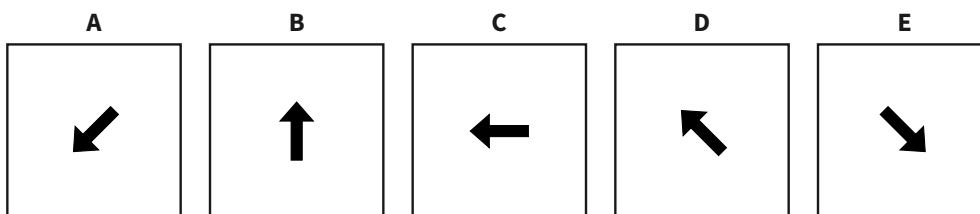


### Question 7

Which figure comes next in the series?

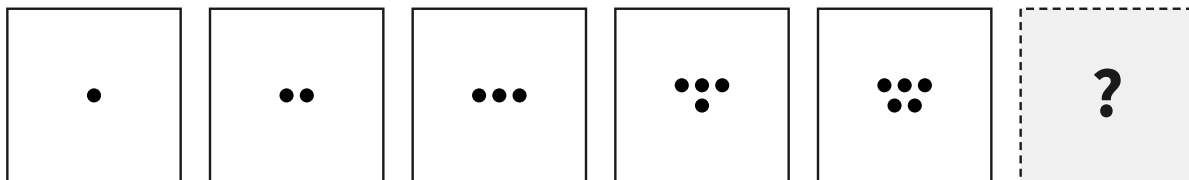


Answer options

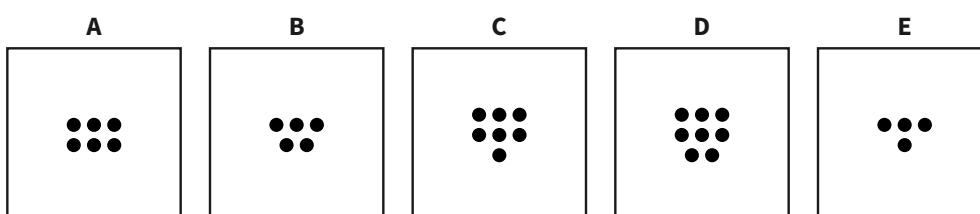


### Question 8

Which figure comes next in the series?

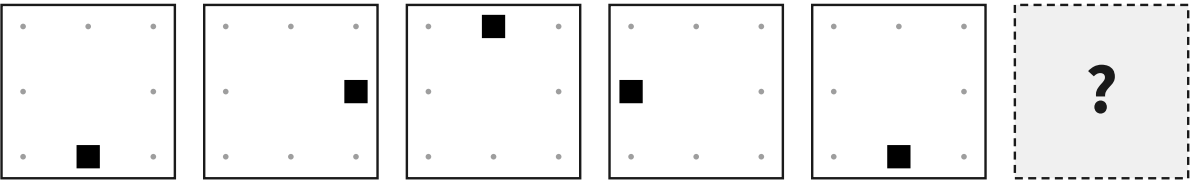


Answer options

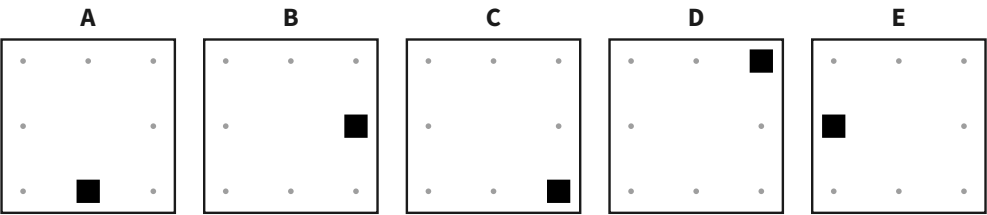


Question 9

Which figure comes next in the series?

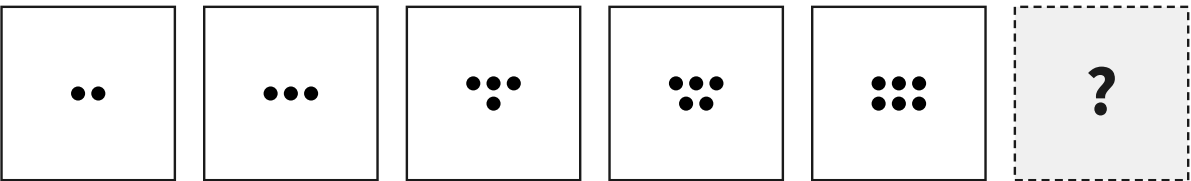


Answer options

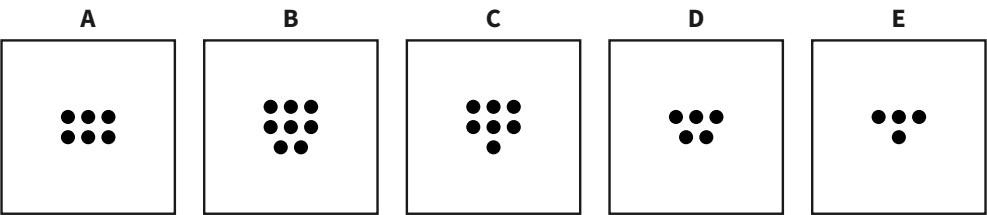


Question 10

Which figure comes next in the series?

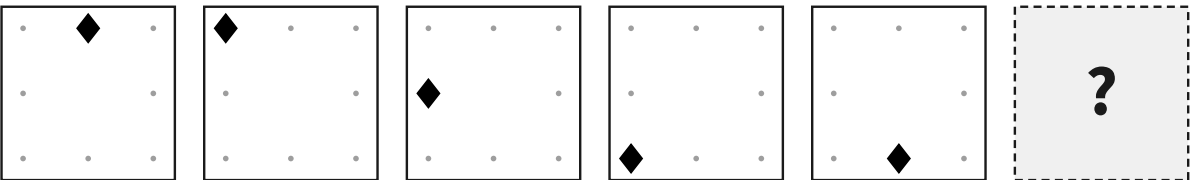


Answer options

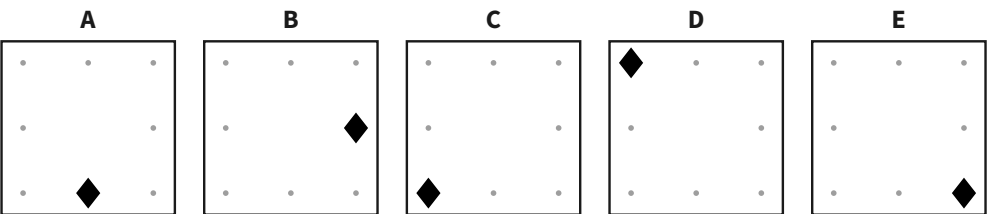


Question 11

Which figure comes next in the series?

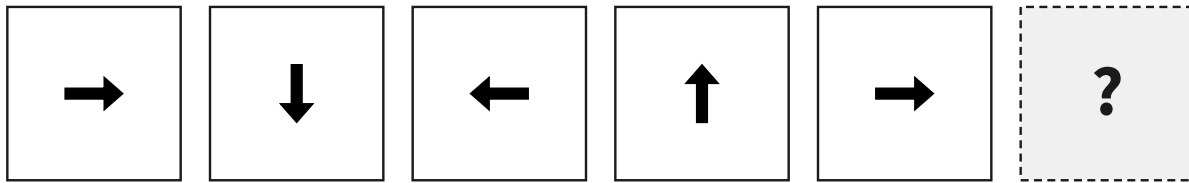


Answer options

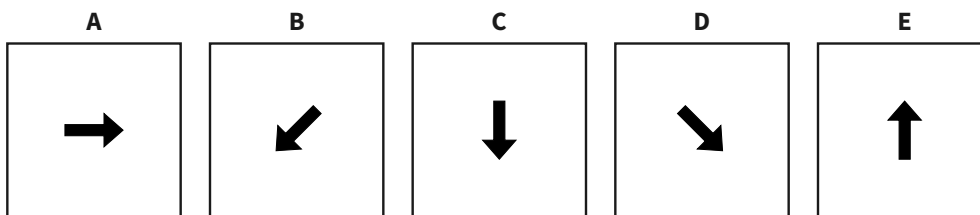


### Question 12

Which figure comes next in the series?

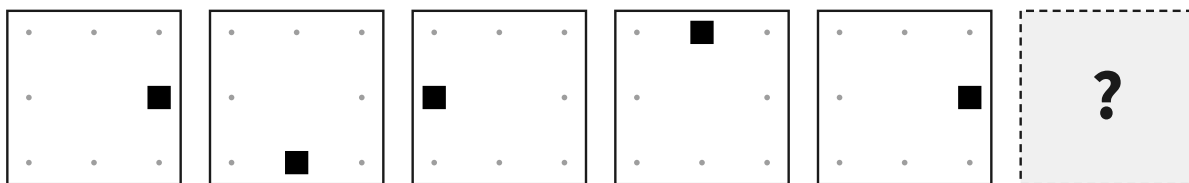


Answer options

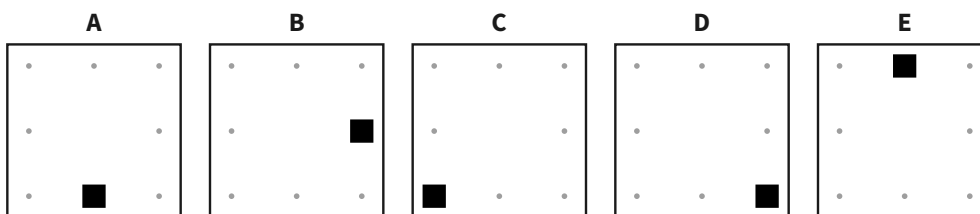


### Question 13

Which figure comes next in the series?

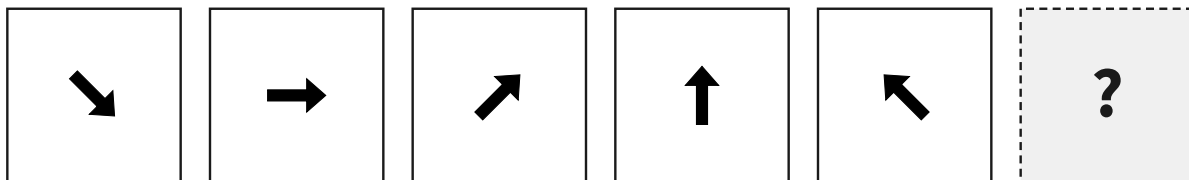


Answer options

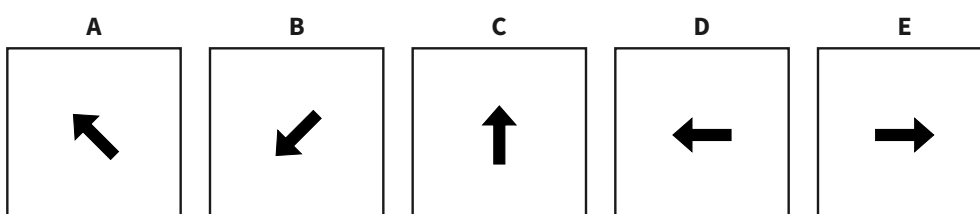


### Question 14

Which figure comes next in the series?

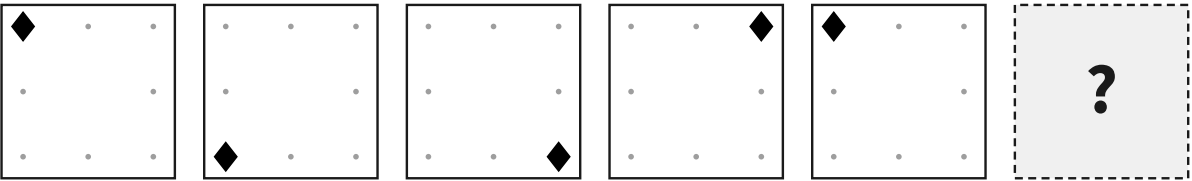


Answer options

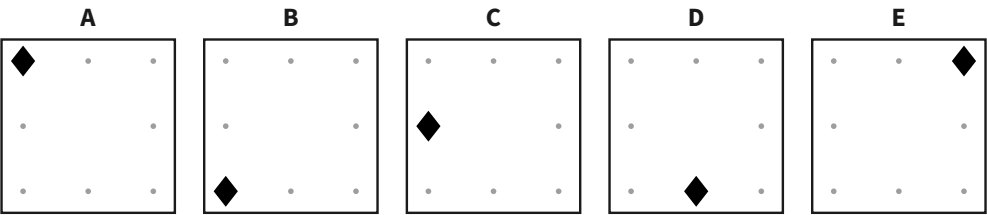


Question 15

Which figure comes next in the series?

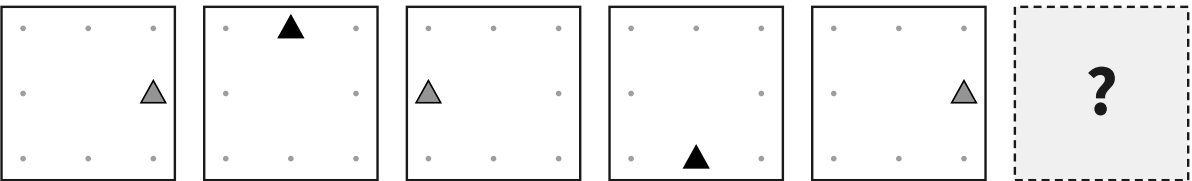


Answer options

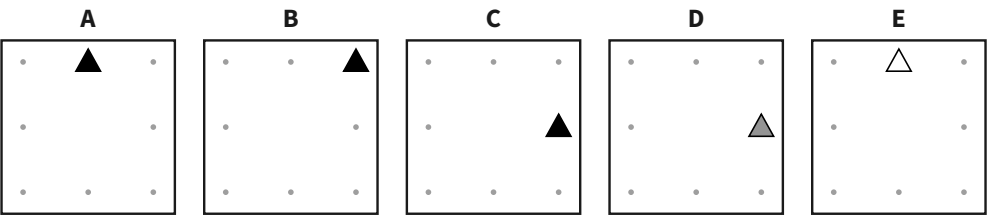


Question 16

Which figure comes next in the series?

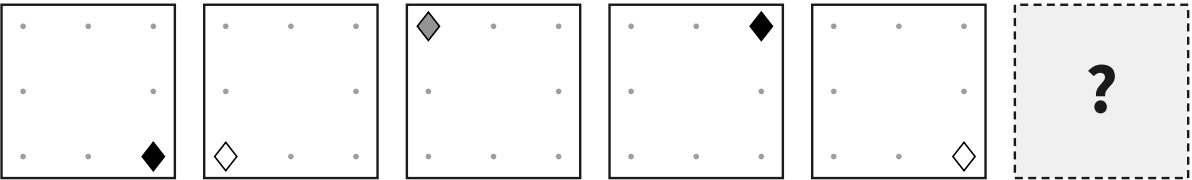


Answer options

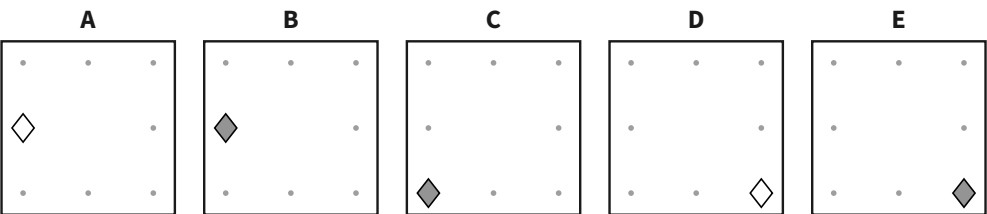


Question 17

Which figure comes next in the series?



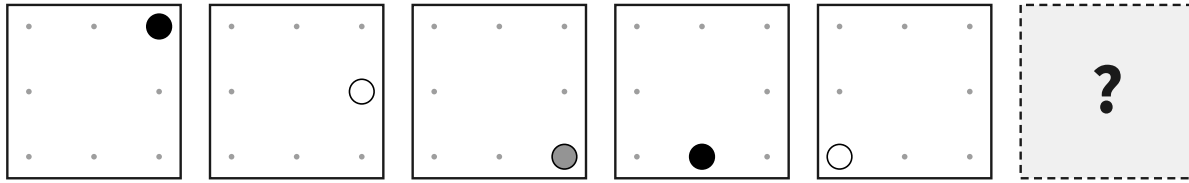
Answer options



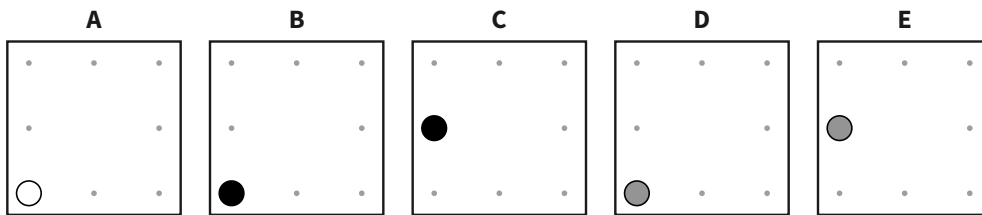


### Question 18

Which figure comes next in the series?

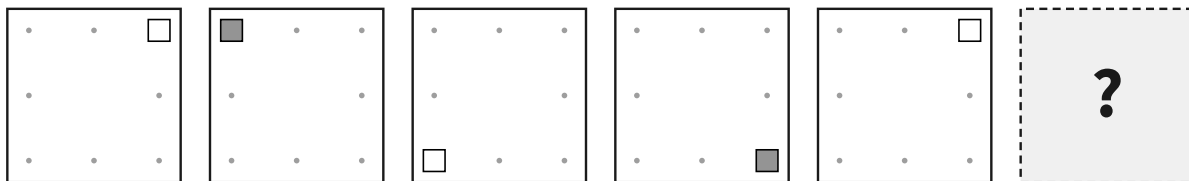


Answer options

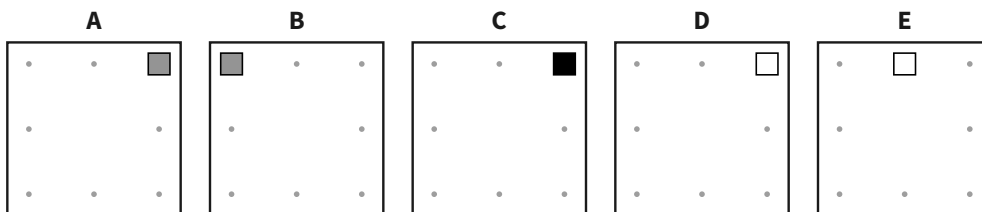


### Question 19

Which figure comes next in the series?

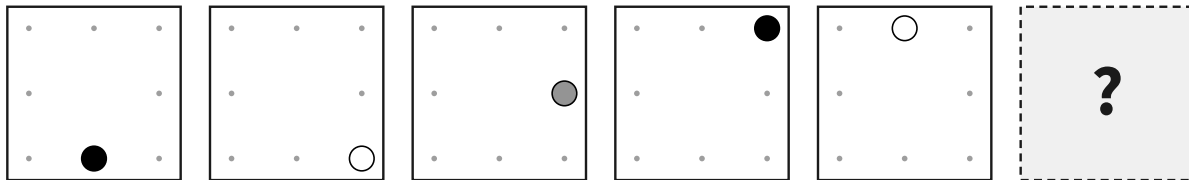


Answer options

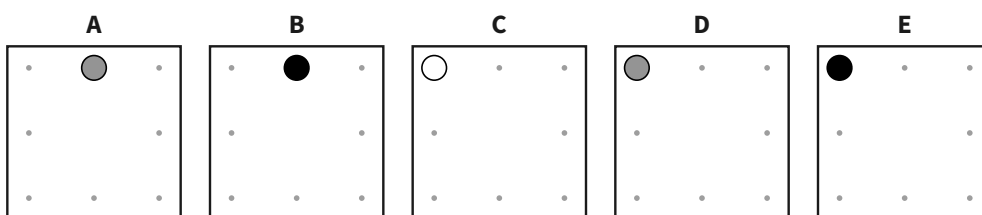


### Question 20

Which figure comes next in the series?



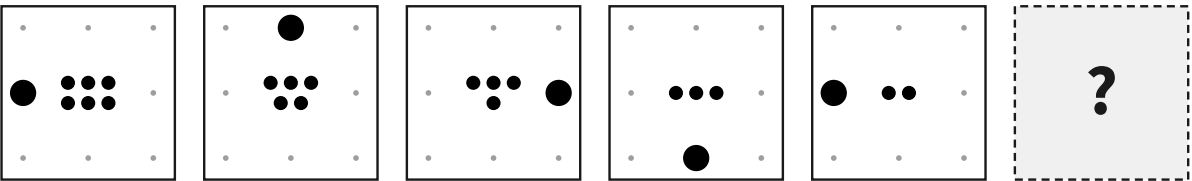
Answer options



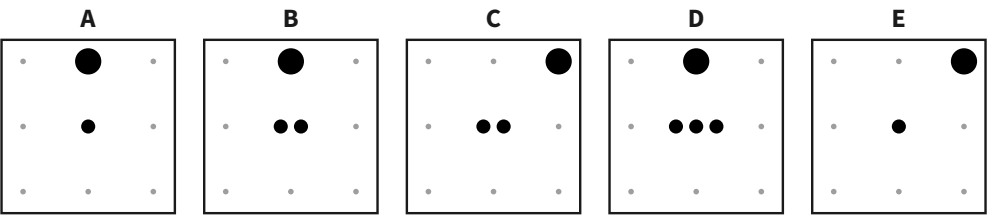
Intermediate

Question 21

Which figure comes next in the series?

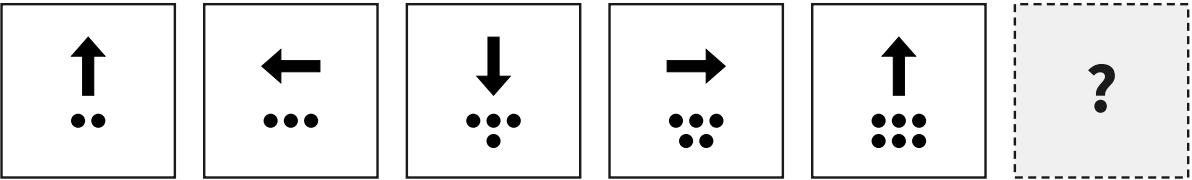


Answer options

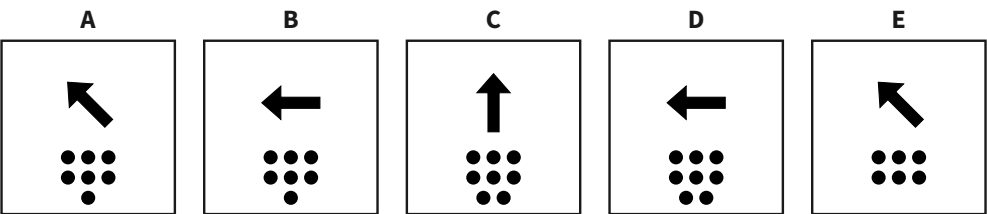


Question 22

Which figure comes next in the series?

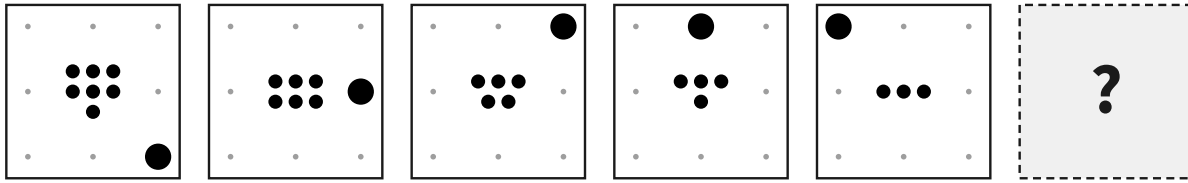


Answer options

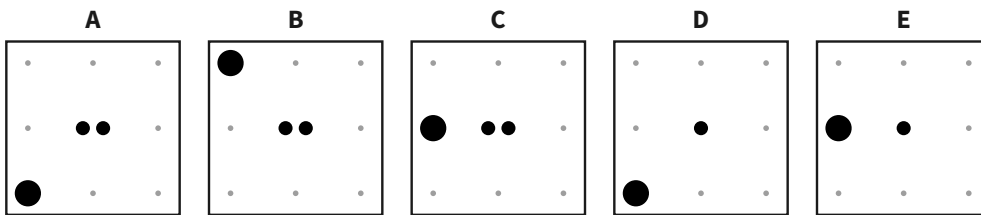


### Question 23

Which figure comes next in the series?

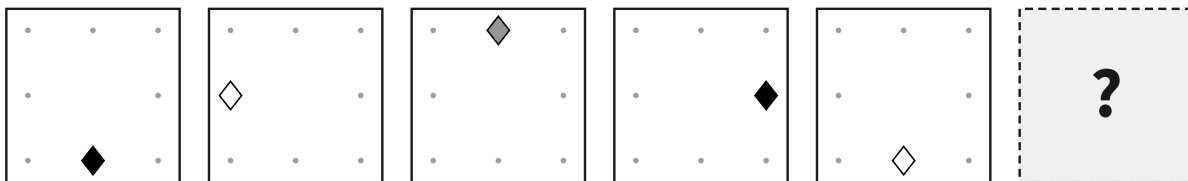


Answer options

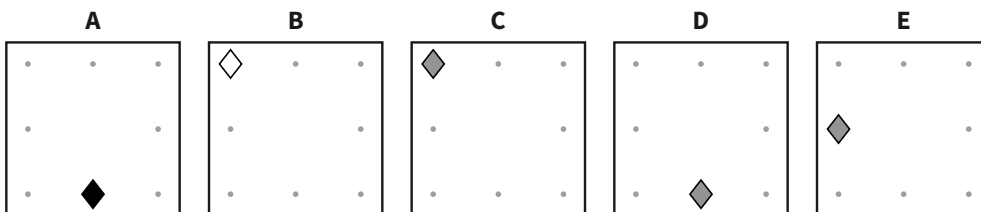


### Question 24

Which figure comes next in the series?

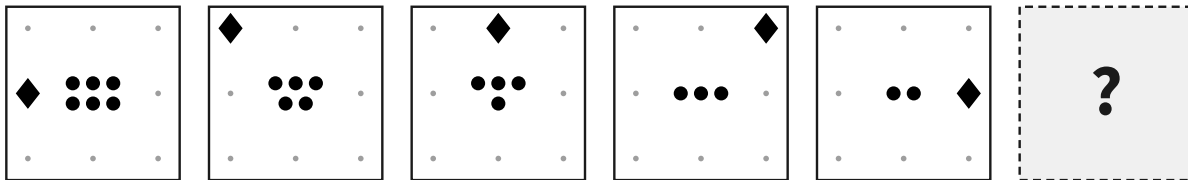


Answer options

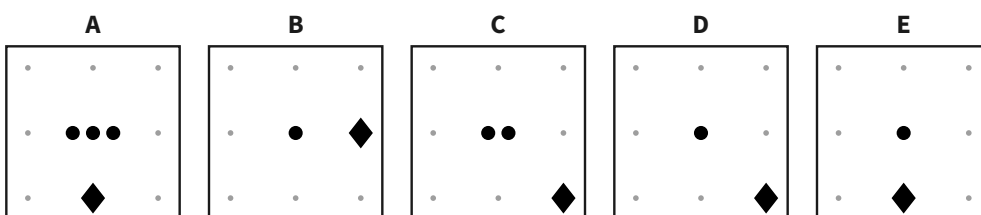


### Question 25

Which figure comes next in the series?

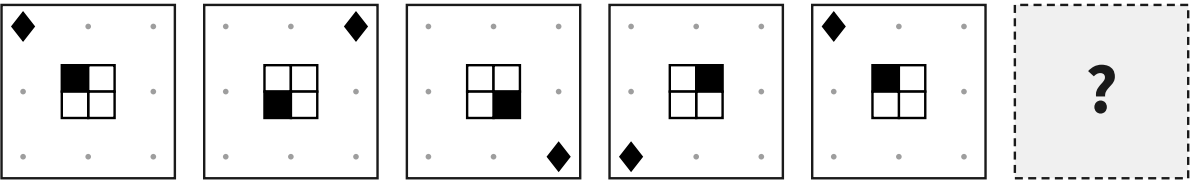


Answer options

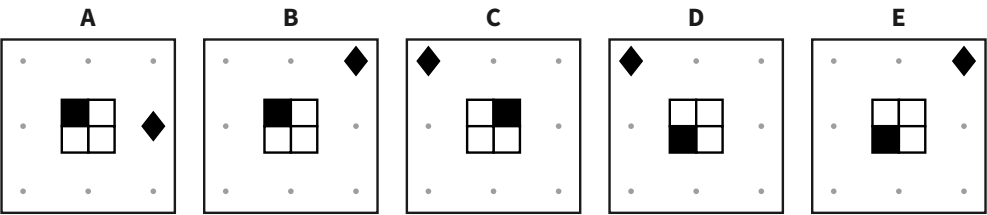


Question 26

Which figure comes next in the series?

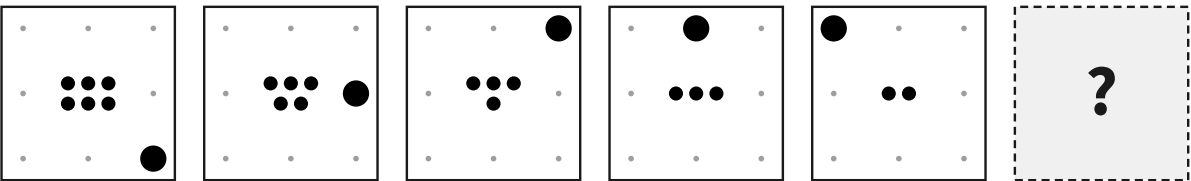


Answer options

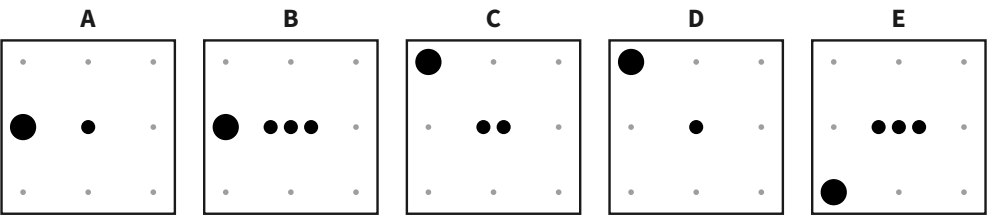


Question 27

Which figure comes next in the series?

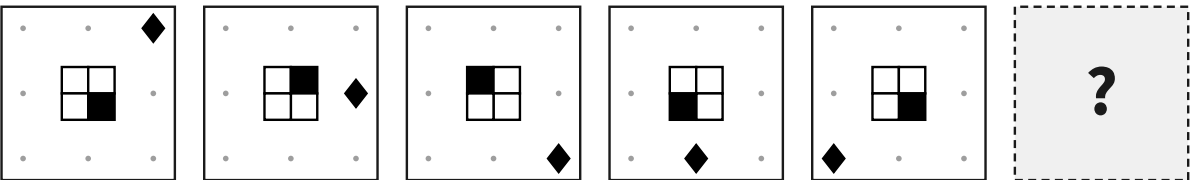


Answer options

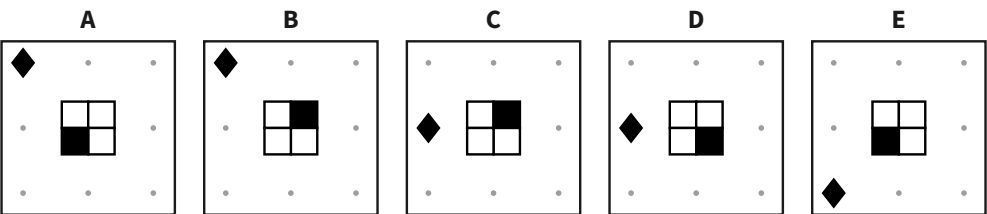


Question 28

Which figure comes next in the series?

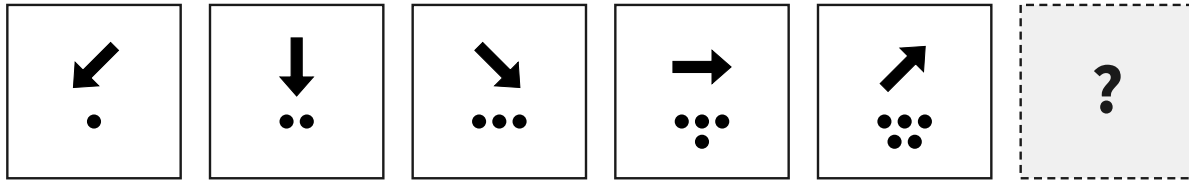


Answer options

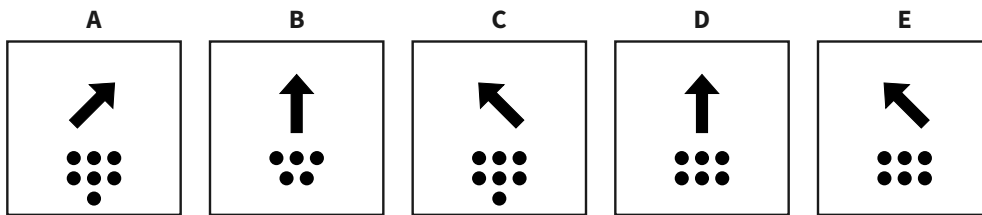


### Question 29

Which figure comes next in the series?

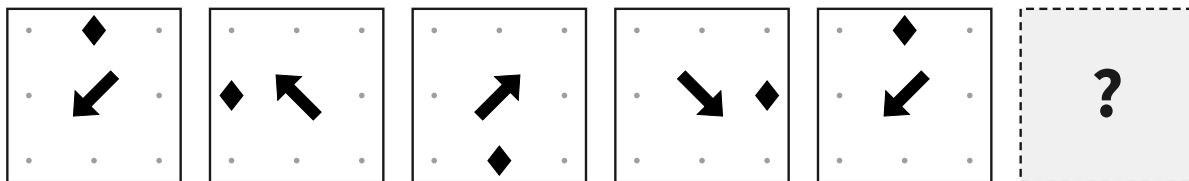


Answer options

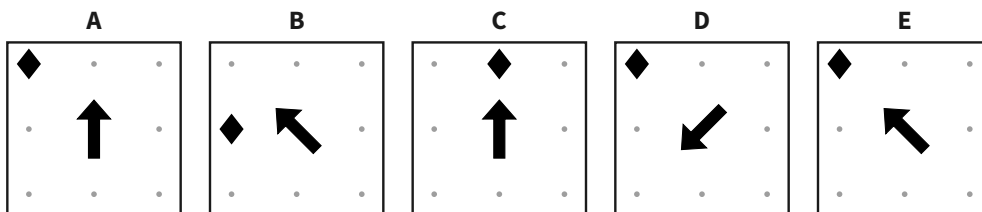


### Question 30

Which figure comes next in the series?

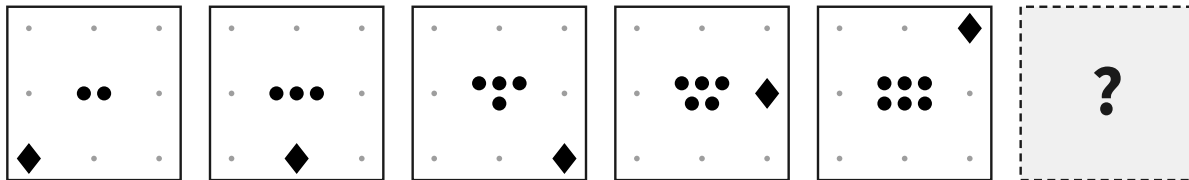


Answer options

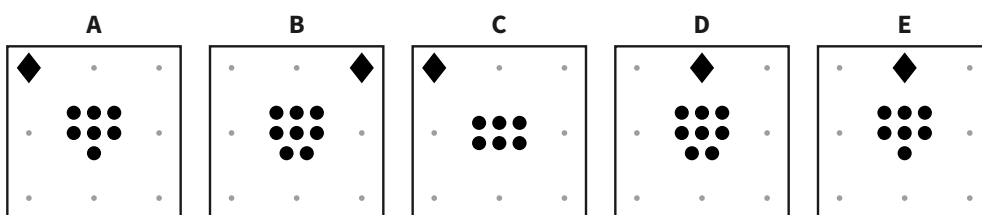


### Question 31

Which figure comes next in the series?

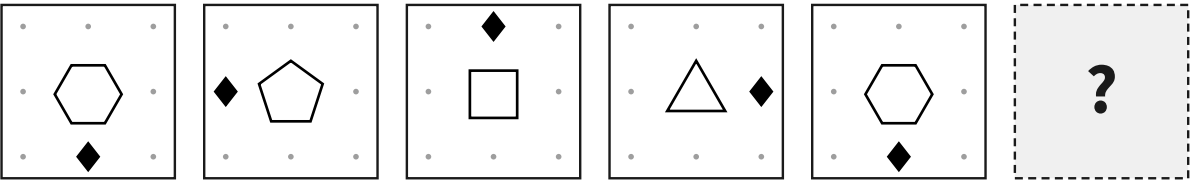


Answer options

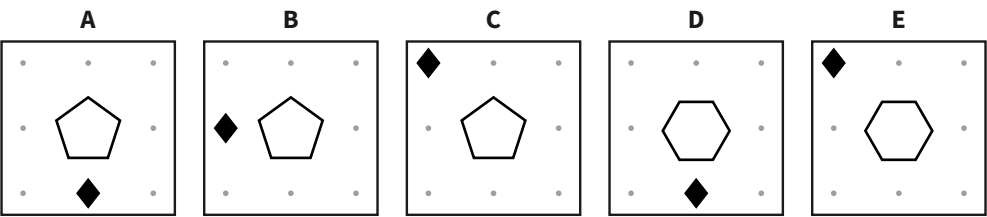


Question 32

Which figure comes next in the series?

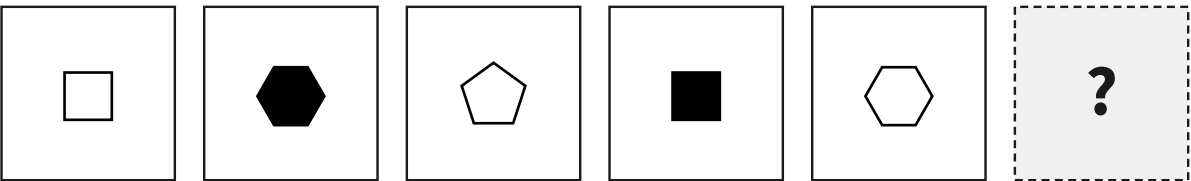


Answer options

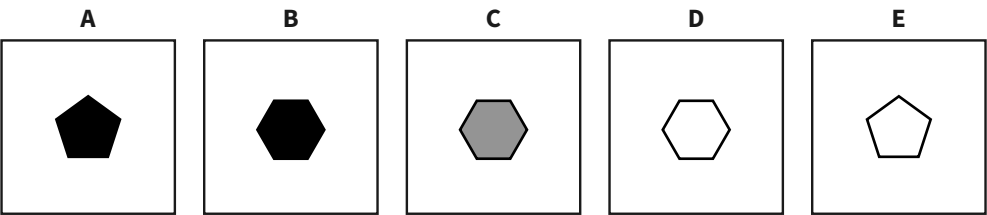


Question 33

Which figure comes next in the series?

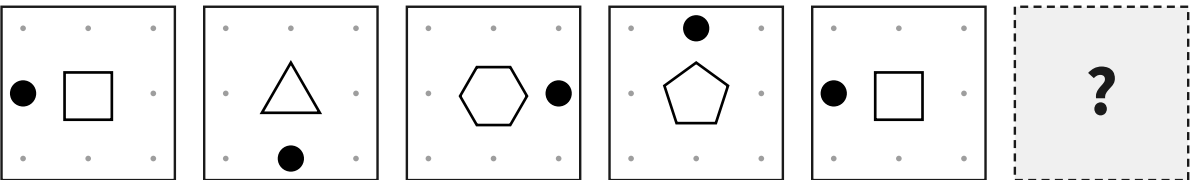


Answer options

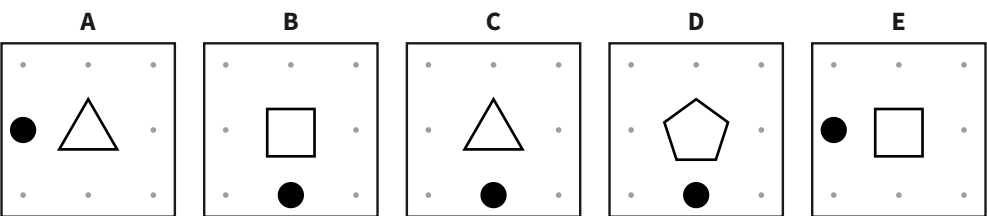


Question 34

Which figure comes next in the series?

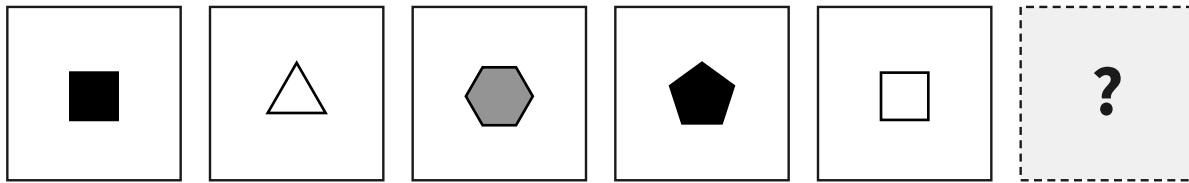


Answer options

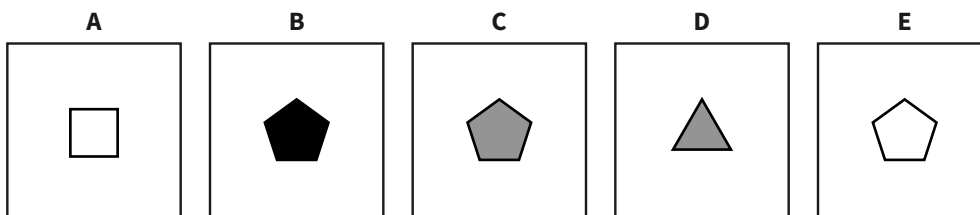


### Question 35

Which figure comes next in the series?

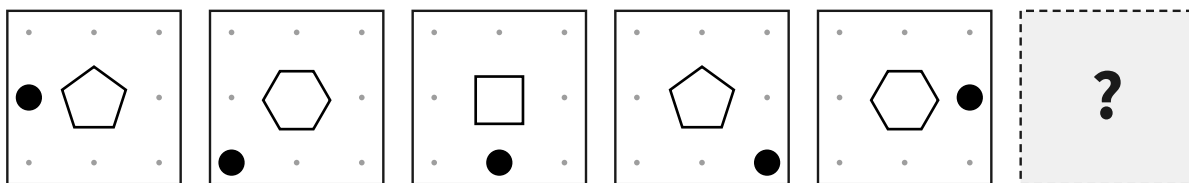


Answer options

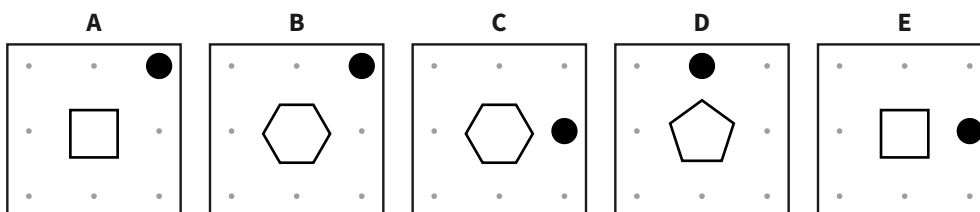


### Question 36

Which figure comes next in the series?

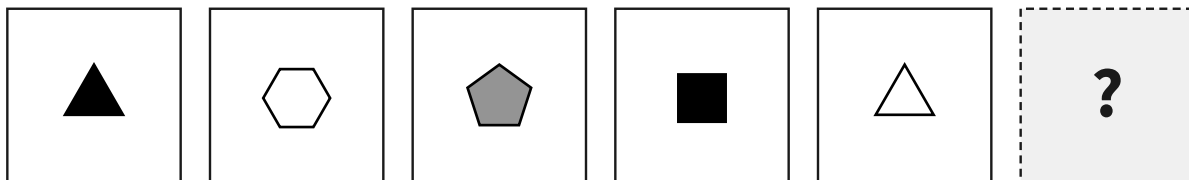


Answer options

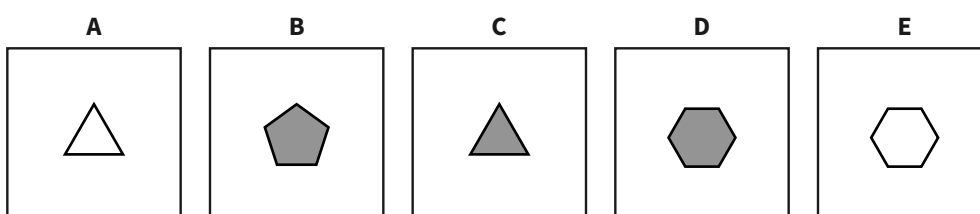


### Question 37

Which figure comes next in the series?

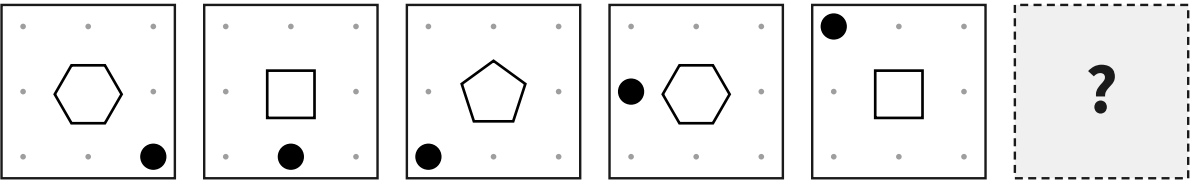


Answer options

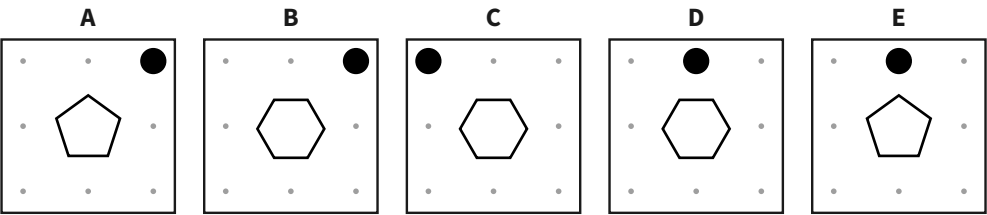


Question 38

Which figure comes next in the series?

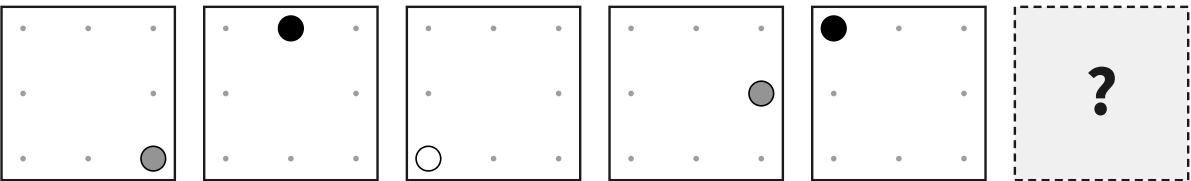


Answer options

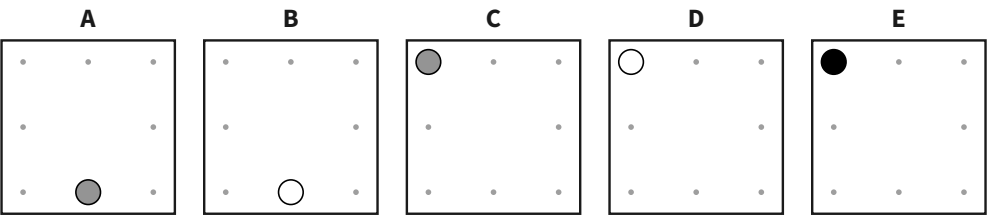


Question 39

Which figure comes next in the series?

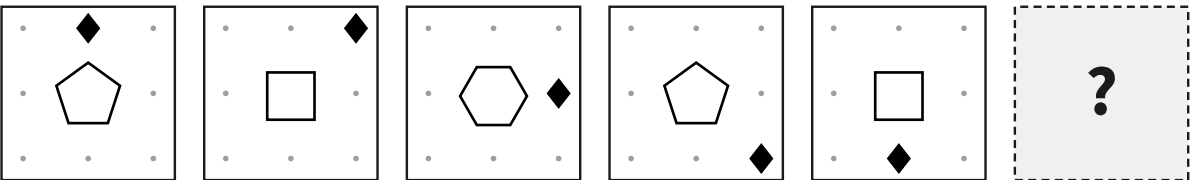


Answer options

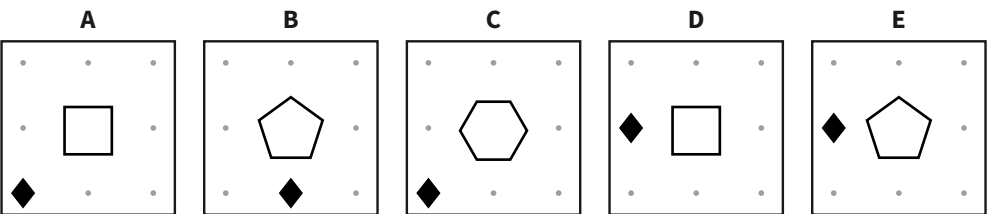


Question 40

Which figure comes next in the series?



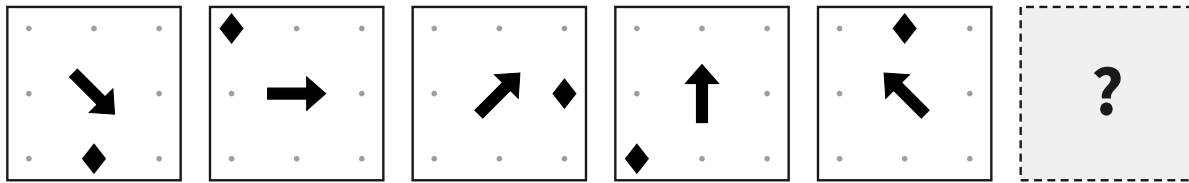
Answer options



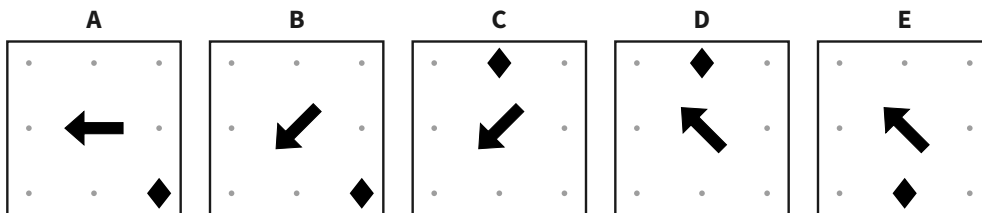


### Question 41

Which figure comes next in the series?

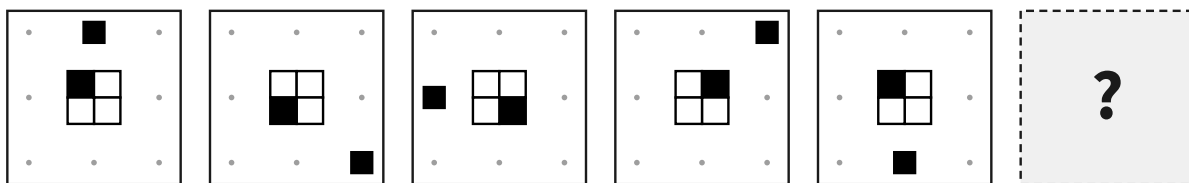


Answer options

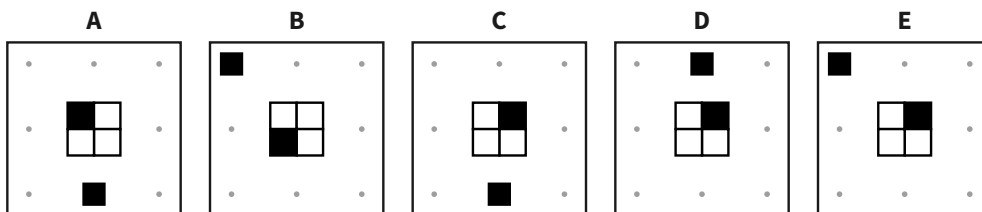


### Question 42

Which figure comes next in the series?

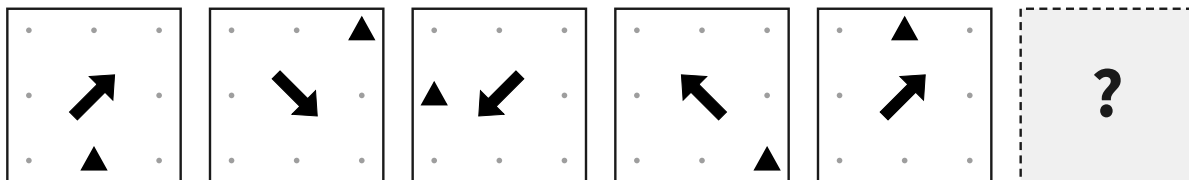


Answer options

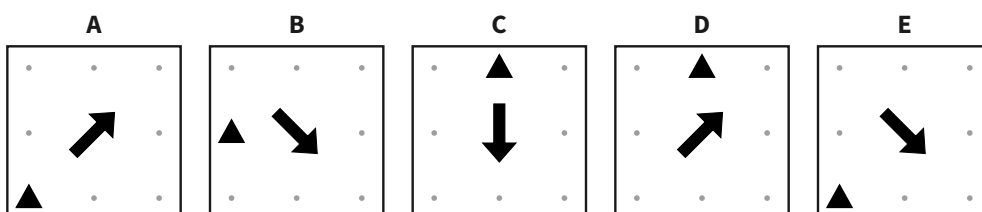


### Question 43

Which figure comes next in the series?

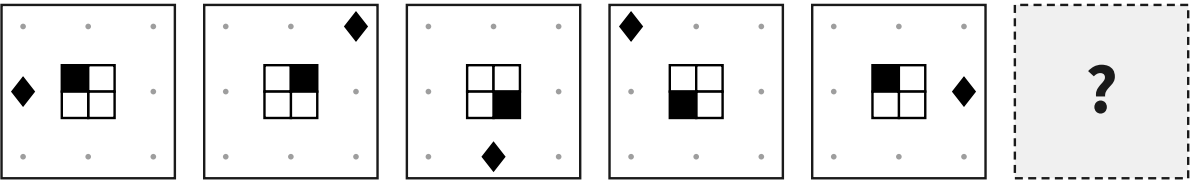


Answer options

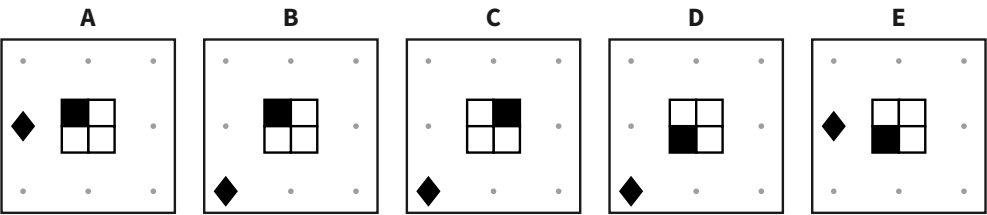


Question 44

Which figure comes next in the series?

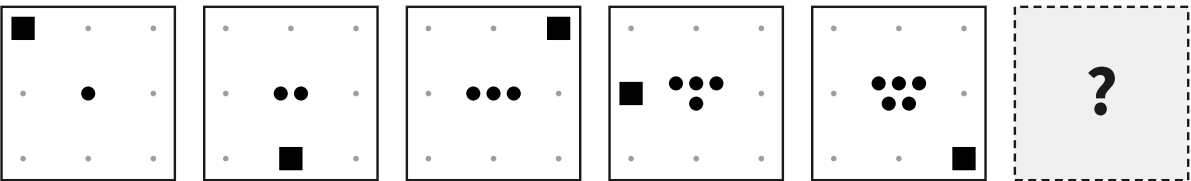


Answer options

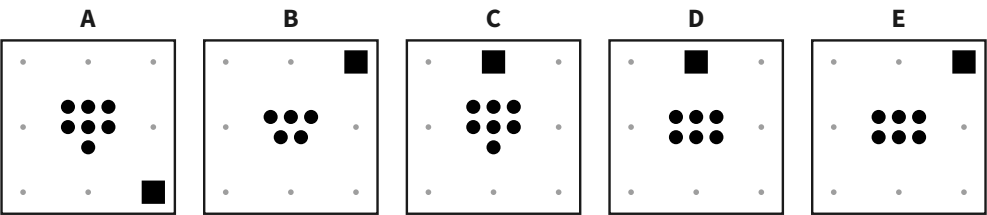


Question 45

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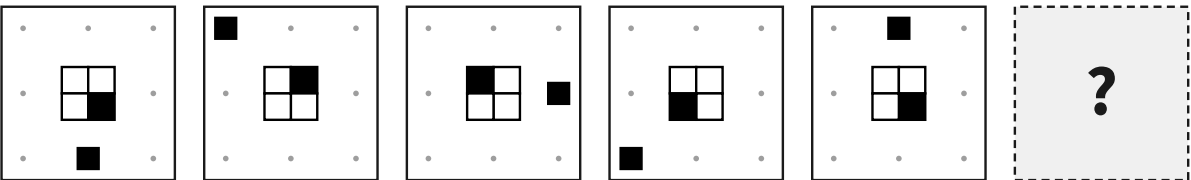


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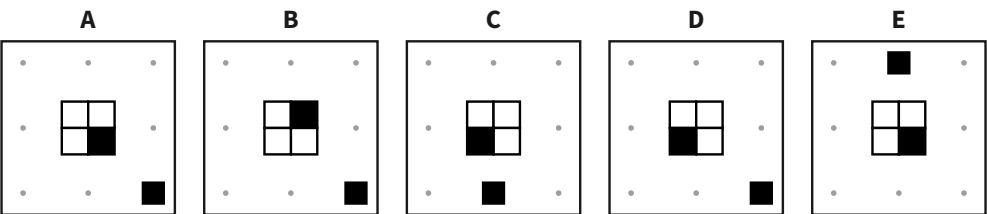


Question 46

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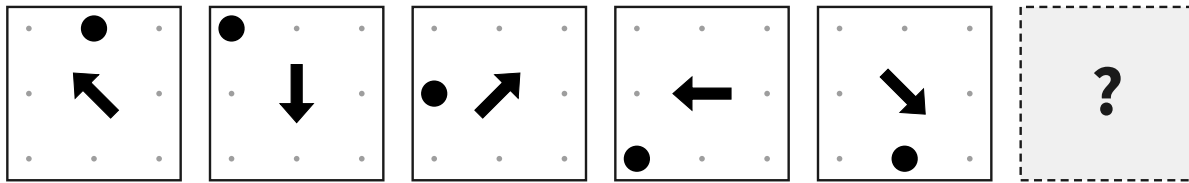


Answer options

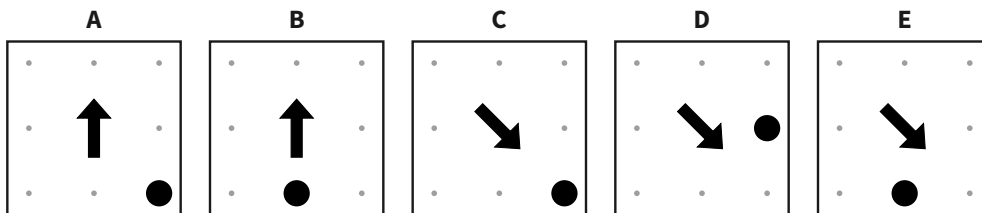


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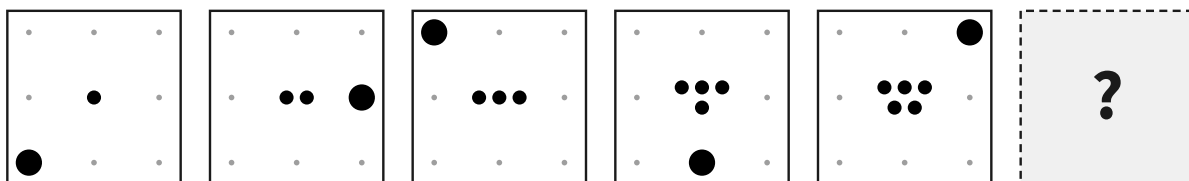


Answer options

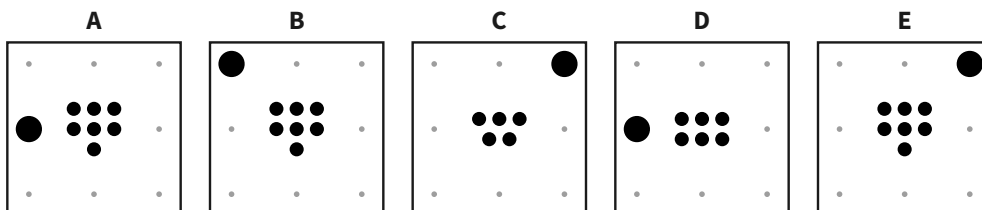


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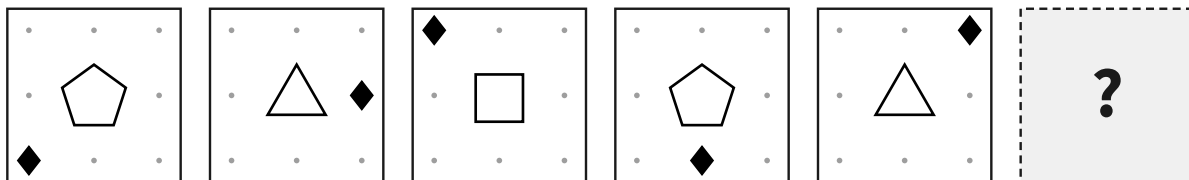


Answer options

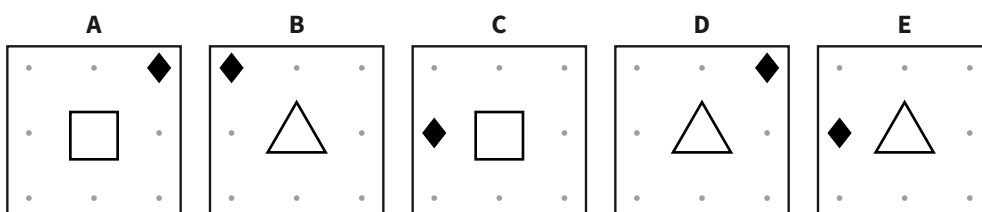


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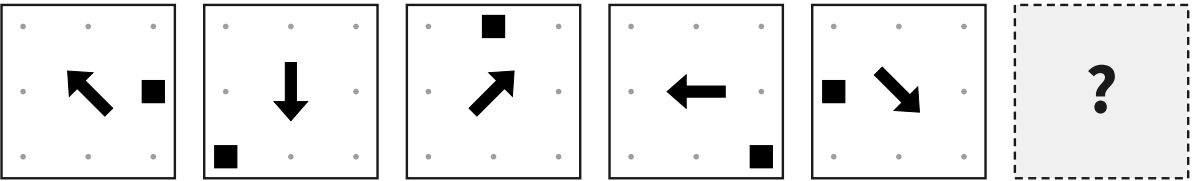


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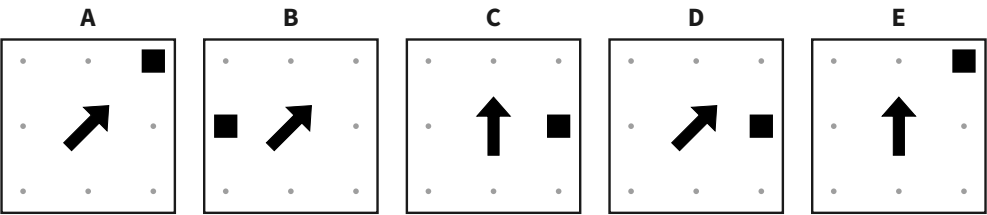


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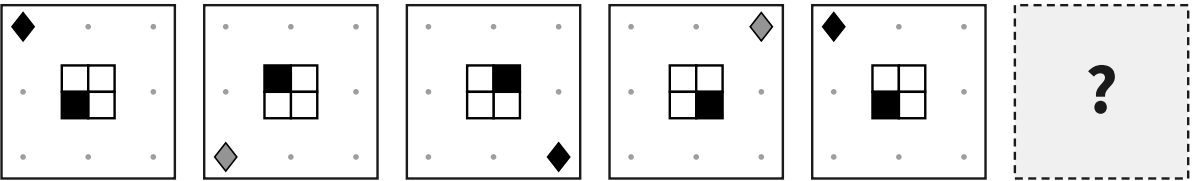
Answer options



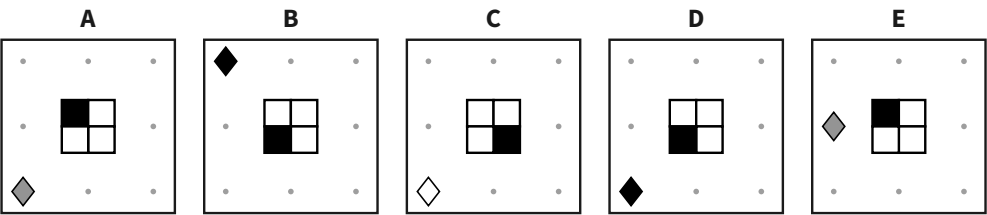
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Question 51

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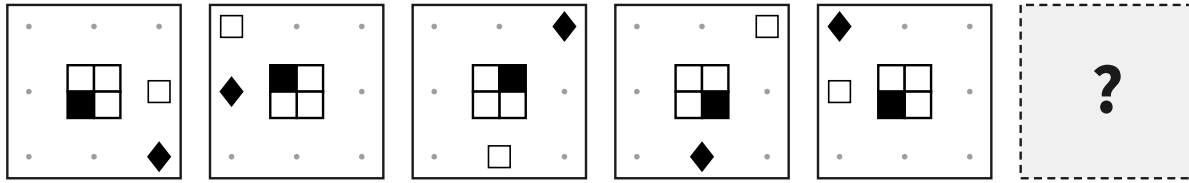


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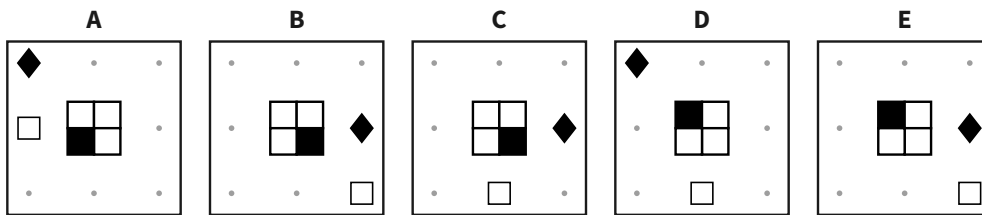


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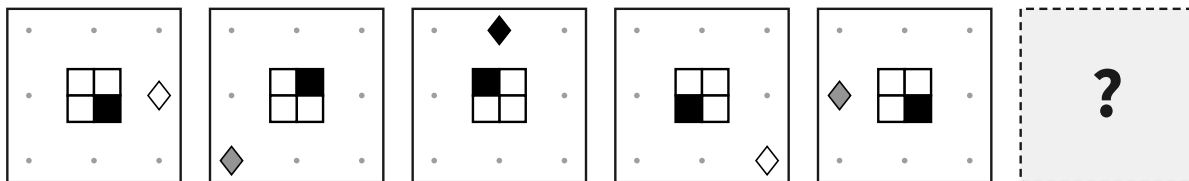


Answer options

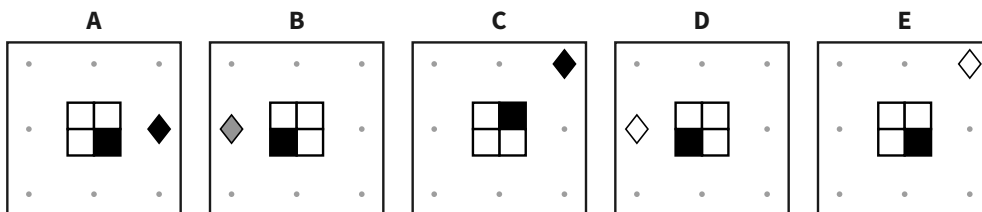


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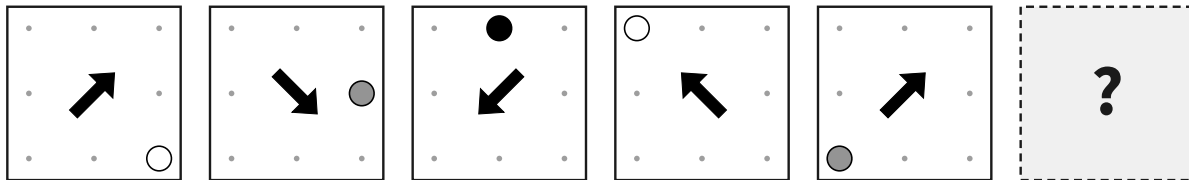


Answer options

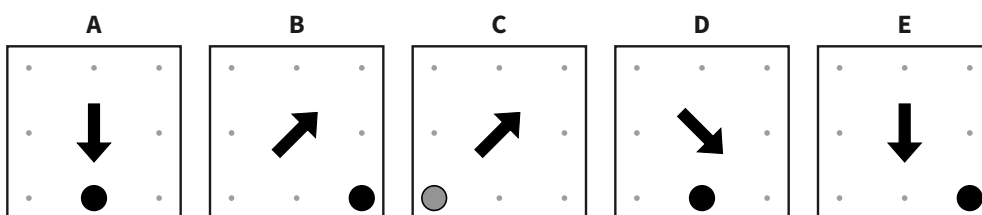


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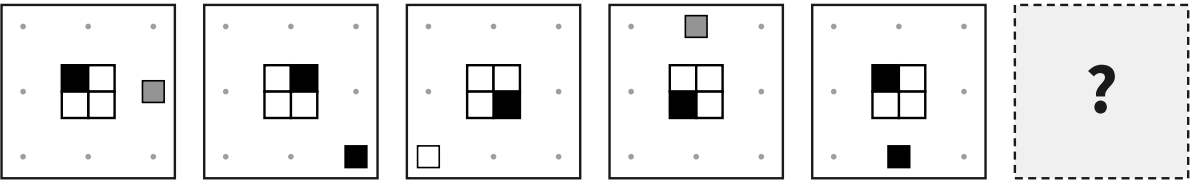


Answer options

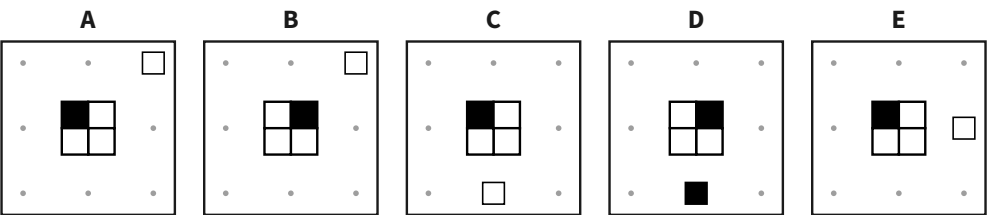


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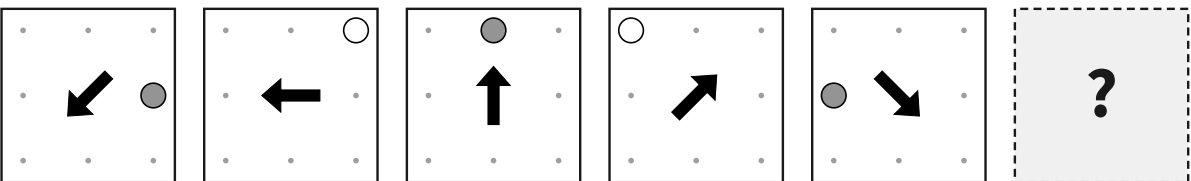


Answer options

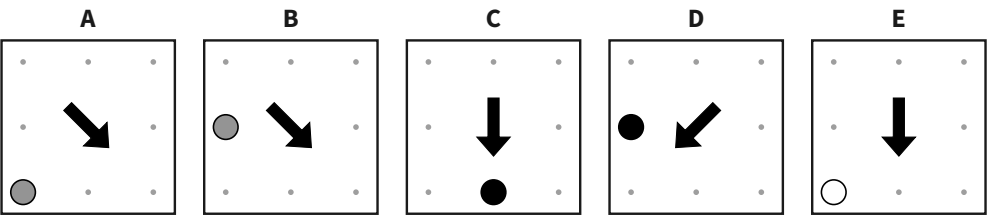


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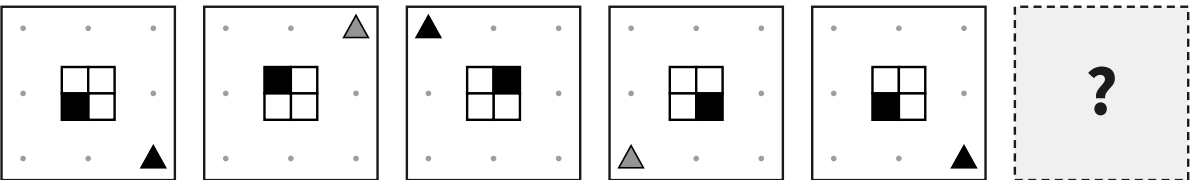


Answer options

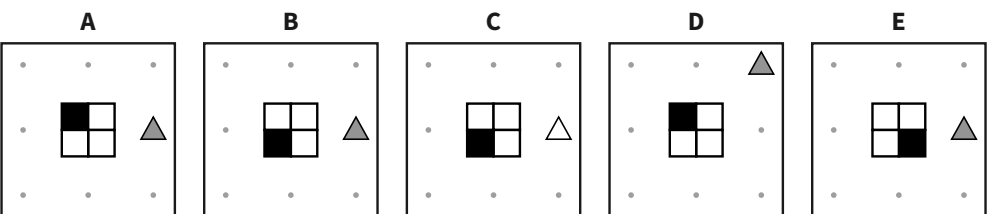


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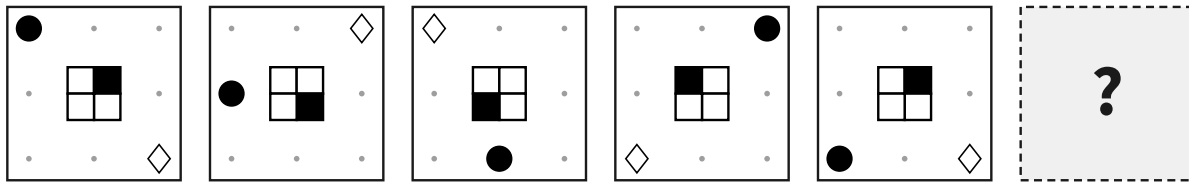


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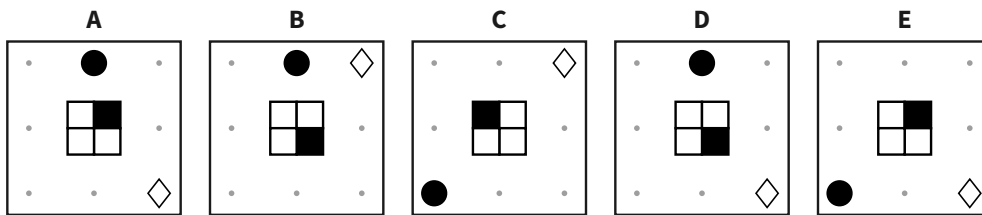


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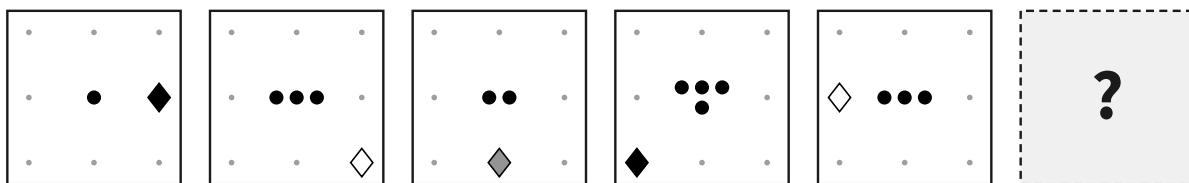


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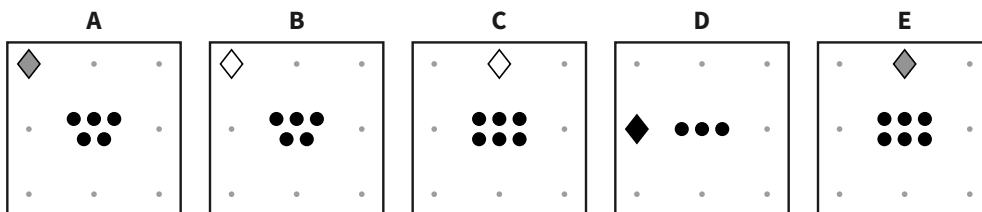


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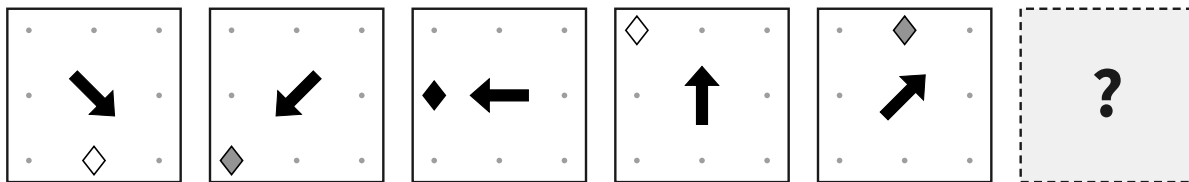


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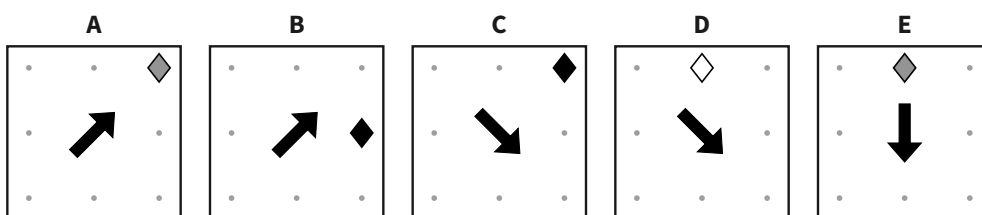


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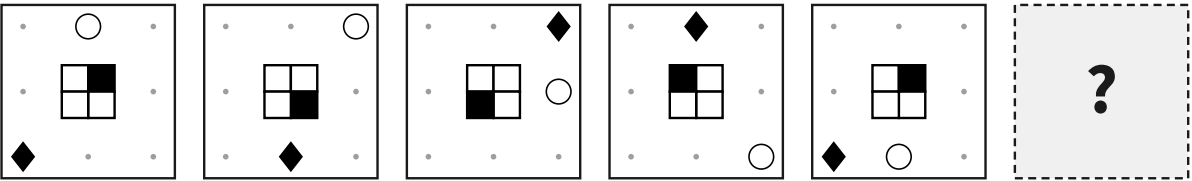


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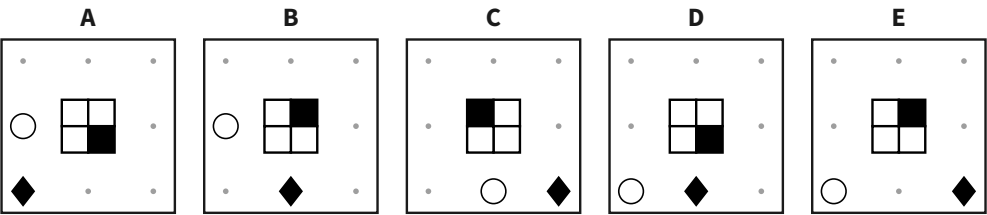


Question 61

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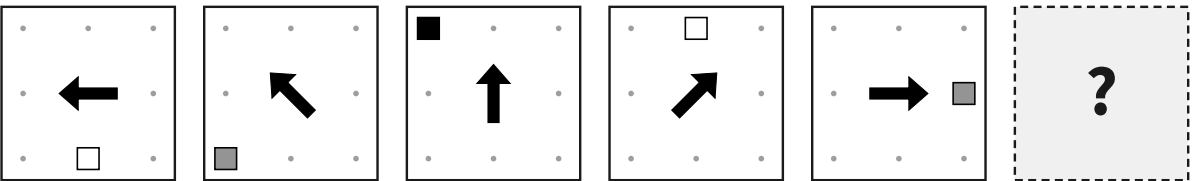


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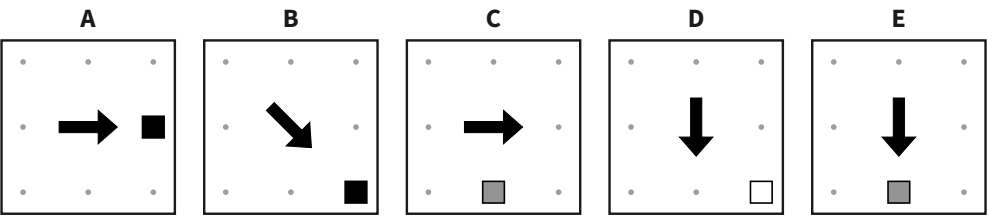


Question 62

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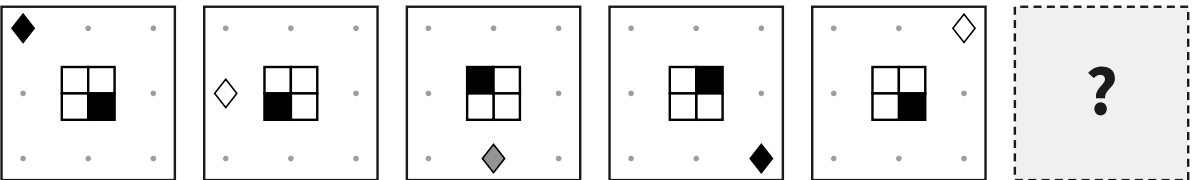


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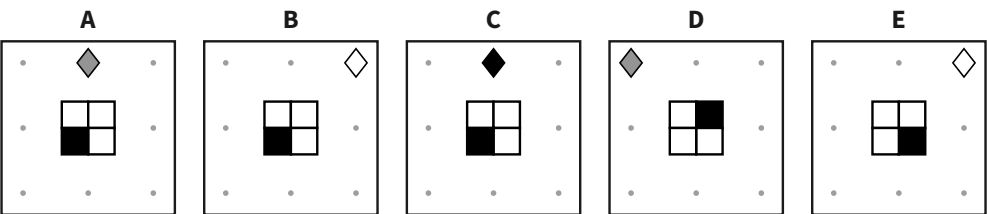


Question 63

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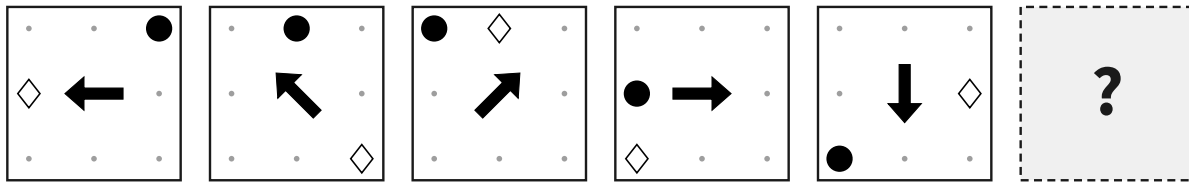
Answer options



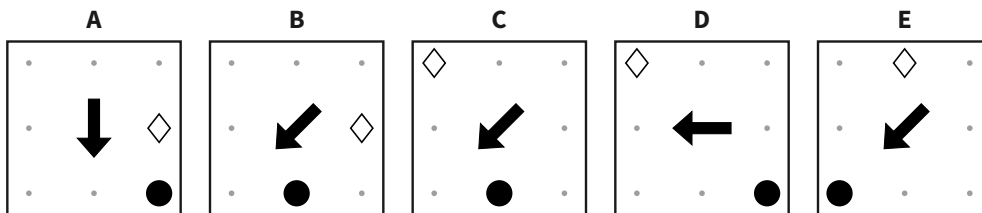


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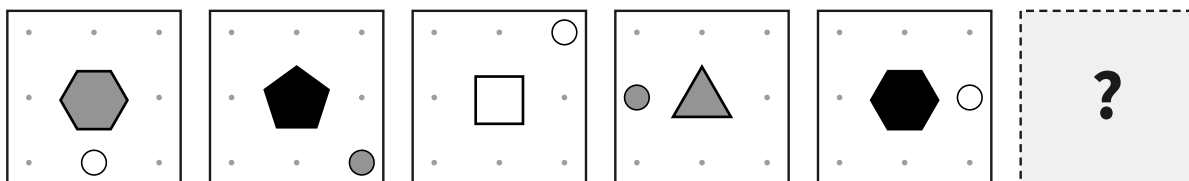


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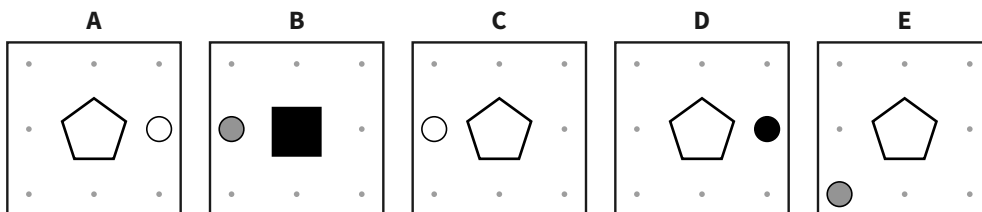


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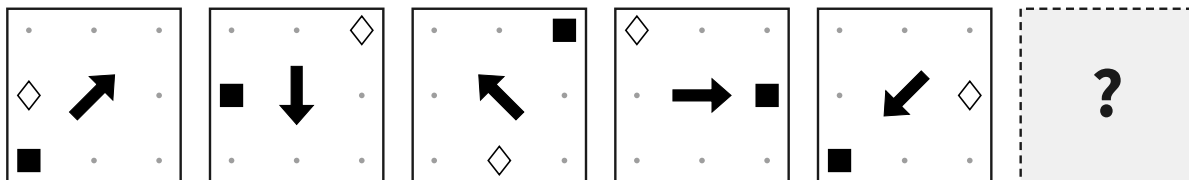


Answer options

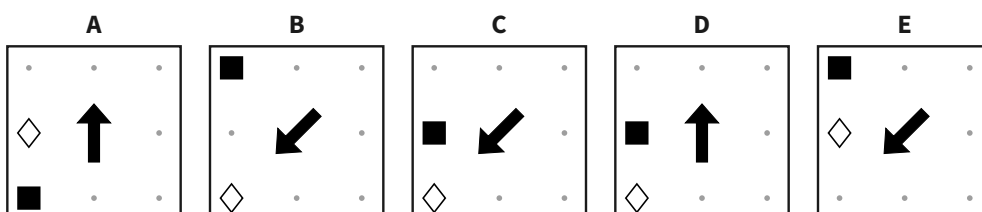


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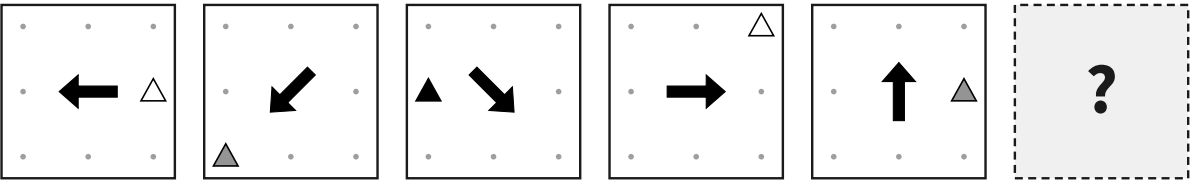


Answer options

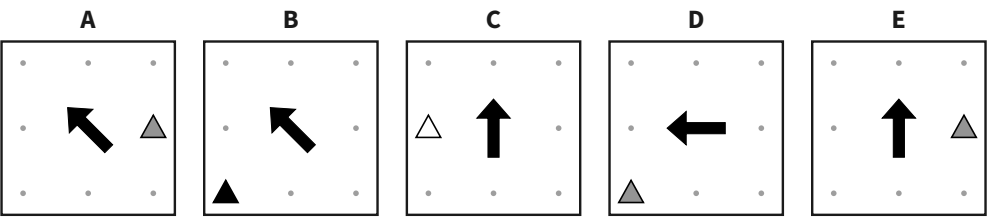


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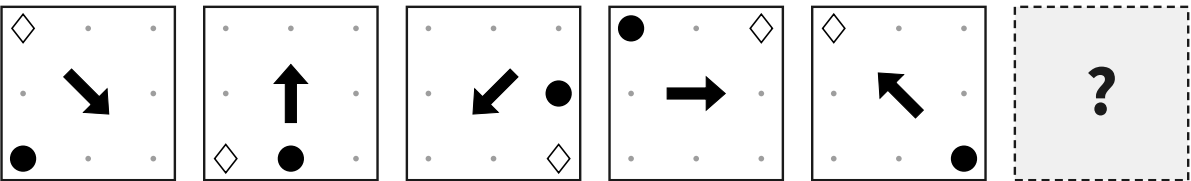


Answer options

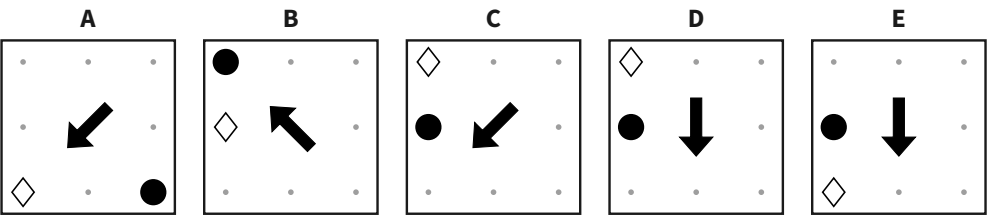


Question 68

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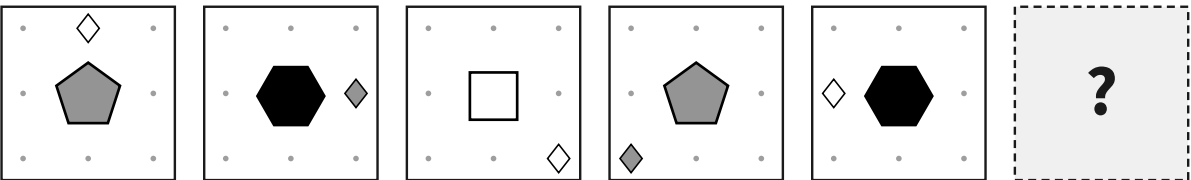


Answer options

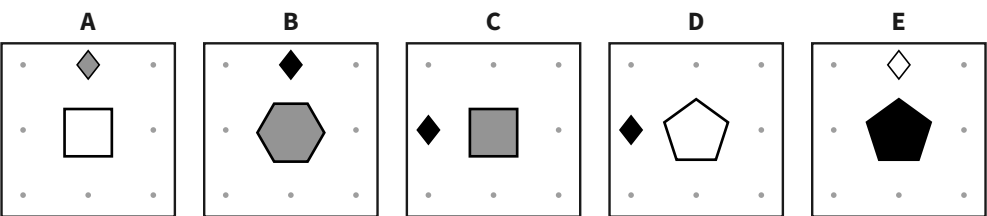


Question 69

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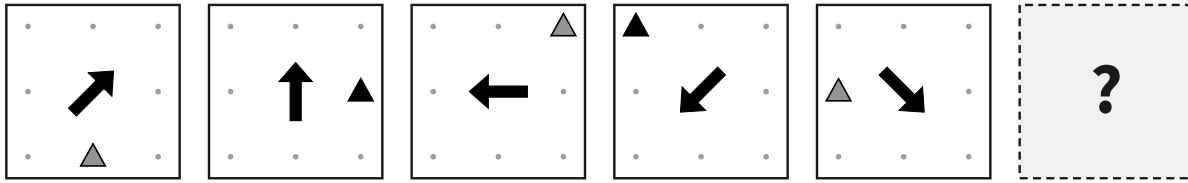


Answer options

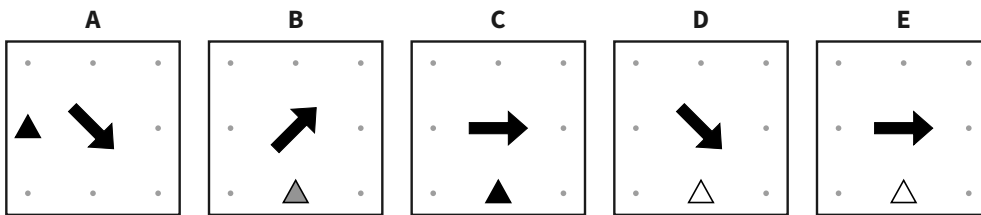


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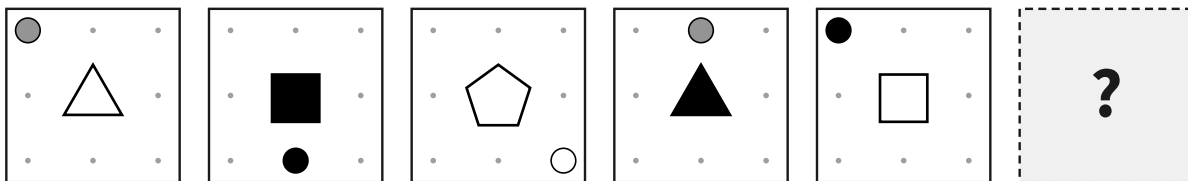


Answer options

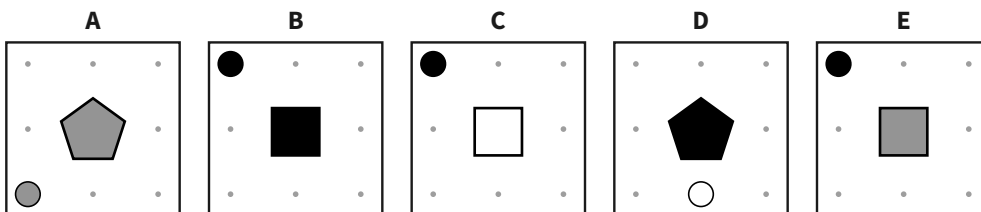


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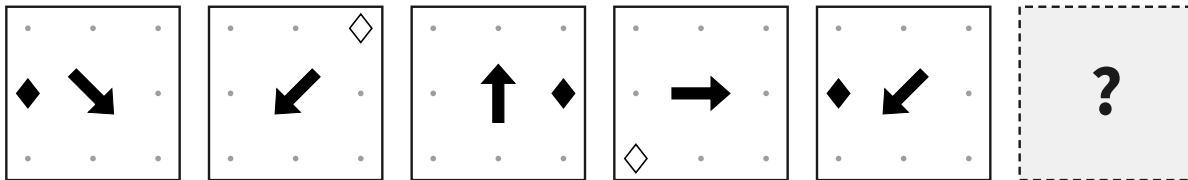


Answer options

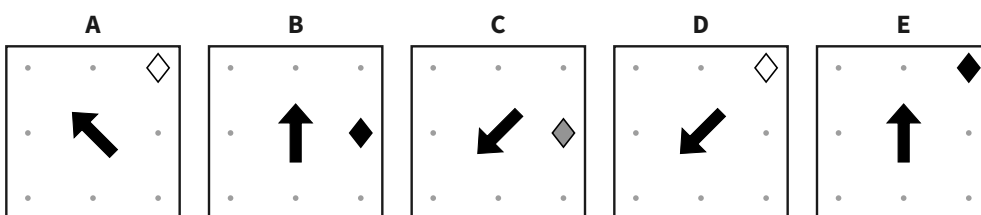


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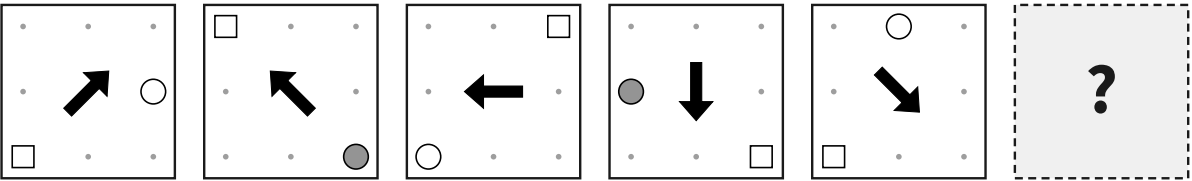


Answer options

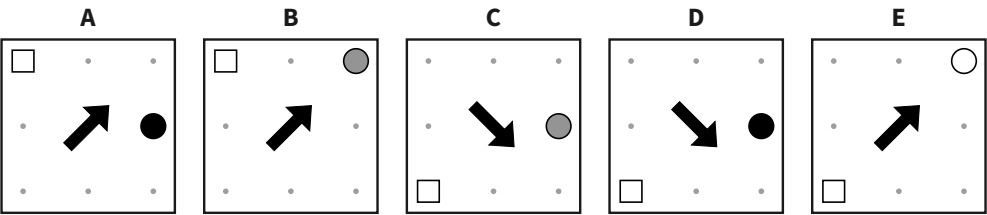


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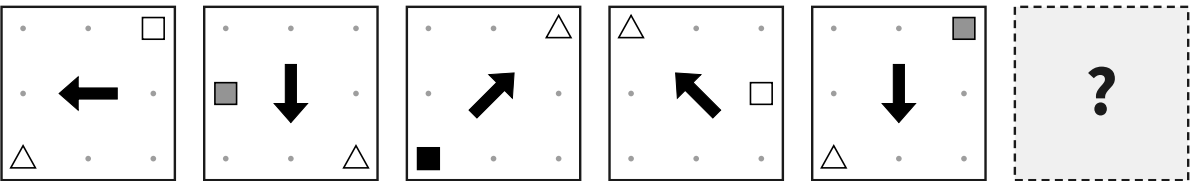


Answer options

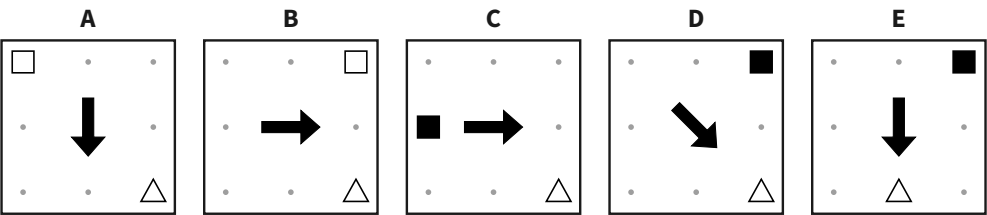


Question 74

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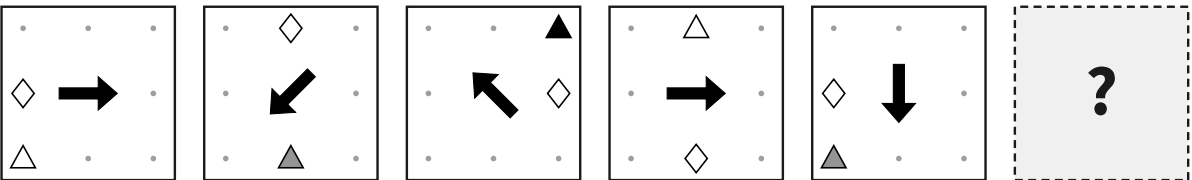


Answer options

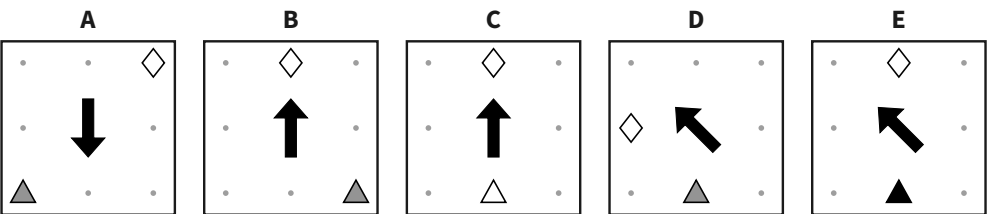


Question 75

Which figure comes next in the series?



Answer options



# Practice Set — Situational Judgement

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For each scenario, choose the MOST effective and the LEAST effective action. Answer as an effective EU staff member would. Suggested pace: about 2 minutes per scenario.

*24 questions · answers and worked solutions in Part IV.*

## Question 1

You are an assistant in a unit of a Directorate-General and share responsibility for the unit's functional mailbox with your colleague Aoife, following an agreed weekly rota. Over the past three weeks, requests that arrive on Aoife's days have often gone unanswered, and several external stakeholders have now written to you directly to complain about delays. You know that Aoife recently took on extra work for a high-profile file and seems stretched. Your head of unit is not aware of the problem. Your own workload is also full, but the backlog in the mailbox keeps growing.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Reply to the stakeholders who complained, explaining that the delays occurred on days when the mailbox was covered by a colleague, so they know you are not responsible.
- B** Clear the most urgent pending requests yourself this week so that stakeholders are not kept waiting any longer, and then let your head of unit know that the mailbox is under strain.
- C** Talk privately with Aoife, acknowledging her extra workload, and propose a temporary rota adjustment that the two of you then put to your head of unit.
- D** Leave the requests that arrive on Aoife's days for her to deal with, since the rota is clear, and concentrate on keeping your own days up to date.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 2**

You are a policy officer preparing a briefing on the uptake of an EU funding programme for small businesses. Monitoring data show that applications from three Member States fell sharply last year. Your colleague Jonas says the reason is obvious: businesses in those countries lost interest after the eligibility rules were tightened. Your director needs the briefing in ten days and wants to know whether the programme design should be revised. While looking at the data, you notice that the same three countries also replaced their national contact points for the programme during the same period.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Compare the timing of both changes with the monthly application figures and ask the new national contact points for their view before concluding on the cause.
- B** Present Jonas's explanation as the most likely cause in the briefing, adding a footnote that other factors may also have played a role.
- C** Report the fall in applications and set out both possible explanations neutrally, recommending that an external evaluation be commissioned to establish the cause before any change is made to the programme design.
- D** Recommend that the eligibility rules be relaxed again, since that change is well documented and reversing it is the quickest way to restore applications from those countries.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 3**

You work as a secretary in a unit of an EU agency. At 9:00 on Monday your inbox contains, in order of arrival: several routine meeting invitations to file; a reminder that last month's mission expense claims must be submitted by Friday; a message from your colleague Marie asking for help formatting a long report 'whenever you have time'; and, received last, a request from your head of unit to book a meeting room for a 40-person stakeholder workshop on Thursday. Rooms of that size are usually booked several days ahead. You have no other fixed commitments this week.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Complete the mission expense claims first, as they have a firm deadline, then book the workshop room and help Marie with her report.
- B** Start with Marie's report, as it is the largest task, and leave both the room booking and the expense claims until the formatting is finished.
- C** Deal with the messages strictly in the order in which they arrived, finishing each one before moving on to the next, so that nothing is overlooked and every request is treated equally.
- D** Book the room now, schedule the expense claims for midweek, give Marie a realistic time for her report and file the invitations when convenient.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 4**

You are a policy officer asked to present the results of a technical evaluation of water-quality monitoring to a meeting of national officials. Many of them are not specialists, and most are working in their second or third language. The evaluation report runs to 120 pages and is full of statistical tables. You have 15 minutes on the agenda. Your colleague Piotr, who co-authored the report, suggests showing as many of the tables as possible 'so nobody can say we hid anything'.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Read out the executive summary of the report word for word, since its wording has already been approved and this avoids any risk of misrepresenting the findings.
- B** Present the three main findings and what they mean for national authorities in plain language, with a simple chart or two, and point to the full report for detail.
- C** Follow Piotr's suggestion and show as many of the statistical tables as fit into the 15 minutes, since completeness protects the team from any later criticism.
- D** Circulate the full report a few days before the meeting and use the 15 minutes to answer questions on it.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 5**

You work in the finance team of an EU agency, processing payments to contractors. During the busy year-end closing period, the payment system is down for a full day. When it comes back, you face a backlog of 60 invoices, several of them close to their legal payment deadline. A contractor, Henrik, phones you and speaks aggressively, saying his invoice is late and threatening to make a formal complaint. You feel tired and stressed after several weeks of overtime.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Stay calm, acknowledge his frustration, check his invoice and its legal deadline, give him a realistic payment date, and then sort the backlog by deadline.
- B** Process his invoice straight away so that he stops calling and does not make a formal complaint, even though other invoices are closer to their deadlines.
- C** Explain firmly that the delay was caused by a system failure that is not your responsibility, and that a complaint will not make the payment go any faster.
- D** Tell him politely that you are dealing with a large backlog and will call him back, then ask your team leader whether extra help can be found.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 6**

You were appointed team leader of six policy officers in a Directorate-General a month ago. You notice that Luca, an experienced member of the team, regularly dominates meetings and dismisses ideas from newer colleagues; two of them have since stopped contributing, and one has told you privately that she no longer feels her views are welcome. Luca's technical work is excellent and he is well respected by external stakeholders. The team must soon prepare a joint proposal that needs input from everyone.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Hold back from intervening for now, as Luca's expertise is essential for the proposal and challenging him so early could harm your working relationship.
- B** Introduce a meeting format in which each member presents their input in turn before open discussion, without raising the issue with Luca directly.
- C** Ask the newer colleagues to send you their ideas in writing so that you can make sure they are reflected in the proposal.
- D** Speak to Luca privately: recognise his expertise, explain the effect his behaviour is having and ask him to help bring out newer colleagues' ideas.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 7**

You are an assistant responsible for compiling a register of grant agreements that will be published on the agency's website next week. During a final check, you find that around ten entries show beneficiary names or amounts that do not match the signed agreements. The data were entered by another team several months ago. Your head of unit, Elena, has stressed that publication must go ahead on schedule. Correcting the ten entries would take you about half a day.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Correct the ten entries yourself against the signed agreements and publish the register on schedule without mentioning the issue, since the problem has now been solved.
- B** Correct the ten entries, spot-check other entries for similar errors, and tell Elena and the other team what you found and corrected.
- C** Tell Elena about the errors and suggest postponing publication by two weeks, so that the other team, which entered the data, can recheck the entire register against the signed agreements.
- D** Publish the register as planned, since the data came from another team and its accuracy is their responsibility.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_



**Question 8**

You are a policy officer who has just moved to a unit dealing with digital policy, an area you know little about. Your new head of unit, Katarina, has asked you to represent the unit at an inter-service working group in six weeks' time. Colleagues often use technical terms you do not yet understand, and you are reluctant to keep asking questions because you feel you ought to know the answers already.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Read the unit's key policy documents and recent working-group minutes in your own time, including evenings, over the coming weeks so that you are well prepared for the first meeting.
- B** Avoid asking questions so as not to appear underqualified, and prepare speaking notes from the relevant documents the evening before each meeting.
- C** Set yourself a six-week learning plan combining reading, a short weekly catch-up with an experienced colleague and relevant training, and check its priorities with Katarina.
- D** Ask Katarina whether an experienced colleague could accompany you to the first working-group meetings, so that the unit is properly represented while you are still getting up to speed.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 9**

You work on the IT service desk of an EU agency. Over the past two days you have received eleven tickets from staff who cannot open shared files in the document management system. Each time, the standard fix of clearing the cache and restarting works for a while, but several users have called back the next day with the same problem. From the ticket details, it looks as if most of the affected users work on the same floor, although you have not checked this systematically.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Advise users on that floor to save their files locally rather than in the document management system until the problem disappears, to avoid further disruption.
- B** Review the eleven tickets together for common factors such as location or device type, and pass your findings to the second-level infrastructure team.
- C** Escalate all eleven tickets to the second-level infrastructure team straight away, explaining that the standard fix only works temporarily and that users are becoming frustrated with the repeated interruptions.
- D** Continue applying the standard fix and add a note to the knowledge base that users may need to repeat it.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

### Question 10

You are a policy officer coordinating the drafting of a joint report with colleagues from two other Directorates-General. Ana, who represents one of them, regularly sends her comments after the agreed deadlines, and her unit's written positions sometimes contradict what was agreed in meetings. The representatives of the third Directorate-General are becoming frustrated. Several sections of the report still depend on input from Ana's unit, and the final report is due to senior management in three weeks.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Send all contributors a revised timetable with firm deadlines and circulate written conclusions after each meeting, making clear that comments received late cannot be taken into account.
- B** Ask your head of unit to raise the problem with Ana's head of unit.
- C** Finalise the report on the basis of what was agreed in meetings, set aside Ana's late comments and submit it to senior management by the agreed date.
- D** Call Ana to understand her unit's constraints, and agree with her a realistic timetable and a written confirmation of positions after each meeting.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

### Question 11

You work in an HR support team. Mihai, a newly recruited colleague who has moved from another country, emails you with several questions about his installation allowance and the documents he must provide. The rules are set out in a long internal guide written in legal language. Mihai says he has read the guide but found it confusing, and he is worried that he may miss the deadline for submitting his documents, which is in three weeks.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Send him the relevant pages of the guide with the key passages highlighted.
- B** Phone him the same day to walk him through the procedure step by step and answer all his questions directly, since a conversation is quicker and more reassuring than another email.
- C** Reply with a short email listing the documents needed, the deadline and next steps, linking to the relevant part of the guide and offering a call if needed.
- D** Reply that all the information he needs is in the guide, which applies to everyone, and that it is his responsibility to read it carefully and meet the deadline.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 12**

You are a policy officer. Your head of unit has asked you to finalise a position paper for a meeting with Member State experts in two weeks. This morning, the director's office asks you for urgent input to a briefing for senior management, needed within 48 hours. At the same time, a colleague asks you to review a draft consultation document 'by the end of the week'. You realise you cannot do all three tasks properly within the time available.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Deliver the briefing input first, tell your head of unit how this affects the position paper, and agree a later review date with your colleague or suggest another reviewer.
- B** Divide your time equally between the three tasks over the coming days so that each of them makes steady progress and nobody is kept waiting.
- C** Explain the situation to your head of unit and ask her to decide which of the three tasks should take priority, since she has the best overall view of the unit's commitments.
- D** Do the briefing input first, then the position paper, and tell your colleague today that you will look at the review only if time allows.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 13**

You are a team leader in a unit managing a call for proposals. One week before the evaluation results must be sent to applicants, an evaluator, Sofia, alerts you that two evaluation reports contain identical paragraphs for different proposals, suggesting that text was copied and pasted. The scores appear consistent with the comments, but you have not checked this in detail. Postponing the notification would mean departing from the timetable published in the call.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Send the results on time as published in the call, and note the issue so that guidance for evaluators on writing comments can be improved for the next call.
- B** Edit the duplicated paragraphs yourself so that the two reports read differently, without changing the scores, and send the results on time.
- C** Ask Sofia, who spotted the problem, to check all the evaluation reports for similar duplicated text before any results are sent, so that applicants receive reliable feedback across the whole call.
- D** Have the two reports checked against the proposals at once, with the evaluators revising comments if needed, and assess whether the timetable can still be met.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 14**

You are the senior assistant coordinating a team of four assistants in a directorate. The director has decided that meeting files will now be managed through a new shared electronic workflow instead of paper folders. Two assistants who have worked in the directorate for many years are openly sceptical and continue to prepare paper files in parallel. This is creating duplicated work and confusion about which version of a file is current, and the director's office has started to notice.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Meet the team to explain the reasons for the change and hear concerns, agree a firm switch-over date, and pair each sceptic with a confident colleague.
- B** Report the two colleagues to the director, explaining the duplication their paper files are causing, so that she can instruct them clearly to stop.
- C** Organise a hands-on training session on the new workflow for the whole team.
- D** Let the two colleagues keep their paper files for now while the rest of the team uses the new workflow, as forcing the change could damage team spirit.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 15**

You are a finance assistant. In your annual appraisal, your reporting officer, Nils, mentions that several of your payment files were returned by the verification team during the year, mostly because supporting documents were missing. The returned files had to be corrected and resubmitted, which caused some payment delays. You were not aware of this pattern. You feel the comment is somewhat unfair, because the verification rules were changed during the year and the change was not well communicated.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Explain during the appraisal that the verification rules changed during the year without proper notice, and ask for the comment to be reworded to reflect this.
- B** Ask the verification team for examples of returned files, identify the recurring gaps and build a checklist based on the current rules, asking about relevant training.
- C** Disregard the feedback, since the errors resulted from a rule change that was outside your control.
- D** Accept the feedback openly in the appraisal and ask Nils to let you know whenever a file is returned in future, so that you can correct any problems as soon as they arise.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 16**

You are a policy officer who has spent four months drafting a legislative proposal on product labelling. At a late stage, senior management decides to change the approach substantially in response to new political priorities, and much of your work will not be used. You have been asked to produce a revised draft within six weeks. You feel disappointed, and some colleagues in the unit are openly complaining about the decision.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Tell your colleagues that the decision shows management does not value the team's work, and do no more than the minimum required on the revised draft.
- B** Wait until the new approach has been confirmed in writing by senior management before starting, in case it changes again.
- C** Acknowledge your disappointment to yourself, identify which parts of your earlier analysis can be reused, and agree a plan for the revised draft with your head of unit.
- D** Start the revised draft from scratch immediately, setting the earlier version aside entirely, so that you are sure of meeting the six-week deadline and are not influenced by the old approach.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 17**

You are a programme officer in an EU delegation in a partner country, assessing whether an EU-funded training programme for local customs officials should be extended. The implementing partner's final report is very positive, citing 400 officials trained and high satisfaction scores. However, an official you meet informally mentions that many of those trained have since moved to posts unrelated to customs. Your recommendation on the extension is due in a month.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Recommend extending the programme, given the strong results in the report, on condition that the next phase includes systematic monitoring of whether trained staff remain in relevant posts.
- B** Ask the implementing partner to clarify how many of the trained officials are still in customs posts.
- C** Recommend against extension on the basis of the official's comment, since it suggests that the training has had no lasting effect on the customs administration.
- D** Request data from the customs administration on where trained staff now work and on relevant performance indicators, and compare them with the report.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 18**

You are a team leader. Your team has just been told that a reorganisation will merge it with another team, but decisions on individual roles and locations will only be taken in two months. Rumours are spreading, and two team members, Elif and Bram, have asked you directly whether their jobs are at risk. Your head of unit has said you may share what has been officially announced but should not speculate about the outcome.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Hold a team meeting to share what is known and what is still undecided, say when more information is expected, and invite questions.
- B** Forward the official announcement to the team by email without further comment.
- C** Answer Elif and Bram individually and in confidence, telling them exactly what has been announced and that you will keep them informed as decisions are taken.
- D** Reassure the team that nobody's job will be affected, to stop the rumours and keep morale up.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 19**

You are the assistant organising a two-day conference for 150 participants run by your unit. With three weeks to go, you realise that the catering contract, the interpretation booking and the participants' travel reimbursements all need attention, and the deadlines for confirming final numbers with the caterer and the interpreters fall in the same week. Your colleague Joana has offered to help, but she is new to the unit and has no experience of organising events.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Deal with each issue as suppliers contact you, since they will remind you of their deadlines.
- B** Hand the interpretation booking over to Joana, as it is a self-contained task, so that you can concentrate fully on the catering contract and the reimbursements.
- C** Draw up a timeline of remaining tasks and deadlines, give Joana well-defined tasks such as collecting travel details, and keep the supplier confirmations yourself.
- D** Handle all the tasks yourself to avoid mistakes at this critical stage, working longer hours over the next three weeks and thanking Joana for her kind offer of help.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 20**

You work on the IT service desk. A major update to the email system was rolled out overnight, and by mid-morning thirty callers are waiting in the phone queue. Your two colleagues are off sick. Many callers have the same known problem, for which there is a simple workaround. Several are impatient, and an assistant to a director insists that her issue must be dealt with first. You realise you will not be able to answer everyone today.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Keep working through the queue as quickly as you can, staying polite and patient with every caller, and stay late if necessary so that as many people as possible are helped today.
- B** Inform your team leader, post an intranet notice on the known issue and workaround to reduce calls, and then handle calls by business impact.
- C** Deal with the director's assistant first because of her director's seniority, then continue with the queue.
- D** Switch the phone line to 'unavailable' for an hour so that you can recover your composure and work through the backlog of email requests instead.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 21**

You work as an assistant in a translation unit, allocating incoming documents to translators according to criteria agreed in the unit. Giulia, a translator, frequently asks you to give her shorter, easier texts, and other assistants often do so because they get on well with her. Kofi, a translator who joined the team recently, has mentioned to you that he always seems to receive the most difficult and urgent documents. Some colleagues are starting to comment on the situation.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Apply the agreed criteria to everyone, explaining them politely to Giulia when she asks, and propose sharing the allocation log.
- B** Continue giving Giulia the texts she prefers, as this keeps relations in the team pleasant.
- C** Give Kofi easier documents for a while to make up for the imbalance.
- D** Tell your head of unit that allocation may not be balanced, mentioning Kofi's remark and colleagues' comments, and suggest that the criteria be reviewed.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 22**

You lead a small team of policy officers working on a public consultation on urban mobility. Nuno, a capable member of the team, has missed two internal deadlines in recent weeks, and his latest draft was well below his usual standard. Colleagues have had to cover for him. You have heard informally that he may be going through a difficult personal situation, but he has not mentioned anything to you.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Remind Nuno firmly of the team's deadlines and the quality expected of his drafts, and arrange a meeting in two weeks to review his progress together.
- B** Raise the missed deadlines at the next team meeting, so that everyone understands that the same standards apply to all.
- C** Meet Nuno privately, describe what you have observed and its effect on the team, ask how he is, and agree on adjustments or support together.
- D** Quietly reassign some of Nuno's tasks to other team members for a few weeks, without discussing it with him, so that he is under less pressure.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_

**Question 23**

You are an assistant who prepares the monthly statistics on processed requests for your unit's management report. The figures come from a shared spreadsheet that several colleagues update. This month the total number of requests is 30% higher than in any previous month, although nothing suggests that the workload has increased significantly. The report is due to your head of unit, Matteo, tomorrow morning, and you have the afternoon available.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Ask Matteo whether the report could be delayed by a few days so that every entry in the spreadsheet can be verified with each of the colleagues who made it.
- B** Send the report on time as requested, adding a note that the total looks unusually high compared with previous months and may need to be checked by the colleagues who entered the data.
- C** Replace this month's total with the average of previous months, since the increase is probably an error.
- D** Check the spreadsheet for duplicates and entry errors, correct what you can verify, and send the report on time, flagging any figures you could not confirm.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_



**Question 24**

You are a team leader. After a high-profile stakeholder conference organised by your team, the feedback forms show that participants found the sessions well organised but felt there was too little time for discussion. Two speakers also told you that the briefing they had received beforehand was not useful. Your director, Ingrid, has praised the event overall, and your team is already starting to plan a similar conference for next year.

**Which action is the MOST effective, and which is the LEAST effective?**

- A** Since the director was satisfied, keep the same format for next year's conference.
- B** Hold a short review with the team, using the feedback forms and speakers' comments, and record specific changes for next year's planning.
- C** Write to the two speakers to thank them for their comments and ask what information they would have found useful in their briefing beforehand.
- D** File the feedback forms and the speakers' comments in the event folder, with a note to the team, so that they can be consulted when next year's planning starts.

Most effective: \_\_\_\_\_ Least effective: \_\_\_\_\_



# Practice Set — Accuracy & Precision

Compare the records with the reference table carefully. Every character counts, including accents and punctuation. Suggested pace: about 1 minute 30 seconds per question.

30 questions · answers and worked solutions in Part IV.

## Question 1

### Reference table

| Row | Reference  | Beneficiary      | Country | Amount (€) |
|-----|------------|------------------|---------|------------|
| 1   | CN-2592/LY | Nieminen, Mika   | FI      | 10,162.90  |
| 2   | MS-4145/KR | Bērziņš, Jānis   | LV      | 66,435.50  |
| 3   | GR-1585/PT | Lindqvist, Sofia | SE      | 50,112.50  |
| 4   | FN-5155/EH | Borg, Joseph     | MT      | 8,225.25   |
| 5   | AD-4814/PX | Ivanov, Ivan     | BG      | 11,873.15  |
| 6   | MS-4911/YW | García, Tomás    | ES      | 66,021.75  |
| 7   | TR-7330/LX | Ferrari, Luca    | IT      | 82,725.15  |
| 8   | FN-7180/EY | Peeters, Emma    | BE      | 17,124.25  |

### Copied record (from row 8)

| Reference  | Beneficiary   | Country | Amount (€) |
|------------|---------------|---------|------------|
| FN-7180/EY | Peeters, Emma | BG      | 17,124.25  |

The record below was copied from row 8 of the reference table. Which part of the copy, if any, contains an error?

- A Reference
- B Amount (€)
- C Country code
- D No error

**Question 2****Reference table**

| Row | Reference  | Beneficiary         | Country | Amount (€) |
|-----|------------|---------------------|---------|------------|
| 1   | FN-7505/LV | Kowalska, Anna      | PL      | 50,441.15  |
| 2   | AD-7388/TG | Ivanov, Ivan        | BG      | 62,708.25  |
| 3   | CN-9071/XW | Papadopoulos, Nikos | EL      | 20,111.50  |
| 4   | TR-1003/DL | Petrović, Ana       | SI      | 65,160.75  |
| 5   | GR-7471/KY | Popescu, Elena      | RO      | 56,254.25  |
| 6   | MS-3986/VA | Novotný, Matej      | SK      | 97,407.50  |
| 7   | CN-2088/GF | Borg, Joseph        | MT      | 51,648.40  |
| 8   | FN-3458/HY | Kovač, Katarina     | HR      | 51,151.25  |

**Copied record (from row 5)**

| Reference  | Beneficiary    | Country | Amount (€) |
|------------|----------------|---------|------------|
| GR-7471/KY | Popescu, Elena | RO      | 56,254.25  |

The record below was copied from row 5 of the reference table. Which part of the copy, if any, contains an error?

- A** Country code
- B** Reference
- C** Amount (€)
- D** No error

**Question 3**

Reference table

| Row | Reference  | Beneficiary     | Country | Amount (€) |
|-----|------------|-----------------|---------|------------|
| 1   | TR-3461/TC | Murphy, Aoife   | IE      | 97,943.50  |
| 2   | CN-6093/XK | Kowalska, Anna  | PL      | 69,764.75  |
| 3   | CN-7186/DH | O'Brien, Seán   | IE      | 50,722.40  |
| 4   | FN-7321/GM | Kask, Liis      | EE      | 54,830.75  |
| 5   | FN-1020/TV | Popescu, Elena  | RO      | 43,427.50  |
| 6   | AD-5413/HJ | García, Tomás   | ES      | 9,629.25   |
| 7   | GR-1279/VF | Silva, Inês     | PT      | 65,206.00  |
| 8   | FN-3880/CT | Janssen, Pieter | BE      | 18,769.50  |

Copied record (from row 8)

| Reference  | Beneficiary     | Country | Amount (€) |
|------------|-----------------|---------|------------|
| FN-3880/TC | Janssen, Pieter | BE      | 18,769.50  |

The record below was copied from row 8 of the reference table. Which part of the copy, if any, contains an error?

- A** Amount (€)
- B** Reference
- C** Country code
- D** No error

# Question 4

## Reference table

| Row | Reference  | Beneficiary      | Country | Amount (€) |
|-----|------------|------------------|---------|------------|
| 1   | FN-8133/TA | Horváth, Zofia   | HU      | 22,221.75  |
| 2   | GR-5501/PG | Georgiou, Maria  | CY      | 74,441.40  |
| 3   | AD-4140/FY | Silva, Inês      | PT      | 35,264.25  |
| 4   | GR-7742/FA | Moreau, Lucas    | FR      | 7,676.00   |
| 5   | FN-1085/EV | Lindqvist, Sofia | SE      | 20,596.25  |
| 6   | MS-2075/YH | Weber, Sophie    | LU      | 42,187.90  |
| 7   | MS-8450/TN | García, Tomás    | ES      | 7,327.00   |
| 8   | TR-4777/MG | Novotný, Matej   | SK      | 53,799.15  |

## Copied record (from row 4)

| Reference  | Beneficiary   | Country | Amount (€) |
|------------|---------------|---------|------------|
| GR-7742/FA | Moreau, Lucas | FR      | 7,766.00   |

The record below was copied from row 4 of the reference table. Which part of the copy, if any, contains an error?

- A Amount (€)
- B Country code
- C Beneficiary
- D No error

**Question 5****Reference table**

| Row | Reference  | Beneficiary     | Country | Amount (€) |
|-----|------------|-----------------|---------|------------|
| 1   | TR-7559/DC | Hansen, Lars    | DK      | 11,234.00  |
| 2   | FN-8710/HU | Georgiou, Maria | CY      | 63,387.90  |
| 3   | FN-1824/KP | Kowalska, Anna  | PL      | 40,193.25  |
| 4   | AD-1588/NP | Janssen, Pieter | BE      | 51,469.90  |
| 5   | MS-7275/XD | Novák, Eva      | CZ      | 48,438.40  |
| 6   | TR-7625/EE | Peeters, Emma   | BE      | 68,495.40  |
| 7   | MS-2837/EU | Popescu, Elena  | RO      | 77,636.15  |
| 8   | FN-5906/AU | Murphy, Aoife   | IE      | 21,658.15  |

**Copied record (from row 6)**

| Reference  | Beneficiary   | Country | Amount (€) |
|------------|---------------|---------|------------|
| TR-7625/EE | Peeters, Emma | BE      | 68,495.40  |

The record below was copied from row 6 of the reference table. Which part of the copy, if any, contains an error?

- A** Amount (€)
- B** Beneficiary
- C** Reference
- D** No error

## Question 6

### Reference table

| Row | Reference  | Beneficiary       | Country | Amount (€) |
|-----|------------|-------------------|---------|------------|
| 1   | GR-9837/WW | Ivanov, Ivan      | BG      | 56,918.90  |
| 2   | AD-7718/VH | O'Brien, Seán     | IE      | 75,238.15  |
| 3   | AD-6675/MM | Nieminen, Mika    | FI      | 2,215.40   |
| 4   | FN-2205/PH | Petratis, Jonas   | LT      | 31,485.40  |
| 5   | AD-6809/UU | Dumont, Marie     | BE      | 50,403.00  |
| 6   | AD-1990/XY | Peeters, Emma     | BE      | 26,027.40  |
| 7   | FN-8068/DW | Wiśniewski, Piotr | PL      | 14,324.25  |
| 8   | CN-5255/FJ | Popescu, Elena    | RO      | 59,164.25  |

### Copied record (from row 2)

| Reference  | Beneficiary   | Country | Amount (€) |
|------------|---------------|---------|------------|
| AD-7718/VH | O'Brien, Seán | IE      | 75,328.15  |

The record below was copied from row 2 of the reference table. Which part of the copy, if any, contains an error?

- A Beneficiary
- B Amount (€)
- C Reference
- D No error



**Question 7****Reference table**

| Row | Reference  | Beneficiary       | Country | Amount (€) |
|-----|------------|-------------------|---------|------------|
| 1   | AD-6655/VL | Wiśniewski, Piotr | PL      | 41,755.90  |
| 2   | MS-9201/RM | Fischer, Georg    | AT      | 68,892.75  |
| 3   | MS-4071/TF | Virtanen, Hanna   | FI      | 51,729.90  |
| 4   | GR-2442/XN | García, Tomás     | ES      | 32,352.90  |
| 5   | TR-4931/FY | Kowalska, Anna    | PL      | 32,213.25  |
| 6   | MS-7710/WF | O'Brien, Seán     | IE      | 55,887.75  |
| 7   | FN-3588/NU | Kovač, Katarina   | HR      | 51,406.00  |
| 8   | GR-1577/TX | Horváth, Zofia    | HU      | 39,175.25  |

**Copied record (from row 5)**

| Reference  | Beneficiary    | Country | Amount (€) |
|------------|----------------|---------|------------|
| TR-4391/FY | Kowalska, Anna | PL      | 32,213.25  |

The record below was copied from row 5 of the reference table. Which part of the copy, if any, contains an error?

- A** Amount (€)
- B** Country code
- C** Reference
- D** No error

**Question 8****Reference table**

| Row | Reference  | Beneficiary         | Country | Amount (€) |
|-----|------------|---------------------|---------|------------|
| 1   | TR-1986/RG | Kask, Liis          | EE      | 65,811.40  |
| 2   | CN-6300/AD | Weber, Sophie       | LU      | 8,792.00   |
| 3   | GR-2619/VD | Silva, Inês         | PT      | 4,699.75   |
| 4   | MS-1251/VD | Bērziņš, Jānis      | LV      | 45,875.75  |
| 5   | CN-8504/TK | Papadopoulos, Nikos | EL      | 53,625.00  |
| 6   | TR-3859/JJ | Kovač, Katarina     | HR      | 82,453.75  |
| 7   | CN-8150/FG | Schmidt, Clara      | DE      | 39,378.90  |
| 8   | TR-7311/FU | Borg, Joseph        | MT      | 44,575.40  |

**Copied record (from row 4)**

| Reference  | Beneficiary    | Country | Amount (€) |
|------------|----------------|---------|------------|
| MS-1251/VD | Brēziņš, Jānis | LV      | 45,875.75  |

The record below was copied from row 4 of the reference table. Which part of the copy, if any, contains an error?

- A** Beneficiary
- B** Reference
- C** Amount (€)
- D** No error

**Question 9****Reference table**

| Row | Reference  | Beneficiary         | Country | Amount (€) |
|-----|------------|---------------------|---------|------------|
| 1   | FN-5841/MP | Janssen, Pieter     | BE      | 62,036.40  |
| 2   | MS-8688/AU | Hansen, Lars        | DK      | 77,966.75  |
| 3   | FN-4801/VD | Papadopoulos, Nikos | EL      | 1,481.50   |
| 4   | FN-6155/RC | García, Tomás       | ES      | 28,045.15  |
| 5   | MS-7389/PD | Kask, Liis          | EE      | 94,535.25  |
| 6   | MS-1617/LX | Georgiou, Maria     | CY      | 30,937.40  |
| 7   | CN-9245/GN | Moreau, Lucas       | FR      | 74,409.25  |
| 8   | CN-1316/GT | Nowak, Mateusz      | PL      | 51,325.15  |

**Copied record (from row 4)**

| Reference  | Beneficiary   | Country | Amount (€) |
|------------|---------------|---------|------------|
| FN-6155/RC | García, Tomás | EE      | 28,045.15  |

The record below was copied from row 4 of the reference table. Which part of the copy, if any, contains an error?

- A** Beneficiary
- B** Country code
- C** Reference
- D** No error

**Question 10****Reference table**

| Row | Reference  | Beneficiary       | Country | Amount (€) |
|-----|------------|-------------------|---------|------------|
| 1   | GR-8640/DY | Janssen, Pieter   | BE      | 51,857.40  |
| 2   | MS-6338/WN | O'Brien, Seán     | IE      | 17,848.40  |
| 3   | MS-8915/JG | Dumont, Marie     | BE      | 13,615.40  |
| 4   | FN-7651/LR | Weber, Sophie     | LU      | 33,497.15  |
| 5   | FN-5175/PN | Kovač, Katarina   | HR      | 79,209.50  |
| 6   | FN-7424/WP | Andersson, Henrik | SE      | 7,043.00   |
| 7   | FN-4800/LK | Fischer, Georg    | AT      | 71,981.25  |
| 8   | FN-4769/HW | Novák, Eva        | CZ      | 535.15     |

**Copied record (from row 4)**

| Reference  | Beneficiary   | Country | Amount (€) |
|------------|---------------|---------|------------|
| FN-7651/LR | Wbeer, Sophie | LU      | 33,497.15  |

The record below was copied from row 4 of the reference table. Which part of the copy, if any, contains an error?

- A** Beneficiary
- B** Reference
- C** Amount (€)
- D** No error

**Question 11****Reference table**

| Row | Reference  | Beneficiary      | Country | Amount (€) |
|-----|------------|------------------|---------|------------|
| 1   | TR-4013/MC | Novotný, Matej   | SK      | 69,597.40  |
| 2   | CN-1023/TJ | Silva, Inês      | PT      | 10,821.00  |
| 3   | GR-2836/UL | Peeters, Emma    | BE      | 4,417.15   |
| 4   | AD-5515/KA | Popescu, Elena   | RO      | 96,555.75  |
| 5   | FN-7130/PM | Moreau, Lucas    | FR      | 58,914.90  |
| 6   | FN-7475/PM | Lindqvist, Sofia | SE      | 9,900.25   |
| 7   | TR-4161/NK | Nieminen, Mika   | FI      | 37,898.40  |
| 8   | TR-4348/GV | Kask, Liis       | EE      | 33,972.90  |

**Copied record (from row 2)**

| Reference  | Beneficiary | Country | Amount (€) |
|------------|-------------|---------|------------|
| CN-1023/TJ | Silva, Inês | PT      | 10,812.00  |

The record below was copied from row 2 of the reference table. Which part of the copy, if any, contains an error?

- A** Beneficiary
- B** Reference
- C** Amount (€)
- D** No error

**Question 12****Reference table**

| Row | Reference  | Beneficiary       | Country | Amount (€) |
|-----|------------|-------------------|---------|------------|
| 1   | AD-7830/MA | Andersson, Henrik | SE      | 78,696.25  |
| 2   | AD-7216/LN | Borg, Joseph      | MT      | 48,840.40  |
| 3   | FN-4057/VH | Petrakis, Jonas   | LT      | 2,069.00   |
| 4   | MS-3405/RG | Novák, Eva        | CZ      | 55,536.75  |
| 5   | TR-6118/GM | Peeters, Emma     | BE      | 68,704.15  |
| 6   | CN-6922/RK | Hansen, Lars      | DK      | 48,592.50  |
| 7   | GR-3961/RH | Janssen, Pieter   | BE      | 90,714.50  |
| 8   | FN-4131/PN | García, Tomás     | ES      | 62,100.15  |

**Copied record (from row 4)**

| Reference  | Beneficiary | Country | Amount (€) |
|------------|-------------|---------|------------|
| MS-3405/RG | Novák, Eva  | CZ      | 55,536.75  |

The record below was copied from row 4 of the reference table. Which part of the copy, if any, contains an error?

- A** Reference
- B** Beneficiary
- C** Amount (€)
- D** No error

**Question 13**

Reference table

| Row | Reference  | Beneficiary         | Country | Amount (€) |
|-----|------------|---------------------|---------|------------|
| 1   | AD-8503/DM | Papadopoulos, Nikos | EL      | 41,860.50  |
| 2   | MS-9035/PN | Dumont, Marie       | BE      | 67,715.15  |
| 3   | CN-5256/TN | Virtanen, Hanna     | FI      | 37,154.00  |
| 4   | MS-8844/LY | Georgiou, Maria     | CY      | 65,459.75  |
| 5   | TR-1252/HP | Lindqvist, Sofia    | SE      | 18,999.50  |
| 6   | TR-4550/UM | Borg, Joseph        | MT      | 34,531.90  |
| 7   | CN-8683/DD | Kowalska, Anna      | PL      | 6,671.40   |
| 8   | FN-8814/EY | Novotný, Matej      | SK      | 63,376.25  |
| 9   | GR-5367/NK | Petrović, Ana       | SI      | 81,568.50  |
| 10  | FN-9866/JG | Moreau, Lucas       | FR      | 69,071.25  |

Which of the following records appears in the reference table EXACTLY as shown?

| Option | Reference  | Beneficiary     | Country | Amount (€) |
|--------|------------|-----------------|---------|------------|
| A      | CN-5256/TN | Virtanen, Hanna | FI      | 37,154.00  |
| B      | GR-5367/NK | Petrović, Ana   | SK      | 81,568.50  |
| C      | TR-4850/UM | Borg, Joseph    | MT      | 34,531.90  |
| D      | FN-8814/YE | Novotný, Matej  | SK      | 63,376.25  |

# Question 14

## Reference table

| Row | Reference  | Beneficiary       | Country | Amount (€) |
|-----|------------|-------------------|---------|------------|
| 1   | GR-7709/PX | Petrakis, Jonas   | LT      | 85,201.15  |
| 2   | CN-3025/VM | Fischer, Georg    | AT      | 14,926.40  |
| 3   | CN-8811/MK | Kask, Liis        | EE      | 63,726.75  |
| 4   | TR-4063/UJ | Dumont, Marie     | BE      | 67,594.75  |
| 5   | CN-1410/HM | Kovač, Katarina   | HR      | 53,230.40  |
| 6   | GR-7092/LJ | Kowalska, Anna    | PL      | 40,994.15  |
| 7   | TR-2312/GA | Virtanen, Hanna   | FI      | 44,490.50  |
| 8   | AD-1424/WJ | Ferrari, Luca     | IT      | 55,375.50  |
| 9   | TR-7019/DP | Andersson, Henrik | SE      | 7,450.90   |
| 10  | MS-8156/LM | Nowak, Mateusz    | PL      | 18,640.25  |

Which of the following records appears in the reference table EXACTLY as shown?

| Option | Reference  | Beneficiary      | Country | Amount (€) |
|--------|------------|------------------|---------|------------|
| A      | CN-1410/HM | Kovač, Katharina | HR      | 53,230.40  |
| B      | MS-8156/LM | Noawk, Mateusz   | PL      | 18,640.25  |
| C      | AD-1424/WJ | Ferrari, Luca    | IE      | 55,375.50  |
| D      | GR-7092/LJ | Kowalska, Anna   | PL      | 40,994.15  |



**Question 15****Reference table**

| Row | Reference  | Beneficiary     | Country | Amount (€) |
|-----|------------|-----------------|---------|------------|
| 1   | FN-4050/VW | Virtanen, Hanna | FI      | 46,589.90  |
| 2   | FN-8391/VW | Novotný, Matej  | SK      | 51,619.90  |
| 3   | MS-5201/AT | Borg, Joseph    | MT      | 32,871.00  |
| 4   | CN-8886/VM | Schmidt, Clara  | DE      | 28,370.25  |
| 5   | AD-4122/YY | Bērziņš, Jānis  | LV      | 49,787.40  |
| 6   | FN-8871/RJ | O'Brien, Seán   | IE      | 33,240.75  |
| 7   | TR-3826/VA | Nieminen, Mika  | FI      | 26,024.25  |
| 8   | FN-8208/PU | Kovač, Katarina | HR      | 1,825.90   |
| 9   | GR-2880/ED | Janssen, Pieter | BE      | 69,735.50  |
| 10  | CN-7050/VA | Popescu, Elena  | RO      | 83,328.15  |

Which of the following records appears in the reference table EXACTLY as shown?

| Option | Reference  | Beneficiary    | Country | Amount (€) |
|--------|------------|----------------|---------|------------|
| A      | TR-3826/VA | Nieminen, Mika | FI      | 26,024.25  |
| B      | CN-7055/VA | Popescu, Elena | RO      | 83,328.15  |
| C      | CN-8868/VM | Schmidt, Clara | DE      | 28,370.25  |
| D      | FN-8391/VW | Novotný, Matej | SK      | 51,169.90  |

## Question 16

### Reference table

| Row | Reference  | Beneficiary       | Country | Amount (€) |
|-----|------------|-------------------|---------|------------|
| 1   | AD-5963/RR | Kask, Liis        | EE      | 85,565.15  |
| 2   | AD-7778/LV | Silva, Inês       | PT      | 30,940.75  |
| 3   | CN-8384/KJ | Ivanov, Ivan      | BG      | 3,966.75   |
| 4   | AD-7833/EM | Wiśniewski, Piotr | PL      | 88,161.90  |
| 5   | AD-9162/XY | Popescu, Elena    | RO      | 23,059.40  |
| 6   | FN-3419/RE | Petrakis, Jonas   | LT      | 54,510.40  |
| 7   | TR-4185/TA | Virtanen, Hanna   | FI      | 49,190.40  |
| 8   | GR-6610/RE | Fischer, Georg    | AT      | 66,279.15  |
| 9   | FN-5818/GN | Ferrari, Luca     | IT      | 17,121.25  |
| 10  | GR-3500/XC | Weber, Sophie     | LU      | 97,110.40  |

Which of the following records appears in the reference table EXACTLY as shown?

| Option | Reference  | Beneficiary       | Country | Amount (€) |
|--------|------------|-------------------|---------|------------|
| A      | AD-5963/RR | Kask, Lis         | EE      | 85,565.15  |
| B      | FN-3419/RE | Petrakis, Jonas   | LT      | 54,510.04  |
| C      | AD-7833/EM | Wiśniewski, Piotr | PT      | 88,161.90  |
| D      | TR-4185/TA | Virtanen, Hanna   | FI      | 49,190.40  |

**Question 17****Reference table**

| Row | Reference  | Beneficiary       | Country | Amount (€) |
|-----|------------|-------------------|---------|------------|
| 1   | MS-1168/YE | Peeters, Emma     | BE      | 32,110.90  |
| 2   | MS-7122/AM | Virtanen, Hanna   | FI      | 97,843.15  |
| 3   | TR-1850/YG | Hansen, Lars      | DK      | 93,037.40  |
| 4   | FN-4457/KW | García, Tomás     | ES      | 31,800.00  |
| 5   | FN-1191/YK | Nieminen, Mika    | FI      | 1,796.75   |
| 6   | GR-3021/UU | Moreau, Lucas     | FR      | 5,121.00   |
| 7   | FN-1146/NC | Wiśniewski, Piotr | PL      | 40,655.15  |
| 8   | TR-1918/XL | Novák, Eva        | CZ      | 94,868.90  |
| 9   | FN-3467/YY | Andersson, Henrik | SE      | 52,093.00  |
| 10  | FN-9188/JN | Ivanov, Ivan      | BG      | 13,561.75  |

Which of the following records appears in the reference table EXACTLY as shown?

| Option | Reference  | Beneficiary      | Country | Amount (€) |
|--------|------------|------------------|---------|------------|
| A      | TR-1850/YG | Hansen, Lars     | DK      | 93,307.40  |
| B      | TR-1918/XL | Novák, Eva       | CZ      | 94,868.90  |
| C      | FN-4457/KW | García, Tomás    | EE      | 31,800.00  |
| D      | FN-3467/YY | Anderson, Henrik | SE      | 52,093.00  |

## Question 18

Reference table

| Row | Reference  | Beneficiary       | Country | Amount (€) |
|-----|------------|-------------------|---------|------------|
| 1   | CN-2450/CU | García, Tomás     | ES      | 38,505.40  |
| 2   | TR-5173/XK | Wiśniewski, Piotr | PL      | 76,491.75  |
| 3   | CN-1448/GN | Ferrari, Luca     | IT      | 69,985.75  |
| 4   | AD-1791/VM | Georgiou, Maria   | CY      | 88,581.50  |
| 5   | GR-8811/KV | Peeters, Emma     | BE      | 77,915.40  |
| 6   | MS-7145/KE | Novotný, Matej    | SK      | 15,137.15  |
| 7   | AD-1889/GU | Rossi, Matteo     | IT      | 31,128.75  |
| 8   | AD-4495/KE | O'Brien, Seán     | IE      | 87,601.15  |
| 9   | MS-1670/LK | Horváth, Zofia    | HU      | 94,020.40  |
| 10  | GR-5711/WJ | Petratis, Jonas   | LT      | 42,883.40  |

Which of the following records appears in the reference table EXACTLY as shown?

| Option | Reference  | Beneficiary     | Country | Amount (€) |
|--------|------------|-----------------|---------|------------|
| A      | AD-1889/GU | Rossi, Matteo   | IT      | 31,128.75  |
| B      | AD-1791/VM | Georigou, Maria | CY      | 88,581.50  |
| C      | GR-8911/KV | Peeters, Emma   | BE      | 77,915.40  |
| D      | MS-1670/LK | Horvath, Zofia  | HU      | 94,020.40  |

**Question 19****Reference table**

| Row | Reference  | Beneficiary       | Country | Amount (€) |
|-----|------------|-------------------|---------|------------|
| 1   | GR-8891/RX | Wiśniewski, Piotr | PL      | 65,015.00  |
| 2   | CN-5765/WD | Silva, Inês       | PT      | 72,332.90  |
| 3   | MS-8905/RC | Kask, Liis        | EE      | 8,161.40   |
| 4   | CN-3820/EJ | Ferrari, Luca     | IT      | 18,106.90  |
| 5   | AD-7551/PT | Petrović, Ana     | SI      | 33,739.50  |
| 6   | FN-8319/XR | Peeters, Emma     | BE      | 80,368.00  |
| 7   | AD-1010/DT | O'Brien, Seán     | IE      | 2,460.15   |
| 8   | CN-8702/MR | Borg, Joseph      | MT      | 60,696.25  |
| 9   | GR-6805/HH | Kowalska, Anna    | PL      | 9,273.15   |
| 10  | GR-3395/KA | Murphy, Aoife     | IE      | 38,284.40  |

Which of the following records appears in the reference table EXACTLY as shown?

| Option | Reference  | Beneficiary   | Country | Amount (€) |
|--------|------------|---------------|---------|------------|
| A      | CN-8702/MR | Borg, Joseph  | MT      | 60,696.52  |
| B      | CN-3820/EJ | Ferrai, Luca  | IT      | 18,106.90  |
| C      | MS-8905/RC | Kask, Liis    | EE      | 1,861.40   |
| D      | AD-1010/DT | O'Brien, Seán | IE      | 2,460.15   |

## Question 20

### Reference table

| Row | Reference  | Beneficiary      | Country | Amount (€) |
|-----|------------|------------------|---------|------------|
| 1   | FN-9072/FV | Dumont, Marie    | BE      | 35,190.50  |
| 2   | AD-7816/TH | Moreau, Lucas    | FR      | 65,043.90  |
| 3   | AD-6297/EN | Hansen, Lars     | DK      | 78,795.50  |
| 4   | CN-5201/TF | Novotný, Matej   | SK      | 64,834.00  |
| 5   | FN-8064/ND | Bērziņš, Jānis   | LV      | 64,813.25  |
| 6   | GR-7719/MR | Weber, Sophie    | LU      | 35,955.25  |
| 7   | TR-1024/KD | Horváth, Zofia   | HU      | 70,239.50  |
| 8   | GR-4632/KK | Rossi, Matteo    | IT      | 28,710.75  |
| 9   | GR-8904/FT | Kovač, Katarina  | HR      | 25,668.90  |
| 10  | FN-8185/XV | Lindqvist, Sofia | SE      | 27,450.15  |

Which of the following records appears in the reference table EXACTLY as shown?

| Option | Reference  | Beneficiary      | Country | Amount (€) |
|--------|------------|------------------|---------|------------|
| A      | FN-8064/ND | Bērziņš, Jānis   | LT      | 64,813.25  |
| B      | FN-9702/FV | Dumont, Marie    | BE      | 35,190.50  |
| C      | FN-8185/XV | Lindqvist, Sofia | SE      | 27,450.15  |
| D      | GR-8904/FT | Kovač, Katarina  | HR      | 52,668.90  |

**Question 21**

Reference table

| Row | Reference  | Beneficiary         | Country | Amount (€) |
|-----|------------|---------------------|---------|------------|
| 1   | FN-3011/ML | Petraitis, Jonas    | LT      | 42,822.75  |
| 2   | MS-7203/XJ | Wiśniewski, Piotr   | PL      | 80,567.90  |
| 3   | FN-8514/AT | Weber, Sophie       | LU      | 52,537.90  |
| 4   | FN-7305/XL | Novotný, Matej      | SK      | 58,545.15  |
| 5   | AD-7120/NV | Popescu, Elena      | RO      | 31,937.25  |
| 6   | TR-4346/DU | Papadopoulos, Nikos | EL      | 29,067.50  |
| 7   | MS-9889/PA | Novák, Eva          | CZ      | 15,917.00  |
| 8   | MS-7550/PF | Petrović, Ana       | SI      | 24,718.15  |
| 9   | GR-1211/YN | Nieminen, Mika      | FI      | 36,075.40  |
| 10  | FN-1103/YY | Andersson, Henrik   | SE      | 21,719.40  |

Which of the following records appears in the reference table EXACTLY as shown?

| Option | Reference  | Beneficiary       | Country | Amount (€) |
|--------|------------|-------------------|---------|------------|
| A      | FN-3011/ML | Petraitis, Jonas  | LT      | 42,822.75  |
| B      | MS-7023/XJ | Wiśniewski, Piotr | PL      | 80,567.90  |
| C      | FN-8514/AT | Weber, Sophie     | LT      | 52,537.90  |
| D      | MS-7550/PF | Pertović, Ana     | SI      | 24,718.15  |

## Question 22

### Reference table

| Row | Reference  | Beneficiary      | Country | Amount (€) |
|-----|------------|------------------|---------|------------|
| 1   | CN-9597/XH | Dumont, Marie    | BE      | 85,558.75  |
| 2   | AD-8713/XG | Lindqvist, Sofia | SE      | 94,389.40  |
| 3   | GR-4941/HG | Rossi, Matteo    | IT      | 10,504.00  |
| 4   | CN-7284/FE | Horváth, Zofia   | HU      | 60,915.90  |
| 5   | GR-7008/GE | Hansen, Lars     | DK      | 81,012.40  |
| 6   | MS-8504/VP | Weber, Sophie    | LU      | 22,727.00  |
| 7   | GR-1500/KF | Silva, Inês      | PT      | 72,251.75  |
| 8   | TR-4859/GV | Ferrari, Luca    | IT      | 18,412.40  |
| 9   | TR-5008/UD | O'Brien, Seán    | IE      | 48,531.00  |
| 10  | FN-6701/LF | Peeters, Emma    | BE      | 11,557.40  |

Which of the following records appears in the reference table EXACTLY as shown?

| Option | Reference  | Beneficiary      | Country | Amount (€) |
|--------|------------|------------------|---------|------------|
| A      | AD-8713/XG | Lindqvist, Sofia | SE      | 94,389.40  |
| B      | GR-1500/KF | Silva, Inês      | PT      | 72,251.75  |
| C      | GR-7008/GE | Hansen, Lars     | DK      | 80,112.40  |
| D      | GR-4941/HG | Rossi, Matteo    | IT      | 10,504.00  |



**Question 23****Reference table**

| Row | Reference  | Beneficiary     | Country | Amount (€) |
|-----|------------|-----------------|---------|------------|
| 1   | GR-1894/DN | Kask, Liis      | EE      | 76,736.90  |
| 2   | AD-3311/NF | Moreau, Lucas   | FR      | 31,077.90  |
| 3   | MS-8861/KJ | Ferrari, Luca   | IT      | 76,023.90  |
| 4   | MS-2958/CL | García, Tomás   | ES      | 88,506.40  |
| 5   | AD-8454/VX | Petratis, Jonas | LT      | 14,388.90  |
| 6   | AD-2085/XH | Kowalska, Anna  | PL      | 92,907.40  |
| 7   | TR-4692/PV | Georgiou, Maria | CY      | 77,823.40  |
| 8   | CN-1798/LK | Schmidt, Clara  | DE      | 12,049.40  |
| 9   | FN-3581/NT | Petrović, Ana   | SI      | 16,488.00  |
| 10  | CN-4182/JV | Ivanov, Ivan    | BG      | 80,388.15  |

**Copied records**

| Copy | Source row | Reference  | Beneficiary   | Country | Amount (€) |
|------|------------|------------|---------------|---------|------------|
| 1    | 9          | FN-3581/NT | Petrović, Ana | SI      | 16,488.00  |
| 2    | 1          | GR-1894/DN | Kask, Liis    | EE      | 76,736.90  |
| 3    | 2          | AD-331I/NF | Moreau, Lucas | FR      | 31,077.90  |
| 4    | 3          | MS-8861/KJ | Ferrari, Luka | IT      | 76,023.90  |
| 5    | 4          | MS-3958/CL | García, Tomás | ES      | 88,506.40  |

Five records were copied from the reference table (the source row is shown for each). How many of the copies contain at least one error?

- A** 2
- B** 3
- C** 4
- D** 5

## Question 24

### Reference table

| Row | Reference  | Beneficiary      | Country | Amount (€) |
|-----|------------|------------------|---------|------------|
| 1   | TR-1606/DL | García, Tomás    | ES      | 97,960.00  |
| 2   | MS-3971/LU | Petrović, Ana    | SI      | 17,963.75  |
| 3   | AD-8407/DF | Moreau, Lucas    | FR      | 24,673.90  |
| 4   | MS-3001/VC | Horváth, Zofia   | HU      | 70,149.15  |
| 5   | FN-7885/TW | Lindqvist, Sofia | SE      | 55,446.15  |
| 6   | TR-1826/FH | Nowak, Mateusz   | PL      | 46,834.00  |
| 7   | GR-5305/NF | Georgiou, Maria  | CY      | 69,607.90  |
| 8   | FN-5440/UF | Nieminen, Mika   | FI      | 72,737.90  |
| 9   | GR-7598/KC | Weber, Sophie    | LU      | 93,255.90  |
| 10  | AD-7611/VG | Dumont, Marie    | BE      | 64,413.75  |

### Copied records

| Copy | Source row | Reference  | Beneficiary     | Country | Amount (€) |
|------|------------|------------|-----------------|---------|------------|
| 1    | 1          | TR-1606/DL | Garíca, Tomás   | ES      | 97,960.00  |
| 2    | 2          | MS-3971/LU | Petroivć, Ana   | SI      | 17,963.75  |
| 3    | 7          | GR-5305/NF | Georgiou, Maria | CY      | 69,607.90  |
| 4    | 8          | FN-5440/UF | Nieminen, Mika  | FI      | 72,737.90  |
| 5    | 3          | AD-8407/DF | Moreau, Lucas   | FR      | 24,673.90  |

Five records were copied from the reference table (the source row is shown for each). How many of the copies contain at least one error?

- A 1
- B 2
- C 3
- D 4

**Question 25****Reference table**

| Row | Reference  | Beneficiary         | Country | Amount (€) |
|-----|------------|---------------------|---------|------------|
| 1   | AD-1117/AT | Janssen, Pieter     | BE      | 14,998.90  |
| 2   | MS-9058/NK | García, Tomás       | ES      | 35,137.15  |
| 3   | GR-5895/EH | Georgiou, Maria     | CY      | 94,396.15  |
| 4   | FN-2316/ND | Novotný, Matej      | SK      | 46,626.40  |
| 5   | GR-8115/XC | Papadopoulos, Nikos | EL      | 65,936.00  |
| 6   | MS-7297/TH | O'Brien, Seán       | IE      | 32,031.00  |
| 7   | MS-3004/VF | Bērziņš, Jānis      | LV      | 69,232.40  |
| 8   | MS-2159/WC | Rossi, Matteo       | IT      | 56,863.40  |
| 9   | AD-3879/UV | Murphy, Aoife       | IE      | 61,776.25  |
| 10  | CN-1163/HY | Peeters, Emma       | BE      | 4,068.00   |

**Copied records**

| Copy | Source row | Reference  | Beneficiary         | Country | Amount (€) |
|------|------------|------------|---------------------|---------|------------|
| 1    | 1          | AD-1117/AT | Janssen, Pieter     | BE      | 14,998.90  |
| 2    | 6          | MS-7297/TH | O'Brien, Seán       | IE      | 32,031.00  |
| 3    | 7          | MS-3004/VF | Bērziņš, Jānis      | LT      | 69,232.40  |
| 4    | 3          | GR-5395/EH | Georgiou, Maria     | CY      | 94,396.15  |
| 5    | 5          | GR-8115/XC | Papadopoulos, Nikos | EL      | 65,936.00  |

Five records were copied from the reference table (the source row is shown for each). How many of the copies contain at least one error?

- A** 2
- B** 3
- C** 4
- D** 5

## Question 26

### Reference table

| Row | Reference  | Beneficiary         | Country | Amount (€) |
|-----|------------|---------------------|---------|------------|
| 1   | MS-9311/TA | Georgiou, Maria     | CY      | 39,509.15  |
| 2   | FN-1183/KU | Fischer, Georg      | AT      | 32,628.50  |
| 3   | FN-8129/EU | Petratis, Jonas     | LT      | 56,319.40  |
| 4   | MS-1704/HL | Papadopoulos, Nikos | EL      | 12,068.50  |
| 5   | CN-5831/HJ | Bērziņš, Jānis      | LV      | 5,336.50   |
| 6   | TR-3384/GP | Lindqvist, Sofia    | SE      | 9,036.75   |
| 7   | CN-5608/AJ | Rossi, Matteo       | IT      | 10,013.90  |
| 8   | MS-3991/FC | Andersson, Henrik   | SE      | 2,368.75   |
| 9   | CN-8688/VE | Moreau, Lucas       | FR      | 45,825.75  |
| 10  | FN-5281/JD | Schmidt, Clara      | DE      | 78,433.00  |

### Copied records

| Copy | Source row | Reference  | Beneficiary         | Country | Amount (€) |
|------|------------|------------|---------------------|---------|------------|
| 1    | 10         | FN-5281/JD | Schmidt, Klara      | DE      | 78,433.00  |
| 2    | 4          | MS-1704/HL | Papadopoulos, Nikos | EL      | 12,068.50  |
| 3    | 7          | CN-5608/AJ | Rossi, Matteo       | IT      | 10,013.90  |
| 4    | 1          | MS-9311/TA | Georgiou, Maria     | CY      | 39,509.15  |
| 5    | 9          | CN-8688/VE | Moreau, Lucas       | FR      | 45,825.57  |

Five records were copied from the reference table (the source row is shown for each). How many of the copies contain at least one error?

- A 2
- B 3
- C 4
- D 5

**Question 27****Reference table**

| Row | Reference  | Beneficiary       | Country | Amount (€) |
|-----|------------|-------------------|---------|------------|
| 1   | TR-5166/NE | Wiśniewski, Piotr | PL      | 12,760.75  |
| 2   | GR-1705/FY | Popescu, Elena    | RO      | 88,119.25  |
| 3   | FN-5351/RW | Janssen, Pieter   | BE      | 38,592.90  |
| 4   | AD-7849/HM | Nieminen, Mika    | FI      | 64,290.40  |
| 5   | TR-4242/RC | Petraitis, Jonas  | LT      | 69,525.50  |
| 6   | TR-1266/HM | Rossi, Matteo     | IT      | 27,132.15  |
| 7   | CN-8608/LV | Lindqvist, Sofia  | SE      | 3,652.25   |
| 8   | CN-4838/UV | Fischer, Georg    | AT      | 95,305.25  |
| 9   | MS-5470/AC | Hansen, Lars      | DK      | 83,254.90  |
| 10  | AD-1710/FP | Kask, Liis        | EE      | 71,092.15  |

**Copied records**

| Copy | Source row | Reference  | Beneficiary      | Country | Amount (€) |
|------|------------|------------|------------------|---------|------------|
| 1    | 5          | TR-4242/CR | Petraitis, Jonas | LT      | 69,525.50  |
| 2    | 3          | FN-5351/RW | Janssen, Pieter  | BE      | 38,592.90  |
| 3    | 8          | CN-4838/UV | Fischer, Georg   | AT      | 95,305.52  |
| 4    | 10         | AD-1710/FP | Kask, Liis       | EE      | 71,092.15  |
| 5    | 7          | CN-8608/LV | Lindqvist, Sofia | SE      | 3,652.25   |

Five records were copied from the reference table (the source row is shown for each). How many of the copies contain at least one error?

- A** 0
- B** 1
- C** 2
- D** 3

## Question 28

### Reference table

| Row | Reference  | Beneficiary         | Country | Amount (€) |
|-----|------------|---------------------|---------|------------|
| 1   | AD-5558/RC | Fischer, Georg      | AT      | 26,958.25  |
| 2   | CN-5817/DK | Silva, Inês         | PT      | 28,099.50  |
| 3   | AD-2851/AK | Dumont, Marie       | BE      | 20,132.40  |
| 4   | AD-7020/TK | Popescu, Elena      | RO      | 53,087.50  |
| 5   | AD-5358/LC | Novák, Eva          | CZ      | 10,125.75  |
| 6   | FN-9015/TC | Papadopoulos, Nikos | EL      | 979.75     |
| 7   | AD-6000/NG | Novotný, Matej      | SK      | 79,170.40  |
| 8   | AD-8782/FH | Georgiou, Maria     | CY      | 19,536.15  |
| 9   | GR-5950/ET | O'Brien, Seán       | IE      | 45,997.25  |
| 10  | MS-4897/CA | Petratis, Jonas     | LT      | 26,699.15  |

### Copied records

| Copy | Source row | Reference  | Beneficiary         | Country | Amount (€) |
|------|------------|------------|---------------------|---------|------------|
| 1    | 6          | FN-9015/TC | Papadopoulos, Nikos | EL      | 979.75     |
| 2    | 9          | GR-5950/ET | O'Brien, Seán       | IE      | 45,997.25  |
| 3    | 5          | AD-5358/LC | Novák, Eva          | CZ      | 10,125.75  |
| 4    | 3          | AD-2851/AK | Dumont, Marie       | BE      | 20,132.40  |
| 5    | 2          | CN-5817/DK | Silva, Inês         | PT      | 28,099.50  |

Five records were copied from the reference table (the source row is shown for each). How many of the copies contain at least one error?

- A 0
- B 1
- C 2
- D 3

**Question 29****Reference table**

| Row | Reference  | Beneficiary      | Country | Amount (€) |
|-----|------------|------------------|---------|------------|
| 1   | TR-7107/AU | Borg, Joseph     | MT      | 95,711.75  |
| 2   | GR-7878/FK | Peeters, Emma    | BE      | 49,374.00  |
| 3   | MS-2985/CG | Schmidt, Clara   | DE      | 86,119.00  |
| 4   | CN-5583/MY | O'Brien, Seán    | IE      | 18,602.40  |
| 5   | AD-1806/NA | Kowalska, Anna   | PL      | 57,333.40  |
| 6   | MS-7983/KM | Lindqvist, Sofia | SE      | 96,030.25  |
| 7   | TR-3817/JH | Virtanen, Hanna  | FI      | 53,085.00  |
| 8   | CN-2815/LK | Fischer, Georg   | AT      | 50,487.50  |
| 9   | CN-7110/HV | Popescu, Elena   | RO      | 81,551.25  |
| 10  | TR-4393/CR | Nowak, Mateusz   | PL      | 10,971.90  |

**Copied records**

| Copy | Source row | Reference  | Beneficiary      | Country | Amount (€) |
|------|------------|------------|------------------|---------|------------|
| 1    | 5          | AD-1806/NA | Kowalska, Anna   | PL      | 57,333.40  |
| 2    | 3          | MS-2985/CG | Schmidt, Clara   | DE      | 86,119.00  |
| 3    | 6          | MS-7983/KM | Lindqvist, Sofia | SE      | 96,030.25  |
| 4    | 10         | TR-4393/CR | Nowak, Mateusz   | PT      | 10,971.90  |
| 5    | 2          | GR-7878/FK | Peeters, Emma    | BE      | 49,374.00  |

Five records were copied from the reference table (the source row is shown for each). How many of the copies contain at least one error?

- A** 0
- B** 1
- C** 2
- D** 3

### Question 30

#### Reference table

| Row | Reference  | Beneficiary       | Country | Amount (€) |
|-----|------------|-------------------|---------|------------|
| 1   | FN-9671/TW | Peeters, Emma     | BE      | 4,838.15   |
| 2   | MS-8224/WM | Novotný, Matej    | SK      | 17,837.50  |
| 3   | FN-7087/UN | Bērziņš, Jānis    | LV      | 42,661.90  |
| 4   | TR-4135/YW | O'Brien, Seán     | IE      | 1,545.25   |
| 5   | TR-4018/TN | Moreau, Lucas     | FR      | 64,934.00  |
| 6   | AD-1580/LR | Kask, Liis        | EE      | 4,750.50   |
| 7   | CN-7811/MU | Petraitis, Jonas  | LT      | 41,915.40  |
| 8   | MS-6113/JK | Ferrari, Luca     | IT      | 17,415.75  |
| 9   | AD-7835/NW | Nowak, Mateusz    | PL      | 84,845.15  |
| 10  | MS-1640/AD | Andersson, Henrik | SE      | 37,828.75  |

#### Copied records

| Copy | Source row | Reference  | Beneficiary      | Country | Amount (€) |
|------|------------|------------|------------------|---------|------------|
| 1    | 3          | FN-7087/UN | Bērziņš, Jānis   | LT      | 42,661.90  |
| 2    | 9          | AD-7835/NW | Nowak, Mateusz   | PL      | 84,845.15  |
| 3    | 7          | CN-7811/MU | Petraitis, Jonas | LT      | 41,915.40  |
| 4    | 1          | FN-6971/TW | Peeters, Emma    | BE      | 4,838.15   |
| 5    | 2          | MS-8224/WM | Novotný, Matej   | SK      | 17,837.50  |

Five records were copied from the reference table (the source row is shown for each). How many of the copies contain at least one error?

- A 0
- B 1
- C 2
- D 3



# Practice Set — Prioritising & Organising

Use the information given to plan, schedule or allocate work. Read every condition before choosing. Suggested pace: about 2 minutes 30 seconds per question.

24 questions · answers and worked solutions in Part IV.

## Question 1

Who is busy, by day and start time

| Start       | Mon         | Tue     | Wed   | Thu | Fri   |
|-------------|-------------|---------|-------|-----|-------|
| 09:00       | C E         | B F     | B C E | F   | B C F |
| 10:00       | C E         | B C     | —     | B F | C F   |
| 11:00       | C E         | B C E F | E F   | B   | B E   |
| 12:00–14:00 | lunch break |         |       |     |       |
| 14:00       | E           | B C     | B E F | B C | E F   |
| 15:00       | E F         | B C E   | B C   | —   | B C E |
| 16:00       | B C E F     | B C     | C F   | C F | C E F |

The table shows who is BUSY (unavailable) during the one-hour slot starting at the time shown; — means that all four colleagues are free. No meetings are held between 12:00 and 14:00. Key: B = Bruno, C = Chiara, E = Eleni, F = Filip.

**A one-hour meeting must be attended by Bruno, Chiara and Eleni; Filip's attendance is optional, so Filip's availability can be ignored. The room is unavailable on Wednesday, and the meeting must start no later than 15:00. What is the earliest suitable slot in the week?**

- A** Thursday 09:00
- B** Wednesday 10:00
- C** Tuesday 09:00
- D** Thursday 15:00

**Question 2**

Who is busy, by day and start time

| Start       | Mon         | Tue   | Wed   | Thu | Fri   |
|-------------|-------------|-------|-------|-----|-------|
| 09:00       | —           | F C B | E     | C E | F C B |
| 10:00       | F B         | C     | C E   | E   | C     |
| 11:00       | C B E       | F E   | F C B | E   | C     |
| 12:00–14:00 | lunch break |       |       |     |       |
| 14:00       | —           | F C B | B E   | B   | C     |
| 15:00       | —           | F C B | C     | F   | F C E |
| 16:00       | C E         | F B   | F B E | F C | —     |

The table shows who is BUSY (unavailable) during the one-hour slot starting at the time shown; — means that all four colleagues are free. No meetings are held between 12:00 and 14:00. Key: F = Filip, C = Chiara, B = Bruno, E = Eleni.

**A one-hour meeting must be attended by Filip, Chiara and Bruno; Eleni's attendance is optional, so Eleni's availability can be ignored. The room is unavailable on Monday, and the meeting must start no later than 14:00. What is the earliest suitable slot in the week?**

- A** Monday 09:00
- B** Tuesday 10:00
- C** Thursday 10:00
- D** Wednesday 09:00

**Question 3**

Who is busy, by day and start time

| Start       | Mon         | Tue   | Wed   | Thu | Fri   |
|-------------|-------------|-------|-------|-----|-------|
| 09:00       | B           | F     | B D C | D F | B     |
| 10:00       | —           | D F   | —     | C   | C     |
| 11:00       | B D F       | B D C | B D F | F C | D F C |
| 12:00–14:00 | lunch break |       |       |     |       |
| 14:00       | D C         | B F C | B C   | F   | D     |
| 15:00       | B D F       | C     | D C   | D   | D F C |
| 16:00       | —           | B D   | D F C | C   | C     |

The table shows who is BUSY (unavailable) during the one-hour slot starting at the time shown; — means that all four colleagues are free. No meetings are held between 12:00 and 14:00. Key: B = Bruno, D = Dawid, F = Filip, C = Chiara.

**A one-hour meeting must be attended by Bruno, Dawid and Filip; Chiara's attendance is optional, so Chiara's availability can be ignored. The room is unavailable on Monday, and the meeting must start no later than 15:00. What is the earliest suitable slot in the week?**

- A** Monday 10:00
- B** Tuesday 09:00
- C** Tuesday 15:00
- D** Thursday 10:00

### Question 4

Who is busy, by day and start time

| Start       | Mon         | Tue   | Wed   | Thu   | Fri |
|-------------|-------------|-------|-------|-------|-----|
| 09:00       | F E C D     | C     | F     | E D   | E C |
| 10:00       | E           | E D   | —     | F D   | E C |
| 11:00       | F D         | F E D | F E   | D     | E D |
| 12:00–14:00 | lunch break |       |       |       |     |
| 14:00       | C D         | F C D | F C D | F D   | C D |
| 15:00       | F C         | F E D | F     | F E C | F D |
| 16:00       | —           | F E   | F E   | —     | F E |

The table shows who is BUSY (unavailable) during the one-hour slot starting at the time shown; — means that all four colleagues are free. No meetings are held between 12:00 and 14:00. Key: F = Filip, E = Eleni, C = Chiara, D = Dawid.

**A one-hour meeting must be attended by Filip, Eleni and Chiara; Dawid's attendance is optional, so Dawid's availability can be ignored. The room is unavailable on Wednesday, and the meeting must start no later than 15:00. What is the earliest suitable slot in the week?**

- A** Wednesday 10:00
- B** Thursday 11:00
- C** Monday 16:00
- D** Monday 11:00

### Question 5

Who is busy, by day and start time

| Start       | Mon         | Tue   | Wed   | Thu   | Fri     |
|-------------|-------------|-------|-------|-------|---------|
| 09:00       | E           | F     | C F   | C B   | C B F   |
| 10:00       | B E         | C F E | —     | B F E | F E     |
| 11:00       | B F E       | E     | B     | C F E | B F E   |
| 12:00–14:00 | lunch break |       |       |       |         |
| 14:00       | C F E       | C F E | C F   | C F   | C B F   |
| 15:00       | C B         | C     | B F E | B     | C B F E |
| 16:00       | F           | —     | C B E | F E   | F E     |

The table shows who is BUSY (unavailable) during the one-hour slot starting at the time shown; — means that all four colleagues are free. No meetings are held between 12:00 and 14:00. Key: C = Chiara, B = Bruno, F = Filip, E = Eleni.

**A one-hour meeting must be attended by Chiara, Bruno and Filip; Eleni's attendance is optional, so Eleni's availability can be ignored. The room is unavailable on Monday, and the meeting must start no later than 15:00. What is the earliest suitable slot in the week?**

- A** Monday 09:00
- B** Tuesday 09:00
- C** Tuesday 11:00
- D** Wednesday 10:00

### Question 6

Who is busy, by day and start time

| Start       | Mon         | Tue   | Wed   | Thu     | Fri   |
|-------------|-------------|-------|-------|---------|-------|
| 09:00       | F C         | C A   | D F C | —       | D F A |
| 10:00       | C           | D F C | A     | D F C A | F C   |
| 11:00       | A           | D C A | D C A | D C A   | C A   |
| 12:00–14:00 | lunch break |       |       |         |       |
| 14:00       | D F C A     | C     | C A   | D F C   | D F C |
| 15:00       | A           | D A   | F C A | D C A   | —     |
| 16:00       | F           | F C A | C     | F C A   | D F C |

The table shows who is BUSY (unavailable) during the one-hour slot starting at the time shown; — means that all four colleagues are free. No meetings are held between 12:00 and 14:00. Key: D = Dawid, F = Filip, C = Chiara, A = Anna.

**A one-hour meeting must be attended by Dawid, Filip and Chiara; Anna's attendance is optional, so Anna's availability can be ignored. The room is unavailable on Monday, and the meeting must start no later than 15:00. What is the earliest suitable slot in the week?**

- A Monday 11:00
- B Thursday 09:00
- C Tuesday 09:00
- D Wednesday 10:00

### Question 7

Who is busy, by day and start time

| Start       | Mon         | Tue | Wed | Thu | Fri   |
|-------------|-------------|-----|-----|-----|-------|
| 09:00       | A           | D   | A E | —   | A E D |
| 10:00       | A B         | E D | E   | E B | B     |
| 11:00       | B           | E B | —   | A E | D     |
| 12:00–14:00 | lunch break |     |     |     |       |
| 14:00       | A           | E D | —   | E   | E D   |
| 15:00       | E B         | —   | —   | —   | A E D |
| 16:00       | D           | —   | D   | D   | A B   |

The table shows who is BUSY (unavailable) during the one-hour slot starting at the time shown; — means that all four colleagues are free. No meetings are held between 12:00 and 14:00. Key: A = Anna, E = Eleni, B = Bruno, D = Dawid.

**A one-hour meeting must be attended by Anna, Eleni and Bruno; Dawid's attendance is optional, so Dawid's availability can be ignored. The room is unavailable on Wednesday, and the meeting must start no later than 15:00. What is the earliest suitable slot in the week?**

- A Thursday 15:00
- B Tuesday 09:00
- C Monday 16:00
- D Monday 14:00

**Question 8**

Who is busy, by day and start time

| Start       | Mon         | Tue     | Wed     | Thu   | Fri     |
|-------------|-------------|---------|---------|-------|---------|
| 09:00       | B C D       | C D     | A       | D     | A       |
| 10:00       | B C         | D       | B A C D | D     | B A C D |
| 11:00       | A C D       | B A C   | A D     | B C   | B       |
| 12:00–14:00 | lunch break |         |         |       |         |
| 14:00       | B C D       | B A D   | A C D   | A     | B A C   |
| 15:00       | B C D       | D       | A D     | B C D | B       |
| 16:00       | B C D       | B A C D | B A C D | B     | B A D   |

The table shows who is BUSY (unavailable) during the one-hour slot starting at the time shown; — means that all four colleagues are free. No meetings are held between 12:00 and 14:00. Key: B = Bruno, A = Anna, C = Chiara, D = Dawid.

**A one-hour meeting must be attended by Bruno, Anna and Chiara; Dawid's attendance is optional, so Dawid's availability can be ignored. The room is unavailable on Tuesday, and the meeting must start no later than 15:00. What is the earliest suitable slot in the week?**

- A** Thursday 09:00
- B** Tuesday 10:00
- C** Wednesday 09:00
- D** Thursday 10:00

**Question 9**

Who is busy, by day and start time

| Start       | Mon         | Tue     | Wed     | Thu | Fri   |
|-------------|-------------|---------|---------|-----|-------|
| 09:00       | D           | B F A   | B D     | F A | B F A |
| 10:00       | —           | B F A D | A D     | —   | F A   |
| 11:00       | B F D       | B F D   | F A     | F A | B A D |
| 12:00–14:00 | lunch break |         |         |     |       |
| 14:00       | B A         | B F A D | B F A D | D   | A     |
| 15:00       | B F         | B F A   | D       | F   | B F D |
| 16:00       | F A D       | B F     | A       | F A | A D   |

The table shows who is BUSY (unavailable) during the one-hour slot starting at the time shown; — means that all four colleagues are free. No meetings are held between 12:00 and 14:00. Key: B = Bruno, F = Filip, A = Anna, D = Dawid.

**A one-hour meeting must be attended by Bruno, Filip and Anna; Dawid's attendance is optional, so Dawid's availability can be ignored. The room is unavailable on Monday, and the meeting must start no later than 15:00. What is the earliest suitable slot in the week?**

- A** Monday 09:00
- B** Thursday 10:00
- C** Wednesday 15:00
- D** Wednesday 09:00

### Question 10

#### Project tasks

| Task           | Working days | Can start only after |
|----------------|--------------|----------------------|
| Collect data   | 6            | —                    |
| Draft analysis | 6            | Collect data         |
| Legal check    | 4            | Draft analysis       |
| Translation    | 6            | Draft analysis       |
| Layout         | 6            | Translation          |
| Final approval | 5            | Legal check, Layout  |

A unit must publish a policy report. Tasks can run in parallel as soon as all their prerequisites are complete, and there are enough staff to work on any number of tasks at once. What is the minimum number of working days needed to complete everything?

- A 33
- B 24
- C 21
- D 29

### Question 11

#### Project tasks

| Task              | Working days | Can start only after               |
|-------------------|--------------|------------------------------------|
| Book venue        | 4            | —                                  |
| Draft programme   | 1            | —                                  |
| Invite speakers   | 3            | Draft programme                    |
| Open registration | 4            | Book venue, Invite speakers        |
| Print materials   | 6            | Draft programme                    |
| Final briefing    | 4            | Open registration, Print materials |

A team is organising a stakeholder conference. Tasks can run in parallel as soon as all their prerequisites are complete, and there are enough staff to work on any number of tasks at once. What is the minimum number of working days needed to complete everything?

- A 12
- B 18
- C 22
- D 8

**Question 12****Project tasks**

| Task                | Working days | Can start only after                       |
|---------------------|--------------|--------------------------------------------|
| Gather requirements | 3            | —                                          |
| Configure system    | 3            | Gather requirements                        |
| Security review     | 2            | Configure system                           |
| Migrate data        | 6            | Configure system                           |
| Train users         | 5            | Configure system                           |
| Go-live check       | 3            | Security review, Migrate data, Train users |

An IT team is rolling out a new case-management tool. Tasks can run in parallel as soon as all their prerequisites are complete, and there are enough staff to work on any number of tasks at once. What is the minimum number of working days needed to complete everything?

- A** 22
- B** 15
- C** 11
- D** 12

**Question 13****Project tasks**

| Task                | Working days | Can start only after               |
|---------------------|--------------|------------------------------------|
| Publish vacancy     | 6            | —                                  |
| Screen applications | 2            | Publish vacancy                    |
| Prepare tests       | 4            | —                                  |
| Run tests           | 6            | Screen applications, Prepare tests |
| Hold interviews     | 4            | Run tests                          |
| Final decision      | 5            | Hold interviews                    |

A selection board is running a recruitment exercise. Tasks can run in parallel as soon as all their prerequisites are complete, and there are enough staff to work on any number of tasks at once. What is the minimum number of working days needed to complete everything?

- A** 18
- B** 27
- C** 19
- D** 23

**Question 14****Project tasks**

| Task           | Working days | Can start only after |
|----------------|--------------|----------------------|
| Collect data   | 1            | —                    |
| Draft analysis | 2            | Collect data         |
| Legal check    | 4            | Draft analysis       |
| Translation    | 2            | Draft analysis       |
| Layout         | 6            | Translation          |
| Final approval | 6            | Legal check, Layout  |

A unit must publish a policy report. Tasks can run in parallel as soon as all their prerequisites are complete, and there are enough staff to work on any number of tasks at once. What is the minimum number of working days needed to complete everything?

- A 13
- B 11
- C 17
- D 21

**Question 15****Project tasks**

| Task              | Working days | Can start only after               |
|-------------------|--------------|------------------------------------|
| Book venue        | 2            | —                                  |
| Draft programme   | 3            | —                                  |
| Invite speakers   | 5            | Draft programme                    |
| Open registration | 3            | Book venue, Draft programme        |
| Print materials   | 4            | Invite speakers                    |
| Final briefing    | 6            | Open registration, Print materials |

A team is organising a stakeholder conference. Tasks can run in parallel as soon as all their prerequisites are complete, and there are enough staff to work on any number of tasks at once. What is the minimum number of working days needed to complete everything?

- A 23
- B 18
- C 12
- D 11



**Question 16****Project tasks**

| Task                | Working days | Can start only after                       |
|---------------------|--------------|--------------------------------------------|
| Gather requirements | 1            | —                                          |
| Configure system    | 1            | Gather requirements                        |
| Security review     | 2            | Configure system                           |
| Migrate data        | 6            | Configure system                           |
| Train users         | 4            | Configure system                           |
| Go-live check       | 3            | Security review, Migrate data, Train users |

An IT team is rolling out a new case-management tool. Tasks can run in parallel as soon as all their prerequisites are complete, and there are enough staff to work on any number of tasks at once. What is the minimum number of working days needed to complete everything?

- A 11
- B 17
- C 9
- D 7

**Question 17****Project tasks**

| Task                | Working days | Can start only after               |
|---------------------|--------------|------------------------------------|
| Publish vacancy     | 6            | —                                  |
| Screen applications | 4            | Publish vacancy                    |
| Prepare tests       | 2            | Publish vacancy                    |
| Run tests           | 5            | Screen applications, Prepare tests |
| Hold interviews     | 4            | Run tests                          |
| Final decision      | 6            | Hold interviews                    |

A selection board is running a recruitment exercise. Tasks can run in parallel as soon as all their prerequisites are complete, and there are enough staff to work on any number of tasks at once. What is the minimum number of working days needed to complete everything?

- A 19
- B 25
- C 23
- D 27

**Question 18**

Four colleagues — Bruno, Chiara, Dawid, Filip — must each take exactly one of four tasks: the budget report, the event logistics, the website update and the press summary. Taking all the constraints listed into account, who must take the budget report?

**Constraints**

- Dawid cannot take the event logistics.
- Dawid cannot take the website update.
- Dawid cannot take the budget report.
- Bruno must take either the press summary or the event logistics.
- Chiara cannot take the website update.

- A** Bruno
- B** Dawid
- C** Filip
- D** Chiara

**Question 19**

Four colleagues — Bruno, Chiara, Dawid, Filip — must each take exactly one of four tasks: the budget report, the event logistics, the website update and the press summary. Taking all the constraints listed into account, who must take the press summary?

**Constraints**

- Filip cannot take the event logistics.
- Dawid cannot take the event logistics.
- Chiara cannot take the press summary.
- Dawid cannot take the budget report.
- Chiara cannot take the event logistics.
- Chiara cannot take the budget report.

- A** Dawid
- B** Chiara
- C** Bruno
- D** Filip

**Question 20**

Four colleagues — Dawid, Filip, Eleni, Chiara — must each take exactly one of four tasks: the budget report, the event logistics, the website update and the press summary. Taking all the constraints listed into account, who must take the press summary?

**Constraints**

- Filip cannot take the budget report.
- Dawid cannot take the press summary.
- Eleni must take either the press summary or the budget report.
- Dawid cannot take the budget report.
- Filip cannot take the event logistics.
- Dawid cannot take the event logistics.

- A** Chiara
- B** Dawid
- C** Filip
- D** Eleni

**Question 21**

Four colleagues — Dawid, Anna, Bruno, Filip — must each take exactly one of four tasks: the budget report, the event logistics, the website update and the press summary. Taking all the constraints listed into account, who must take the event logistics?

**Constraints**

- Filip cannot take the event logistics.
- Filip cannot take the press summary.
- Anna cannot take the press summary.
- Bruno cannot take the press summary.
- Bruno cannot take the website update.
- Bruno cannot take the event logistics.

- A** Filip
- B** Bruno
- C** Dawid
- D** Anna

**Question 22**

Four colleagues — Chiara, Dawid, Filip, Anna — must each take exactly one of four tasks: the budget report, the event logistics, the website update and the press summary. Taking all the constraints listed into account, who must take the budget report?

**Constraints**

- Anna cannot take the press summary.
- Anna cannot take the budget report.
- Chiara cannot take the press summary.
- Chiara cannot take the website update.
- Dawid must take either the budget report or the press summary.
- Anna cannot take the website update.

- A** Anna
- B** Dawid
- C** Chiara
- D** Filip

**Question 23**

Four colleagues — Dawid, Filip, Bruno, Anna — must each take exactly one of four tasks: the budget report, the event logistics, the website update and the press summary. Taking all the constraints listed into account, who must take the press summary?

**Constraints**

- Bruno cannot take the website update.
- Bruno cannot take the press summary.
- Filip cannot take the website update.
- Filip cannot take the budget report.
- Filip cannot take the press summary.
- Anna cannot take the website update.

- A** Filip
- B** Anna
- C** Bruno
- D** Dawid

**Question 24**

Four colleagues — Eleni, Filip, Anna, Dawid — must each take exactly one of four tasks: the budget report, the event logistics, the website update and the press summary. Taking all the constraints listed into account, who must take the budget report?

**Constraints**

- Anna cannot take the event logistics.
- Anna cannot take the website update.
- Filip cannot take the event logistics.
- Eleni cannot take the event logistics.
- Filip cannot take the budget report.
- Anna cannot take the budget report.

- A** Eleni
- B** Anna
- C** Dawid
- D** Filip



PART III

# Mock Exams

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*Three timed mock exams combining verbal, numerical and abstract reasoning.  
Sit each one in a single session.*





# Mock Exam 1

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**40 questions • suggested time: 65 minutes**

- Questions 1–20: verbal reasoning (about 35 minutes)
- Questions 21–30: numerical reasoning (about 20 minutes)
- Questions 31–40: abstract reasoning (about 10 minutes)

These timings reflect typical past EPSO formats; check the Notice of Competition for your procedure. Use the answer sheet at the back of the book, work without interruptions, and answer every question: in typical formats there is no penalty for a wrong answer.

**Score guide (out of 40).** 34–40: excellent, test-ready. 28–33: strong; polish weaker areas. 22–27: developing; revisit the method chapters and redo your errors. Below 22: build accuracy first with the practice sets. Real pass marks differ between procedures, so treat these bands as a guide only.

## Verbal reasoning (questions 1–20)

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### Question 1

Many organisms, from cyanobacteria to humans, display circadian rhythms, internal cycles of roughly 24 hours that regulate processes such as sleep, body temperature and hormone release. In humans, a small region of the brain called the suprachiasmatic nucleus acts as a master clock, but it does not keep perfect time on its own. Instead, it is continually reset by external cues, of which light is the most powerful. Light reaching the eyes in the morning tends to advance the clock, whereas bright light late in the evening tends to delay it. Experiments in which volunteers lived for weeks without clocks or daylight showed that their rhythms persisted but drifted, typically settling into cycles slightly longer than 24 hours. Problems arise when internal time and external demands diverge, as in night-shift work or rapid travel across several time zones. Workers on rotating night shifts frequently report poor sleep and digestive complaints, and researchers continue to investigate whether longer-term health problems can be attributed to the disruption itself or to other features of shift work.

Which of the following statements is correct?

- A Without time cues, the body clock keeps running but drifts from the 24-hour day.
- B Exposure to bright light late in the evening tends to make the body clock run earlier.
- C Research has established that rotating night shifts cause long-term illness.
- D Only humans and other mammals possess circadian rhythms.

### Question 2

Most plants fix carbon dioxide from the air using an enzyme known as Rubisco. Despite its central role, the enzyme is inefficient in one important respect: it can react with oxygen instead of carbon dioxide, triggering a process called photorespiration that wastes energy and releases carbon the plant has already fixed. Photorespiration becomes more frequent as temperatures rise and when plants close the pores in their leaves to conserve water, because the concentration of carbon dioxide inside the leaf then falls. Some plants, including maize and sugar cane, have evolved a mechanism known as C4 photosynthesis. These plants first capture carbon dioxide in outer leaf cells and then pump it into inner cells where Rubisco is located, keeping the concentration of carbon dioxide around the enzyme high and largely suppressing photorespiration. As a result, C4 plants tend to perform well in hot, sunny and dry conditions. The mechanism carries an energy cost, however, so in cool, moist climates plants without it are often equally or more productive.

Which of the following statements is correct?

- A C4 plants avoid photorespiration because they do not use the enzyme Rubisco at all.
- B Closing its leaf pores to save water can make a plant fix carbon less efficiently.
- C C4 plants are more productive than other plants in every climate.
- D Photorespiration becomes less frequent as temperatures rise.

**Question 3**

Peatlands are wetlands in which waterlogged conditions slow the decomposition of dead plant material, so that it accumulates over thousands of years as peat. Although they cover only a small share of the world's land surface, peatlands store a very large quantity of carbon – more, by many estimates, than all the world's forests combined. When peatlands are drained for agriculture, for forestry or for the extraction of peat as fuel or for horticulture, the peat is exposed to air and begins to decompose, releasing carbon dioxide. Drained peat is also more vulnerable to fire. Restoring peatlands usually involves blocking drainage channels to raise the water table again, a process known as rewetting. Rewetting can greatly reduce emissions from degraded sites, but it does not immediately return a peatland to its original condition, and the recovery of characteristic vegetation such as sphagnum mosses may take many years. In some cases, rewetting may also cause a temporary increase in emissions of methane, another greenhouse gas.

**Which of the following statements is correct?**

- A** Peatlands cover a large proportion of the earth's land surface.
- B** Rewetting a drained peatland quickly restores its original vegetation.
- C** Peat forms because waterlogging speeds up the decay of plant material.
- D** Draining a peatland can turn it into a source of carbon dioxide emissions.

**Question 4**

The presumption of innocence is a cornerstone of criminal justice in democratic societies. It means that a person accused of an offence is treated as innocent until proven guilty according to law. Its most important practical consequence concerns the burden of proof: it is for the prosecution to establish guilt, not for the accused to prove innocence. In most systems, the prosecution must do so to a high standard, often expressed as proof “beyond reasonable doubt”. Where doubt remains, it must benefit the accused. The presumption also has implications outside the courtroom. Public officials are expected to refrain from statements that present a suspect as guilty before a verdict, since such statements may prejudice the proceedings. Civil disputes between private parties, by contrast, are usually decided on a lower standard, under which a claim succeeds if it is shown to be more likely than not. As a result, the same facts may lead to an acquittal in criminal proceedings yet give rise to liability in civil proceedings.

**Which of the following statements is correct?**

- A** An accused person must present evidence of innocence in order to be acquitted.
- B** A criminal acquittal does not rule out civil liability for the same facts.
- C** Civil courts usually apply the same standard of proof as criminal courts.
- D** Public officials are prohibited from making any comment on a criminal investigation.

**Question 5**

Psychologists use the term anchoring to describe the tendency of people to rely heavily on the first piece of numerical information they encounter when making an estimate. In a classic type of experiment, participants are first asked whether a quantity – the height of a tree, say, or the price of a bottle of wine – is higher or lower than a given figure, and are then asked for their own estimate. Their answers tend to move towards the initial figure, even when participants have been told that it was chosen at random. The effect has been observed among experts as well as among laypeople, although its strength varies from one study to another. Researchers have proposed several explanations. One is that people start from the anchor and then adjust insufficiently; another is that the anchor brings to mind information consistent with it. Practical advice for decision-makers follows from these findings: in negotiations or budgeting, it may be useful to form an independent estimate before looking at any figure proposed by the other side.

**Which of the following statements is correct?**

- A** An initial figure can influence people's estimates even when they know that it has no meaningful basis.
- B** Experts are protected from anchoring because their specialist knowledge allows them to ignore irrelevant figures.
- C** Anchoring has been found to have the same strength in every study that has tested it.
- D** Researchers have firmly established that insufficient adjustment from the anchor is the only cause of the effect.

**Question 6**

A regional capital has adopted a ten-year plan to increase the share of trips made by bicycle. The plan's first phase, now completed, created cycle lanes physically separated from motor traffic along four main radial roads. Counts on these routes show that cycling increased substantially after the lanes opened, with the largest growth among women and older residents – groups that earlier surveys had identified as particularly deterred by mixed traffic. The second phase, which the council has approved but not yet funded, would link the radial routes with an orbital lane. Officials acknowledge that part of the increase on the new lanes may reflect cyclists switching from parallel streets rather than new cycling trips. A citywide household travel survey, to be repeated every two years, is intended to clarify this point. The plan also provided secure bicycle parking at the main railway station, which is reported to be operating at full capacity on weekdays.

**Which of the following statements is NOT supported by the passage?**

- A** Concern about sharing the road with motor vehicles had discouraged some residents from cycling.
- B** Funding for the orbital lane has not yet been secured.
- C** The rise in counts on the radial routes may overstate the growth in cycling across the city as a whole.
- D** The household travel survey has confirmed that the new lanes generated additional cycling trips.

**Question 7**

Research over several decades suggests that sleep plays an active role in memory, rather than simply protecting newly learned information from interference. In typical experiments, participants learn a task or a list of word pairs and are tested after an interval that contains either sleep or an equivalent period of wakefulness. Those who have slept generally perform better, and the benefit is often linked to specific stages of sleep. Deep, slow-wave sleep appears particularly important for remembering facts and events, while some studies associate rapid eye movement (REM) sleep with emotional memories and certain skills. During slow-wave sleep, patterns of brain activity recorded during learning are replayed, which is thought to help transfer memories from temporary storage in the hippocampus to longer-term networks in the cortex. The findings have practical implications for students, although researchers caution that most experiments involve simple laboratory tasks and short intervals, and that the effect of sleep on complex learning over weeks or months is less well understood.

**Which of the following statements is correct?**

- A** Sleep benefits memory only by shielding new information from interference.
- B** REM sleep appears to be the stage most important for remembering facts and events.
- C** Findings from short, simple laboratory tasks may not fully apply to complex learning over several months.
- D** Students who sleep more than their peers will achieve better examination results.

**Question 8**

Languages constantly borrow words from one another, and the pattern of borrowing often reflects historical contact between communities. English offers a well-known example. After the Norman Conquest, French became for several centuries the language of the court, the law and much of the aristocracy, while English continued to be spoken by the great majority of the population. A trace of this division is often seen in pairs of words for animals and their meat: the living animals generally kept English names – cow, sheep, swine – while the meat served at table acquired names of French origin, such as beef, mutton and pork. Linguists caution, however, against treating such neat examples as a complete explanation, since some patterns have more than one cause. Borrowing is also not limited to vocabulary for new objects or ideas; it may replace existing native words, especially in prestigious domains. A large share of the English vocabulary is of French or Latin origin, yet the most frequently used words, such as articles, pronouns and common verbs, remain overwhelmingly Germanic.

**Which of the following statements is correct?**

- A** After the Norman Conquest, most people in England switched to speaking French.
- B** A borrowed word does not necessarily name something that was previously unknown to the borrowing community.
- C** Because so much of its vocabulary comes from French or Latin, English is classified as a Romance language.
- D** The animal and meat word pairs prove that every difference in English vocabulary can be traced to social class.

**Question 9**

The so-called paradox of thrift describes a situation in which a decision that is sensible for each household becomes harmful when everyone takes it at once. If a single family decides to spend less and save more, its own financial position improves and the wider economy is scarcely affected. If, however, most households cut their spending simultaneously – for example, because they fear losing their jobs – businesses face a fall in demand for their goods and services. Firms may respond by reducing output and laying off workers, which lowers incomes. With lower incomes, households may be unable to save as much as they had planned, so that total saving in the economy does not rise and may even fall. The argument is most closely associated with economists who stress the role of demand in determining output in the short run. Critics point out that the paradox applies chiefly when an economy has unused capacity; when resources are fully employed, higher saving can finance investment and support growth over the longer term.

**Which of the following statements is correct?**

- A** Higher saving is always damaging to the economy as a whole.
- B** A single household that decides to save more will cause firms to cut their output.
- C** Critics of the paradox of thrift maintain that it is invalid under all economic conditions.
- D** When many households try to save more at once, total saving may fail to rise.

**Question 10**

In many parts of the world, children begin school in a language they do not speak at home. Research on early-years education suggests that this can impede learning, since pupils must master a new language and new content simultaneously. Programmes that teach literacy first in the home language and introduce the official language gradually have generally produced stronger reading skills – including, after several years, in the second language itself. Researchers attribute this to the transfer of underlying skills: a child who understands how written symbols represent sounds in one language can apply that understanding to another. Such programmes are not without difficulty. They require teaching materials in each language, teachers who are proficient in it and, in linguistically diverse areas, decisions about which of many home languages to support. Some parents also resist them, fearing that time spent on the home language will slow their children's acquisition of the language needed for further education and employment – a concern that the research largely does not bear out.

**Which of the following statements is correct?**

- A** Learning to read first in the home language can eventually strengthen reading in a second language.
- B** Parents generally prefer home-language programmes to instruction in the official language.
- C** Home-language programmes are straightforward to implement in linguistically diverse areas.
- D** Children taught first in their home language show stronger reading skills in the official language from their first year of school.

**Question 11**

In a clinical trial, patients who receive an inactive substance often report some improvement in their symptoms. This placebo response can arise from patients' expectations, from the natural tendency of many conditions to fluctuate or improve over time, and from the extra attention that trial participants receive. To separate the genuine effect of a treatment from these influences, researchers compare a group receiving the treatment with a control group receiving a placebo that looks identical. In a double-blind trial, neither the patients nor the researchers assessing them know who is in which group until the results are analysed. This matters because researchers who know which patients received the active treatment may, without intending to, interpret ambiguous outcomes more favourably. Placebo responses tend to be largest for outcomes that patients report themselves, such as pain or fatigue, and smallest for objective measurements, such as blood test results. Blinding is not always possible: in trials of surgical procedures or exercise programmes, for instance, participants usually know which intervention they have received.

**Which of the following statements is correct?**

- A** Improvements reported by patients given a placebo are caused entirely by their expectations.
- B** Blinding assessors reduces the risk that their knowledge of the allocation leads them to favour the treatment.
- C** Placebo responses tend to be greatest for objective measurements such as blood tests.
- D** Trials of exercise programmes cannot produce reliable results.

**Question 12**

When an organisation moves a large share of its staff to remote working, the effects on productivity are often measured through output indicators such as tasks completed or files processed. These indicators may capture routine work well, but they tend to overlook the informal exchanges through which much organisational knowledge circulates: the quick question to a colleague, the conversation overheard in a corridor, the advice offered to a newcomer. Studies of knowledge work suggest that such exchanges matter disproportionately for less experienced employees, who rely on them to learn tacit aspects of the job that are rarely written down. Experienced staff, whose networks are already established, may find their productivity unchanged or even improved when working remotely, since they face fewer interruptions. The resulting risk is that an organisation judging remote work solely by short-term output may conclude that it has no costs, while the learning of its junior staff quietly suffers. Some organisations have responded by designating regular office days for teams or by formalising mentoring.

**Which of the following statements is correct?**

- A** Remote working reduces the productivity of experienced employees.
- B** Output indicators may give an incomplete picture of the effects of remote working.
- C** Organisations have generally responded to remote working by introducing compulsory office days.
- D** Junior employees are less productive than experienced ones because they are interrupted more often.

**Question 13**

A regional education authority reduced class sizes in the first two years of primary school from an average of around twenty-eight pupils to around twenty in a sample of schools, while comparable schools continued with larger classes. Schools were assigned to the two groups by lottery, so that differences in outcomes could more confidently be attributed to class size. After two years, pupils in the smaller classes scored moderately higher in reading tests, and the difference was larger for pupils from low-income families. In mathematics, the difference between the two groups was small and not statistically significant. Teachers in smaller classes reported spending less time managing behaviour, although classroom observations found that many continued to teach in much the same way as before. The authority estimated that extending smaller classes to all schools would require several hundred additional teachers, as well as extra classroom space. It has not yet decided whether to do so, and some advisers have suggested that the same money might achieve more if spent on targeted support for struggling readers.

**Which of the following statements is correct?**

- A** Smaller classes produced significantly better results in both reading and mathematics.
- B** Teachers in the smaller classes substantially changed their teaching methods.
- C** The reading gap in favour of smaller classes was wider among pupils from low-income families.
- D** The authority has decided to reduce class sizes in all of its primary schools.

**Question 14**

An agency's code of conduct regulates gifts and hospitality offered to officials in connection with their duties. Gifts of purely symbolic value – a pen, a calendar, a modest book – may be accepted without formality. Any other gift must be declined, unless refusing it would cause offence, as may happen during official visits abroad. In that case the official may accept the gift on behalf of the agency, but must register it within ten working days and hand it over; it then becomes the property of the agency. Hospitality is treated differently. Officials may accept a meal or an invitation to an event where their attendance serves the agency's interests, but not where the host is a party to a procedure that the official is currently handling, such as a tender or an inspection. In cases of doubt, officials are expected to consult the ethics adviser before accepting, and a written opinion from the adviser protects them from later disciplinary action, provided they have followed it. A gift offered through a member of the official's family is treated as if it had been offered to the official directly.

**Which of the following statements is correct?**

- A** Officials may keep a gift of more than symbolic value if refusing it would cause offence.
- B** An official handling a tender may accept a dinner from a bidding company if attending serves the agency's interests.
- C** A gift accepted to avoid causing offence ends up belonging to the agency, not to the official.
- D** Officials are required to consult the ethics adviser before accepting any form of hospitality whatsoever.



**Question 15**

A regional tax authority has begun using an automated system to select returns for detailed audit. The system assigns each return a risk score based on patterns found in past cases in which errors or fraud were identified. Returns above a certain score are referred to human inspectors, who decide whether to open an audit; the system itself cannot impose penalties or reach conclusions about a taxpayer's liability. Supporters argue that the approach allows limited staff to concentrate on the cases most likely to yield results. An internal review found that audits selected with the system's help recovered more unpaid tax per inspector hour than those selected by the previous method. The review cautioned, however, that because the system learns from past audit decisions, it may reproduce any biases in how cases were chosen in the past, and that taxpayers in groups rarely audited before may be under-represented in the data. It recommended regular independent testing of the system's outputs. The authority has also been asked by an oversight body to explain how taxpayers can find out whether an automated score contributed to their selection.

**Which of the following statements is NOT supported by the passage?**

- A** The decision to open an audit rests with human inspectors rather than with the automated system.
- B** Audits picked with the system's assistance brought in more unpaid tax per hour of inspectors' time than those picked by the old method.
- C** Groups that were seldom audited in the past may be poorly represented in the data on which the system relies.
- D** The authority already has a procedure for telling taxpayers whether an automated score contributed to their selection.

**Question 16**

A coastal municipality has been debating how to respond to repeated flooding of low-lying farmland near an estuary. Engineers initially recommended raising the existing sea wall, but a study commissioned by the council suggested that “managed realignment” – deliberately allowing the sea to reclaim a strip of farmland behind a new, lower embankment further inland – would be cheaper to maintain over several decades. The study added that the resulting salt marsh could absorb wave energy and provide habitat for wading birds, provided that the land was not subsequently drained. In its formal response, the council approved funding for a detailed feasibility assessment of realignment on one section of the coast, while agreeing to carry out urgent repairs to the sea wall elsewhere. Landowners whose fields would be affected have asked for compensation; the council has stated that it will consider compensation arrangements only if the feasibility assessment recommends proceeding. Environmental groups have welcomed the study but cautioned that a salt marsh can take many years to become fully established.

**Which of the following statements is correct?**

- A** So far, the council’s only step towards realignment is to fund a feasibility study for part of the coastline.
- B** The council has agreed to compensate the landowners whose fields would be affected by realignment.
- C** The study concluded that managed realignment would be cheaper to construct than raising the sea wall.
- D** Once the farmland is flooded, the new salt marsh will immediately provide habitat for wading birds.

**Question 17**

Superconductivity is a state in which certain materials conduct electricity with no resistance at all. It was discovered in 1911, when mercury cooled to a few degrees above absolute zero was found to lose its electrical resistance abruptly. For most of the twentieth century, known superconductors worked only at such extremely low temperatures, which required cooling with costly liquid helium. In 1986, researchers found a class of ceramic materials that became superconducting at much higher temperatures, and soon afterwards related materials were found that could be cooled with liquid nitrogen, which is far cheaper and easier to handle. Even these “high-temperature” superconductors, however, must still be kept far below the freezing point of water. Superconducting magnets are already used routinely in hospital scanners for magnetic resonance imaging and in particle accelerators. Wider uses, such as loss-free power transmission, have been demonstrated in limited trials, but cooling costs and the brittleness of ceramic materials have so far restricted their adoption. A material that is superconducting at room temperature would transform many technologies, but claims of such a discovery have repeatedly failed to be confirmed.

**Which of the following statements is correct?**

- A** Superconductors known before 1986 needed a costlier coolant than some found later.
- B** Every superconductor discovered since 1986 can be cooled with liquid nitrogen.
- C** Brittleness is the only obstacle to the wider use of superconductors for power transmission.
- D** A room-temperature superconductor has been discovered but has not yet been put to practical use.

**Question 18**

The plague epidemic that swept across Europe in the mid-fourteenth century killed a very large proportion of the population; historians' estimates vary, but many regions appear to have lost between a third and a half of their inhabitants. One consequence was a severe shortage of labour. With fewer workers available to till the land and practise crafts, surviving labourers were often able to demand higher wages or better terms from landlords. Authorities in several kingdoms responded with laws that attempted to hold wages at their pre-epidemic levels and to restrict the movement of workers seeking better pay. The effectiveness of these measures appears to have varied considerably, and in many places wages rose despite them. Many historians argue that, over the following century, the changed balance between land and labour contributed to the weakening of serfdom in much of western Europe. They caution, however, that the epidemic was not the sole cause of this change, and point out that in parts of eastern Europe the obligations imposed on peasants actually became heavier in later centuries.

**Which of the following statements is correct?**

- A** The labour shortage caused by the epidemic brought serfdom to an end throughout Europe within the following century.
- B** Laws limiting wages succeeded in keeping pay at pre-epidemic levels in most places.
- C** Many historians see the epidemic as one of several causes of serfdom's decline in much of western Europe.
- D** Historians agree that about half of the population of Europe as a whole died during the mid-fourteenth-century epidemic.

**Question 19**

Evaluations of wage subsidies – payments to employers who hire people from target groups such as the long-term unemployed – often distinguish between gross and net effects. The gross effect is simply the number of subsidised hires. The net effect is smaller, because some of those hires would have taken place anyway, a problem known as deadweight, and because subsidised workers may displace unsubsidised workers at the same or competing firms. An evaluation of a scheme run by a national employment service estimated that deadweight accounted for roughly half of all subsidised hires. It also found that participants were more likely than a comparable group of non-participants to be in work two years later, suggesting that the subsidy may have helped them to regain skills and contacts. The evaluators cautioned, however, that this finding could partly reflect the way employers selected participants, since firms may have chosen the most employable candidates. They concluded that the scheme's cost per additional job was considerably higher than its cost per subsidised hire, and recommended that future schemes be more narrowly targeted.

**Which of the following statements is correct?**

- A** The evaluation demonstrated that the subsidy was the cause of participants' better employment prospects two years later.
- B** Roughly half of the subsidised workers took the place of unsubsidised workers at competing firms.
- C** Because of deadweight, each genuinely additional job cost the scheme more than each subsidised hire.
- D** The evaluators recommended that wage subsidy schemes be discontinued because of their high cost.

**Question 20**

Before the middle of the twentieth century, general cargo was loaded onto ships piece by piece: sacks, barrels and crates were handled individually by large teams of dockworkers, and a vessel could spend as long in port as at sea. The introduction of standardised steel containers changed this. Because containers of agreed dimensions could be lifted by crane directly between ships, lorries and railway wagons, goods no longer had to be unpacked and repacked at each stage. Loading times fell sharply, theft and damage declined, and ships could spend a far greater proportion of their time travelling. The benefits depended on standardisation: early container systems used incompatible sizes, and the advantages multiplied only once international dimensions were agreed. The change also had costs. Employment in traditional docks fell steeply, and many old city-centre ports, lacking space for container cranes and storage yards, lost traffic to new terminals built elsewhere. Economists continue to debate how much of the post-war growth in world trade should be attributed to containerisation as opposed to falling tariffs and other factors.

**Which of the following statements is correct?**

- A** The full benefits of containers depended on agreeing standard international sizes.
- B** Containerisation was the main cause of the growth in world trade after the Second World War.
- C** Old city-centre ports lost traffic because their dockworkers resisted the introduction of containers.
- D** Once containers were introduced, ships spent more of their time in port than at sea.

## Numerical reasoning (questions 21–30)

### Question 21

Visitors to national science museums (thousands)

| Country | Year 1 | Year 2 | Year 3 |
|---------|--------|--------|--------|
| Sweden  | 180    | 183    | 223    |
| France  | 851    | 798    | 830    |
| Greece  | 341    | 364    | 379    |
| Denmark | 259    | 292    | 301    |
| Austria | 487    | 553    | 530    |

*All data are fictitious.*

By approximately what percentage did the number of visitors to science museums in Sweden increase between Year 1 and Year 3?

- A 19.3%
- B 23.9%
- C 21.9%
- D 28.5%

### Question 22

Translation requests received by a language unit in one month, by source language

| Source language | Requests |
|-----------------|----------|
| Italian         | 75       |
| Spanish         | 125      |
| Swedish         | 80       |
| Dutch           | 59       |

*All data are fictitious.*

What is the ratio of requests from Spanish to requests from Italian, in its simplest form?

- A 5:3
- B 5:4
- C 7:4
- D 3:5

**Question 23****Grant applications received and approval rate**

| Country | Applications | Approval rate |
|---------|--------------|---------------|
| Croatia | 900          | 24%           |
| Germany | 1,400        | 15%           |
| Denmark | 1,800        | 25%           |
| Latvia  | 2,300        | 15%           |

*All data are fictitious.***How many applications from Latvia were rejected?**

- A** 2,128
- B** 2,300
- C** 1,955
- D** 1,610

**Question 24****Rail routes: distance and average train speed**

| Route                 | Distance (km) | Average speed (km/h) |
|-----------------------|---------------|----------------------|
| Westfield–Eastbay     | 222           | 120                  |
| Fairhaven–Kingsford   | 121           | 120                  |
| Southmere–Riverton    | 336           | 110                  |
| Stonebridge–Hillcrest | 229           | 90                   |

*All data are fictitious.***If the average speed on the Southmere–Riverton route rose to 150 km/h, approximately how many minutes would the journey save?**

- A** 134 minutes
- B** 73 minutes
- C** 67 minutes
- D** 49 minutes

**Question 25**

Staff numbers and average monthly salary by unit

| Unit   | Staff | Average monthly salary (€) |
|--------|-------|----------------------------|
| Unit D | 34    | 5,400                      |
| Unit C | 43    | 7,200                      |
| Unit E | 45    | 8,400                      |
| Unit B | 15    | 3,800                      |

*All data are fictitious.*

What is the average monthly salary across all staff in the four units combined (to the nearest euro)?

- A €6,775
- B €7,814
- C €6,200
- D €5,980

**Question 26**

Population and public spending on culture in five regions of one country

| Region         | Population (million) | Spending (€ million) |
|----------------|----------------------|----------------------|
| Western region | 2.37                 | 270.4                |
| Island region  | 5.62                 | 647.8                |
| Central region | 4.13                 | 834.3                |
| Coastal region | 1.15                 | 265.7                |
| Alpine region  | 2.13                 | 434.1                |

*All data are fictitious.*

Which of the following regions had the highest spending on culture per inhabitant?

- A Central region
- B Alpine region
- C Coastal region
- D Island region

**Question 27****Cases closed per month by three teams**

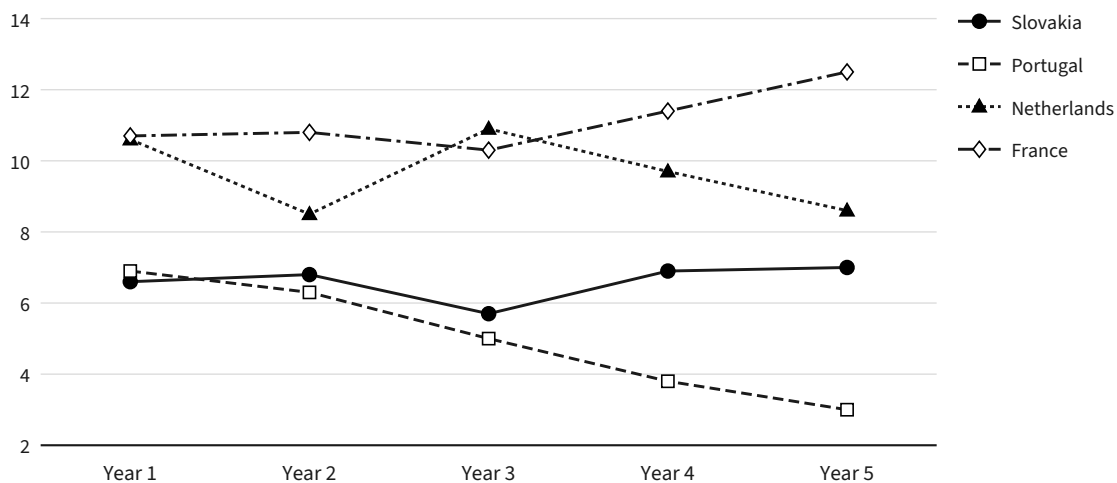
| Team       | Jan | Feb | Mar | Apr | May | Jun |
|------------|-----|-----|-----|-----|-----|-----|
| Team East  | 43  | 39  | 59  | 22  | 37  | —   |
| Team North | 49  | 23  | 40  | 26  | 33  | 59  |
| Team South | 31  | 54  | 18  | 28  | 51  | 19  |

— = figure not yet available

*All data are fictitious.***How many cases would Team East need to close in June for its average over January to June to be exactly 50 cases per month?**

- A** 120
- B** 100
- C** 50
- D** 40



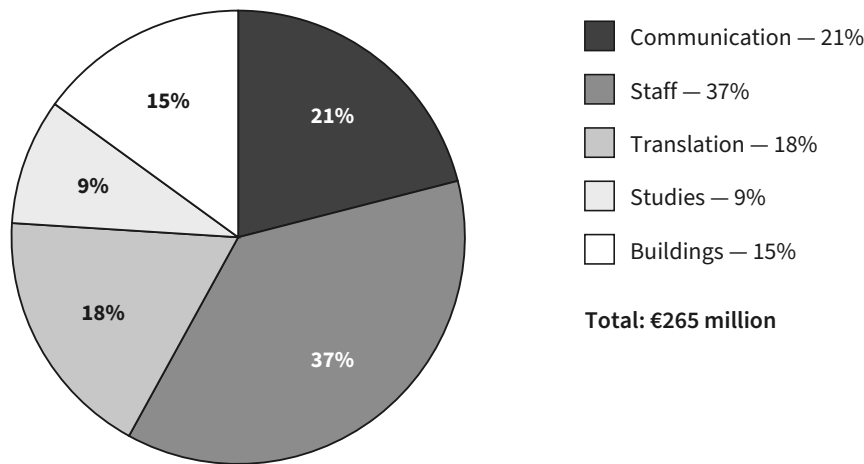
**Question 28****Unemployment rate (% of labour force)***Unit: %**All data are fictitious.*

**By approximately what percentage did the unemployment rate in the Netherlands fall between Year 1 and Year 2?**

- A** 2.8%
- B** 8.5%
- C** 24.7%
- D** 19.8%

Question 29

Breakdown of an agency’s annual budget (total €265 million)



All data are fictitious.

Next year the Staff budget is to rise by 10% while all other items stay unchanged. By how much will the Staff budget then exceed the Communication budget?

- A €52.2 million
- B €46.6 million
- C €42.4 million
- D €68.9 million

Question 30

Plastic collected for recycling by four municipal schemes this year, and expected annual growth

| Municipality | Collected this year (tonnes) | Expected annual growth |
|--------------|------------------------------|------------------------|
| Eastbay      | 2,790                        | 15%                    |
| Riverton     | 2,630                        | 8%                     |
| Kingsford    | 4,010                        | 10%                    |
| Hillcrest    | 4,820                        | 10%                    |

All data are fictitious.

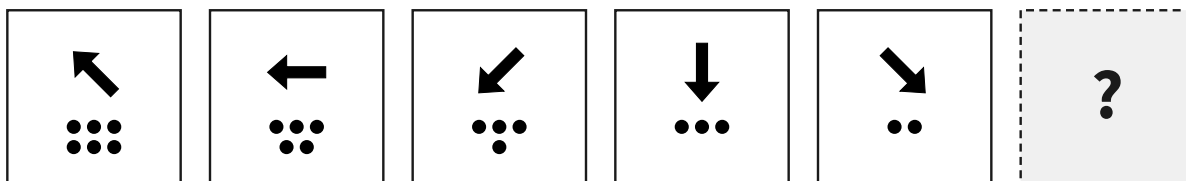
If the amount collected in Eastbay grows at its expected annual rate, compounded, approximately how many tonnes will be collected three years from now?

- A 4,880 t
- B 3,690 t
- C 4,243 t
- D 4,046 t

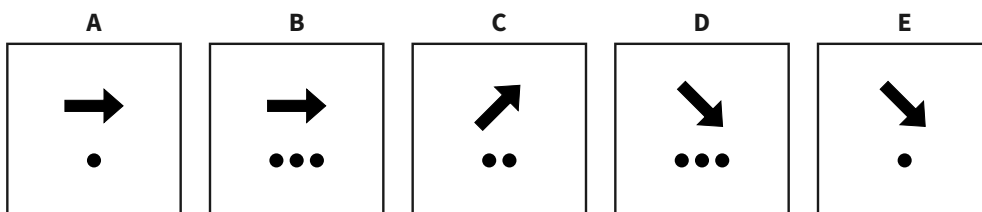
## Abstract reasoning (questions 31–40)

### Question 31

Which figure comes next in the series?

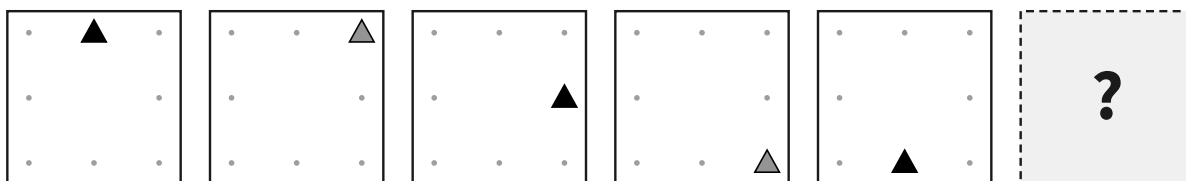


Answer options

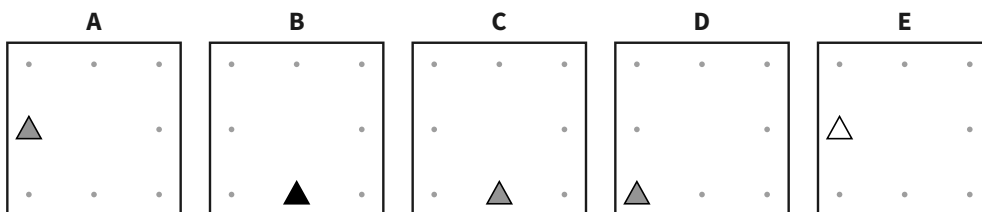


### Question 32

Which figure comes next in the series?

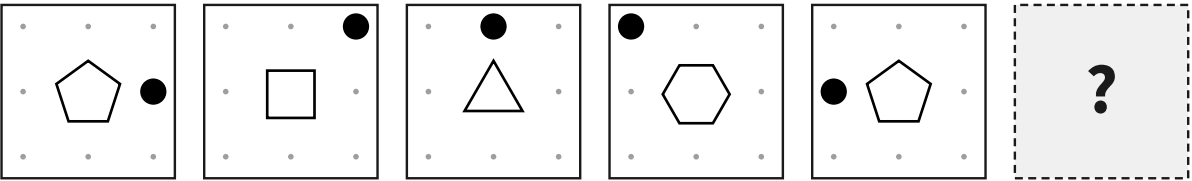


Answer options

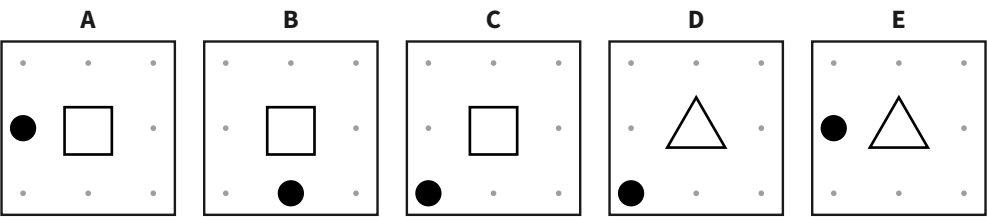


Question 33

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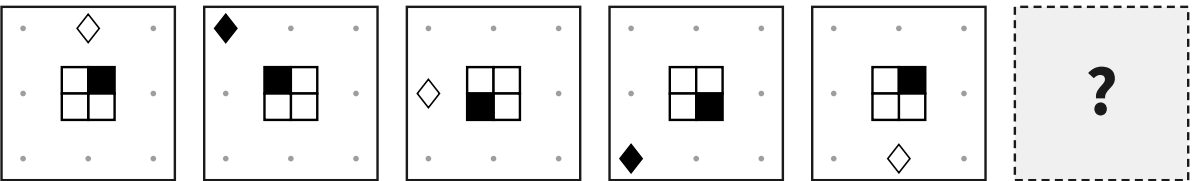


Answer options

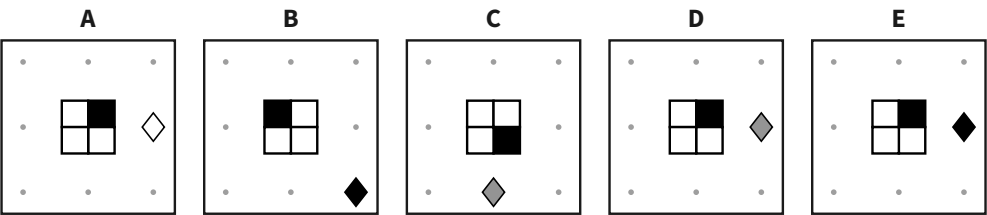


Question 34

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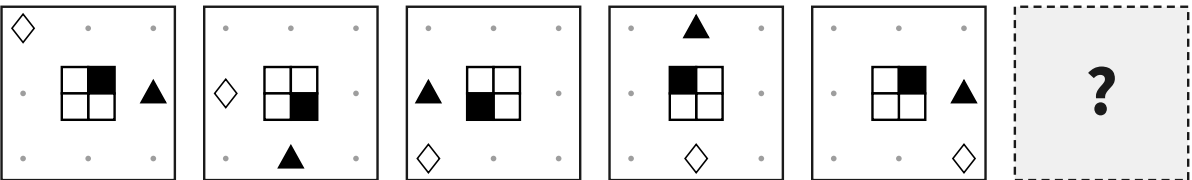


Answer options

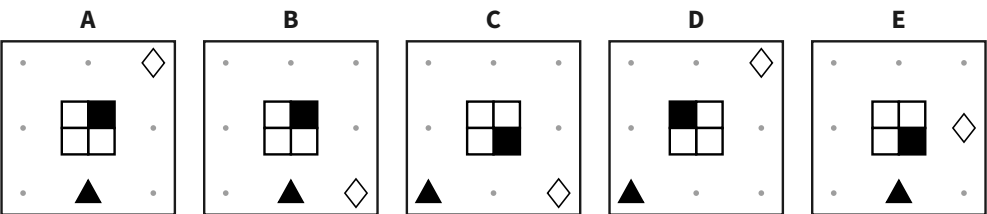


Question 35

Which figure comes next in the series?



Answer options

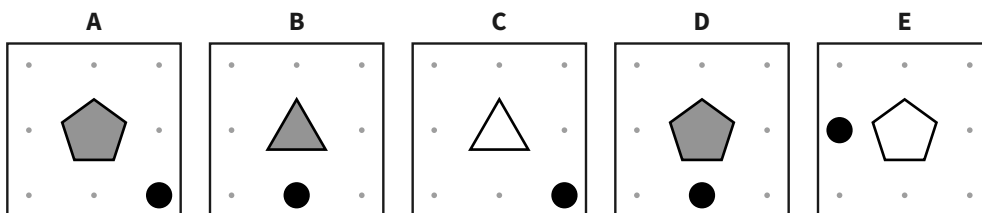


**Question 36**

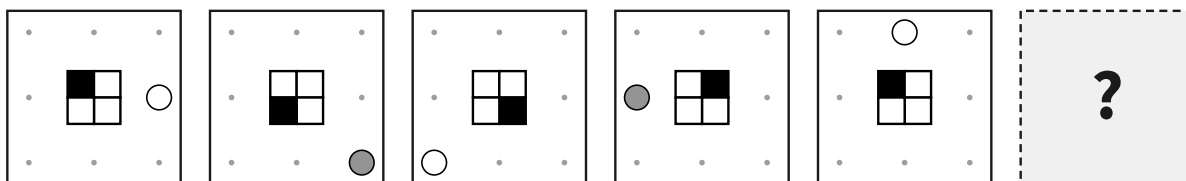
Which figure comes next in the series?



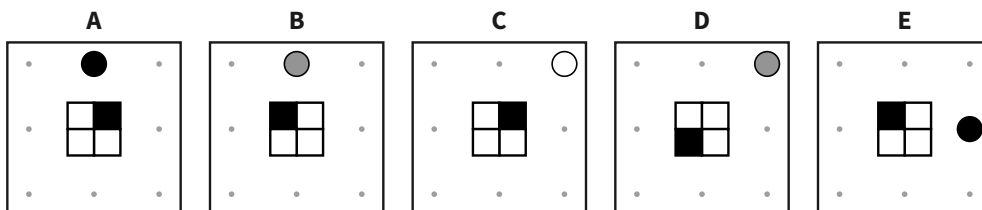
Answer options

**Question 37**

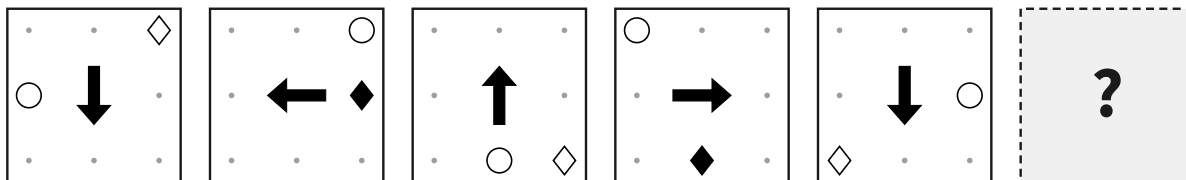
Which figure comes next in the series?



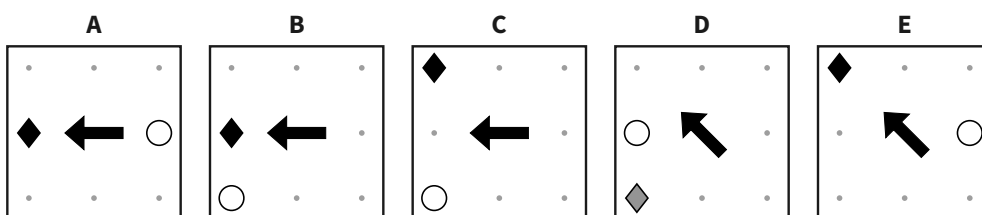
Answer options

**Question 38**

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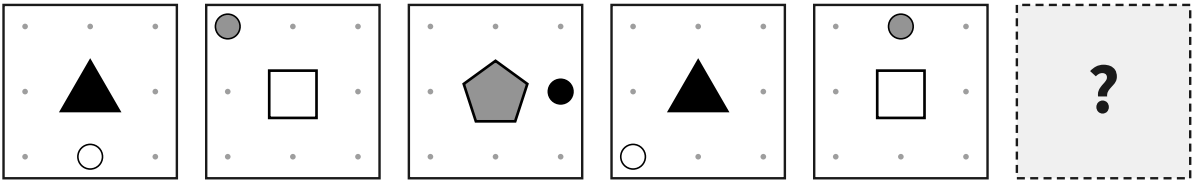


Answer options

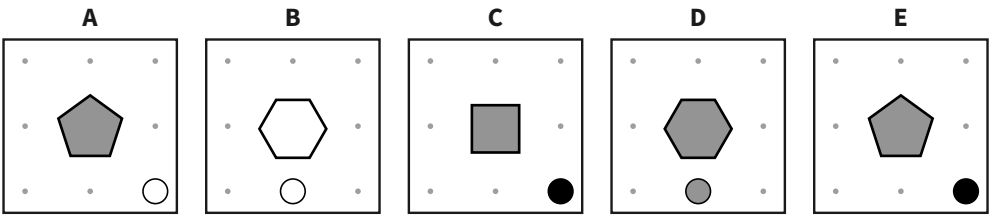


Question 39

Which figure comes next in the series?

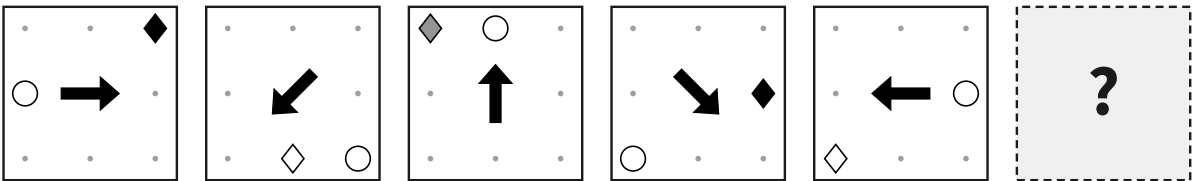


Answer options

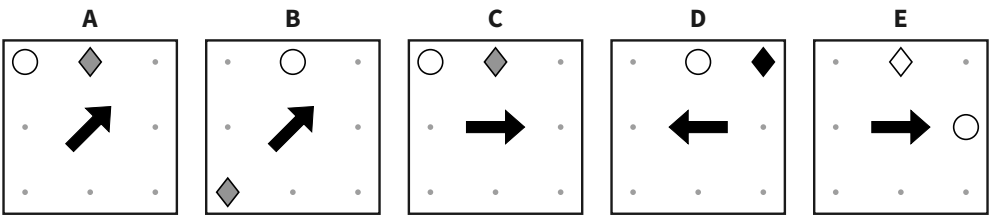


Question 40

Which figure comes next in the series?



Answer options



# Mock Exam 2

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**40 questions • suggested time: 65 minutes**

- Questions 1–20: verbal reasoning (about 35 minutes)
- Questions 21–30: numerical reasoning (about 20 minutes)
- Questions 31–40: abstract reasoning (about 10 minutes)

These timings reflect typical past EPSO formats; check the Notice of Competition for your procedure. Use the answer sheet at the back of the book, work without interruptions, and answer every question: in typical formats there is no penalty for a wrong answer.

**Score guide (out of 40).** 34–40: excellent, test-ready. 28–33: strong; polish weaker areas. 22–27: developing; revisit the method chapters and redo your errors. Below 22: build accuracy first with the practice sets. Real pass marks differ between procedures, so treat these bands as a guide only.

## Verbal reasoning (questions 1–20)

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### Question 1

In a matrix organisation, employees report to two managers at once: typically a functional manager, responsible for a specialist field such as engineering or finance, and a project manager, responsible for delivering a particular piece of work. The structure is intended to combine the depth of expertise found in functional departments with the flexibility needed to deliver projects that cut across them. Specialists can be assigned to several projects in turn without leaving their home department, which helps to keep their skills up to date and allows scarce expertise to be shared. The arrangement has well-known drawbacks. When the two managers disagree about priorities, employees may receive conflicting instructions, and decisions can slow down as issues are escalated for resolution. Clear rules on which manager decides what – for example, the project manager setting deadlines and the functional manager overseeing technical standards – are therefore often regarded as essential. Matrix structures are most common in organisations that run many complex projects at the same time.

Which of the following statements is correct?

- A** Staff in a matrix structure may be given competing instructions by their two managers.
- B** In a matrix organisation, specialists must leave their functional department in order to join a project.
- C** Matrix structures speed up decision-making by giving every employee two sources of authority.
- D** Matrix structures are found mainly in small organisations that run one project at a time, where lines of authority are easy to establish.



**Question 2**

A provincial city is considering whether to expand its district heating network, which currently supplies hot water to around a fifth of households through insulated underground pipes. The heat comes mainly from a combined heat and power plant on the edge of the city. Conventional power stations release a large share of the energy in their fuel as waste heat, typically through cooling towers; a combined heat and power plant captures much of this heat and puts it to use, so that a greater proportion of the fuel's energy is used overall. Supporters of expansion argue that a single central source is easier to switch to cleaner fuels than thousands of individual boilers, and that the network could also take in heat from other sources, such as industrial processes or large heat pumps. Opponents point to the high cost of laying new pipes in streets already crowded with other utilities, and to the disruption that the works would cause. They also note that a network is most economical in densely built areas, where many customers can be connected per kilometre of pipe. The city council has commissioned a feasibility study before taking a decision.

**Which of the following statements is correct?**

- A** Most households in the city already receive heat from the district network.
- B** Combined heat and power plants make use of heat that conventional power stations generally waste.
- C** The council has decided to extend the network to all parts of the city.
- D** District heating networks are equally economical in sparsely and densely built areas, since the cost of laying pipe is the same everywhere.

**Question 3**

Code-switching, the practice of alternating between two or more languages within a single conversation or even a single sentence, was long regarded by teachers and commentators as evidence that a speaker had not fully mastered either language. Research in sociolinguistics has largely overturned this view. Studies of bilingual communities show that switches tend to follow regular grammatical constraints: speakers rarely switch at points where the word orders of the two languages would clash. Switching also performs social functions. A speaker may change language to signal solidarity with a listener, to quote someone, to mark a shift in topic, or to express an emotion for which one language seems better suited. Fluent bilinguals, rather than learners, are often the most skilful code-switchers, because effective switching requires a command of the grammar of both languages. Attitudes nevertheless vary considerably. In some communities code-switching is an unremarkable feature of everyday speech, while in others it remains stigmatised, particularly in formal settings such as schools and public administration.

**Which of the following statements is correct?**

- A** Code-switching occurs mainly among speakers who have not yet mastered either of their languages.
- B** Code-switching is now widely accepted in formal settings such as schools and public administration.
- C** Where the two languages order words differently, bilinguals seldom switch from one to the other.
- D** Speakers switch language chiefly because one of their languages lacks the vocabulary needed to express certain emotions.

**Question 4**

In many jurisdictions, consumers who buy goods or services at a distance – online, by telephone or by mail order – are entitled to a cooling-off period during which they may withdraw from the contract without giving a reason. The rationale is that a buyer who cannot inspect a product before purchase is at a disadvantage compared with someone shopping in a physical store. Under a typical scheme of this kind, the period for goods runs for a fixed number of days from the moment they are delivered, rather than from the date of the order, so that the consumer has a genuine opportunity to examine them. The seller must refund the price, normally including the standard delivery charge, although the consumer may be asked to bear the direct cost of returning the item. Certain categories are usually excluded, such as perishable food, goods made to the buyer's personal specifications and sealed audio or video recordings that have been unsealed after delivery. If the seller fails to inform the consumer of the right to withdraw, the period is commonly extended considerably.

**Which of the following statements is correct?**

- A** Under the typical scheme described, the cooling-off period for goods starts when the consumer receives them.
- B** Withdrawing consumers never have to bear any costs.
- C** The cooling-off period applies equally to purchases made in physical shops.
- D** Consumers must explain their reasons for withdrawing before they can obtain a refund.

**Question 5**

A regional authority has launched a three-year pilot scheme to strengthen links between local farmers and consumers. Under the scheme, producers located within 80 kilometres of the regional capital may apply for a grant towards the cost of setting up stalls at weekly markets or establishing online ordering systems. Grants cover up to half of eligible costs, with the remainder to be met by the producers themselves. Participating farms must sell at least a third of the relevant products directly to consumers or through no more than one intermediary. The authority's stated aims are to increase the share of the sale price retained by farmers and to give consumers clearer information about where their food comes from. Officials have acknowledged that shorter supply chains do not automatically reduce the environmental footprint of food: a large number of small deliveries by van may generate more emissions per kilogram than a single lorry delivering to a supermarket. For this reason, transport emissions will be monitored during the pilot, and the authority will decide at its end whether to extend the scheme across the whole region.

**Which of the following statements is correct?**

- A** Producers receiving a grant must themselves fund at least half of the eligible costs of their project.
- B** The authority expects the scheme to reduce the emissions associated with transporting food.
- C** Participating farms are required to sell all their products without any intermediary.
- D** The scheme has already been extended to producers throughout the region.

**Question 6**

As the share of electricity generated from wind and solar power grows, grid operators face the challenge of matching supply, which varies with the weather, to demand, which follows its own daily and seasonal patterns. Energy storage is one response. The oldest large-scale storage technology is pumped-storage hydroelectricity. When electricity is plentiful and cheap, water is pumped from a lower reservoir to an upper one; when demand rises, the water is released through turbines to generate power. Some energy is lost in each cycle, so a plant returns less electricity than it consumes, yet the arrangement can still be economically attractive because it buys power when prices are low and sells it when they are high. Pumped storage can respond within minutes and can supply power for many hours. Its main limitation is geographical: it requires two reservoirs at substantially different heights, together with suitable geology and water resources, and new sites often face environmental objections. Batteries, by contrast, can be installed almost anywhere and react even faster, but they are generally used to supply power over shorter periods.

**Which of the following statements is correct?**

- A** Pumped-storage plants generate more electricity than they consume.
- B** Batteries are replacing pumped storage as the principal form of grid storage.
- C** Pumped-storage plants can be built wherever an adequate supply of water is available, whatever the surrounding terrain.
- D** A pumped-storage plant can be profitable even though it loses energy in each cycle of pumping and generation.

**Question 7**

Comparing prices in a supermarket is less straightforward than it appears. Products are sold in packages of differing sizes, and a larger pack is not always cheaper per kilogram or per litre than a smaller one, particularly when the smaller pack is on promotion. To help shoppers, many consumer-protection regimes require retailers to display a unit price – the price per standard quantity – alongside the selling price. A consumer organisation in one country examined how effectively this information was being used. It found that shoppers who reported noticing unit prices were more likely to choose the option that was cheaper per unit than those who did not. However, the organisation cautioned that this association did not prove that the labels themselves changed behaviour, since shoppers who are already price-conscious may be more inclined to look for such information. It also found that unit prices were sometimes expressed in different units for similar products – per 100 grams for one brand and per kilogram for another – which made comparison harder. The organisation recommended clearer and more consistent presentation, and suggested that unit prices be printed in a font size comparable to that of the selling price.

**Which of the following statements is correct?**

- A** The study demonstrated that unit-price labels lead shoppers to choose cheaper products.
- B** Bigger packs always cost less per kilogram.
- C** Retailers in every country are legally obliged to display unit prices.
- D** Inconsistent units made some unit prices hard to compare.

**Question 8**

Because light travels at a finite speed, astronomers never observe celestial objects as they are at the present moment, but as they were when the light left them. Light from the Sun takes roughly eight minutes to reach the Earth; light from the nearest stars beyond the Sun takes several years; and light from distant galaxies may have travelled for millions or even billions of years. Distances on this scale are therefore often expressed in light-years – a unit of distance, not of time, equal to the distance that light covers in one year. The delay has a consequence that astronomers exploit: by observing objects at increasing distances, they are in effect looking further into the past, and can compare young galaxies with older ones nearer to us. It also means that an object seen in the night sky may have changed considerably, or even ceased to exist in its observed form, by the time its light is detected. The principle applies equally to nearby objects, although there the delay is too short to matter for most purposes.

**Which of the following statements is correct?**

- A** A light-year is a measure of the time that light takes to travel from a distant star to the Earth.
- B** Distant galaxies show astronomers an earlier stage of cosmic history than nearby ones do.
- C** Most of the stars that can be seen in the night sky have already ceased to exist.
- D** The delay in the arrival of light affects only objects lying outside our own galaxy, not nearby ones.

**Question 9**

The capacity of a lithium-ion battery declines gradually over its life, a process known as degradation. Two broad mechanisms are usually distinguished. Calendar ageing occurs even when a battery is not in use and is accelerated by high temperatures and by storing the cell at a high state of charge. Cycle ageing results from repeated charging and discharging; it tends to be more severe when cells are routinely charged to full capacity and drained almost completely, and when very high charging currents are used. For this reason, many manufacturers of electric vehicles allow drivers to set a charging limit below 100 per cent for everyday use, reserving full charges for long journeys. Cold conditions pose a different problem: they temporarily reduce the energy a battery can deliver and slow down charging, but much of this lost performance returns once the battery warms up. Battery management systems monitor the temperature and voltage of individual cells and can restrict charging power to protect them.

**Which of the following statements is correct?**

- A** A battery that is never used will not lose any capacity.
- B** Cold weather permanently reduces the capacity of a lithium-ion battery.
- C** Keeping a fully charged battery in a hot place is likely to speed up the loss of its capacity.
- D** Manufacturers of electric vehicles require drivers to keep everyday charging below 100 per cent.

**Question 10**

Public research funding has traditionally been allocated through two broad channels. In the first, researchers propose projects on topics of their own choosing, and independent experts select the most promising through peer review. This “bottom-up” model is valued for protecting scientific freedom and for supporting curiosity-driven research whose practical applications may emerge only decades later. In the second, funders define priority areas in advance and invite proposals that address them. A variant of this approach, sometimes called mission-oriented funding, goes further by setting a concrete, measurable objective – for example, a specified reduction in the energy consumption of buildings – and mobilising researchers, firms and public authorities from several disciplines to achieve it. Advocates argue that missions give research a clearer sense of direction and make its contribution to society more visible. Critics respond that governments are not necessarily well placed to pick the right targets, and that an excessive focus on missions could crowd out fundamental research. Many funding systems therefore combine both approaches, reserving a substantial share of their budgets for researcher-led projects.

**Which of the following statements is correct?**

- A** Missions differ from ordinary priority areas in having a quantified target.
- B** Bottom-up funding has been shown to yield more useful applications than mission-oriented funding.
- C** Missions draw on a single discipline only.
- D** Critics of mission-oriented funding consider that missions should replace fundamental research.

**Question 11**

Until the nineteenth century, most towns set their clocks by the position of the sun, so that noon occurred when the sun was highest in the sky. Because the sun’s position changes as one travels east or west, towns only a short distance apart kept slightly different times. This caused few problems while journeys were slow, but the spread of railways made the variations inconvenient: timetables had to list arrival and departure times, and differing local times confused passengers and staff alike. Railway companies in several countries therefore adopted a single standard time across their networks, often based on the time of a capital city or a major observatory. At first, many towns continued to use local time for civil purposes, and some public clocks were fitted with two minute hands, one showing local time and the other railway time. Legal recognition of standard time followed in many countries only some years later. Towards the end of the century, an international conference recommended a single prime meridian from which time zones around the world could be calculated, although countries adopted the system at different speeds.

**Which of the following statements is correct?**

- A** Standard time was first introduced by governments, which then required railway companies to adopt it.
- B** For a time, some towns displayed railway time alongside their own local time.
- C** Neighbouring towns kept different times because their clocks were poorly maintained.
- D** After the international conference, every country immediately adopted the recommended system of time zones.

**Question 12**

Fisheries scientists often use the concept of maximum sustainable yield (MSY) to guide catch limits. The idea rests on the observation that a fish population grows most rapidly at an intermediate size: when a stock is very small, there are too few adults to produce many young, while a stock close to the carrying capacity of its environment grows slowly because individuals compete for food and space. In theory, by keeping a stock at the size at which growth is fastest and harvesting only the surplus each year, a fishery could take the largest catch that can be sustained indefinitely. In practice, the approach faces several difficulties. Estimates of stock size are uncertain, and natural fluctuations in recruitment – the number of young fish entering the population – can be large. Single-species models also overlook interactions within the ecosystem, for instance when the target species is food for other fish, seabirds or marine mammals. For these reasons, many scientists advise setting catches somewhat below the theoretical maximum, as a buffer against error.

**Which of the following statements is NOT supported by the passage?**

- A** A fish population close to the carrying capacity of its environment tends to grow slowly.
- B** Uncertainty about the size of stocks is one of the obstacles to applying MSY in practice.
- C** Many scientists favour a safety margin below the theoretical maximum catch.
- D** Single-species MSY models fully account for effects on the target species' predators.

**Question 13**

Antibiotic resistance arises through natural selection. Within any large population of bacteria, a small number of cells may carry genetic variations that make them less susceptible to a particular drug. When an antibiotic is used, susceptible bacteria are killed or prevented from multiplying, while resistant cells survive and reproduce, so that over time they come to make up a larger share of the population. Bacteria can also acquire resistance genes from other bacteria, including those of different species, through the exchange of small pieces of DNA. It is the bacteria, not the patients, that become resistant. Resistance can develop wherever antibiotics are used, in agriculture as well as in human medicine. Stopping a prescribed course early has long been discouraged, although some researchers now argue that the optimal length of treatment varies between infections and that shorter courses may be appropriate in certain cases. There is broad agreement, however, that unnecessary use of antibiotics – for example, against viral infections, on which they have no effect – increases the selective pressure that favours resistant strains.

**Which of the following statements is correct?**

- A** Patients who take antibiotics often become resistant to them.
- B** Antibiotics produce the genetic variations that make bacteria resistant.
- C** Genes conferring resistance can be transferred between bacteria of different species.
- D** Researchers now agree that shorter courses of antibiotics are preferable for all infections.

**Question 14**

Heat pumps are often described as devices that generate heat, but this is misleading: they move heat from one place to another. A domestic air-source heat pump extracts thermal energy from outdoor air, even when the air is cold, and transfers it into a building by means of a refrigerant that is repeatedly compressed and expanded. Because the electricity consumed powers the transfer rather than producing the heat directly, a well-designed unit can deliver several units of heat for each unit of electricity used. This ratio, called the coefficient of performance, is not fixed. It declines as the temperature difference between the heat source and the building's heating system increases, which means that performance tends to fall on the coldest days, precisely when demand is highest. Engineers therefore recommend pairing heat pumps with well-insulated buildings and with radiators or underfloor systems designed to operate at lower water temperatures than conventional boilers typically use. Ground-source models, which draw heat from below the surface, where temperatures are more stable through the year, usually achieve a higher average coefficient, although their installation costs are generally greater because of the excavation required.

**Which of the following statements is correct?**

- A** Air-source heat pumps cannot operate when the outdoor air is cold.
- B** Because ground-source heat pumps are more efficient on average, they are the most cost-effective choice for households.
- C** A heat pump's coefficient of performance is at its highest on the coldest days, when demand for heating peaks.
- D** Lower heating-water temperatures should raise a heat pump's coefficient of performance.

**Question 15**

A mid-sized engineering firm conducted a six-month trial in which employees in two of its four departments worked four days a week with no reduction in pay, while the other two departments continued on a five-day schedule. Participation was voluntary at departmental level. Over the trial, the participating departments maintained their output, and staff reported lower levels of fatigue. Absence through sickness also fell, although it fell slightly in the non-participating departments as well. Managers noted that meetings had been shortened and that some tasks previously considered essential had been discontinued without apparent loss. The firm's board has welcomed the results but has not yet decided whether to extend the arrangement. In its report, the trial's steering group observed that the departments which volunteered may have been those best suited to a shorter working week, and recommended a second trial in the remaining departments before any company-wide decision is taken. Client-facing teams, it noted, would need cover arrangements to ensure availability throughout the five-day week.

**Which of the following statements is NOT supported by the passage?**

- A** The departments taking part in the trial did not experience a fall in output.
- B** The reduction in sickness absence cannot confidently be attributed to the four-day week alone.
- C** The steering group considered that the trial's results might not apply equally to every department.
- D** The board has decided to extend the four-day week to the whole company once a second trial has been held.

**Question 16**

Historians of nineteenth-century transatlantic migration have long cautioned against picturing it as a single, permanent departure by isolated individuals. Much of it followed the pattern now called chain migration: early arrivals sent letters home describing conditions and wages, and frequently paid for the passage of relatives or neighbours, who in turn settled close to them. As a result, emigrants from a particular valley or district often clustered in a small number of destinations rather than dispersing across the continent. The replacement of sailing vessels by steamships during the century shortened the crossing from several weeks to a matter of days and made it markedly safer. Some historians argue that this change, by lowering the cost and risk of travel, encouraged a growing share of migrants to treat the journey as temporary, working abroad for a few seasons before returning home. Others caution that rates of return varied so widely between groups that no general rule can be drawn from shipping technology alone. What is not disputed is that letters and money sent home linked sending and receiving communities for decades.

Which of the following statements is correct?

- A Migrants who had already settled abroad often helped to pay for the journeys of people from their home area.
- B Chain migration developed only after steamships had made the crossing cheaper and safer.
- C Emigrants from the same district typically spread out widely across the continent in search of work.
- D Historians agree that the introduction of steamships made temporary migration more common.

**Question 17**

Education systems differ in the age at which pupils are separated into different types of school according to perceived ability. In some systems this occurs as early as ten; in others, all pupils follow a common curriculum until about sixteen. Comparative studies have generally found that early selection is associated with a closer link between pupils' socio-economic background and their attainment. Supporters of early tracking argue that it allows teaching to be tailored to pupils' needs and that high achievers benefit from being taught together. The evidence on this point is less clear-cut: some studies find modest gains for the highest-attaining pupils, but these appear to be offset by losses among lower-attaining pupils, so that average performance is not raised. Researchers caution, however, that countries which select early also differ from others in many respects, including funding and teacher training, which makes it difficult to isolate the effect of selection itself. Systems that have postponed selection have not always seen the expected reduction in inequality.

Which of the following statements is correct?

- A Early selection has been shown to cause greater inequality in pupils' attainment.
- B Early tracking lowers the attainment of high-achieving pupils.
- C Postponing selection reliably weakens the link between socio-economic background and attainment.
- D The evidence described does not suggest that early tracking raises average attainment.



**Question 18**

The principle of legal certainty requires that the law be clear, accessible and reasonably predictable, so that individuals and businesses can plan their affairs. One of its best-known expressions is the rule against retroactivity. In criminal law, this rule is applied strictly: no one may be punished for an act that did not constitute an offence when it was committed, nor receive a heavier penalty than the one applicable at the time. Many systems, however, allow a more lenient penalty introduced later to benefit the offender. Outside criminal law, the position is more nuanced. Retroactive rules in fields such as taxation are not automatically invalid, but courts tend to accept them only exceptionally – for example, where the objective requires it and the legitimate expectations of those concerned are duly respected. Legal certainty must also be balanced against other values: a law that could never be amended would protect expectations at the cost of preventing necessary reform.

**Which of the following statements is correct?**

- A** Retroactive tax rules are prohibited under the principle of legal certainty.
- B** A heavier criminal penalty introduced after an offence may be applied to it if the objective requires it.
- C** Legal certainty requires that laws, once adopted, remain unchanged.
- D** A later, milder penalty may benefit an offender in many systems.

**Question 19**

Modern conservation of historic buildings and artworks rests on a few widely shared principles, though their application is often debated. The first is minimal intervention: conservators should do no more than is necessary to stabilise an object and slow its deterioration. The second is reversibility – any material added should, as far as possible, be removable in future without damaging the original, so that later generations, with better techniques, can revisit today's decisions. A third principle requires that new work be distinguishable from old on close inspection, even if it is designed to blend in at a normal viewing distance. These ideas emerged partly in reaction to nineteenth-century restorations, in which architects sometimes rebuilt medieval structures according to what they believed the building ought to have looked like, removing later additions in the process. Critics argue that perfect reversibility is rarely achievable in practice, since even cleaning removes material irreversibly. Most practitioners therefore treat reversibility as an aspiration that guides the choice between methods rather than as an absolute requirement.

**Which of the following statements is correct?**

- A** Conservators are required to ensure that every intervention can be completely reversed.
- B** A repair that is invisible from a normal distance may still meet the principles described.
- C** Modern conservation principles arose entirely from a desire to undo the work of nineteenth-century restorers.
- D** Most practitioners regard cleaning as a breach of the reversibility principle and therefore avoid it.

**Question 20**

A partnership of universities and transport operators has completed an eighteen-month trial of driverless electric shuttles on a fixed route linking a suburban railway station with a business park. Throughout the trial, a safety operator remained on board with the ability to stop the vehicle, and intervened on average once every few kilometres, most often when parked vehicles or pedestrians obstructed the lane. The shuttles carried passengers without any injury-causing collision. Passenger surveys recorded high satisfaction, although respondents were mostly regular commuters who had volunteered to take part. The partnership's final report concludes that the technology is ready for operation without an on-board operator on fully segregated routes, but not yet in mixed traffic. It recommends that the participating authorities consider a second phase in which a single remote supervisor would monitor several vehicles at once. The authorities have welcomed the report but have not committed to a second phase, citing uncertainty over insurance arrangements and the cost of adapting the route.

**Which of the following statements is correct?**

- A** The trial showed that the shuttles can operate safely in mixed traffic without human supervision.
- B** The participating authorities have agreed to launch a second phase with remote supervision.
- C** The high satisfaction recorded in the surveys can be taken to reflect the views of the general public.
- D** The trial's safety record was achieved with an on-board operator who regularly had to intervene.

## Numerical reasoning (questions 21–30)

### Question 21

Regional development fund allocations by sector (€ million)

| Region  | Health | Research | Education | Digital | Total |
|---------|--------|----------|-----------|---------|-------|
| East    | 420    | 470      | 370       | 950     | 2,210 |
| Central | 730    | 340      | 390       | 150     | 1,610 |
| North   | 200    | 740      | 270       | 220     | 1,430 |
| West    | 620    | 520      | 270       | 400     | 1,810 |

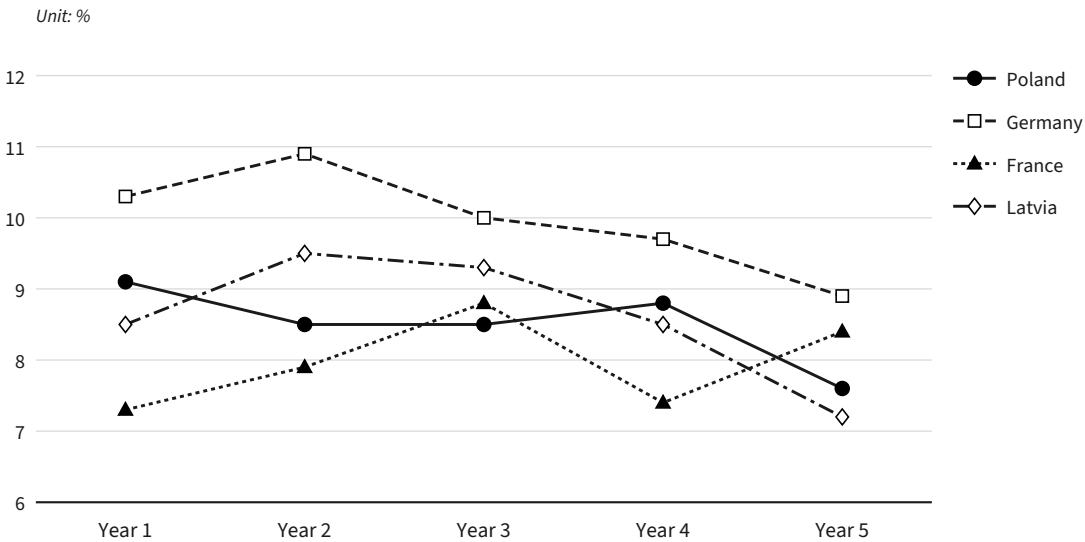
*All data are fictitious.*

What percentage of the North region's total allocation went to Education?

- A** 18.9%
- B** 23.3%
- C** 26.2%
- D** 20.8%

Question 22

Unemployment rate (% of labour force)



| Series  | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|---------|--------|--------|--------|--------|--------|
| Poland  | 9.1    | 8.5    | 8.5    | 8.8    | 7.6    |
| Germany | 10.3   | 10.9   | 10.0   | 9.7    | 8.9    |
| France  | 7.3    | 7.9    | 8.8    | 7.4    | 8.4    |
| Latvia  | 8.5    | 9.5    | 9.3    | 8.5    | 7.2    |

All data are fictitious.

By how many percentage points did the unemployment rate in Latvia fall between Year 2 and Year 5?

- A 2.7 percentage points
- B 2.3 percentage points
- C 3.6 percentage points
- D 1.0 percentage points

**Question 23**

Translation requests received by a language unit in one month, by source language

| Source language | Requests |
|-----------------|----------|
| Dutch           | 18       |
| French          | 24       |
| English         | 81       |
| Italian         | 58       |

*All data are fictitious.*

What is the ratio of requests from Dutch to requests from French, in its simplest form?

- A** 4:5
- B** 4:3
- C** 3:5
- D** 3:4

**Question 24**

Population and public spending on culture in five regions of one country

| Region          | Population (million) | Spending (€ million) |
|-----------------|----------------------|----------------------|
| Eastern region  | 3.72                 | 604.6                |
| Central region  | 3.42                 | 623.6                |
| Northern region | 4.27                 | 417.2                |
| Southern region | 4.88                 | 516.8                |
| Western region  | 0.66                 | 68.2                 |

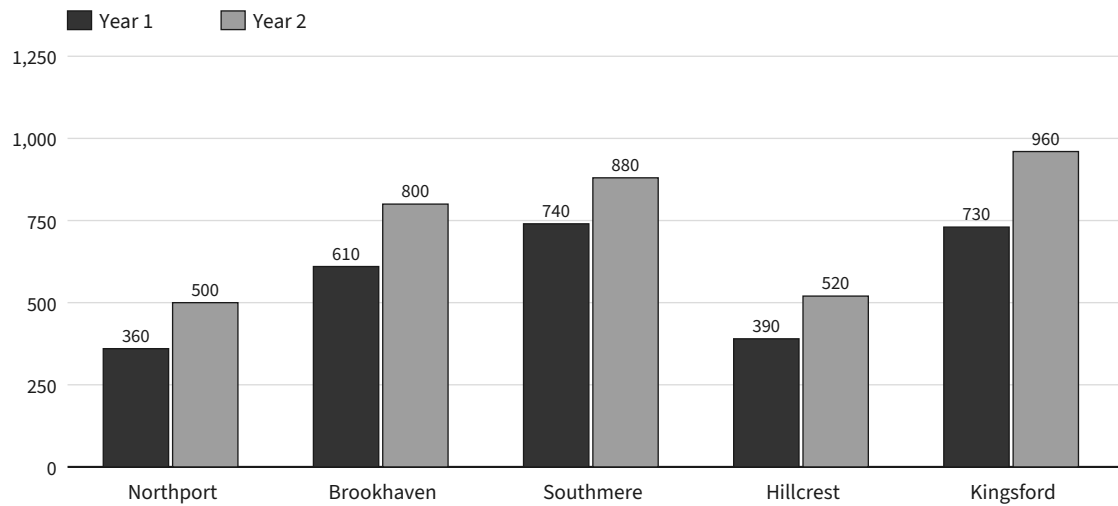
*All data are fictitious.*

Which of the following regions had the lowest spending on culture per inhabitant?

- A** Western region
- B** Northern region
- C** Eastern region
- D** Central region

Question 25

Containers handled at five ports (thousand units)



Unit: thousand containers

All data are fictitious.

Which of the following ports recorded the largest percentage increase in containers handled between Year 1 and Year 2?

- A Hillcrest
- B Brookhaven
- C Kingsford
- D Northport

**Question 26****Table 1 — Participants per training course**

| Course             | Q1 | Q2 | Q3 | Q4 |
|--------------------|----|----|----|----|
| Data analysis      | 28 | 18 | 27 | 17 |
| Project management | 16 | 10 | 11 | 17 |
| Negotiation skills | 24 | 15 | 12 | 4  |
| Financial rules    | 13 | 8  | 8  | 29 |

**Table 2 — Cost per participant (€)**

| Course             | Cost (€) |
|--------------------|----------|
| Data analysis      | 380      |
| Project management | 140      |
| Negotiation skills | 130      |
| Financial rules    | 150      |

*All data are fictitious.*

**What was the total cost of the “Financial rules” course for Q1 to Q2 inclusive?**

- A** €3,150
- B** €4,350
- C** €8,700
- D** €7,980

**Question 27**

**Price of a standard office laptop in four markets (fictitious prices and rates)**

| Currency        | Code | Units per €1 | Local price |
|-----------------|------|--------------|-------------|
| Norwegian krone | NOK  | 11.39        | 6,957       |
| Icelandic króna | ISK  | 145.5        | 75,400      |
| Canadian dollar | CAD  | 1.42         | 1,000       |
| Swiss franc     | CHF  | 0.94         | 558         |

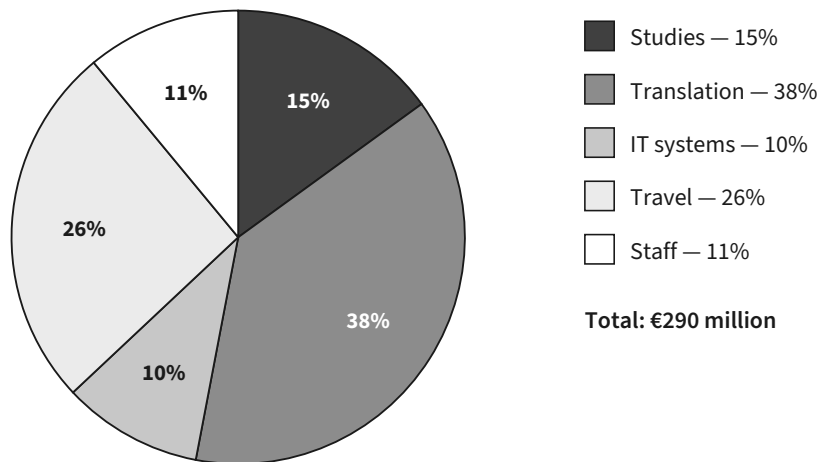
*All data are fictitious.*

**In euros, approximately how much more does a standard office laptop cost in the most expensive of the four markets than in the cheapest?**

- A** €111
- B** €75
- C** €186
- D** €93

Question 28

Breakdown of an agency’s annual budget (total €290 million)



All data are fictitious.

Next year the Travel budget is to rise by 15% while all other items stay unchanged. By how much will the Travel budget then exceed the Staff budget?

- A €87.0 million
- B €50.0 million
- C €43.5 million
- D €54.8 million

Question 29

Budget for this year and change compared with last year

| Programme                | This year (€ thousand) | Change on last year |
|--------------------------|------------------------|---------------------|
| Clean-bus procurement    | 5,300.0                | +25%                |
| Student mobility grants  | 1,836.0                | –15%                |
| Coastal flood defences   | 4,212.0                | +30%                |
| Youth employment schemes | 2,064.0                | +20%                |

All data are fictitious.

What was last year’s budget for the “Clean-bus procurement” programme?

- A €5,050.0k
- B €7,066.7k
- C €4,240.0k
- D €3,975.0k



**Question 30**

Plastic collected for recycling by four municipal schemes this year, and expected annual growth

| Municipality | Collected this year (tonnes) | Expected annual growth |
|--------------|------------------------------|------------------------|
| Riverton     | 3,510                        | 12%                    |
| Kingsford    | 4,220                        | 12%                    |
| Northport    | 4,490                        | 20%                    |
| Stonebridge  | 5,260                        | 12%                    |

*All data are fictitious.*

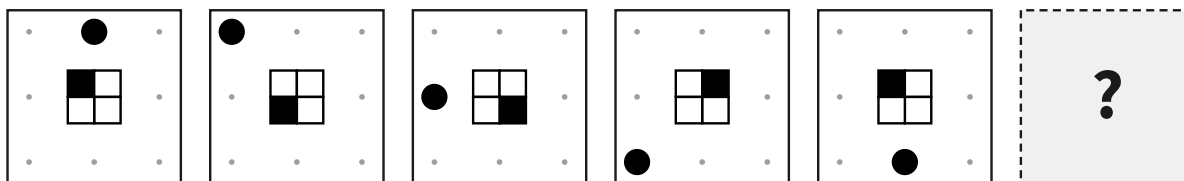
If the amount collected in Northport grows at its expected annual rate, compounded, approximately how many tonnes will be collected three years from now?

- A** 6,466 t
- B** 7,759 t
- C** 9,310 t
- D** 7,184 t

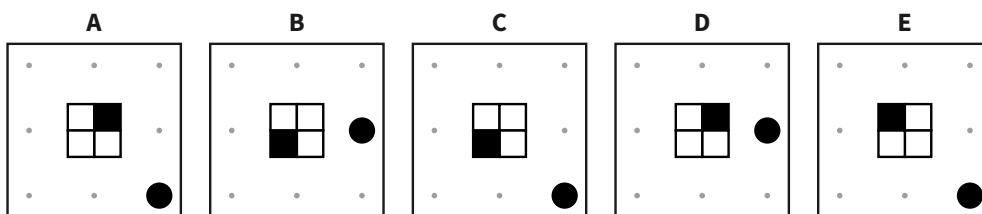
### Abstract reasoning (questions 31–40)

### Question 31

**Which figure comes next in the series?**

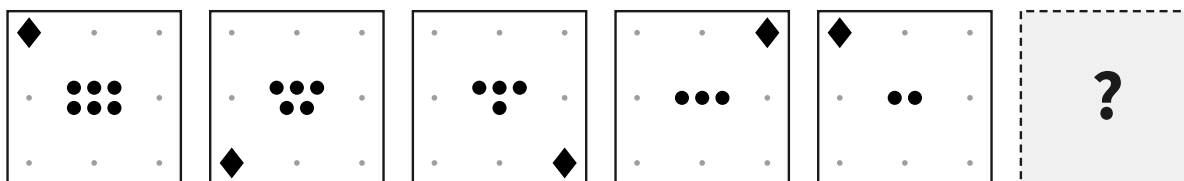


### Answer options

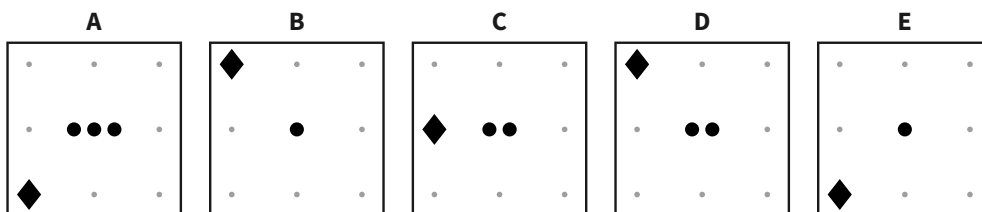


### Question 32

**Which figure comes next in the series?**

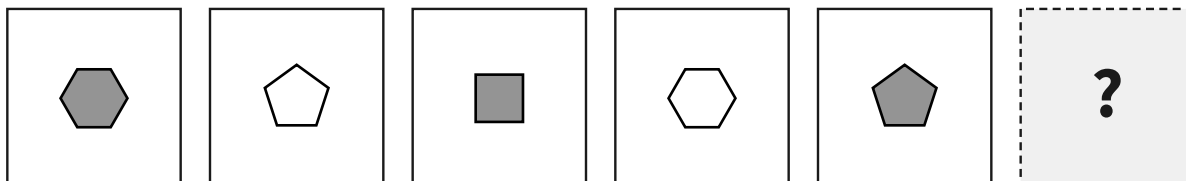


### Answer options

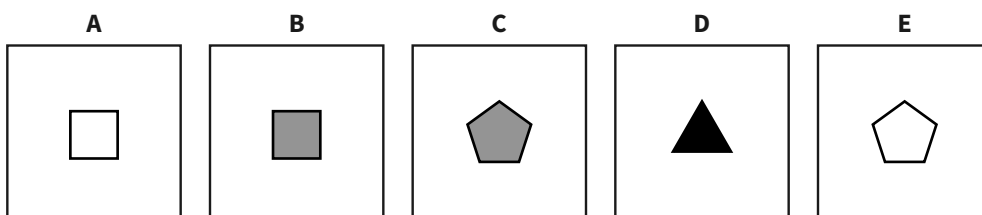


**Question 33**

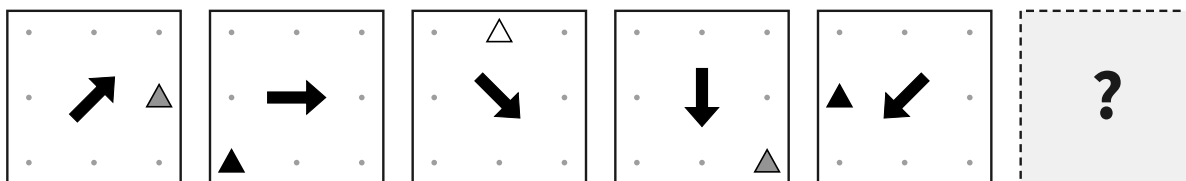
Which figure comes next in the series?



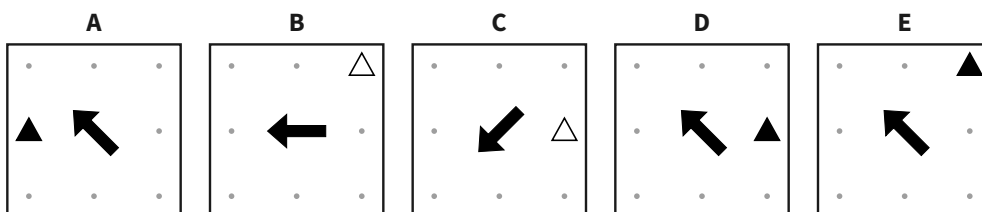
Answer options

**Question 34**

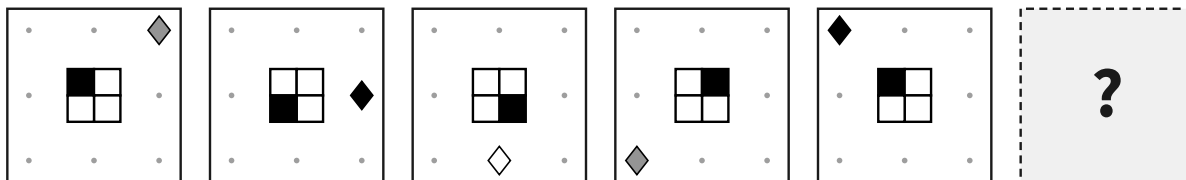
Which figure comes next in the series?



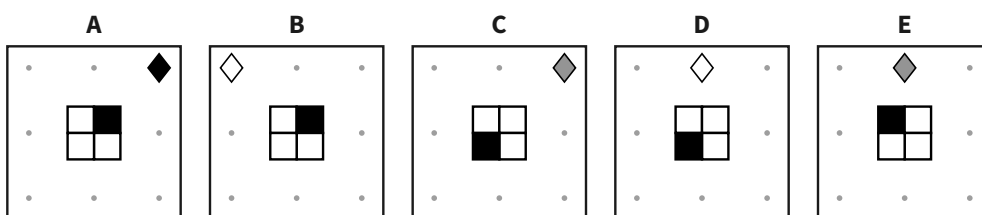
Answer options

**Question 35**

Which figure comes next in the series?

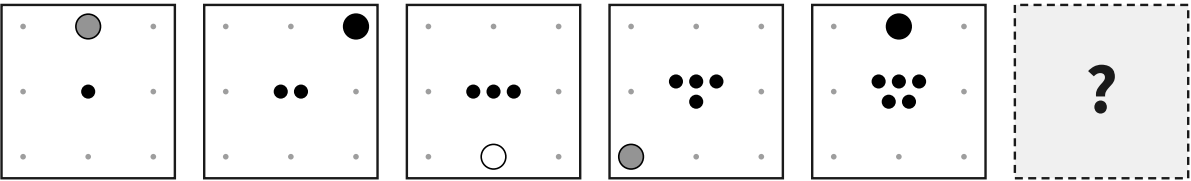


Answer options

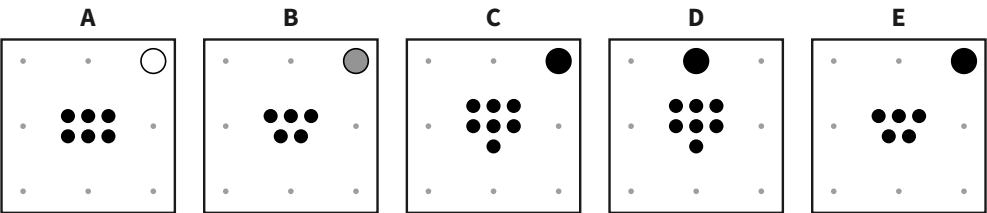


Question 36

Which figure comes next in the series?

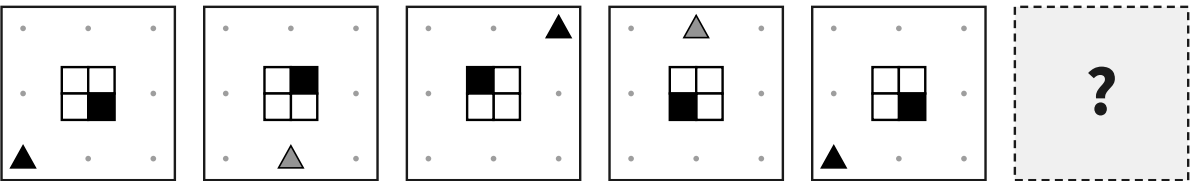


Answer options

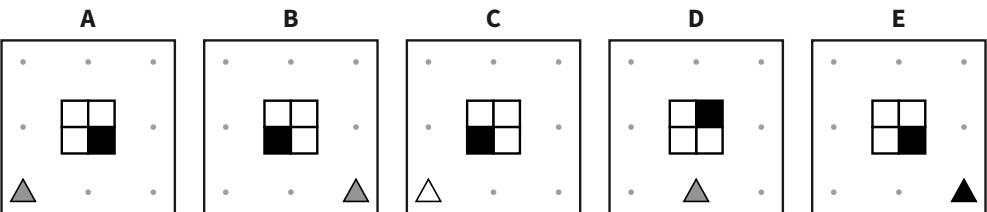


Question 37

Which figure comes next in the series?

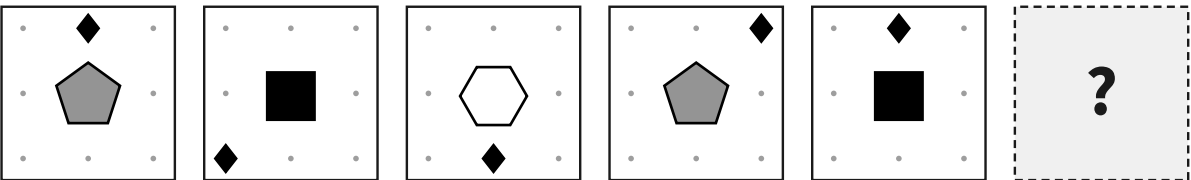


Answer options

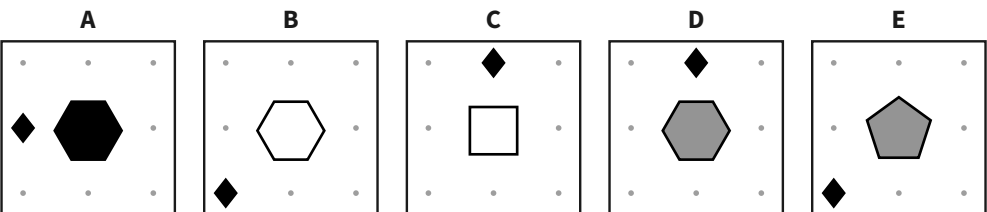


Question 38

Which figure comes next in the series?

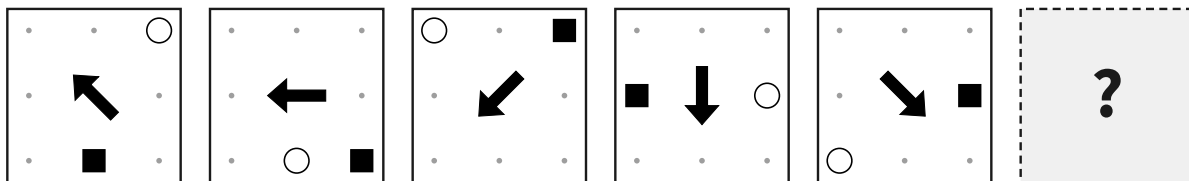


Answer options

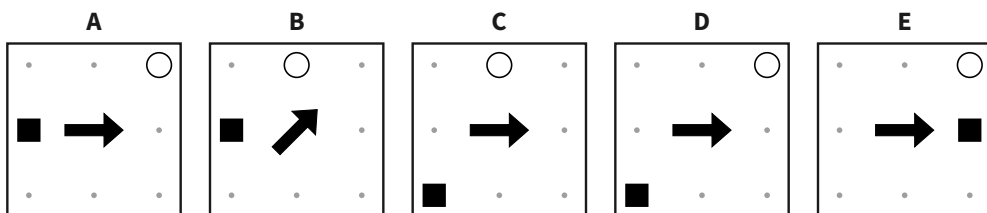


**Question 39**

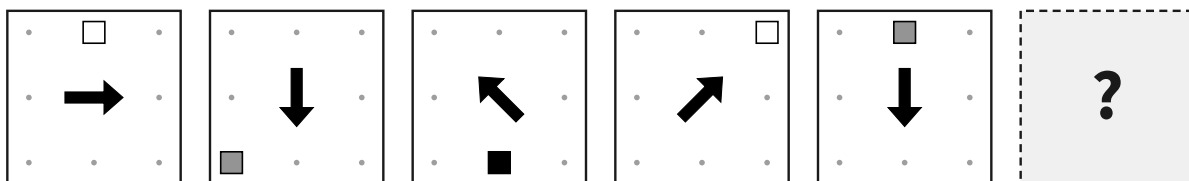
Which figure comes next in the series?



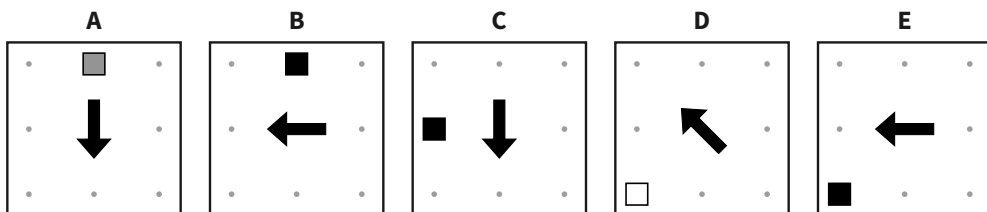
Answer options

**Question 40**

Which figure comes next in the series?



Answer options





# Mock Exam 3

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**40 questions • suggested time: 65 minutes**

- Questions 1–20: verbal reasoning (about 35 minutes)
- Questions 21–30: numerical reasoning (about 20 minutes)
- Questions 31–40: abstract reasoning (about 10 minutes)

These timings reflect typical past EPSO formats; check the Notice of Competition for your procedure. Use the answer sheet at the back of the book, work without interruptions, and answer every question: in typical formats there is no penalty for a wrong answer.

**Score guide (out of 40).** 34–40: excellent, test-ready. 28–33: strong; polish weaker areas. 22–27: developing; revisit the method chapters and redo your errors. Below 22: build accuracy first with the practice sets. Real pass marks differ between procedures, so treat these bands as a guide only.

## Verbal reasoning (questions 1–20)

### Question 1

Before railways, moving heavy goods overland was slow and costly. A horse could pull only a modest load in a cart on the poor roads of the eighteenth century, but the same horse towing a barge on still water could move many times as much. This advantage drove a wave of canal building in several European countries in the late eighteenth and early nineteenth centuries, much of it financed by private investors and promoted by mine owners and manufacturers seeking cheaper access to markets for coal, iron and pottery. Canals were not easy to build in hilly country: changes in level required locks, tunnels or aqueducts, all of which added to costs and, in the case of locks, to journey times. Reliable water supplies were also needed to refill the locks, particularly on the highest sections of a route. From the 1830s onwards, railways offered faster transport that was less constrained by terrain, and many canal companies saw their traffic and income decline, although some canals continued to carry bulk cargoes for decades.

Which of the following statements is correct?

- A Canals were built mainly with money provided by governments.
- B The arrival of railways brought all canal traffic to an end within a few years, even for bulk cargoes such as coal.
- C A key attraction of canals was that a horse could move a far heavier load on water than on land.
- D Locks shortened journey times on canals in hilly areas.

### Question 2

When high-speed rail lines are built between major cities, they often compete directly with short-haul flights. The decisive factor for many travellers is total door-to-door journey time rather than the speed of the vehicle itself. Although an aircraft travels far faster than any train, air passengers must reach airports that are often located outside city centres, pass through security checks and arrive well before departure. Railway stations, by contrast, are typically central, and boarding takes only minutes, so on shorter routes the train can deliver passengers sooner despite its lower speed. As a result, analyses of a number of routes suggest that rail tends to win a large share of the combined market when the train journey takes less than about three hours, while air travel generally remains dominant where the train takes much longer. Other considerations also play a part, including ticket prices, the frequency of services and the ability to work during the journey. Building high-speed lines is very expensive, however, and the case for a new line depends heavily on the number of passengers it is expected to carry.

Which of the following statements is correct?

- A The slower mode of transport can sometimes offer the quicker overall journey.
- B High-speed rail always captures the majority of travellers on routes where the train takes under three hours.
- C Travellers choose between rail and air solely on the basis of journey time.
- D High-speed rail lines are cheaper to build than airports.



**Question 3**

Desalination removes dissolved salts from seawater or brackish groundwater to produce fresh water. The two main approaches are thermal distillation, in which water is evaporated and the vapour condensed, and reverse osmosis, in which water is forced at high pressure through a membrane that holds back most of the salt. Reverse osmosis has become the dominant technology in many regions because it generally requires less energy than distillation, although it still uses considerably more energy than the treatment of fresh water drawn from rivers or aquifers. Membranes must be protected from clogging, so the incoming water is usually pre-treated to remove particles and organic matter. Every desalination plant produces a concentrated salt solution, or brine, as a by-product. If this brine is discharged carelessly, it can harm marine organisms near the outlet, because it is denser than seawater and tends to sink and spread along the sea bed. Plant operators can reduce the impact by diluting the brine before release or by using diffusers that disperse it more widely.

**Which of the following statements is correct?**

- A** Reverse osmosis uses less energy than distillation but more than treating river water.
- B** Reverse osmosis membranes remove all of the dissolved salt from the seawater passing through them.
- C** Brine discharged from desalination plants tends to rise towards the surface.
- D** Desalination is the cheapest way of supplying fresh water to coastal cities.

**Question 4**

Date labels on food are intended to inform consumers, yet they are frequently misunderstood, and confusion between the two most common types is thought to contribute to avoidable household food waste. A “use by” date relates to safety. It appears on highly perishable products, such as fresh meat or ready-prepared salads, which could pose a health risk after a short period. Such food should not be eaten after the date shown, even if it looks and smells acceptable, because some harmful bacteria produce no visible signs of spoilage. A “best before” date, by contrast, relates to quality. It is used on products such as dried pasta, tinned goods and biscuits, which may lose some flavour or texture over time but generally remain safe to eat after the date, provided they have been stored correctly and the packaging is undamaged. Both kinds of date assume that the storage instructions on the label have been followed; a product kept in a warm place may deteriorate before its date. Several public campaigns have therefore sought to explain the difference to consumers.

**Which of the following statements is correct?**

- A** Food past its “use by” date is safe to eat as long as it smells normal.
- B** A product that has not been stored as instructed may deteriorate before the date printed on it.
- C** Products labelled with a “best before” date must be thrown away once that date has passed.
- D** Confusion over date labels is the main cause of the food that households throw away unnecessarily.

**Question 5**

Nitrogen is essential to all living things, yet the nitrogen gas that makes up most of the atmosphere is unusable by most organisms. It must first be “fixed”, that is, converted into compounds such as ammonia that plants can absorb. In nature, this is done chiefly by certain bacteria, some of which live in nodules on the roots of certain plants, and to a lesser extent by lightning. Other bacteria later return nitrogen to the atmosphere through a process called denitrification, completing the cycle. In the early twentieth century, an industrial process was developed for fixing nitrogen from the air on a large scale, making synthetic fertilisers cheap and abundant and greatly increasing crop yields. However, crops take up only part of the nitrogen applied to fields. Much of the remainder, particularly in the form of nitrate, is washed by rain into rivers and coastal waters. There it can stimulate excessive growth of algae; when the algae die and decompose, bacteria consume the dissolved oxygen, sometimes creating zones in which fish and other animals cannot survive. Measures to limit run-off include applying fertiliser in smaller doses timed to crop needs and maintaining vegetated strips along watercourses.

**Which of the following statements is correct?**

- A** Most organisms can make direct use of the nitrogen gas in the atmosphere.
- B** Algal growth fed by run-off can lead indirectly to a loss of oxygen in the water.
- C** Lightning is the main natural means by which nitrogen is fixed.
- D** Nitrogen returns to the atmosphere mainly through lightning.

**Question 6**

In general contract law, an agreement is formed when one party makes an offer and the other accepts it unconditionally. Not every proposal, however, amounts to an offer. Many legal systems distinguish an offer – a statement by which a person indicates willingness to be bound as soon as the terms are accepted – from an invitation to negotiate, sometimes called an invitation to treat. Goods displayed in a shop window with a price label, or advertised in a catalogue, are commonly treated as invitations: it is the customer who makes the offer by proposing to buy, and the seller who may accept or decline. The rationale is practical. If a display were an offer, a shopkeeper with limited stock could be bound to sell to more customers than the stock allowed. Where a reply to an offer introduces new or different terms, it is generally treated not as an acceptance but as a counter-offer, which rejects the original offer and may itself be accepted or refused.

**Which of the following statements is correct?**

- A** A shop displaying goods with a price label is obliged to sell them to any customer who asks.
- B** A reply that accepts an offer while altering one of its terms concludes a contract on the altered terms.
- C** Where displayed goods are treated as an invitation to treat, it is the customer proposing to buy who makes the offer.
- D** All legal systems treat catalogue advertisements as invitations to negotiate.

**Question 7**

Research on decision-making has repeatedly shown that people do not evaluate outcomes in purely absolute terms. Instead, they tend to judge them as gains or losses relative to a reference point, often their current situation, and losses generally weigh more heavily than gains of the same size: losing a sum of money typically causes more dissatisfaction than finding the same sum causes pleasure. This asymmetry, known as loss aversion, helps to explain the framing effect, whereby logically equivalent options lead to different choices depending on whether they are described in terms of what may be gained or what may be lost. In experiments, a treatment described as having a ninety per cent survival rate is often chosen more readily than one described as having a ten per cent mortality rate, although the two descriptions convey the same information. For public bodies, the implication is not that framing can be avoided – every message must be phrased somehow – but that communications should be tested and, where important decisions are at stake, presented in more than one form.

**Which of the following statements is correct?**

- A** Public bodies should aim to communicate important decisions without using any framing at all.
- B** People generally feel a gain more intensely than they feel a loss of exactly the same size.
- C** Doctors are more susceptible to the framing effect than their patients.
- D** Two equivalent descriptions of a treatment can lead people to make different choices.

**Question 8**

Until the eighteenth century, most professional musicians in Europe earned their living in the service of a court, a church or an aristocratic household. Music was composed for particular occasions, and the employer largely determined what was performed. During the eighteenth century, the public concert – an event open to anyone able to buy a ticket or subscription – became increasingly important, first in large commercial cities. This gave composers and performers a new source of income, although few could rely on it alone, and many combined concert work with teaching, publishing or remaining forms of patronage. The growth of a paying public also influenced the music itself. Organisers needed to attract audiences repeatedly, which favoured works that made a strong impression on first hearing, and the use of large halls encouraged larger orchestras. Over the following century, concert programmes increasingly included works by composers who were no longer living, contributing to the idea of a lasting repertoire of “classical” masterpieces.

**Which of the following statements is correct?**

- A** Public concerts allowed most composers to give up teaching and patronage entirely.
- B** The need to win paying audiences had some influence on the kind of music written.
- C** Public concerts first became important in small rural communities before spreading to cities.
- D** Orchestras grew larger mainly because composers of the period demanded greater volume and dramatic contrast in their music.

**Question 9**

At the close of the nineteenth century, the English reformer Ebenezer Howard proposed a model of urban development that he called the garden city. Howard was troubled both by the overcrowding and poor sanitation of large industrial cities and by the depopulation and limited opportunities of the countryside. His solution was to combine the advantages of town and country in planned settlements of limited size, each surrounded by a permanent agricultural belt. Once a garden city reached its intended population, growth would continue through the founding of a new settlement nearby rather than through the expansion of the existing one, producing a cluster of towns linked by rail. Crucially, Howard intended the land to be held collectively, so that the rise in land values resulting from development would benefit the community rather than private landlords. Two English towns were built on garden-city principles in the early twentieth century, and the idea influenced planners in many countries. Critics have observed, however, that communal ownership of land was only partly realised in practice, and that many later “garden suburbs” borrowed the low-density design while abandoning the self-contained character of Howard’s scheme.

**Which of the following statements is correct?**

- A** Howard believed that garden cities should keep expanding outwards as their populations grew.
- B** Howard wanted gains in land value from development to go to residents collectively.
- C** Later garden suburbs kept all the essentials of Howard’s model.
- D** Howard’s proposal was motivated solely by poor living conditions in industrial cities.

**Question 10**

A port town has run a participatory budgeting scheme for several years. Each year, a fixed share of the municipal capital budget – around five per cent – is set aside for projects proposed and selected by residents. Any resident aged sixteen or over may submit a proposal, which municipal officials then check for legal and technical feasibility and for cost. Proposals that pass this check are put to a public vote, and the highest-ranked projects are funded until the allocated sum is exhausted. The council retains a formal right to reject a winning project, although it has never exercised this right. An evaluation commissioned by the town found that turnout in the votes remained modest, typically below one in ten eligible residents, and that participants were disproportionately older and more highly educated than the population as a whole. Neighbourhoods with active residents’ associations tended to submit more proposals and to win a larger share of funding. The evaluation recommended targeted outreach in under-represented districts, but the council has not yet decided whether to adopt this recommendation.

**Which of the following statements is NOT supported by the passage?**

- A** Targeted outreach in under-represented districts has helped to broaden participation in the scheme.
- B** Proposals are reviewed by officials for feasibility before residents vote on them.
- C** Some residents under the age of eighteen are entitled to put forward proposals.
- D** The council has the power to set aside the result of the public vote on a project, even though it has never used it.

**Question 11**

The Encyclopédie, published in France in the middle of the eighteenth century under the editorship of Denis Diderot and, in its early years, Jean le Rond d'Alembert, is often regarded as a defining work of the Enlightenment. Its ambition was to gather and organise the knowledge of the age, from philosophy and theology to the crafts and trades. The attention paid to the practical arts was itself significant: by describing in detail how artisans worked, and illustrating their tools in numerous plates, the editors implied that manual skill deserved the same respect as abstract learning. The work was also a vehicle for criticism. Because open attacks on religious and political authority were dangerous, contributors sometimes placed critical remarks in unexpected articles and used cross-references to guide attentive readers from an orthodox entry to a subversive one. The project met repeated opposition; at one point the royal authorities withdrew permission for its publication, and the remaining volumes of text were later issued with misleading indications of where they had been printed. Despite these obstacles, the work was completed and found readers well beyond France.

**Which of the following statements is correct?**

- A** The Encyclopédie was devoted mainly to the crafts and trades rather than to philosophy.
- B** Diderot and d'Alembert edited the Encyclopédie jointly from its beginning to its completion.
- C** Cross-references were used to steer readers towards criticism.
- D** Losing official permission ended the project prematurely.

**Question 12**

The numerals used throughout Europe today originated in India and reached the continent largely through the works of scholars writing in Arabic. Their decisive feature is not the shapes of the digits but the positional principle: the value of a symbol depends on where it stands, so that the same “3” can mean three, thirty or three hundred. Such a system requires a symbol for an empty position, and the development of zero as a placeholder made it workable. Roman numerals, by contrast, are not positional, which made written arithmetic cumbersome; in practice, calculations were usually performed on a counting board or abacus and only the results were written down. The new notation spread slowly in medieval Europe. It was taken up first by some scholars and by merchants who valued the ability to calculate directly on paper, while for a considerable time other merchants and officials continued to keep their accounts in the older notation. One reason sometimes given is a concern that the new digits could be altered more easily – turning a 0 into a 6 or a 9, for example.

**Which of the following statements is correct?**

- A** It was chiefly the shapes of the Indian digits that made them better suited to calculation than Roman numerals.
- B** Because Roman numerals were not positional, medieval Europeans were unable to carry out arithmetic before adopting the new system.
- C** With Roman numerals, the intermediate steps of a calculation were generally not written down, only its result.
- D** Fears about falsification led authorities throughout Europe to ban the new numerals in accounting.

**Question 13**

Road traffic is the main source of environmental noise in most urban areas, and long-term exposure has been associated with sleep disturbance and stress. A city transport department has assessed several measures for reducing it. Above speeds of roughly 30 to 50 kilometres per hour, noise from cars comes mainly from the contact between tyres and the road surface rather than from engines, so engine-related measures matter most in slow, stop-start traffic. Porous asphalt, which absorbs part of the sound, can reduce noise noticeably on faster roads, but its pores gradually become clogged and it must be cleaned or replaced more often than conventional surfaces. Lowering speed limits reduces tyre noise and is cheap to implement, although its effect depends on enforcement. Noise barriers are effective for buildings directly behind them but are unsuitable for most city streets, where they would block access to homes and shops. The department notes that a shift to electric vehicles will reduce noise mainly at low speeds, because at higher speeds tyre noise dominates regardless of how a vehicle is powered.

Which of the following statements is NOT supported by the passage?

- A Electric vehicles are likely to bring the greatest noise benefit in slow-moving traffic.
- B Porous asphalt requires more maintenance than standard road surfaces.
- C The noise benefit of a lower speed limit may be limited if drivers do not comply with it.
- D Noise barriers are the most effective way of reducing noise on city streets.

**Question 14**

Patents grant inventors a temporary right to prevent others from exploiting their inventions, in exchange for public disclosure of how those inventions work. The system rests on a trade-off. Without some protection, rivals could copy a new product at little cost, and inventors might be unwilling to invest in costly research whose results others could freely appropriate. With protection, however, the patent holder can charge prices above those that would prevail under competition, restricting access to the invention for the duration of the patent. Disclosure is intended to offset this cost: once the patent expires, anyone may use the invention, and even before then, others can learn from the published description and build upon it. Economists disagree about where the balance lies. The importance of patents appears to vary considerably between sectors. In fields such as pharmaceuticals, where development costs are high and products are easy to imitate, firms tend to regard patents as essential, whereas in other industries firms often rely more on secrecy, speed to market or complementary services. Some researchers also argue that overlapping patents in complex technologies can impede, rather than encourage, further innovation.

Which of the following statements is NOT supported by the passage?

- A The disclosure required by the patent system benefits other inventors only once the patent has expired.
- B A patent may allow its holder to set prices higher than would be possible in a competitive market.
- C The value that firms attach to patents is likely to depend partly on how easily their products can be imitated.
- D It has been suggested that patents can, in some circumstances, hinder innovation.

**Question 15**

Social contract theory seeks to explain the legitimacy of political authority by imagining how individuals might agree to form a government. Thomas Hobbes, writing in the mid-seventeenth century amid civil war in England, described life in a hypothetical “state of nature” without government as a condition of constant insecurity. Rational individuals, he argued, would therefore surrender extensive powers to a sovereign capable of keeping the peace; even a harsh ruler was preferable to a return to anarchy. John Locke, writing later in the same century, took a less bleak view of the state of nature, in which people already possessed natural rights to life, liberty and property. For Locke, government was established to protect these rights more securely, and its authority depended on the consent of the governed. If a government persistently violated the rights it had been created to protect, the people were entitled to resist and replace it. Both thinkers thus grounded authority in agreement rather than in divine appointment or hereditary claim, but they drew sharply different conclusions about the limits of that authority.

Which of the following statements is correct?

- A Locke’s account implies that a government’s claim to obedience depends on how it behaves.
- B The passage presents the civil war as the cause of Hobbes’s pessimism.
- C Locke regarded any violation of natural rights by a government as sufficient grounds for replacing it.
- D Hobbes held that a sovereign’s harshness justified a return to the state of nature.

**Question 16**

For more than two thousand years, Euclid’s geometry was regarded not merely as a useful system but as a description of the necessary structure of space. Most of its postulates seemed self-evident, but the fifth – the parallel postulate, which in one common formulation states that through a point not on a given line exactly one parallel to that line can be drawn – appeared less obvious, and many mathematicians tried to prove it from the others. In the nineteenth century, several mathematicians working independently showed that this effort was bound to fail: consistent geometries could be constructed in which the parallel postulate is replaced by an alternative, allowing either no parallel lines or infinitely many. In such geometries, the angles of a triangle no longer add up to exactly two right angles. The discovery was at first regarded by many as a mathematical curiosity. Its significance grew considerably when physicists later found that curved geometries of this kind describe space and time on very large scales better than Euclid’s does, although Euclidean geometry remains an excellent approximation for everyday purposes.

Which of the following statements is NOT supported by the passage?

- A Attempts to derive the parallel postulate from Euclid’s other postulates could not have succeeded.
- B The importance of the new geometries for physics was not recognised as soon as they were discovered.
- C In some geometries, the angles of a triangle do not add up to two right angles.
- D Physicists’ use of non-Euclidean geometry has shown that Euclidean geometry gives unreliable results in everyday applications.

**Question 17**

The term “Green Revolution” refers to the spread, from the mid-twentieth century onwards, of high-yielding varieties of staple cereals, notably wheat and rice, together with synthetic fertilisers, pesticides and irrigation. The new varieties were bred to be shorter and sturdier, so that they could carry heavier grain without falling over, and to respond strongly to fertiliser. In regions where these inputs were available, yields rose dramatically, and the threat of famine receded in several countries that had previously depended on food imports. Historians and economists nonetheless offer a mixed assessment. The gains were concentrated in irrigated areas; farmers in rain-fed regions, where the new varieties performed less reliably, benefited far less. Because the full package required purchased inputs, wealthier farmers with access to credit were often able to adopt it more quickly than smallholders. Heavy use of irrigation and agrochemicals has also been linked, in some districts, to falling groundwater levels and soil degradation. Supporters reply that without higher yields, far more forest and grassland would have had to be converted to farmland to feed growing populations.

**Which of the following statements is correct?**

- A** The Green Revolution raised yields to a similar extent in irrigated and rain-fed regions.
- B** Smallholders were unable to adopt the new package of varieties and inputs.
- C** Supporters of the Green Revolution maintain that it lessened the need to turn natural land over to farming.
- D** The Green Revolution caused groundwater levels to fall and soils to degrade in every region where the new package was adopted.

**Question 18**

In many wholesale electricity markets, the price for a given hour is set through an auction. Generators submit offers stating how much electricity they can supply and at what price, and the offers are ranked from the cheapest upwards – the so-called merit order. Plants are called upon in this sequence until demand is met, and in a common design all successful generators then receive the price offered by the last, most expensive plant needed. Because wind and solar installations have almost no fuel costs, they can offer their output at very low prices and are usually placed at the front of the queue. When they produce a great deal of electricity, they push more expensive plants, typically those burning gas, out of the merit order, and the price paid to all generators tends to fall. This is known as the merit-order effect. It has a paradoxical consequence: the more wind and solar capacity is installed, the lower the market prices those same installations may receive when they are generating strongly. Conversely, when renewable output is low and gas-fired plants set the price, a rise in the cost of gas can raise the price of all electricity traded in that hour, whatever its source.

**Which of the following statements is correct?**

- A** Under the design described, each successful generator is paid the price that it offered.
- B** Wind and solar generators always determine the wholesale price of electricity.
- C** The merit-order effect guarantees that expanding renewable capacity will lower households' electricity bills.
- D** A wind farm can be paid the price set by a gas-fired plant.



**Question 19**

Until well into the nineteenth century, screws, nuts and bolts were largely made according to the individual practice of each workshop. The angle of the thread, its depth and the number of turns per inch varied from maker to maker, so that a bolt from one factory would rarely fit a nut from another. Repairing a machine far from where it had been built could therefore mean cutting replacement parts by hand. As mechanised production expanded, engineers began to argue that a limited set of agreed thread forms would reduce costs and make parts interchangeable. Proposals of this kind were taken up first by individual firms and engineering societies, and only gradually by governments and by major purchasers such as navies. Standardisation within countries did not end the problem. Different nations adopted different systems, and machinery assembled from parts made in more than one country could still require several sets of tools. Efforts to reconcile national systems continued for decades. The history of the screw thread is often cited as an example of how a seemingly minor technical detail can carry large economic consequences.

Which of the following statements is NOT supported by the passage?

- A** Before standardisation, parts made by different workshops were often not interchangeable.
- B** Agreed thread forms were promoted partly on the grounds that they would help to lower costs.
- C** Governments were the first to adopt standard thread forms, and private firms followed their lead.
- D** Standardisation within individual countries did not by itself remove every difficulty in combining parts.

**Question 20**

In many legal systems, courts reviewing measures that restrict individual rights apply a test of proportionality. Although formulations differ, the test usually involves several stages. First, the measure must pursue a legitimate aim, such as the protection of public health or public order. Second, it must be suitable for achieving that aim; a measure that does nothing to advance its stated purpose will fail at this stage. Third, it must be necessary, meaning that the aim could not be achieved by a less restrictive means that would be equally effective. Some courts add a final stage, sometimes called proportionality in the strict sense, at which they weigh the burden on the individual against the importance of the aim, even where no less restrictive alternative exists. Courts generally allow public authorities a margin of discretion, particularly in complex technical or economic matters, and the intensity of review may vary with the importance of the right affected. A measure that fails any one stage is regarded as disproportionate, however well it performs at the others.

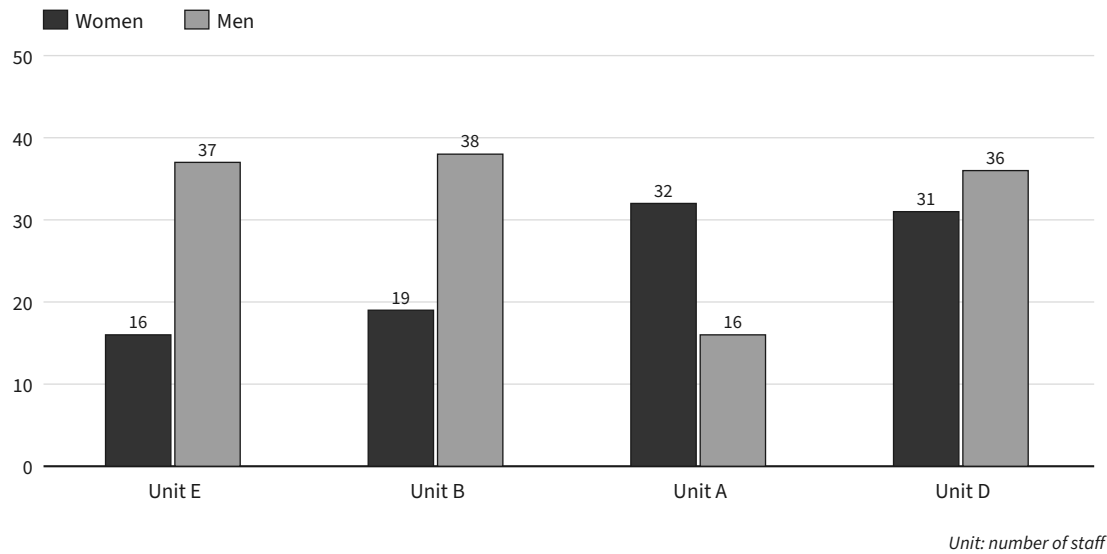
Which of the following statements is correct?

- A** A suitable measure with a legitimate aim may still fail the test if an equally effective, less restrictive option existed.
- B** Under the test, a measure that pursues a legitimate aim will be upheld provided that it actually helps to achieve that aim.
- C** All courts applying the test weigh the burden on the individual against the importance of the aim pursued.
- D** Courts review measures that restrict rights with the same intensity, whichever right is affected and whatever the subject matter.

## Numerical reasoning (questions 21–30)

### Question 21

Staff by gender in four units



All data are fictitious.

If the total staff of Unit A increases by 25% and the proportion of women stays the same, how many women will Unit A employ?

- A 40
- B 20
- C 44
- D 57

### Question 22

Visitors to national art galleries (thousands)

| Country  | Year 1 | Year 2 | Year 3 |
|----------|--------|--------|--------|
| Slovenia | 326    | 304    | 285    |
| Poland   | 842    | 863    | 869    |
| Austria  | 421    | 451    | 432    |
| Greece   | 839    | 849    | 1,004  |
| Czechia  | 695    | 753    | 826    |

All data are fictitious.

By approximately what percentage did the number of visitors to art galleries in Slovenia decrease between Year 1 and Year 3?

- A 6.2%
- B 12.6%
- C 6.7%
- D 14.4%

**Question 23**

Grant applications received and approval rate

| Country     | Applications | Approval rate |
|-------------|--------------|---------------|
| Hungary     | 1,800        | 12%           |
| Netherlands | 3,000        | 30%           |
| Romania     | 500          | 18%           |
| Spain       | 2,400        | 24%           |

*All data are fictitious.*

How many applications from Romania were rejected?

- A** 500
- B** 320
- C** 410
- D** 455

**Question 24**

Staff numbers and average monthly salary by unit

| Unit   | Staff | Average monthly salary (€) |
|--------|-------|----------------------------|
| Unit A | 20    | 4,500                      |
| Unit B | 48    | 4,200                      |
| Unit D | 15    | 8,900                      |
| Unit E | 20    | 9,000                      |

*All data are fictitious.*

What is the average monthly salary across all staff in the four units combined (to the nearest euro)?

- A** €6,650
- B** €4,288
- C** €7,336
- D** €5,875

### Question 25

Index of average house prices (Year 1 = 100)

| Country   | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|-----------|--------|--------|--------|--------|--------|
| Lithuania | 100.0  | 107.3  | 115.2  | 118.4  | 120.7  |
| Czechia   | 100.0  | 107.1  | 111.8  | 112.3  | 122.1  |
| Denmark   | 100.0  | 106.7  | 109.8  | 114.7  | 120.4  |
| Hungary   | 100.0  | 103.1  | 103.9  | 110.8  | 118.0  |

*All data are fictitious.*

By approximately what percentage did average house prices in Czechia rise between Year 3 and Year 5?

- A 9.2%
- B 8.4%
- C 22.1%
- D 10.3%

### Question 26

Rail routes: distance and average train speed

| Route               | Distance (km) | Average speed (km/h) |
|---------------------|---------------|----------------------|
| Midvale–Stonebridge | 122           | 100                  |
| Riverton–Brookhaven | 193           | 120                  |
| Westfield–Fairhaven | 326           | 80                   |
| Kingsford–Hillcrest | 140           | 140                  |

*All data are fictitious.*

If the average speed on the Riverton–Brookhaven route rose to 150 km/h, approximately how many minutes would the journey save?

- A 10 minutes
- B 19 minutes
- C 24 minutes
- D 29 minutes

**Question 27**

Cases closed per month by three teams

| Team       | Jan | Feb | Mar | Apr | May | Jun |
|------------|-----|-----|-----|-----|-----|-----|
| Team East  | 40  | 25  | 52  | 57  | 64  | 33  |
| Team South | 24  | 41  | 55  | 54  | 40  | —   |
| Team West  | 21  | 63  | 39  | 43  | 55  | 53  |

— = figure not yet available

*All data are fictitious.*

How many cases would Team South need to close in June for its average over January to June to be exactly 50 cases per month?

- A 50
- B 36
- C 43
- D 86

**Question 28**

Staff numbers and average monthly salary by unit

| Unit   | Staff | Average monthly salary (€) |
|--------|-------|----------------------------|
| Unit B | 33    | 5,300                      |
| Unit E | 19    | 7,600                      |
| Unit A | 7     | 9,200                      |
| Unit C | 16    | 7,300                      |

*All data are fictitious.*

What is the average monthly salary across all staff in the four units combined (to the nearest euro)?

- A €7,752
- B €6,140
- C €6,673
- D €7,350

Question 29

Population and public spending on culture in five regions of one country

| Region         | Population (million) | Spending (€ million) |
|----------------|----------------------|----------------------|
| Coastal region | 5.75                 | 701.0                |
| Western region | 0.43                 | 82.9                 |
| Central region | 3.54                 | 555.3                |
| Alpine region  | 1.72                 | 277.0                |
| Eastern region | 3.92                 | 373.1                |

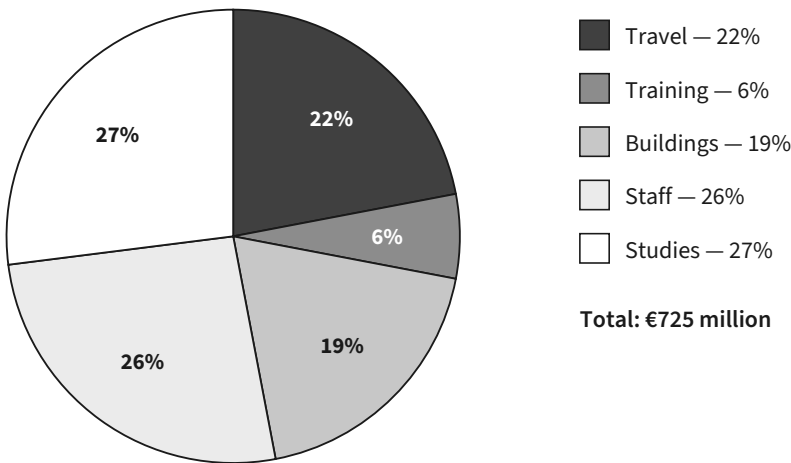
All data are fictitious.

Which of the following regions had the highest spending on culture per inhabitant?

- A Coastal region
- B Central region
- C Western region
- D Eastern region

Question 30

Breakdown of an agency’s annual budget (total €725 million)



All data are fictitious.

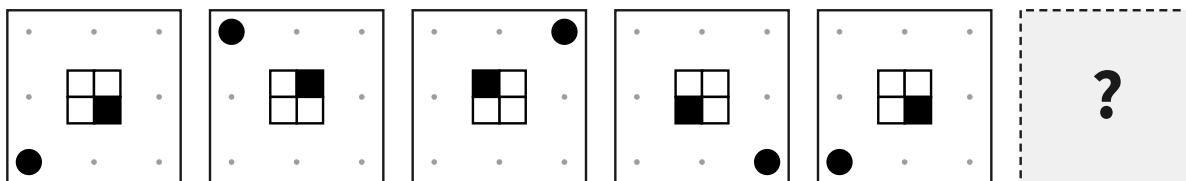
Next year the Studies budget is to rise by 15% while all other items stay unchanged. By how much will the Studies budget then exceed the Buildings budget?

- A €87.4 million
- B €166.8 million
- C €58.0 million
- D €66.7 million

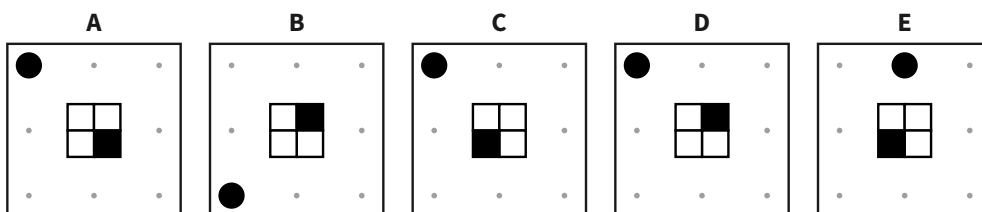
## Abstract reasoning (questions 31–40)

### Question 31

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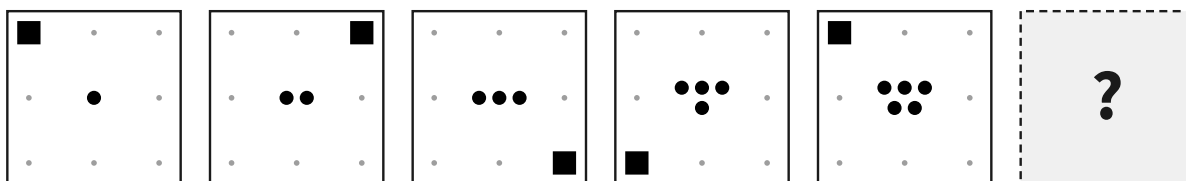


Answer options

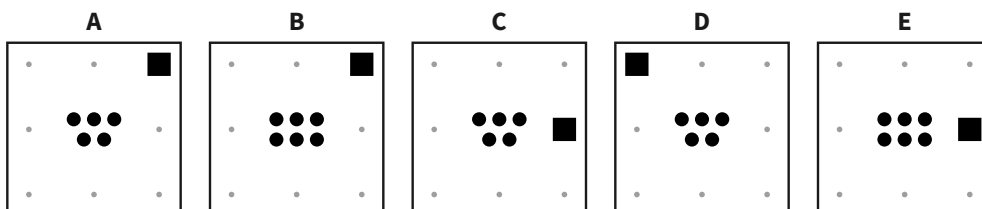


### Question 32

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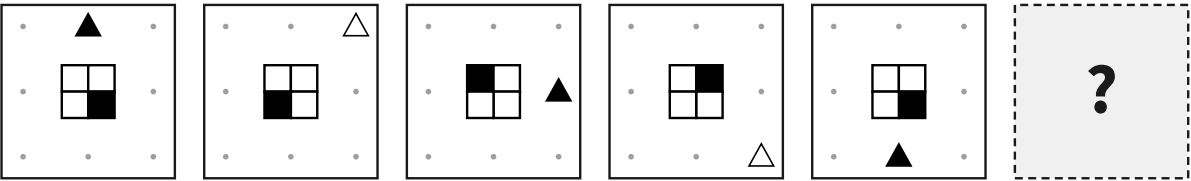


Answer options

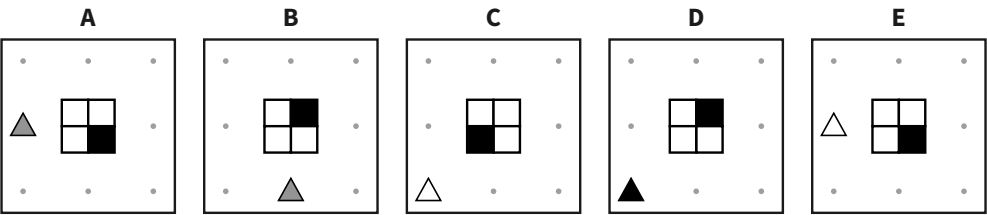


Question 33

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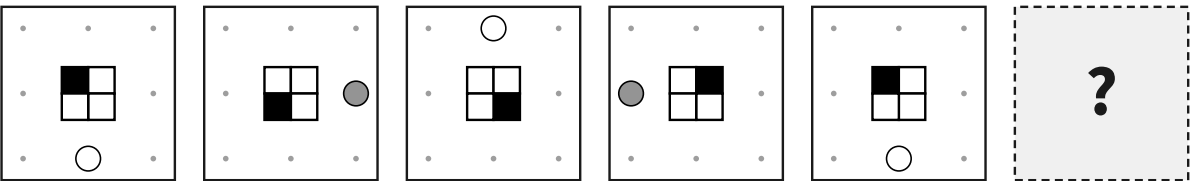


Answer options

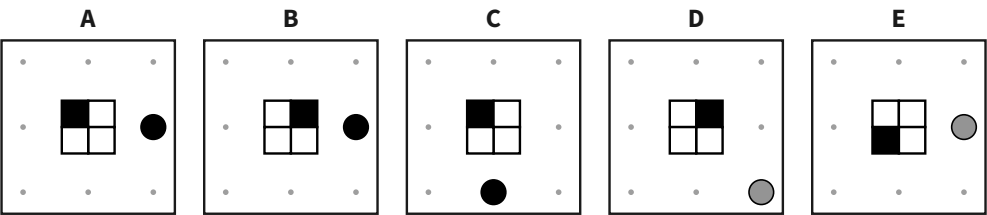


Question 34

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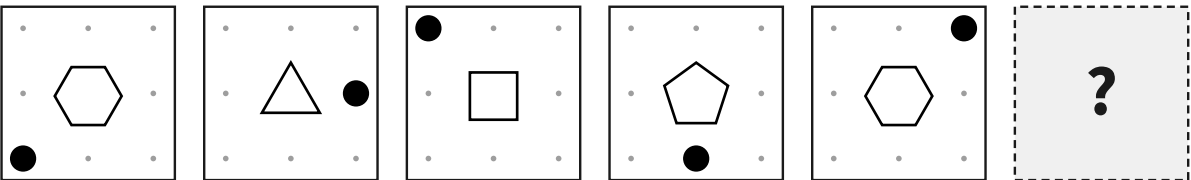


Answer options

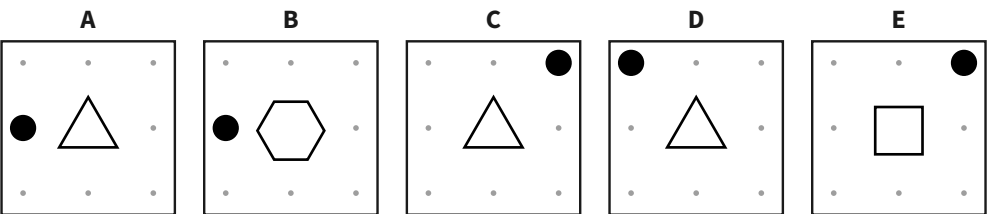


Question 35

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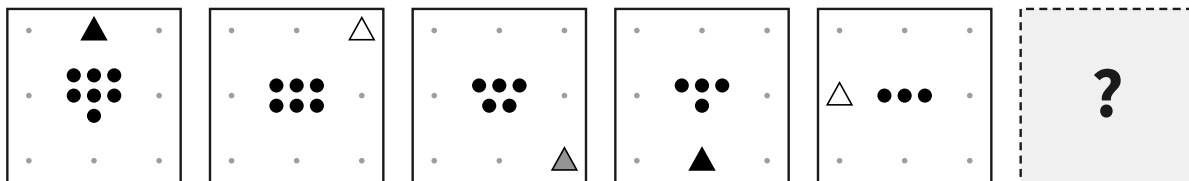
Answer options



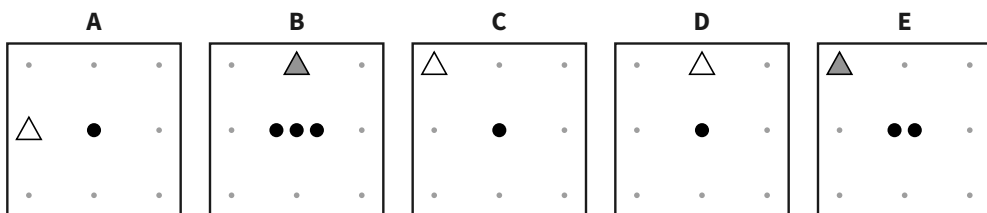


**Question 36**

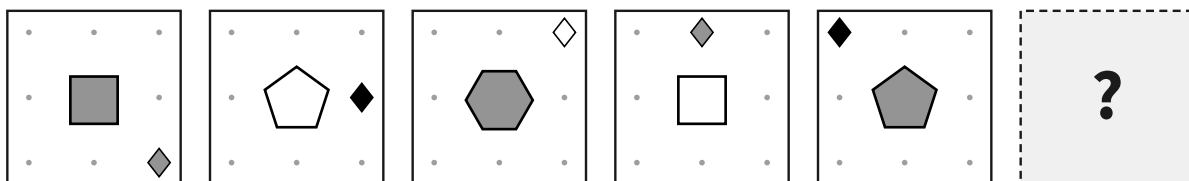
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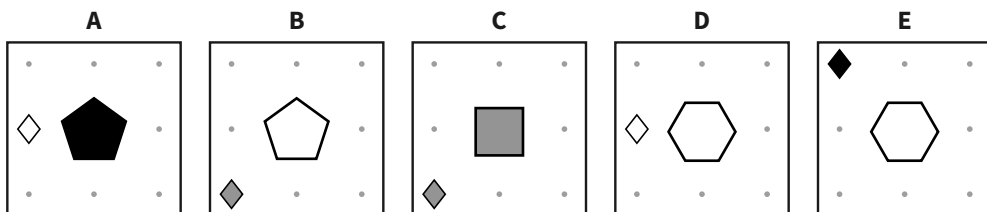
Answer options

**Question 37**

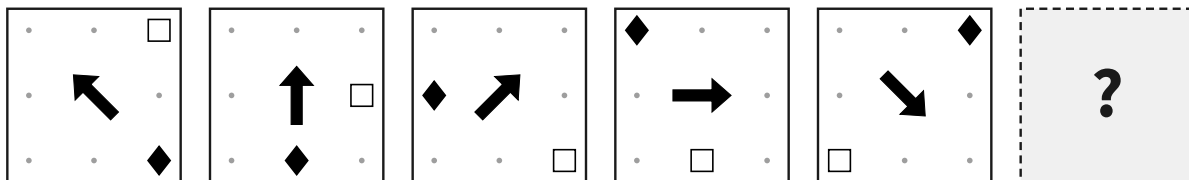
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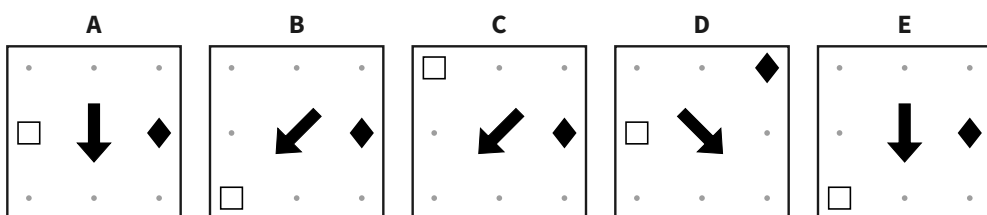
Answer options

**Question 38**

Which figure comes next in the series?

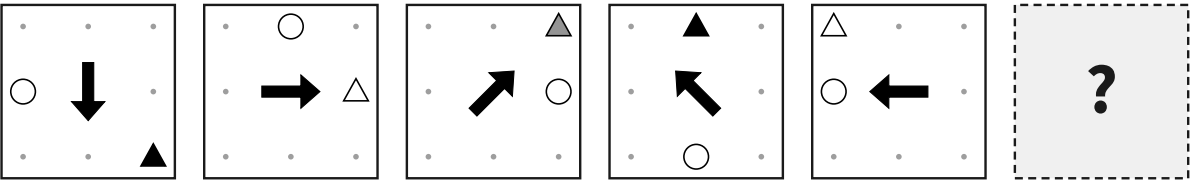


Answer options

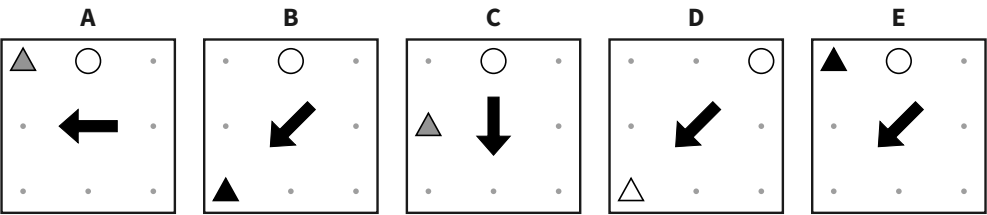


Question 39

Which figure comes next in the series?

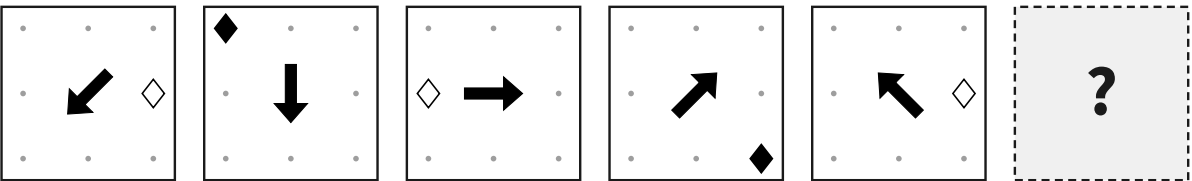


Answer options

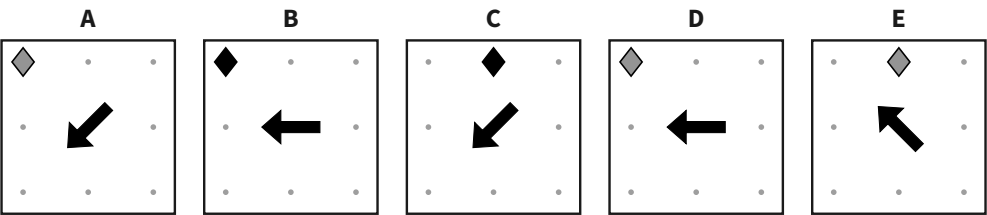


Question 40

Which figure comes next in the series?



Answer options



PART IV

# Answers and Worked Solutions

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*A quick answer key for every set, followed by a full explanation of every question.*



# Solutions — Verbal Reasoning

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## Answer key

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 1 C  | 2 B  | 3 B  | 4 A  | 5 B  | 6 C  | 7 A  | 8 B  |
| 9 D  | 10 D | 11 B | 12 C | 13 B | 14 C | 15 C | 16 A |
| 17 B | 18 D | 19 C | 20 A | 21 C | 22 A | 23 B | 24 A |
| 25 A | 26 B | 27 C | 28 C | 29 A | 30 B | 31 B | 32 D |
| 33 D | 34 C | 35 A | 36 D | 37 D | 38 B | 39 C | 40 C |
| 41 D | 42 B | 43 A | 44 A | 45 C | 46 C | 47 A | 48 D |
| 49 C | 50 D | 51 B | 52 D | 53 A | 54 D | 55 A | 56 B |
| 57 C | 58 C | 59 D | 60 B | 61 D | 62 C | 63 C | 64 D |
| 65 B |      |      |      |      |      |      |      |

## Worked solutions

### Question 1 — Answer: C

- A.** Incorrect – supported. The apprentice “receives a wage”, lower than a qualified worker’s but rising during training.
- B.** Incorrect – supported. Qualifications are designed to be recognised “across a whole sector rather than by a single firm”.
- C.** Correct – not supported. The passage reports only an association and notes that low youth unemployment may also reflect other economic features.
- D.** Incorrect – supported. Transplanting the model often depends on employers being willing to invest in training young people they may not retain.

### Question 2 — Answer: B

- A.** Incorrect – extreme wording. The passage says neither measure is inherently superior and that the mean is the right choice when totals matter.
- B.** Correct. A small number of very high earners pull the mean upwards, while the median is hardly affected, which is why the median better describes a typical household.
- C.** Incorrect – extreme wording. Offices “frequently prefer” the median for a typical household, and the mean “remains valuable”; nothing has been abandoned.
- D.** Incorrect – cause and effect. The passage attributes divergence between the two measures to skewed distributions, not to errors in the data.

### Question 3 — Answer: B

- A.** Incorrect – contradiction. A pidgin has no native speakers; when children acquire it as a first language, it becomes a creole.
- B.** Correct. A creole results when children acquire a pidgin as their first language, whereas a pidgin has no native speakers.
- C.** Incorrect – extreme wording. Not every pidgin becomes a creole, and many disappear.
- D.** Incorrect – partial truth. A pidgin draws most of its vocabulary from the dominant language, while its grammar is reduced to essentials.

**Question 4 — Answer: A**

- A.** Correct. Countries can gain from trade even when one of them is more efficient at producing everything, as Country Y illustrates.
- B.** Incorrect – contradiction. The principle does not imply that every individual benefits, and some workers may lose their jobs.
- C.** Incorrect – contradiction. What matters is opportunity cost, not absolute efficiency: X uses fewer workers for both goods, yet Y should specialise in wine.
- D.** Incorrect – contradiction. Many supporters of open trade favour policies to help affected workers retrain.

**Question 5 — Answer: B**

- A.** Incorrect – extreme wording. The one-third fall is an average; complex commercial projects showed little change.
- B.** Correct. Staff said the main benefit was the end of duplicate data entry, not any change in how applications are assessed.
- C.** Incorrect – contradiction. A paper option has been kept for applicants without internet access.
- D.** Incorrect – contradiction. Staff attributed the benefit to removing duplicate data entry rather than to changes in assessment.

**Question 6 — Answer: C**

- A.** Incorrect – contradiction. The Earth is slightly closer to the Sun during the northern winter.
- B.** Incorrect – extreme wording. The passage names two effects, the steeper angle of sunlight and longer days.
- C.** Correct. Because of the tilt, each hemisphere leans towards the Sun at a different time of year, so their seasons are reversed.
- D.** Incorrect – contradiction. Seasonal temperature differences near the Equator are “generally small”.

**Question 7 — Answer: A**

- A.** Correct. The gravitational effect (running fast) outweighs the speed effect (running slow), so uncorrected satellite clocks “gain a small amount of time each day”.
- B.** Incorrect – partial truth. Speed makes the satellites’ clocks run slow; it is the weaker gravity that makes them run fast.
- C.** Incorrect – contradiction. A receiver needs signals from “at least four satellites”.
- D.** Incorrect – outside knowledge. The passage says nothing about categories of user; it presents the corrections as necessary for the system as a whole.

**Question 8 — Answer: B**

- A.** Incorrect – contradiction. Such efforts can lengthen the season but “rarely eliminate the peak”.
- B.** Correct. Capacity built for peak demand stands idle for much of the year, raising the average cost of each visitor served.
- C.** Incorrect – outside knowledge. The passage presents temporary employment as a problem for workers seeking a stable income and says nothing about preferences for flexibility.
- D.** Incorrect – contradiction. The timing of the peak is “largely determined by climate and by the dates of school holidays”.

**Question 9 — Answer: D**

- A.** Incorrect – contradiction. The roads were built primarily for moving troops and officials; trade was a wider use that developed over time.
- B.** Incorrect – contradiction. The courier stations were generally reserved for those carrying official authorisation.
- C.** Incorrect – extreme wording. Engineers favoured direct routes only “where possible” and adapted alignments in hilly or marshy areas.
- D.** Correct. The roads were built primarily for troops and officials, yet merchants and private travellers used them extensively.

**Question 10 — Answer: D**

- A.** Incorrect – contradiction. Adoption was “neither immediate nor universal”, and traditional units persisted in France for decades.
- B.** Incorrect – extreme wording. Some countries “never fully abandoned their customary measures”.
- C.** Incorrect – contradiction. The original definition was based on the distance from the North Pole to the Equator; the speed of light is never mentioned.
- D.** Correct. The system was designed with units linked by multiples of ten, replacing an inconsistent patchwork of local measures.

**Question 11 — Answer: B**

- A.** Incorrect – contradiction. Jacobs criticised the separation of functions and argued that mixed use makes streets safest.
- B.** Correct. Her idea of “eyes on the street” holds that residents and shopkeepers who notice what happens informally discourage crime.
- C.** Incorrect – contradiction. Her ideas were at first dismissed by many professionals and only later became influential.
- D.** Incorrect – contradiction. She defended older buildings because their low rents help new businesses.

**Question 12 — Answer: C**

- A.** Incorrect – extreme wording. The passage stresses that the appropriate span depends on the nature of the work and that wide spans can overload managers.
- B.** Incorrect – contradiction. Tall structures have narrow spans, meaning fewer direct reports per manager.
- C.** Correct. Where tasks are routine and staff experienced, “a manager can oversee a large team effectively”, which is what a wide span means.
- D.** Incorrect – scope shift. The passage says “some” organisations reintroduced intermediate posts, not most.

**Question 13 — Answer: B**

- A.** Incorrect – supported. Open ocean water absorbs “the great majority” of sunlight, whereas sea ice reflects most of it.
- B.** Correct – not supported. The feedback only “helps to explain” Arctic warming, and the passage names other contributing processes, so “alone” is unjustified.
- C.** Incorrect – supported. The passage states that “positive” refers to reinforcement rather than to any beneficial effect.
- D.** Incorrect – supported. The feedback operates mainly in spring and summer, as there is little or no sunlight to absorb during the polar night.

**Question 14 — Answer: C**

- A.** Incorrect – extreme wording. A cereal sown after a legume “often requires less” fertiliser; the need is reduced, not removed.
- B.** Incorrect – unsupported inference. The four-field system spread across parts of northern Europe, but the passage says nothing about where crop rotation began.
- C.** Correct. In the four-field system a fodder crop replaced the fallow year, in which land had previously been left unplanted.
- D.** Incorrect – contradiction. The inability to specialise in the most lucrative crop is presented as a drawback of rotation.

**Question 15 — Answer: C**

- A.** Incorrect – contradiction. The charge applies only on weekdays between seven in the morning and seven in the evening.
- B.** Incorrect – extreme wording. Only “some retailers” reported fewer customers, and overall retail spending was broadly unchanged.
- C.** Correct. The revenue is reserved by law for public transport and cycling infrastructure.
- D.** Incorrect – contradiction. Traffic on the ring road around the zone increased slightly.

**Question 16 — Answer: A**

- A.** Correct. The ash reacted with lime to give a mortar that could set even under water.
- B.** Incorrect – contradiction. Modern concrete uses “a different kind of cement” and steel reinforcement; only the basic principle is shared.
- C.** Incorrect – extreme wording. The dome is “one of the largest unreinforced concrete domes”, not the largest ever built.
- D.** Incorrect – contradiction. The builders used “progressively lighter materials towards the top” to reduce weight.

**Question 17 — Answer: B**

- A.** Incorrect – contradiction. Pasteurisation does not aim to kill all micro-organisms, and some heat-resistant bacteria and spores survive.
- B.** Correct. Pasteur developed the technique initially to prevent the spoilage of wine, and only afterwards was it applied to milk.
- C.** Incorrect – contradiction. “Many consumers notice a difference in taste”.
- D.** Incorrect – extreme wording. Pasteurisation “contributed to” the decline; the passage does not present it as the sole cause.

**Question 18 — Answer: D**

- A.** Incorrect – contradiction. Subscription libraries charged an annual fee and “still excluded most working people”.
- B.** Incorrect – cause and effect. The passage says only that “some claimed” such an effect; nothing indicates that it was demonstrated.
- C.** Incorrect – contradiction. Early public libraries often kept books behind counters; open shelves “became common only later”.
- D.** Correct. Opponents questioned whether public money should be spent on novels, then often regarded as frivolous.



**Question 19 — Answer: C**

- A.** Incorrect – contradiction. The League was not a state with a single government, and decisions were not always applied uniformly.
- B.** Incorrect – contradiction. Decline was gradual, the League was never formally dissolved, and its last assembly took place in the seventeenth century.
- C.** Correct. Merchants at the kontors enjoyed privileges such as reduced tolls.
- D.** Incorrect – cause and effect. The passage attributes the decline to stronger monarchies and Dutch and English competition, not to falling demand.

**Question 20 — Answer: A**

- A.** Correct. Returned containers are collected separately, so the material tends to be cleaner and easier to recycle.
- B.** Incorrect – contradiction. Retailers may need storage space or machines to accept returned containers.
- C.** Incorrect – scope shift. Deposits apply only to the containers a scheme covers, and its design determines which types are included.
- D.** Incorrect – unsupported inference. Very high return rates are typical where schemes are well established, not necessarily when they are first introduced.

**Question 21 — Answer: C**

- A.** Incorrect – supported. Legal advisers reviewed every revised letter “to confirm that its legal effect was unchanged”.
- B.** Incorrect – supported. Among recipients of the revised letter, fewer “telephoned for clarification”.
- C.** Correct – not supported. Only payment reminders are to be revised; extending the approach to other letters will merely be considered after further tests, so a possible future step is presented as an adopted measure.
- D.** Incorrect – supported. The working group cautions that results for more complex notices “may differ”.

**Question 22 — Answer: A**

- A.** Correct. The passage states that a language with hundreds of thousands of speakers can be in serious danger if they are mostly elderly, because the key criterion is acquisition by children.
- B.** Incorrect – extreme wording. The passage says immersion programmes produced new speakers “in several cases”, not wherever they were introduced.
- C.** Incorrect – contradiction. Documentation is said to “preserve valuable knowledge”, even though it does not create new speakers.
- D.** Incorrect – outside knowledge. The passage attributes the breakdown to parents’ belief that another language offers better prospects; school prohibitions are not mentioned.

**Question 23 — Answer: B**

- A.** Incorrect – supported. Visitor numbers “usually rise, sometimes sharply” when charges are abolished.
- B.** Correct – not supported. Much of the increase comes from people who already visited museums, rather than from groups that previously stayed away.
- C.** Incorrect – supported. Price is “only one barrier among several”, alongside time, distance and a sense of exclusion.
- D.** Incorrect – supported. Lost ticket revenue must be replaced through subsidy, donations or commercial income.

**Question 24 — Answer: A**

- A.** Correct – not supported. Several studies found that face-to-face conversation “fell rather than rose” after such moves.
- B.** Incorrect – supported. The cost-saving claim is “generally accepted”, since more employees fit into a given floor area.
- C.** Incorrect – supported. Lacking privacy, employees may wear headphones, avoid conversations or seek quiet spaces elsewhere.
- D.** Incorrect – supported. Complaints about noise and interruptions are especially common among those whose tasks require sustained concentration.

**Question 25 — Answer: A**

- A.** Correct. Each hour of difference between the two times equals fifteen degrees of longitude, so knowing both times gives the ship’s longitude.
- B.** Incorrect – contradiction. Latitude could be found “with reasonable accuracy”, while longitude “posed a far harder problem”.
- C.** Incorrect – extreme wording. Early chronometers were costly, and many ships kept using the astronomical method for decades.
- D.** Incorrect – partial truth. One hour equals fifteen degrees, so two hours correspond to thirty degrees.

**Question 26 — Answer: B**

- A.** Incorrect – scope shift. Transcriptions are produced only “for a selection of registers”; images are made for every page.
- B.** Correct. Access to the originals is restricted to cases where the image is insufficient, such as examining the paper or ink.
- C.** Incorrect – contradiction. Digitisation is “not a substitute for preservation”, and digital files need continuous funding.
- D.** Incorrect – unsupported inference. Volunteers have only been invited, and the consortium “hopes” accuracy will improve over time.

**Question 27 — Answer: C**

- A.** Incorrect – supported. The algae “supply the coral with much of the energy it needs”.
- B.** Incorrect – supported. Pollution and unusually low tides can also trigger bleaching locally.
- C.** Correct – not supported. A bleached coral “is not necessarily dead” and may recover if temperatures return to normal.
- D.** Incorrect – supported. Extensive coral mortality can affect the many fish and invertebrates that reefs shelter.

**Question 28 — Answer: C**

- A.** Incorrect – unsupported inference. Proprietors are described only as posting shipping information; the passage says nothing about them paying claims.
- B.** Incorrect – contradiction. Backers each stated the share they accepted, and “no single backer usually took on the whole risk”.
- C.** Correct. Because the risk was shared among several backers, a loss could be absorbed by many people instead of ruining one.
- D.** Incorrect – extreme wording. Only “some of these informal arrangements” developed into formal institutions.

**Question 29 — Answer: A**

- A.** Correct. The board accepted the first part of the recommendation, extending anonymised sifting, but asked for further cost analysis before deciding on the training.
- B.** Incorrect – contradiction. Anonymisation had no measurable effect on who was eventually appointed.
- C.** Incorrect – extreme wording. Interview invitations rose in four of the six ministries and were unchanged in the other two.
- D.** Incorrect – cause and effect. The evaluators suggested that the scoring grids may have limited the scope for bias, not introduced it.

**Question 30 — Answer: B**

- A.** Incorrect – contradiction. The circulation is contrasted with wind-driven surface currents; it is driven by differences in density.
- B.** Correct. Sea-ice formation leaves salt behind, and saltier water is denser and tends to sink.
- C.** Incorrect – extreme wording. The warm northward flow is “one of several factors”, not the sole cause of the mild climate.
- D.** Incorrect – unsupported inference. Scientists have studied whether fresh water could weaken the circulation; no established finding is reported.

**Question 31 — Answer: B**

- A.** Incorrect – unsupported inference. Heat from vehicles and air-conditioning is one factor that “adds to the total”, not the main cause.
- B.** Correct. The difference is usually most pronounced at night and in calm, clear weather.
- C.** Incorrect – contradiction. Trees need an adequate supply of water to provide cooling during dry periods.
- D.** Incorrect – contradiction. Lighter-coloured materials reflect more sunlight, whereas dark surfaces absorb a large proportion of it.

**Question 32 — Answer: D**

- A.** Incorrect – contradiction. The method lets anyone send a confidential message “without first agreeing a secret”.
- B.** Incorrect – partial truth. The owner signs with his or her own private key; others verify the signature with the public key.
- C.** Incorrect – extreme wording. Such problems are only “believed to be extremely hard”; this is neither a proof nor a claim of impossibility.
- D.** Correct. Because public-key operations are slow, they are typically used to exchange a shorter secret key, which encrypts the bulk of the data “by faster methods”.

**Question 33 — Answer: D**

- A.** Incorrect – outside knowledge. The passage gives the French name but says nothing about where or when the principle was first formulated.
- B.** Incorrect – contradiction. Records from one creator are “not merged” with those of other sources, even when their subjects overlap.
- C.** Incorrect – scope shift. Original order is to be kept only “wherever it can still be identified”, not in every case.
- D.** Correct. Because context gives documents meaning, rearranging them by topic can destroy evidence of how an organisation worked; keeping the original arrangement preserves it.

**Question 34 — Answer: C**

- A.** Incorrect – extreme wording. On a test taken within minutes, rereading can produce equal or even superior results.
- B.** Incorrect – contradiction. The benefit appears “even when the recall attempt ... receives no feedback”.
- C.** Correct. Students prefer methods that feel effective, and researchers point to a “mismatch between perceived and actual effectiveness”.
- D.** Incorrect – scope shift. Researchers recommend building low-stakes quizzes into ordinary lessons; nothing is said about replacing examinations.

**Question 35 — Answer: A**

- A.** Correct – not supported. The researchers said the data were “consistent with a causal effect” but could not establish one; they did not conclude that there is no effect.
- B.** Incorrect – supported. High-membership districts were the most affluent, and income is associated with both library use and results.
- C.** Incorrect – supported. The design compares pupils where new branches open with pupils in similar districts “where no branch is planned”.
- D.** Incorrect – supported. The researchers acknowledged that the design would not be perfect because the districts had not been selected at random.

**Question 36 — Answer: D**

- A.** Incorrect – extreme wording. Decisions belong at the lowest level “capable of handling them effectively”, and higher levels may act where lower ones cannot.
- B.** Incorrect – contradiction. The idea was originally applied to relations between the state and smaller associations such as families.
- C.** Incorrect – partial truth. It is critics, not courts, who argue that the principle is too vague to be enforced.
- D.** Correct. The passage says the answer may change with circumstances, citing cross-boundary problems that a higher authority may handle more effectively.

**Question 37 — Answer: D**

- A.** Incorrect – contradiction. A programme’s value depends on more than test accuracy, including false positives and overdiagnosis.
- B.** Incorrect – partial truth. That describes a false positive; overdiagnosis is the detection of genuine conditions that would never have caused harm.
- C.** Incorrect – contradiction. Recommendations are often restricted to particular age groups.
- D.** Correct. Overdiagnosis detects genuine conditions that would never cause symptoms, and treating them brings side-effects without benefit.

**Question 38 — Answer: B**

- A.** Incorrect – supported. Borrowers with fixed debts benefit because the real value of what they owe falls as prices rise.
- B.** Correct – not supported. The passage states that even anticipated inflation imposes costs, such as more frequent price changes and effort spent avoiding holding cash.
- C.** Incorrect – supported. Where inflation is anticipated, wages and benefits can be linked to prices, reducing its effects.
- D.** Incorrect – supported. High and variable inflation makes it harder to tell relative price changes from changes in the general price level.

**Question 39 — Answer: C**

- A.** Incorrect – extreme wording. Staff are only “encouraged” to go to their manager first and may report directly where that is inappropriate.
- B.** Incorrect – contradiction. Good-faith reporters are protected “even if the concern proves unfounded”.
- C.** Correct. Where an internal report has had no response within three months, the matter may be taken to an external oversight body.
- D.** Incorrect – contradiction. Anonymous reports are allowed; they are only “harder to investigate”.

**Question 40 — Answer: C**

- A.** Incorrect – contradiction. Chains of visual signalling stations already allowed faster transmission where they had been built.
- B.** Incorrect – contradiction. Because telegrams were expensive, most private correspondence “continued to go by post” for decades.
- C.** Correct. Railway companies used the telegraph to coordinate movements on single-track lines, improving safety and allowing more trains to run.
- D.** Incorrect – partial truth. The telegraph spread from the 1840s, but the first lasting transatlantic cable was completed only in 1866.

**Question 41 — Answer: D**

- A.** Incorrect – scope shift. The 95 per cent figure is given for measles only, and the passage says herd immunity does not apply at all to diseases such as tetanus.
- B.** Incorrect – contradiction. Herd immunity “can be undermined locally even when national coverage appears high”.
- C.** Incorrect – contradiction. Tetanus is acquired from environmental spores, so herd immunity “offers no protection” against it.
- D.** Correct. The threshold rises with contagiousness, and for less transmissible diseases it is “considerably lower”.

**Question 42 — Answer: B**

- A.** Incorrect – contradiction. Seawater “remains slightly alkaline”; the term describes a shift along the pH scale, not acidic oceans.
- B.** Correct. Ocean uptake “slows the rise of atmospheric concentrations” but reduces carbonate, so some shell-builders grow more slowly or form weaker structures.
- C.** Incorrect – extreme wording. Only “some of them” are affected, and responses “vary considerably between species”.
- D.** Incorrect – contradiction. Colder water absorbs carbon dioxide more readily, which is why polar seas are expected to be affected early.

**Question 43 — Answer: A**

- A.** Correct. The passage concludes that stability varies “according to the purpose for which a text is preserved”, with special techniques used where precise wording mattered.
- B.** Incorrect – contradiction. Singers did not memorise a fixed text but recomposed the poem at each performance.
- C.** Incorrect – unsupported comparison. The passage makes no comparison with written transmission and rejects any uniform verdict on oral reliability.
- D.** Incorrect – contradiction. The singers insisted that they were repeating the song exactly, even though performances differed.

**Question 44 — Answer: A**

- A.** Correct. A theory that can accommodate any conceivable result is, on Popper's view, rendered unscientific rather than strengthened.
- B.** Incorrect – contradiction. Popper “did not claim that falsifiable theories are true”.
- C.** Incorrect – contradiction. Popper did not hold that unfalsifiable claims are meaningless; he sought only to mark the boundary of empirical science.
- D.** Incorrect – unsupported inference. The point about logical guarantees is raised by later philosophers; no such claim is attributed to Bacon.

**Question 45 — Answer: C**

- A.** Incorrect – extreme wording. Sellers may raise prices so that “part of the burden passes to buyers”.
- B.** Incorrect – contradiction. Where substitutes are available, sellers find it harder to raise prices and absorb a larger share themselves.
- C.** Correct. Legal and economic incidence “frequently differ”, and who bears the cost depends on how buyers and sellers react in each market.
- D.** Incorrect – contradiction. Deciding whom to tax formally “does not, by itself, determine who pays”.

**Question 46 — Answer: C**

- A.** Incorrect – contradiction. The evaluation “could not determine” whether sales had increased because data were missing.
- B.** Incorrect – contradiction. The foundation will consult publishers' associations before deciding on this point.
- C.** Correct. The foundation has agreed to require sales reporting from the next funding round, as the evaluators recommended.
- D.** Incorrect – extreme wording. The passage attributes the concentration partly to the higher number of applications from those markets and does not mention selection criteria.

**Question 47 — Answer: A**

- A.** Correct. In posts exposed to outside pressure, rotation can limit relationships that might compromise impartiality, yet moving post-holders too soon may lose specialist knowledge built up over years.
- B.** Incorrect – extreme wording. For such posts, organisations tend to rotate at longer intervals or rely on documentation and hand-over periods; rotation is not ruled out.
- C.** Incorrect – unsupported comparison. A broader understanding is “considered valuable” for future managers, but no performance comparison is reported.
- D.** Incorrect – unsupported inference. Management preparation is one of several benefits listed; the passage does not identify any one as the main purpose.

**Question 48 — Answer: D**

- A.** Incorrect – unsupported inference. A clearly defined membership is only one of several conditions that make success more likely, and the research did not claim that community management always works.
- B.** Incorrect – contradiction. In the earlier view, the costs of depletion are spread across everyone, while the benefits accrue to the individual.
- C.** Incorrect – contradiction. Successful arrangements used graduated sanctions rather than immediately severe ones.
- D.** Correct. Users' ability to take part in changing the rules is listed among the conditions under which such arrangements tended to succeed.

**Question 49 — Answer: C**

- A.** Incorrect – contradiction. The rest of the contract continues to apply if it can operate without the unfair term.
- B.** Incorrect – contradiction. The adequacy of the price is, in many such regimes, excluded from review where it is expressed in plain and intelligible language.
- C.** Correct. The exclusion of core terms such as the price applies only if they are written in plain and intelligible language, so an obscurely drafted price term loses that protection.
- D.** Incorrect – extreme wording. In many such regimes, plainly worded terms defining the main subject matter are excluded from review.

**Question 50 — Answer: D**

- A.** Incorrect – partial truth. Critics did reject the theory, but “chiefly” because Wegener lacked a mechanism; the passage never says they disputed his coastline evidence.
- B.** Incorrect – contradiction. Sea-floor spreading was revealed by surveys decades later, after the Second World War, and supplied the mechanism Wegener had lacked.
- C.** Incorrect – extreme wording. Plate movement explains the location of “most” earthquakes and volcanoes, not all of them.
- D.** Correct. The chief objection was the lack of a mechanism, and ocean-floor surveys revealed sea-floor spreading, “providing the kind of mechanism that Wegener had lacked”.

**Question 51 — Answer: B**

- A.** Incorrect – cause and effect. Missed appointments were less frequent in the video group, but because patients chose their format, differences “cannot be attributed with confidence” to it.
- B.** Correct. Mental health and diabetes clinicians rated video as generally adequate, whereas dermatologists often found image quality insufficient and had to recall a notable share of patients.
- C.** Incorrect – contradiction. Only about two-fifths of patients chose video.
- D.** Incorrect – contradiction. Patients who chose video lived, on average, further from the hospital.

**Question 52 — Answer: D**

- A.** Incorrect – partial truth. To halve the margin of error the sample must be “roughly quadrupled”, not doubled.
- B.** Incorrect – contradiction. The margin of error “says nothing about systematic bias”.
- C.** Incorrect – extreme wording. The passage says only that the result will be distorted if employed people’s usage differs; it gives no direction.
- D.** Correct. Enlarging a biased sample “merely produces a more precise estimate of the wrong quantity”.

**Question 53 — Answer: A**

- A.** Correct. Articles must currently be made freely available within twelve months of publication, so a delay of some months complies with the rules.
- B.** Incorrect – contradiction. Mandatory data deposit is only a recommendation; at present, data handling is left to each project team.
- C.** Incorrect – partial truth. Any new rules would apply only to projects that begin after the board’s vote, not to ongoing ones.
- D.** Incorrect – extreme wording. The proposal allows justified exceptions, such as personal-data protection or agreed commercial confidentiality.

**Question 54 — Answer: D**

- A.** Incorrect – scope shift. The policy covers only regulations adopted after it came into force, exemptions exist, and the period is “normally” seven years.
- B.** Incorrect – unsupported inference. The extension was only proposed by a committee; the government will merely consider it later.
- C.** Incorrect – contradiction. Critics worry that regulations will be renewed routinely without scrutiny, not that too many will be abolished.
- D.** Correct. Exemptions may be granted where a regulation implements an international obligation.

**Question 55 — Answer: A**

- A.** Correct. Linguists can partly reconstruct an ancestor’s words “even if no written record of it survives”.
- B.** Incorrect – contradiction. Such terms are borrowed more often than basic vocabulary, so they are weaker evidence, not the strongest.
- C.** Incorrect – contradiction. A shared word may arise by chance or borrowing, and “neither case demonstrates common descent”.
- D.** Incorrect – scope shift. The family includes “most languages of Europe”, not every one.

**Question 56 — Answer: B**

- A.** Incorrect – supported. Baiting will take place outside the breeding season, when the birds are at sea and less likely to be affected.
- B.** Correct – not supported. The managers have not ruled out fencing as a temporary measure; their objections are that fencing would protect only part of the island and need continual maintenance.
- C.** Incorrect – supported. The management board approved the programme after a public consultation.
- D.** Incorrect – supported. If even a few rats survive, the population can recover within a few years.

**Question 57 — Answer: C**

- A.** Incorrect – cause and effect. Some studies find repeaters “more likely” to leave early, and the passage warns that such comparisons are hard to interpret.
- B.** Incorrect – contradiction. Studies often find an “initial improvement” that later fades.
- C.** Correct. Teachers select pupils for repetition on characteristics that researchers may not observe, which makes comparisons difficult.
- D.** Incorrect – unsupported inference. Researchers say only that the resources “might achieve more” if spent on targeted support; this is a tentative conclusion, not an established finding.

**Question 58 — Answer: C**

- A.** Incorrect – unsupported inference. Hydrogen is one of two options still under study for the longest line; the board will decide only after the further study.
- B.** Incorrect – contradiction. Battery trains performed well on the shorter lines and are recommended there.
- C.** Correct. With current battery capacities, they “would be unable to complete the longest line without an additional charging point midway”.
- D.** Incorrect – partial truth. Hydrogen trains are less efficient, but the appraisal keeps them as a possible option for the longest line rather than rejecting them.



**Question 59 — Answer: D**

- A.** Incorrect – contradiction. Cutting the deficit does not reduce the debt; while any deficit remains, the debt keeps growing.
- B.** Incorrect – unsupported inference. The ratio can fall “even while the debt rises in absolute terms” if output grows faster, so a falling ratio does not prove repayment.
- C.** Incorrect – cause and effect. A recession can raise the ratio “without any additional borrowing”, simply by shrinking output.
- D.** Correct. Annual deficits add to the absolute debt, but the ratio can still fall if the economy grows faster than the debt.

**Question 60 — Answer: B**

- A.** Incorrect – contradiction. The passage states that degradation “is nonetheless not always irreversible” and gives examples of recovery.
- B.** Correct. Land under pressure from overgrazing, clearing or poor irrigation is “far less able to recover” after drought.
- C.** Incorrect – unsupported comparison. The passage says both factors contribute but does not rank them.
- D.** Incorrect – extreme wording. Recovery has occurred “in some areas” and depends heavily on local conditions.

**Question 61 — Answer: D**

- A.** Incorrect – contradiction. Hydrologists stress that the sustainable rate is not simply equal to recharge, because rivers, wetlands and springs also depend on part of the inflow.
- B.** Incorrect – contradiction. Compaction of clay aquifers is “largely permanent”, so a recovery in water levels need not reverse it.
- C.** Incorrect – scope shift. The passage links seawater intrusion to falling water tables in coastal areas only.
- D.** Correct. Some deep aquifers are effectively not being replenished; when abstraction persistently exceeds recharge, the water table falls and pumping costs rise because water must be lifted further.

**Question 62 — Answer: C**

- A.** Incorrect – contradiction. Floating sea ice makes “almost no direct contribution”; the main drivers are thermal expansion and land-based ice.
- B.** Incorrect – scope shift. A global average need not hold at every coast: relative change also reflects land movement, and some regions are rising.
- C.** Correct. Relative sea-level change combines changes in water level with vertical land movement, so uplift offsets part of the rise.
- D.** Incorrect – partial truth. Compaction of sediments is “sometimes” the cause of subsidence; it is not identified as the main one.

**Question 63 — Answer: C**

- A.** Incorrect – supported. His explanation “did not fit prevailing theories of disease”.
- B.** Incorrect – supported. The high-mortality clinic was staffed by doctors and students coming from autopsies, the source Semmelweis identified; the midwives’ clinic was not.
- C.** Correct – not supported. The germ theory was established only “in the following decades”, after Semmelweis had reached his conclusion, so the chronology is reversed.
- D.** Incorrect – supported. Some doctors “were offended by the suggestion that they themselves transmitted illness”.

**Question 64 — Answer: D**

- A.** Incorrect – contradiction. Online publication was only recommended; the board has deferred its decision pending legal advice.
- B.** Incorrect – contradiction. The board has not rejected the claims procedure; it has asked the panel to draft it in more detail before considering it.
- C.** Incorrect – contradiction. The current policy covers documentation before purchase and imposes no obligation to research items already in the collection.
- D.** Correct. The board accepted the first recommendation and funded research posts, while the other two remain deferred or under development.

**Question 65 — Answer: B**

- A.** Incorrect – supported. The label was “applied much later”, so it was not the builders’ own term.
- B.** Correct – not supported. The techniques “were not entirely new” and pointed arches appear in earlier buildings, so Gothic builders did not invent them.
- C.** Incorrect – supported. Flying buttresses “receive the remaining thrust and transfer it to the ground”.
- D.** Incorrect – supported. Rib vaults carry loads to particular points, so the walls no longer bore the whole load and could be opened up for stained glass.

# Solutions — Numerical Reasoning

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## Answer key

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 1 D  | 2 B  | 3 D  | 4 C  | 5 B  | 6 C  | 7 B  | 8 B  |
| 9 C  | 10 A | 11 C | 12 C | 13 D | 14 A | 15 C | 16 D |
| 17 C | 18 C | 19 B | 20 A | 21 C | 22 B | 23 A | 24 B |
| 25 C | 26 D | 27 A | 28 D | 29 A | 30 C | 31 D | 32 B |
| 33 A | 34 A | 35 D | 36 D | 37 C | 38 A | 39 D | 40 C |
| 41 D | 42 B | 43 D | 44 A | 45 B | 46 D | 47 A | 48 D |
| 49 B | 50 C | 51 A | 52 C | 53 B | 54 A | 55 B | 56 D |
| 57 D | 58 C | 59 D | 60 B | 61 A | 62 C | 63 A | 64 C |
| 65 B | 66 C | 67 B | 68 A | 69 B | 70 A | 71 B | 72 B |
| 73 D | 74 A | 75 D |      |      |      |      |      |

## Worked solutions

### Question 1 — Answer: D

- > A change in percentage points is the simple difference between the two rates:  $13.7 - 11.3 = +2.4$  points, i.e. a rise of 2.4 percentage points.
- > Typical error: computing the relative (percentage) change instead.

**Correct option: D** (2.4 percentage points)

### Question 2 — Answer: B

- > Unit A has 10 women and 8 men (18 staff).
- > New total =  $18 \times 1.5 = 27$ ; women =  $27 \times 10/18 = 15$ .
- > Shortcut: if the proportion is unchanged, the number of women also grows by 50%:  $10 \times 1.5 = 15$ .

**Correct option: B** (15)

### Question 3 — Answer: D

- > Percentage change uses the earlier value as the base:  $(414 - 363) \div 363 \times 100$ .
- >  $= 51 \div 363 \times 100 \approx 14.0\%$ , i.e. an increase of about 14.0%.
- > Typical errors: dividing by the Year 3 value (wrong base), or adding the two year-on-year percentage changes (percentages do not add).

**Correct option: D** (14.0%)

### Question 4 — Answer: C

- > English : Italian = 40 : 32. Both numbers divide by 8, giving 5:4.
- > Typical error: reversing the order of the ratio (4:5).

**Correct option: C** (5:4)

**Question 5 — Answer: B**

- › The South region's total allocation is €2,640 million, of which Education received €660 million.
- ›  $660 \div 2,640 \times 100 \approx 25.0\%$ .
- › Typical errors: dividing by the Education column total (the region's share of that sector) or by the remainder instead of the total.

**Correct option: B** (25.0%)

**Question 6 — Answer: C**

- › Percentage change uses the earlier value as the base:  $(491 - 399) \div 399 \times 100$ .
- ›  $= 92 \div 399 \times 100 \approx 23.1\%$ , i.e. an increase of about 23.1%.
- › Typical errors: dividing by the Year 3 value (wrong base), or adding the two year-on-year percentage changes (percentages do not add).

**Correct option: C** (23.1%)

**Question 7 — Answer: B**

- › The North region's total allocation is €1,300 million, of which Culture received €250 million.
- ›  $250 \div 1,300 \times 100 \approx 19.2\%$ .
- › Typical errors: dividing by the Culture column total (the region's share of that sector) or by the remainder instead of the total.

**Correct option: B** (19.2%)

**Question 8 — Answer: B**

- › German : French = 45 : 75. Both numbers divide by 15, giving 3:5.
- › Typical error: reversing the order of the ratio (5:3).

**Correct option: B** (3:5)

**Question 9 — Answer: C**

- › Approved =  $2,200 \times 40\% = 880$ .
- › Rejected =  $2,200 - 880 = 1,320$ .

**Correct option: C** (1,320)

**Question 10 — Answer: A**

- › Unit A has 50 women and 60 men (110 staff).
- › New total =  $110 \times 1.1 = 121$ ; women =  $121 \times 50/110 = 55$ .
- › Shortcut: if the proportion is unchanged, the number of women also grows by 10%:  $50 \times 1.1 = 55$ .

**Correct option: A** (55)

**Question 11 — Answer: C**

- › French : Spanish = 40 : 16. Both numbers divide by 8, giving 5:2.
- › Typical error: reversing the order of the ratio (2:5).

**Correct option: C** (5:2)

**Question 12 — Answer: C**

- › Approved =  $600 \times 40\% = 240$ .
- › Rejected =  $600 - 240 = 360$ .

**Correct option: C** (360)

**Question 13 — Answer: D**

- › Percentage change uses the earlier value as the base:  $(464 - 390) \div 390 \times 100$ .
- ›  $= 74 \div 390 \times 100 \approx 19.0\%$ , i.e. an increase of about 19.0%.
- › Typical errors: dividing by the Year 3 value (wrong base), or adding the two year-on-year percentage changes (percentages do not add).

**Correct option: D** (19.0%)**Question 14 — Answer: A**

- › The North region's total allocation is €2,110 million, of which Transport received €220 million.
- ›  $220 \div 2,110 \times 100 \approx 10.4\%$ .
- › Typical errors: dividing by the Transport column total (the region's share of that sector) or by the remainder instead of the total.

**Correct option: A** (10.4%)**Question 15 — Answer: C**

- › Unit B has 30 women and 40 men (70 staff).
- › New total =  $70 \times 1.2 = 84$ ; women =  $84 \times 30/70 = 36$ .
- › Shortcut: if the proportion is unchanged, the number of women also grows by 20%:  $30 \times 1.2 = 36$ .

**Correct option: C** (36)**Question 16 — Answer: D**

- › The South region's total allocation is €2,740 million, of which Digital received €940 million.
- ›  $940 \div 2,740 \times 100 \approx 34.3\%$ .
- › Typical errors: dividing by the Digital column total (the region's share of that sector) or by the remainder instead of the total.

**Correct option: D** (34.3%)**Question 17 — Answer: C**

- › Percentage change uses the earlier value as the base:  $(749 - 629) \div 629 \times 100$ .
- ›  $= 120 \div 629 \times 100 \approx 19.1\%$ , i.e. an increase of about 19.1%.
- › Typical errors: dividing by the Year 3 value (wrong base), or adding the two year-on-year percentage changes (percentages do not add).

**Correct option: C** (19.1%)**Question 18 — Answer: C**

- › A change in percentage points is the simple difference between the two rates:  $6.4 - 8.5 = -2.1$  points, i.e. a fall of 2.1 percentage points.
- › Typical error: computing the relative (percentage) change instead.

**Correct option: C** (2.1 percentage points)**Question 19 — Answer: B**

- › Polish : English = 26 : 65. Both numbers divide by 13, giving 2:5.
- › Typical error: reversing the order of the ratio (5:2).

**Correct option: B** (2:5)**Question 20 — Answer: A**

- › Unit C has 10 women and 20 men (30 staff).
- › New total =  $30 \times 1.4 = 42$ ; women =  $42 \times 10/30 = 14$ .
- › Shortcut: if the proportion is unchanged, the number of women also grows by 40%:  $10 \times 1.4 = 14$ .

**Correct option: A** (14)

**Question 21 — Answer: C**

- › Unit B has 10 women and 25 men (35 staff).
- › New total =  $35 \times 1.2 = 42$ ; women =  $42 \times 10/35 = 12$ .
- › Shortcut: if the proportion is unchanged, the number of women also grows by 20%:  $10 \times 1.2 = 12$ .

**Correct option: C** (12)

**Question 22 — Answer: B**

- › The Central region's total allocation is €1,910 million, of which Digital received €880 million.
- ›  $880 \div 1,910 \times 100 \approx 46.1\%$ .
- › Typical errors: dividing by the Digital column total (the region's share of that sector) or by the remainder instead of the total.

**Correct option: B** (46.1%)

**Question 23 — Answer: A**

- › Spending per inhabitant = spending  $\div$  population (€ million  $\div$  million inhabitants = € per inhabitant):
- › Coastal region  $\approx$  €235; Central region  $\approx$  €206; Island region  $\approx$  €165; Eastern region  $\approx$  €182.
- › The highest figure is for the Coastal region. Trap: the Central region has the largest total spending, but not the highest spending per inhabitant.

**Correct option: A** (Coastal region)

**Question 24 — Answer: B**

- › With index numbers, the percentage change between two years is (later index – earlier index)  $\div$  earlier index  $\times 100$ .
- ›  $(121.9 - 109.8) \div 109.8 \times 100 \approx 11.0\%$ .
- › Typical errors: subtracting the indices (12.1 is a change in index points, not per cent) or measuring from the base year (index – 100).

**Correct option: B** (11.0%)

**Question 25 — Answer: C**

- › Participants Q1 to Q3 inclusive:  $6 + 8 + 16 = 30$ .
- › Total cost =  $30 \times \text{€}140 = \text{€}4,200$ .
- › Typical errors: including the wrong quarters, using the whole year, or reading the wrong row of either table.

**Correct option: C** (€4,200)

**Question 26 — Answer: D**

- › Convert every price to euros by DIVIDING the local price by the number of units per €1:
- › JPY:  $69,950 \div 153.9 \approx \text{€}454.52$ ; CHF:  $437 \div 0.97 \approx \text{€}450.52$ ; ISK:  $52,120 \div 146.6 \approx \text{€}355.53$ ; USD:  $406 \div 1.08 \approx \text{€}375.93$ .
- › Most expensive (JPY) – cheapest (ISK)  $\approx \text{€}454.52 - \text{€}355.53 = \text{€}98.99$ , i.e. about €99.
- › Typical errors: ranking the markets by local price (the currencies are not comparable) or comparing the wrong pair of markets.

**Correct option: D** (€99)

**Question 27 — Answer: A**

- › For an average of 50 over 6 months, the total must be  $50 \times 6 = 300$ .
- › January–May total =  $52 + 34 + 48 + 26 + 60 = 220$ .
- › June must therefore be  $300 - 220 = 80$ .
- › Typical errors: answering with the target itself (50) or with the current average (about 44).

**Correct option: A** (80)

**Question 28 — Answer: D**

- › Total salary bill =  $44 \times 9,100 + 29 \times 8,700 + 16 \times 5,800 + 18 \times 7,800 = €885,900$ .
- › Total staff = 107. Weighted average =  $€885,900 \div 107 \approx €8,279$ .
- › Typical error: the simple average of the four unit averages (€7,850) ignores the different unit sizes.

**Correct option: D** (€8,279)**Question 29 — Answer: A**

- › Percentage change =  $(\text{Year 2} - \text{Year 1}) \div \text{Year 1} \times 100$ :
- › Southmere:  $(500 - 350) \div 350 \approx 42.9\%$ ; Riverton:  $(410 - 390) \div 390 \approx 5.1\%$ ; Kingsford:  $(1120 - 870) \div 870 \approx 28.7\%$ ; Hillcrest:  $(800 - 700) \div 700 \approx 14.3\%$ .
- › Largest: Southmere. Trap: Kingsford has the largest increase in absolute terms (+250 thousand), but from a larger base.

**Correct option: A** (Southmere)**Question 30 — Answer: C**

- › Approvals needed =  $1,900 \times 45\% = 855$ .
- › Currently approved =  $1,900 \times 18\% = 342$ . Additional approvals = 513.

**Correct option: C** (513)**Question 31 — Answer: D**

- › Time = distance  $\div$  speed. Now:  $153 \div 110 \times 60 = 83.5$  minutes. New:  $153 \div 160 \times 60 = 57.4$  minutes.
- › Saving =  $83.5 - 57.4 = 26.1$ , i.e. about 26 minutes.
- › Typical error: assuming journey time falls by the same percentage as speed rises.

**Correct option: D** (26 minutes)**Question 32 — Answer: B**

- › Participants Q1 to Q2 inclusive:  $27 + 10 = 37$ .
- › Total cost =  $37 \times €450 = €16,650$ .
- › Typical errors: including the wrong quarters, using the whole year, or reading the wrong row of either table.

**Correct option: B** (€16,650)**Question 33 — Answer: A**

- › Time = distance  $\div$  speed. Now:  $299 \div 90 \times 60 = 199.3$  minutes. New:  $299 \div 110 \times 60 = 163.1$  minutes.
- › Saving =  $199.3 - 163.1 = 36.2$ , i.e. about 36 minutes.
- › Typical error: assuming journey time falls by the same percentage as speed rises.

**Correct option: A** (36 minutes)**Question 34 — Answer: A**

- › With index numbers, the percentage change between two years is  $(\text{later index} - \text{earlier index}) \div \text{earlier index} \times 100$ .
- ›  $(123.7 - 111.2) \div 111.2 \times 100 \approx 11.2\%$ .
- › Typical errors: subtracting the indices (12.5 is a change in index points, not per cent) or measuring from the base year (index - 100).

**Correct option: A** (11.2%)**Question 35 — Answer: D**

- › Relative change =  $(\text{new} - \text{old}) \div \text{old} \times 100 = (10.6 - 8.9) \div 8.9 \times 100 \approx +19.1\%$ , i.e. a rise of about 19.1%.
- › Typical errors: 1.7 is the change in percentage points, not a percentage change; another is dividing by the later value.

**Correct option: D** (19.1%)

**Question 36 — Answer: D**

- › For an average of 40 over 6 months, the total must be  $40 \times 6 = 240$ .
- › January–May total =  $40 + 45 + 21 + 46 + 32 = 184$ .
- › June must therefore be  $240 - 184 = 56$ .
- › Typical errors: answering with the target itself (40) or with the current average (about 37).

**Correct option: D** (56)**Question 37 — Answer: C**

- › Spending per inhabitant = spending  $\div$  population (€ million  $\div$  million inhabitants = € per inhabitant):
- › Southern region  $\approx$  €220; Northern region  $\approx$  €252; Central region  $\approx$  €160; Western region  $\approx$  €207.
- › The lowest figure is for the Central region. Trap: the Southern region has the smallest total spending, but not the lowest spending per inhabitant.

**Correct option: C** (Central region)**Question 38 — Answer: A**

- › Percentage change =  $(\text{Year 2} - \text{Year 1}) \div \text{Year 1} \times 100$ :
- › Fairhaven:  $(350 - 240) \div 240 \approx 45.8\%$ ; Northport:  $(330 - 280) \div 280 \approx 17.9\%$ ; Southmere:  $(800 - 570) \div 570 \approx 40.4\%$ ; Kingsford:  $(890 - 660) \div 660 \approx 34.8\%$ .
- › Largest: Fairhaven. Trap: Southmere has the largest increase in absolute terms (+230 thousand), but from a larger base.

**Correct option: A** (Fairhaven)**Question 39 — Answer: D**

- › Total salary bill =  $27 \times 7,200 + 47 \times 8,800 + 24 \times 4,900 + 9 \times 7,100 = \text{€}789,500$ .
- › Total staff = 107. Weighted average =  $\text{€}789,500 \div 107 \approx \text{€}7,379$ .
- › Typical error: the simple average of the four unit averages (€7,000) ignores the different unit sizes.

**Correct option: D** (€7,379)**Question 40 — Answer: C**

- › Convert every price to euros by DIVIDING the local price by the number of units per €1:
- › CAD:  $709 \div 1.42 \approx \text{€}499.30$ ; GBP:  $403 \div 0.83 \approx \text{€}485.54$ ; USD:  $680 \div 1.08 \approx \text{€}629.63$ ; ISK:  $100,810 \div 153.1 \approx \text{€}658.46$ .
- › Most expensive (ISK) – cheapest (GBP)  $\approx \text{€}658.46 - \text{€}485.54 = \text{€}172.92$ , i.e. about €173.
- › Typical errors: ranking the markets by local price (the currencies are not comparable) or comparing the wrong pair of markets.

**Correct option: C** (€173)**Question 41 — Answer: D**

- › With index numbers, the percentage change between two years is  $(\text{later index} - \text{earlier index}) \div \text{earlier index} \times 100$ .
- ›  $(118.3 - 106.0) \div 106.0 \times 100 \approx 11.6\%$ .
- › Typical errors: subtracting the indices (12.3 is a change in index points, not per cent) or measuring from the base year (index – 100).

**Correct option: D** (11.6%)



**Question 42 — Answer: B**

- Percentage change =  $(\text{Year 2} - \text{Year 1}) \div \text{Year 1} \times 100$ :
- Brookhaven:  $(850 - 650) \div 650 \approx 30.8\%$ ; Kingsford:  $(460 - 340) \div 340 \approx 35.3\%$ ; Hillcrest:  $(400 - 360) \div 360 \approx 11.1\%$ ; Westfield:  $(340 - 290) \div 290 \approx 17.2\%$ .
- Largest: Kingsford. Trap: Brookhaven has the largest increase in absolute terms (+200 thousand), but from a larger base.

**Correct option: B** (Kingsford)**Question 43 — Answer: D**

- Convert every price to euros by DIVIDING the local price by the number of units per €1:
- JPY:  $90,720 \div 164.7 \approx €550.82$ ; CHF:  $646 \div 0.92 \approx €702.17$ ; USD:  $576 \div 1.10 \approx €523.64$ ; GBP:  $545 \div 0.86 \approx €633.72$ .
- Most expensive (CHF) – cheapest (USD)  $\approx €702.17 - €523.64 = €178.54$ , i.e. about €179.
- Typical errors: ranking the markets by local price (the currencies are not comparable) or comparing the wrong pair of markets.

**Correct option: D** (€179)**Question 44 — Answer: A**

- For an average of 40 over 6 months, the total must be  $40 \times 6 = 240$ .
- January–May total =  $33 + 50 + 24 + 27 + 44 = 178$ .
- June must therefore be  $240 - 178 = 62$ .
- Typical errors: answering with the target itself (40) or with the current average (about 36).

**Correct option: A** (62)**Question 45 — Answer: B**

- Spending per inhabitant = spending  $\div$  population (€ million  $\div$  million inhabitants = € per inhabitant):
- Alpine region  $\approx €137$ ; Central region  $\approx €245$ ; Western region  $\approx €91$ ; Coastal region  $\approx €107$ .
- The highest figure is for the Central region. Trap: the Alpine region has the largest total spending, but not the highest spending per inhabitant.

**Correct option: B** (Central region)**Question 46 — Answer: D**

- Relative change =  $(\text{new} - \text{old}) \div \text{old} \times 100 = (7.7 - 9.3) \div 9.3 \times 100 \approx -17.2\%$ , i.e. a fall of about 17.2%.
- Typical errors: 1.6 is the change in percentage points, not a percentage change; another is dividing by the later value.

**Correct option: D** (17.2%)**Question 47 — Answer: A**

- Participants Q2 to Q3 inclusive:  $30 + 7 = 37$ .
- Total cost =  $37 \times €160 = €5,920$ .
- Typical errors: including the wrong quarters, using the whole year, or reading the wrong row of either table.

**Correct option: A** (€5,920)**Question 48 — Answer: D**

- Time = distance  $\div$  speed. Now:  $317 \div 120 \times 60 = 158.5$  minutes. New:  $317 \div 150 \times 60 = 126.8$  minutes.
- Saving =  $158.5 - 126.8 = 31.7$ , i.e. about 32 minutes.
- Typical error: assuming journey time falls by the same percentage as speed rises.

**Correct option: D** (32 minutes)

**Question 49 — Answer: B**

- › Total salary bill =  $38 \times 4,000 + 26 \times 5,500 + 11 \times 7,600 + 20 \times 8,900 = €556,600$ .
- › Total staff = 95. Weighted average =  $€556,600 \div 95 \approx €5,859$ .
- › Typical error: the simple average of the four unit averages (€6,500) ignores the different unit sizes.

**Correct option: B** (€5,859)

**Question 50 — Answer: C**

- › Approvals needed =  $2,300 \times 45\% = 1,035$ .
- › Currently approved =  $2,300 \times 20\% = 460$ . Additional approvals = 575.

**Correct option: C** (575)

**Question 51 — Answer: A**

- › Time = distance  $\div$  speed. Now:  $292 \div 120 \times 60 = 146.0$  minutes. New:  $292 \div 140 \times 60 = 125.1$  minutes.
- › Saving =  $146.0 - 125.1 = 20.9$ , i.e. about 21 minutes.
- › Typical error: assuming journey time falls by the same percentage as speed rises.

**Correct option: A** (21 minutes)

**Question 52 — Answer: C**

- › Convert every price to euros by DIVIDING the local price by the number of units per €1:
- › CAD:  $676 \div 1.44 \approx €469.44$ ; JPY:  $96,340 \div 158.4 \approx €608.21$ ; CHF:  $476 \div 0.95 \approx €501.05$ ; GBP:  $420 \div 0.82 \approx €512.20$ .
- › Most expensive (JPY) – cheapest (CAD)  $\approx €608.21 - €469.44 = €138.76$ , i.e. about €139.
- › Typical errors: ranking the markets by local price (the currencies are not comparable) or comparing the wrong pair of markets.

**Correct option: C** (€139)

**Question 53 — Answer: B**

- › Spending per inhabitant = spending  $\div$  population (€ million  $\div$  million inhabitants = € per inhabitant):
- › Island region  $\approx €151$ ; Southern region  $\approx €101$ ; Alpine region  $\approx €255$ ; Northern region  $\approx €219$ .
- › The lowest figure is for the Southern region. Trap: the Island region has the smallest total spending, but not the lowest spending per inhabitant.

**Correct option: B** (Southern region)

**Question 54 — Answer: A**

- › Total salary bill =  $13 \times 8,200 + 16 \times 8,500 + 44 \times 4,700 + 17 \times 7,300 = €573,500$ .
- › Total staff = 90. Weighted average =  $€573,500 \div 90 \approx €6,372$ .
- › Typical error: the simple average of the four unit averages (€7,175) ignores the different unit sizes.

**Correct option: A** (€6,372)

**Question 55 — Answer: B**

- › With index numbers, the percentage change between two years is (later index – earlier index)  $\div$  earlier index  $\times 100$ .
- ›  $(116.9 - 102.9) \div 102.9 \times 100 \approx 13.6\%$ .
- › Typical errors: subtracting the indices (14.0 is a change in index points, not per cent) or measuring from the base year (index – 100).

**Correct option: B** (13.6%)

**Question 56 — Answer: D**

- › Percentage change =  $(\text{Year 2} - \text{Year 1}) \div \text{Year 1} \times 100$ :
- › Westfield:  $(450 - 470) \div 470 \approx -4.3\%$ ; Kingsford:  $(590 - 450) \div 450 \approx 31.1\%$ ; Stonebridge:  $(1090 - 860) \div 860 \approx 26.7\%$ ; Fairhaven:  $(340 - 240) \div 240 \approx 41.7\%$ .
- › Largest: Fairhaven. Trap: Stonebridge has the largest increase in absolute terms (+230 thousand), but from a larger base.

**Correct option: D** (Fairhaven)**Question 57 — Answer: D**

- › For an average of 45 over 6 months, the total must be  $45 \times 6 = 270$ .
- › January–May total =  $50 + 20 + 41 + 24 + 38 = 173$ .
- › June must therefore be  $270 - 173 = 97$ .
- › Typical errors: answering with the target itself (45) or with the current average (about 35).

**Correct option: D** (97)**Question 58 — Answer: C**

- › Participants Q2 to Q4 inclusive:  $15 + 10 + 12 = 37$ .
- › Total cost =  $37 \times \text{€}500 = \text{€}18,500$ .
- › Typical errors: including the wrong quarters, using the whole year, or reading the wrong row of either table.

**Correct option: C** (€18,500)**Question 59 — Answer: D**

- › Participants Q2 to Q4 inclusive:  $10 + 29 + 17 = 56$ .
- › Total cost =  $56 \times \text{€}580 = \text{€}32,480$ .
- › Typical errors: including the wrong quarters, using the whole year, or reading the wrong row of either table.

**Correct option: D** (€32,480)**Question 60 — Answer: B**

- › With index numbers, the percentage change between two years is  $(\text{later index} - \text{earlier index}) \div \text{earlier index} \times 100$ .
- ›  $(127.3 - 108.0) \div 108.0 \times 100 \approx 17.9\%$ .
- › Typical errors: subtracting the indices (19.3 is a change in index points, not per cent) or measuring from the base year (index - 100).

**Correct option: B** (17.9%)**Question 61 — Answer: A**

- › Convert every price to euros by DIVIDING the local price by the number of units per €1:
- › CHF:  $690 \div 0.96 \approx \text{€}718.75$ ; ISK:  $107,260 \div 148.6 \approx \text{€}721.80$ ; GBP:  $609 \div 0.84 \approx \text{€}725.00$ ; JPY:  $95,660 \div 158.2 \approx \text{€}604.68$ .
- › Most expensive (GBP) – cheapest (JPY)  $\approx \text{€}725.00 - \text{€}604.68 = \text{€}120.32$ , i.e. about €120.
- › Typical errors: ranking the markets by local price (the currencies are not comparable) or comparing the wrong pair of markets.

**Correct option: A** (€120)

**Question 62 — Answer: C**

- Percentage change =  $(\text{Year 2} - \text{Year 1}) \div \text{Year 1} \times 100$ :
- Midvale:  $(590 - 570) \div 570 \approx 3.5\%$ ; Riverton:  $(820 - 840) \div 840 \approx -2.4\%$ ; Hillcrest:  $(570 - 420) \div 420 \approx 35.7\%$ ; Stonebridge:  $(1030 - 880) \div 880 \approx 17.0\%$ .
- Largest: Hillcrest. Trap: Stonebridge has the largest increase in absolute terms (+150 thousand), but from a larger base.

**Correct option: C** (Hillcrest)**Question 63 — Answer: A**

- Communication now =  $28\% \times \text{€}550 \text{ million} = \text{€}154 \text{ million}$ ; after +10%:  $\text{€}169.40 \text{ million}$ .
- Buildings =  $11\% \times \text{€}550 \text{ million} = \text{€}60.5 \text{ million}$ .
- Difference =  $\text{€}108.90 \text{ million} \approx \text{€}108.9 \text{ million}$ . Typical error: adding 10 percentage points to the share instead of raising the amount by 10%.

**Correct option: A** (€108.9 million)**Question 64 — Answer: C**

- Compound growth:  $2,070 \times 1.08^3 = 2,070 \times 1.2597 \approx 2,608 \text{ tonnes}$ .
- Typical error: simple growth,  $2,070 \times (1 + 3 \times 0.08) = 2,567 \text{ tonnes}$ , ignores compounding; others apply the rate for the wrong number of years.

**Correct option: C** (2,608 t)**Question 65 — Answer: B**

- This year = last year  $\times 0.85$ , so last year = this year  $\div 0.85$ .
- $2,210.0 \div 0.85 = 2,600.0$  (€ thousand).
- Typical error: increasing this year's figure by 15% applies the percentage to the wrong base.

**Correct option: B** (€2,600.0k)**Question 66 — Answer: C**

- This year = last year  $\times 0.88$ , so last year = this year  $\div 0.88$ .
- $2,499.2 \div 0.88 = 2,840.0$  (€ thousand).
- Typical error: increasing this year's figure by 12% applies the percentage to the wrong base.

**Correct option: C** (€2,840.0k)**Question 67 — Answer: B**

- Compound growth:  $2,060 \times 1.15^3 = 2,060 \times 1.5209 \approx 3,133 \text{ tonnes}$ .
- Typical error: simple growth,  $2,060 \times (1 + 3 \times 0.15) = 2,987 \text{ tonnes}$ , ignores compounding; others apply the rate for the wrong number of years.

**Correct option: B** (3,133 t)**Question 68 — Answer: A**

- Training now =  $41\% \times \text{€}280 \text{ million} = \text{€}114.8 \text{ million}$ ; after +20%:  $\text{€}137.76 \text{ million}$ .
- Staff =  $9\% \times \text{€}280 \text{ million} = \text{€}25.2 \text{ million}$ .
- Difference =  $\text{€}112.56 \text{ million} \approx \text{€}112.6 \text{ million}$ . Typical error: adding 20 percentage points to the share instead of raising the amount by 20%.

**Correct option: A** (€112.6 million)

**Question 69 — Answer: B**

- › Staff now =  $13\% \times \text{€}740 \text{ million} = \text{€}96.2 \text{ million}$ ; after +20%:  $\text{€}115.44 \text{ million}$ .
- › Translation =  $11\% \times \text{€}740 \text{ million} = \text{€}81.4 \text{ million}$ .
- › Difference =  $\text{€}34.04 \text{ million} \approx \text{€}34.0 \text{ million}$ . Typical error: adding 20 percentage points to the share instead of raising the amount by 20%.

**Correct option: B** (€34.0 million)**Question 70 — Answer: A**

- › Compound growth:  $1,610 \times 1.1^3 = 1,610 \times 1.3310 \approx 2,143 \text{ tonnes}$ .
- › Typical error: simple growth,  $1,610 \times (1 + 3 \times 0.1) = 2,093 \text{ tonnes}$ , ignores compounding; others apply the rate for the wrong number of years.

**Correct option: A** (2,143 t)**Question 71 — Answer: B**

- › This year = last year  $\times 0.92$ , so last year = this year  $\div 0.92$ .
- ›  $3,606.4 \div 0.92 = 3,920.0$  (€ thousand).
- › Typical error: increasing this year's figure by 8% applies the percentage to the wrong base.

**Correct option: B** (€3,920.0k)**Question 72 — Answer: B**

- › Compound growth:  $8,020 \times 1.15^3 = 8,020 \times 1.5209 \approx 12,197 \text{ tonnes}$ .
- › Typical error: simple growth,  $8,020 \times (1 + 3 \times 0.15) = 11,629 \text{ tonnes}$ , ignores compounding; others apply the rate for the wrong number of years.

**Correct option: B** (12,197 t)**Question 73 — Answer: D**

- › IT systems now =  $24\% \times \text{€}755 \text{ million} = \text{€}181.2 \text{ million}$ ; after +10%:  $\text{€}199.32 \text{ million}$ .
- › Communication =  $11\% \times \text{€}755 \text{ million} = \text{€}83.05 \text{ million}$ .
- › Difference =  $\text{€}116.27 \text{ million} \approx \text{€}116.3 \text{ million}$ . Typical error: adding 10 percentage points to the share instead of raising the amount by 10%.

**Correct option: D** (€116.3 million)**Question 74 — Answer: A**

- › This year = last year  $\times 1.08$ , so last year = this year  $\div 1.08$ .
- ›  $5,076.0 \div 1.08 = 4,700.0$  (€ thousand).
- › Typical error: reducing this year's figure by 8% applies the percentage to the wrong base.

**Correct option: A** (€4,700.0k)**Question 75 — Answer: D**

- › This year = last year  $\times 0.88$ , so last year = this year  $\div 0.88$ .
- ›  $3,977.6 \div 0.88 = 4,520.0$  (€ thousand).
- › Typical error: increasing this year's figure by 12% applies the percentage to the wrong base.

**Correct option: D** (€4,520.0k)



# Solutions — Abstract Reasoning

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## Answer key

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 1 E  | 2 A  | 3 C  | 4 D  | 5 B  | 6 E  | 7 D  | 8 A  |
| 9 B  | 10 C | 11 E | 12 C | 13 A | 14 D | 15 B | 16 A |
| 17 C | 18 E | 19 B | 20 D | 21 A | 22 B | 23 C | 24 E |
| 25 D | 26 E | 27 A | 28 C | 29 D | 30 B | 31 E | 32 B |
| 33 A | 34 C | 35 D | 36 A | 37 D | 38 E | 39 B | 40 C |
| 41 A | 42 B | 43 E | 44 C | 45 D | 46 B | 47 A | 48 D |
| 49 C | 50 E | 51 A | 52 E | 53 C | 54 D | 55 B | 56 E |
| 57 D | 58 B | 59 A | 60 C | 61 D | 62 B | 63 A | 64 C |
| 65 E | 66 D | 67 B | 68 E | 69 A | 70 C | 71 D | 72 A |
| 73 B | 74 C | 75 E |      |      |      |      |      |

## Worked solutions

### Question 1 — Answer: E

**Rule:** the circle moves 1 position clockwise around the frame each time.

**Why the other options are wrong:** **A** — the circle has not moved on from the fifth figure; **B** — the circle is one position too far clockwise; **C** — the circle is two positions too far anticlockwise; **D** — the circle is on the opposite side of the frame.

### Question 2 — Answer: A

**Rule:** the arrow rotates 45° anticlockwise each time.

**Why the other options are wrong:** **B** — the arrow has not rotated on from the fifth figure; **C** — the arrow is rotated 45° too far anticlockwise; **D** — the arrow is rotated 90° too far clockwise; **E** — the arrow points the opposite way.

### Question 3 — Answer: C

**Rule:** the square moves 1 position clockwise around the frame each time.

**Why the other options are wrong:** **A** — the square has not moved on from the fifth figure; **B** — the square is one position too far clockwise; **D** — the square is two positions too far anticlockwise; **E** — the square is on the opposite side of the frame.

### Question 4 — Answer: D

**Rule:** the arrow rotates 45° clockwise each time.

**Why the other options are wrong:** **A** — the arrow has not rotated on from the fifth figure; **B** — the arrow is rotated 45° too far clockwise; **C** — the arrow is rotated 90° too far anticlockwise; **E** — the arrow points the opposite way.

**Question 5 — Answer: B**

**Rule:** the triangle moves 2 positions clockwise around the frame each time.

**Why the other options are wrong:** **A** — the triangle has not moved on from the fifth figure; **C** — the triangle is one position too far clockwise; **D** — the triangle is one position too far anticlockwise; **E** — the triangle is on the opposite side of the frame.

**Question 6 — Answer: E**

**Rule:** the number of dots decreases by 1 each time.

**Why the other options are wrong:** **A** — there are 3 dots instead of 2; **B** — there is 1 dot instead of 2; **C** — there are 4 dots instead of 2; **D** — there are 5 dots instead of 2.

**Question 7 — Answer: D**

**Rule:** the arrow rotates 90° clockwise each time.

**Why the other options are wrong:** **A** — the arrow has not rotated on from the fifth figure; **B** — the arrow is rotated 45° too far clockwise; **C** — the arrow is rotated 45° too far anticlockwise; **E** — the arrow points the opposite way.

**Question 8 — Answer: A**

**Rule:** the number of dots increases by 1 each time.

**Why the other options are wrong:** **B** — there are 5 dots instead of 6; **C** — there are 7 dots instead of 6; **D** — there are 8 dots instead of 6; **E** — there are 4 dots instead of 6.

**Question 9 — Answer: B**

**Rule:** the square moves 2 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the square has not moved on from the fifth figure; **C** — the square is one position too far clockwise; **D** — the square is one position too far anticlockwise; **E** — the square is on the opposite side of the frame.

**Question 10 — Answer: C**

**Rule:** the number of dots increases by 1 each time.

**Why the other options are wrong:** **A** — there are 6 dots instead of 7; **B** — there are 8 dots instead of 7; **D** — there are 5 dots instead of 7; **E** — there are 4 dots instead of 7.

**Question 11 — Answer: E**

**Rule:** the diamond moves 1 position anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the diamond has not moved on from the fifth figure; **B** — the diamond is one position too far anticlockwise; **C** — the diamond is two positions too far clockwise; **D** — the diamond is on the opposite side of the frame.

**Question 12 — Answer: C**

**Rule:** the arrow rotates 90° clockwise each time.

**Why the other options are wrong:** **A** — the arrow has not rotated on from the fifth figure; **B** — the arrow is rotated 45° too far clockwise; **D** — the arrow is rotated 45° too far anticlockwise; **E** — the arrow points the opposite way.

**Question 13 — Answer: A**

**Rule:** the square moves 2 positions clockwise around the frame each time.

**Why the other options are wrong:** **B** — the square has not moved on from the fifth figure; **C** — the square is one position too far clockwise; **D** — the square is one position too far anticlockwise; **E** — the square is on the opposite side of the frame.



**Question 14 — Answer: D**

**Rule:** the arrow rotates 45° anticlockwise each time.

**Why the other options are wrong:** **A** — the arrow has not rotated on from the fifth figure; **B** — the arrow is rotated 45° too far anticlockwise; **C** — the arrow is rotated 90° too far clockwise; **E** — the arrow points the opposite way.

**Question 15 — Answer: B**

**Rule:** the diamond moves 2 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the diamond has not moved on from the fifth figure; **C** — the diamond is one position too far clockwise; **D** — the diamond is one position too far anticlockwise; **E** — the diamond is on the opposite side of the frame.

**Question 16 — Answer: A**

**Rules:** the triangle moves 2 positions anticlockwise around the frame each time; the triangle's shading alternates between grey and black.

**Why the other options are wrong:** **B** — the triangle is one position too far clockwise; **C** — the triangle has not moved on from the fifth figure; **D** — the triangle has not moved on from the fifth figure; the triangle is grey instead of black; **E** — the triangle is white instead of black.

**Question 17 — Answer: C**

**Rules:** the diamond moves 2 positions clockwise around the frame each time; the diamond's shading cycles black → white → grey.

**Why the other options are wrong:** **A** — the diamond is one position too far clockwise; the diamond is white instead of grey; **B** — the diamond is one position too far clockwise; **D** — the diamond has not moved on from the fifth figure; the diamond is white instead of grey; **E** — the diamond has not moved on from the fifth figure.

**Question 18 — Answer: E**

**Rules:** the circle moves 1 position clockwise around the frame each time; the circle's shading cycles black → white → grey.

**Why the other options are wrong:** **A** — the circle has not moved on from the fifth figure; the circle is white instead of grey; **B** — the circle has not moved on from the fifth figure; the circle is black instead of grey; **C** — the circle is black instead of grey; **D** — the circle has not moved on from the fifth figure.

**Question 19 — Answer: B**

**Rules:** the square moves 2 positions anticlockwise around the frame each time; the square's shading alternates between white and grey.

**Why the other options are wrong:** **A** — the square has not moved on from the fifth figure; **C** — the square has not moved on from the fifth figure; the square is black instead of grey; **D** — the square has not moved on from the fifth figure; the square is white instead of grey; **E** — the square is one position too far clockwise; the square is white instead of grey.

**Question 20 — Answer: D**

**Rules:** the circle moves 1 position anticlockwise around the frame each time; the circle's shading cycles black → white → grey.

**Why the other options are wrong:** **A** — the circle has not moved on from the fifth figure; **B** — the circle has not moved on from the fifth figure; the circle is black instead of grey; **C** — the circle is white instead of grey; **E** — the circle is black instead of grey.

**Question 21 — Answer: A**

**Rules:** the number of dots decreases by 1 each time; the circle moves 2 positions clockwise around the frame each time.

**Why the other options are wrong:** **B** — there are 2 dots instead of 1; **C** — there are 2 dots instead of 1; the circle is one position too far clockwise; **D** — there are 3 dots instead of 1; **E** — the circle is one position too far clockwise.

**Question 22 — Answer: B**

**Rules:** the arrow rotates 90° anticlockwise each time; the number of dots increases by 1 each time.

**Why the other options are wrong:** **A** — the arrow is rotated 45° too far clockwise; **C** — the arrow has not rotated on from the fifth figure; there are 8 dots instead of 7; **D** — there are 8 dots instead of 7; **E** — the arrow is rotated 45° too far clockwise; there are 6 dots instead of 7.

**Question 23 — Answer: C**

**Rules:** the number of dots decreases by 1 each time; the circle moves 1 position anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the circle is one position too far anticlockwise; **B** — the circle has not moved on from the fifth figure; **D** — there is 1 dot instead of 2; the circle is one position too far anticlockwise; **E** — there is 1 dot instead of 2.

**Question 24 — Answer: E**

**Rules:** the diamond moves 2 positions clockwise around the frame each time; the diamond's shading cycles black → white → grey.

**Why the other options are wrong:** **A** — the diamond has not moved on from the fifth figure; the diamond is black instead of grey; **B** — the diamond is one position too far clockwise; the diamond is white instead of grey; **C** — the diamond is one position too far clockwise; **D** — the diamond has not moved on from the fifth figure.

**Question 25 — Answer: D**

**Rules:** the number of dots decreases by 1 each time; the diamond moves 1 position clockwise around the frame each time.

**Why the other options are wrong:** **A** — there are 3 dots instead of 1; the diamond is one position too far clockwise; **B** — the diamond has not moved on from the fifth figure; **C** — there are 2 dots instead of 1; **E** — the diamond is one position too far clockwise.

**Question 26 — Answer: E**

**Rules:** the shaded quarter of the square moves one quarter anticlockwise each time; the diamond moves 2 positions clockwise around the frame each time.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; the diamond is one position too far clockwise; **B** — the wrong quarter of the square is shaded; **C** — the wrong quarter of the square is shaded; the diamond has not moved on from the fifth figure; **D** — the diamond has not moved on from the fifth figure.

**Question 27 — Answer: A**

**Rules:** the number of dots decreases by 1 each time; the circle moves 1 position anticlockwise around the frame each time.

**Why the other options are wrong:** **B** — there are 3 dots instead of 1; **C** — there are 2 dots instead of 1; the circle has not moved on from the fifth figure; **D** — the circle has not moved on from the fifth figure; **E** — there are 3 dots instead of 1; the circle is one position too far anticlockwise.

**Question 28 — Answer: C**

**Rules:** the shaded quarter of the square moves one quarter anticlockwise each time; the diamond moves 1 position clockwise around the frame each time.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; the diamond is one position too far clockwise; **B** — the diamond is one position too far clockwise; **D** — the wrong quarter of the square is shaded; **E** — the wrong quarter of the square is shaded; the diamond has not moved on from the fifth figure.

**Question 29 — Answer: D**

**Rules:** the arrow rotates  $45^\circ$  anticlockwise each time; the number of dots increases by 1 each time.

**Why the other options are wrong:** **A** — the arrow has not rotated on from the fifth figure; there are 7 dots instead of 6; **B** — there are 5 dots instead of 6; **C** — the arrow is rotated  $45^\circ$  too far anticlockwise; there are 7 dots instead of 6; **E** — the arrow is rotated  $45^\circ$  too far anticlockwise.

**Question 30 — Answer: B**

**Rules:** the arrow rotates  $90^\circ$  clockwise each time; the diamond moves 2 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the arrow is rotated  $45^\circ$  too far clockwise; the diamond is one position too far clockwise; **C** — the arrow is rotated  $45^\circ$  too far clockwise; the diamond has not moved on from the fifth figure; **D** — the arrow has not rotated on from the fifth figure; the diamond is one position too far clockwise; **E** — the diamond is one position too far clockwise.

**Question 31 — Answer: E**

**Rules:** the number of dots increases by 1 each time; the diamond moves 1 position anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the diamond is one position too far anticlockwise; **B** — there are 8 dots instead of 7; the diamond has not moved on from the fifth figure; **C** — there are 6 dots instead of 7; the diamond is one position too far anticlockwise; **D** — there are 8 dots instead of 7.

**Question 32 — Answer: B**

**Rules:** the central shape cycles hexagon → pentagon → square → triangle and then repeats; the diamond moves 2 positions clockwise around the frame each time.

**Why the other options are wrong:** **A** — the diamond has not moved on from the fifth figure; **C** — the diamond is one position too far clockwise; **D** — the central shape is a hexagon instead of a pentagon; the diamond has not moved on from the fifth figure; **E** — the central shape is a hexagon instead of a pentagon; the diamond is one position too far clockwise.

**Question 33 — Answer: A**

**Rules:** the central shape cycles hexagon → pentagon → square and then repeats; the central shape's shading alternates between white and black.

**Why the other options are wrong:** **B** — the central shape is a hexagon instead of a pentagon; **C** — the central shape is a hexagon instead of a pentagon; the central shape is grey instead of black; **D** — the central shape is a hexagon instead of a pentagon; the central shape is white instead of black; **E** — the central shape is white instead of black.

**Question 34 — Answer: C**

**Rules:** the central shape cycles hexagon → pentagon → square → triangle and then repeats; the circle moves 2 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the circle has not moved on from the fifth figure; **B** — the central shape is a square instead of a triangle; **D** — the central shape is a pentagon instead of a triangle; **E** — the central shape is a square instead of a triangle; the circle has not moved on from the fifth figure.

**Question 35 — Answer: D**

**Rules:** the central shape cycles hexagon → pentagon → square → triangle and then repeats; the central shape's shading cycles black → white → grey.

**Why the other options are wrong:** **A** — the central shape is a square instead of a triangle; the central shape is white instead of grey; **B** — the central shape is a pentagon instead of a triangle; the central shape is black instead of grey; **C** — the central shape is a pentagon instead of a triangle; **E** — the central shape is a pentagon instead of a triangle; the central shape is white instead of grey.

**Question 36 — Answer: A**

**Rules:** the central shape cycles square → pentagon → hexagon and then repeats; the circle moves 1 position anticlockwise around the frame each time.

**Why the other options are wrong:** **B** — the central shape is a hexagon instead of a square; **C** — the central shape is a hexagon instead of a square; the circle has not moved on from the fifth figure; **D** — the central shape is a pentagon instead of a square; the circle is one position too far anticlockwise; **E** — the circle has not moved on from the fifth figure.

**Question 37 — Answer: D**

**Rules:** the central shape cycles hexagon → pentagon → square → triangle and then repeats; the central shape's shading cycles black → white → grey.

**Why the other options are wrong:** **A** — the central shape is a triangle instead of a hexagon; the central shape is white instead of grey; **B** — the central shape is a pentagon instead of a hexagon; **C** — the central shape is a triangle instead of a hexagon; **E** — the central shape is white instead of grey.

**Question 38 — Answer: E**

**Rules:** the central shape cycles square → pentagon → hexagon and then repeats; the circle moves 1 position clockwise around the frame each time.

**Why the other options are wrong:** **A** — the circle is one position too far clockwise; **B** — the central shape is a hexagon instead of a pentagon; the circle is one position too far clockwise; **C** — the central shape is a hexagon instead of a pentagon; the circle has not moved on from the fifth figure; **D** — the central shape is a hexagon instead of a pentagon.

**Question 39 — Answer: B**

**Rules:** the circle moves 3 positions anticlockwise around the frame each time; the circle's shading cycles grey → black → white.

**Why the other options are wrong:** **A** — the circle is grey instead of white; **C** — the circle has not moved on from the fifth figure; the circle is grey instead of white; **D** — the circle has not moved on from the fifth figure; **E** — the circle has not moved on from the fifth figure; the circle is black instead of white.

**Question 40 — Answer: C**

**Rules:** the central shape cycles hexagon → pentagon → square and then repeats; the diamond moves 1 position clockwise around the frame each time.

**Why the other options are wrong:** **A** — the central shape is a square instead of a hexagon; **B** — the central shape is a pentagon instead of a hexagon; the diamond has not moved on from the fifth figure; **D** — the central shape is a square instead of a hexagon; the diamond is one position too far clockwise; **E** — the central shape is a pentagon instead of a hexagon; the diamond is one position too far clockwise.

**Question 41 — Answer: A**

**Rules:** the arrow rotates  $45^\circ$  anticlockwise each time; the diamond moves 3 positions clockwise around the frame each time.

**Why the other options are wrong:** **B** — the arrow is rotated  $45^\circ$  too far anticlockwise; **C** — the arrow is rotated  $45^\circ$  too far anticlockwise; the diamond has not moved on from the fifth figure; **D** — the arrow has not rotated on from the fifth figure; the diamond has not moved on from the fifth figure; **E** — the arrow has not rotated on from the fifth figure; the diamond is one position too far clockwise.

**Question 42 — Answer: B**

**Rules:** the shaded quarter of the square moves one quarter anticlockwise each time; the square moves 3 positions clockwise around the frame each time.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; the square has not moved on from the fifth figure; **C** — the wrong quarter of the square is shaded; the square has not moved on from the fifth figure; **D** — the wrong quarter of the square is shaded; the square is one position too far clockwise; **E** — the wrong quarter of the square is shaded.

**Question 43 — Answer: E**

**Rules:** the arrow rotates  $90^\circ$  clockwise each time; the triangle moves 3 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the arrow has not rotated on from the fifth figure; **B** — the triangle is one position too far clockwise; **C** — the arrow is rotated  $45^\circ$  too far clockwise; the triangle has not moved on from the fifth figure; **D** — the arrow has not rotated on from the fifth figure; the triangle has not moved on from the fifth figure.

**Question 44 — Answer: C**

**Rules:** the shaded quarter of the square moves one quarter clockwise each time; the diamond moves 3 positions clockwise around the frame each time.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; the diamond is one position too far clockwise; **B** — the wrong quarter of the square is shaded; **D** — the wrong quarter of the square is shaded; **E** — the wrong quarter of the square is shaded; the diamond is one position too far clockwise.

**Question 45 — Answer: D**

**Rules:** the number of dots increases by 1 each time; the square moves 3 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — there are 7 dots instead of 6; the square has not moved on from the fifth figure; **B** — there are 5 dots instead of 6; the square is one position too far clockwise; **C** — there are 7 dots instead of 6; **E** — the square is one position too far clockwise.

**Question 46 — Answer: B**

**Rules:** the shaded quarter of the square moves one quarter anticlockwise each time; the square moves 3 positions clockwise around the frame each time.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; **C** — the wrong quarter of the square is shaded; the square is one position too far clockwise; **D** — the wrong quarter of the square is shaded; **E** — the wrong quarter of the square is shaded; the square has not moved on from the fifth figure.

**Question 47 — Answer: A**

**Rules:** the arrow rotates 135° anticlockwise each time; the circle moves 1 position anticlockwise around the frame each time.

**Why the other options are wrong:** **B** — the circle has not moved on from the fifth figure; **C** — the arrow has not rotated on from the fifth figure; **D** — the arrow has not rotated on from the fifth figure; the circle is one position too far anticlockwise; **E** — the arrow has not rotated on from the fifth figure; the circle has not moved on from the fifth figure.

**Question 48 — Answer: D**

**Rules:** the number of dots increases by 1 each time; the circle moves 3 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — there are 7 dots instead of 6; **B** — there are 7 dots instead of 6; the circle is one position too far clockwise; **C** — there are 5 dots instead of 6; the circle has not moved on from the fifth figure; **E** — there are 7 dots instead of 6; the circle has not moved on from the fifth figure.

**Question 49 — Answer: C**

**Rules:** the central shape cycles triangle → square → pentagon and then repeats; the diamond moves 3 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the diamond has not moved on from the fifth figure; **B** — the central shape is a triangle instead of a square; the diamond is one position too far clockwise; **D** — the central shape is a triangle instead of a square; the diamond has not moved on from the fifth figure; **E** — the central shape is a triangle instead of a square.

**Question 50 — Answer: E**

**Rules:** the arrow rotates 135° anticlockwise each time; the square moves 3 positions clockwise around the frame each time.

**Why the other options are wrong:** **A** — the arrow is rotated 45° too far clockwise; **B** — the arrow is rotated 45° too far clockwise; the square has not moved on from the fifth figure; **C** — the square is one position too far clockwise; **D** — the arrow is rotated 45° too far clockwise; the square is one position too far clockwise.

**Question 51 — Answer: A**

**Rules:** the shaded quarter of the square moves one quarter clockwise each time; the diamond moves 2 positions anticlockwise around the frame each time; the diamond's shading alternates between black and grey.

**Why the other options are wrong:** **B** — the wrong quarter of the square is shaded; the diamond has not moved on from the fifth figure; the diamond is black instead of grey; **C** — the wrong quarter of the square is shaded; the diamond is white instead of grey; **D** — the wrong quarter of the square is shaded; the diamond is black instead of grey; **E** — the diamond is one position too far clockwise.

**Question 52 — Answer: E**

**Rules:** the shaded quarter of the square moves one quarter clockwise each time; the diamond moves 3 positions clockwise around the frame each time; the square moves 3 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; the diamond has not moved on from the fifth figure; the square has not moved on from the fifth figure; **B** — the wrong quarter of the square is shaded; **C** — the wrong quarter of the square is shaded; the square is one position too far clockwise; **D** — the diamond has not moved on from the fifth figure; the square is one position too far clockwise.

**Question 53 — Answer: C**

**Rules:** the shaded quarter of the square moves one quarter anticlockwise each time; the diamond moves 3 positions clockwise around the frame each time; the diamond's shading cycles white → grey → black.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; the diamond is one position too far clockwise; **B** — the wrong quarter of the square is shaded; the diamond has not moved on from the fifth figure; the diamond is grey instead of black; **D** — the wrong quarter of the square is shaded; the diamond has not moved on from the fifth figure; the diamond is white instead of black; **E** — the wrong quarter of the square is shaded; the diamond is white instead of black.

**Question 54 — Answer: D**

**Rules:** the arrow rotates 90° clockwise each time; the circle moves anticlockwise around the frame, alternately 1 and 2 positions at a time; the circle's shading cycles white → grey → black.

**Why the other options are wrong:** **A** — the arrow is rotated 45° too far clockwise; **B** — the arrow has not rotated on from the fifth figure; the circle is one position too far anticlockwise; **C** — the arrow has not rotated on from the fifth figure; the circle has not moved on from the fifth figure; the circle is grey instead of black; **E** — the arrow is rotated 45° too far clockwise; the circle is one position too far anticlockwise.

**Question 55 — Answer: B**

**Rules:** the shaded quarter of the square moves one quarter clockwise each time; the square moves clockwise by 1, then 2, then 3, then 4 positions (one more each time); the square's shading cycles grey → black → white.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; **C** — the wrong quarter of the square is shaded; the square has not moved on from the fifth figure; **D** — the square has not moved on from the fifth figure; the square is black instead of white; **E** — the wrong quarter of the square is shaded; the square is one position too far clockwise.

**Question 56 — Answer: E**

**Rules:** the arrow rotates clockwise, alternately by 45° and 90°; the circle moves 1 position anticlockwise around the frame each time; the circle's shading alternates between grey and white.

**Why the other options are wrong:** **A** — the arrow has not rotated on from the fifth figure; the circle is grey instead of white; **B** — the arrow has not rotated on from the fifth figure; the circle has not moved on from the fifth figure; the circle is grey instead of white; **C** — the circle is one position too far anticlockwise; the circle is black instead of white; **D** — the arrow is rotated 45° too far clockwise; the circle has not moved on from the fifth figure; the circle is black instead of white.

**Question 57 — Answer: D**

**Rules:** the shaded quarter of the square moves one quarter clockwise each time; the triangle moves 2 positions anticlockwise around the frame each time; the triangle's shading alternates between black and grey.

**Why the other options are wrong:** **A** — the triangle is one position too far clockwise; **B** — the wrong quarter of the square is shaded; the triangle is one position too far clockwise; **C** — the wrong quarter of the square is shaded; the triangle is one position too far clockwise; the triangle is white instead of grey; **E** — the wrong quarter of the square is shaded; the triangle is one position too far clockwise.

**Question 58 — Answer: B**

**Rules:** the shaded quarter of the square moves one quarter clockwise each time; the circle moves anticlockwise by 1, then 2, then 3, then 4 positions (one more each time); the diamond moves 2 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; the diamond has not moved on from the fifth figure; **C** — the wrong quarter of the square is shaded; the circle has not moved on from the fifth figure; **D** — the diamond has not moved on from the fifth figure; **E** — the wrong quarter of the square is shaded; the circle has not moved on from the fifth figure; the diamond has not moved on from the fifth figure.

**Question 59 — Answer: A**

**Rules:** the number of dots changes by +2, then -1, alternately; the diamond moves 1 position clockwise around the frame each time; the diamond's shading cycles black → white → grey.

**Why the other options are wrong:** **B** — the diamond is white instead of grey; **C** — there are 6 dots instead of 5; the diamond is one position too far clockwise; the diamond is white instead of grey; **D** — there are 3 dots instead of 5; the diamond has not moved on from the fifth figure; the diamond is black instead of grey; **E** — there are 6 dots instead of 5; the diamond is one position too far clockwise.

**Question 60 — Answer: C**

**Rules:** the arrow rotates clockwise, alternately by 90° and 45°; the diamond moves 1 position clockwise around the frame each time; the diamond's shading cycles white → grey → black.

**Why the other options are wrong:** **A** — the arrow has not rotated on from the fifth figure; the diamond is grey instead of black; **B** — the arrow has not rotated on from the fifth figure; the diamond is one position too far clockwise; **D** — the diamond has not moved on from the fifth figure; the diamond is white instead of black; **E** — the arrow is rotated 45° too far clockwise; the diamond has not moved on from the fifth figure; the diamond is grey instead of black.

**Question 61 — Answer: D**

**Rules:** the shaded quarter of the square moves one quarter clockwise each time; the diamond moves anticlockwise around the frame, alternately 1 and 3 positions at a time; the circle moves 1 position clockwise around the frame each time.

**Why the other options are wrong:** **A** — the diamond has not moved on from the fifth figure; the circle is one position too far clockwise; **B** — the wrong quarter of the square is shaded; the circle is one position too far clockwise; **C** — the wrong quarter of the square is shaded; the diamond is one position too far anticlockwise; the circle has not moved on from the fifth figure; **E** — the wrong quarter of the square is shaded; the diamond is one position too far anticlockwise.

**Question 62 — Answer: B**

**Rules:** the arrow rotates 45° clockwise each time; the square moves clockwise around the frame, alternately 1 and 2 positions at a time; the square's shading cycles white → grey → black.

**Why the other options are wrong:** **A** — the arrow has not rotated on from the fifth figure; the square has not moved on from the fifth figure; **C** — the arrow has not rotated on from the fifth figure; the square is one position too far clockwise; the square is grey instead of black; **D** — the arrow is rotated 45° too far clockwise; the square is white instead of black; **E** — the arrow is rotated 45° too far clockwise; the square is one position too far clockwise; the square is grey instead of black.



**Question 63 — Answer: A**

**Rules:** the shaded quarter of the square moves one quarter clockwise each time; the diamond moves anticlockwise around the frame, alternately 1 and 2 positions at a time; the diamond's shading cycles black → white → grey.

**Why the other options are wrong:** **B** — the diamond has not moved on from the fifth figure; the diamond is white instead of grey; **C** — the diamond is black instead of grey; **D** — the wrong quarter of the square is shaded; the diamond is one position too far anticlockwise; **E** — the wrong quarter of the square is shaded; the diamond has not moved on from the fifth figure; the diamond is white instead of grey.

**Question 64 — Answer: C**

**Rules:** the arrow rotates clockwise, alternately by  $45^\circ$  and  $90^\circ$ ; the circle moves 1 position anticlockwise around the frame each time; the diamond moves 3 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the arrow has not rotated on from the fifth figure; the circle is one position too far anticlockwise; the diamond has not moved on from the fifth figure; **B** — the diamond has not moved on from the fifth figure; **D** — the arrow is rotated  $45^\circ$  too far clockwise; the circle is one position too far anticlockwise; **E** — the circle has not moved on from the fifth figure; the diamond is one position too far clockwise.

**Question 65 — Answer: E**

**Rules:** the central shape cycles hexagon → pentagon → square → triangle and then repeats; the central shape's shading cycles grey → black → white; the circle moves anticlockwise by 1, then 2, then 3, then 4 positions (one more each time); the circle's shading alternates between white and grey.

**Why the other options are wrong:** **A** — the circle has not moved on from the fifth figure; the circle is white instead of grey; **B** — the central shape is a square instead of a pentagon; the central shape is black instead of white; the circle is one position too far clockwise; **C** — the circle is one position too far clockwise; the circle is white instead of grey; **D** — the circle has not moved on from the fifth figure; the circle is black instead of grey.

**Question 66 — Answer: D**

**Rules:** the arrow rotates  $135^\circ$  clockwise each time; the square moves clockwise around the frame, alternately 1 and 3 positions at a time; the diamond moves 3 positions clockwise around the frame each time.

**Why the other options are wrong:** **A** — the square has not moved on from the fifth figure; the diamond is one position too far clockwise; **B** — the arrow has not rotated on from the fifth figure; the square is one position too far clockwise; **C** — the arrow has not rotated on from the fifth figure; **E** — the arrow has not rotated on from the fifth figure; the square is one position too far clockwise; the diamond is one position too far clockwise.

**Question 67 — Answer: B**

**Rules:** the arrow rotates anticlockwise, alternately by  $45^\circ$  and  $90^\circ$ ; the triangle moves clockwise around the frame, alternately 3 and 1 positions at a time; the triangle's shading cycles white → grey → black.

**Why the other options are wrong:** **A** — the triangle has not moved on from the fifth figure; the triangle is grey instead of black; **C** — the arrow has not rotated on from the fifth figure; the triangle is one position too far clockwise; the triangle is white instead of black; **D** — the arrow is rotated  $45^\circ$  too far anticlockwise; the triangle is grey instead of black; **E** — the arrow has not rotated on from the fifth figure; the triangle has not moved on from the fifth figure; the triangle is grey instead of black.

**Question 68 — Answer: E**

**Rules:** the arrow rotates 135° anticlockwise each time; the circle moves anticlockwise by 1, then 2, then 3, then 4 positions (one more each time); the diamond moves 2 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the arrow is rotated 45° too far clockwise; the circle has not moved on from the fifth figure; **B** — the arrow has not rotated on from the fifth figure; the circle is one position too far clockwise; the diamond is one position too far clockwise; **C** — the arrow is rotated 45° too far clockwise; the diamond has not moved on from the fifth figure; **D** — the diamond has not moved on from the fifth figure.

**Question 69 — Answer: A**

**Rules:** the central shape cycles square → pentagon → hexagon and then repeats; the central shape's shading cycles grey → black → white; the diamond moves clockwise around the frame, alternately 2 and 1 positions at a time; the diamond's shading alternates between white and grey.

**Why the other options are wrong:** **B** — the central shape is a hexagon instead of a square; the central shape is grey instead of white; the diamond is black instead of grey; **C** — the central shape is grey instead of white; the diamond has not moved on from the fifth figure; the diamond is black instead of grey; **D** — the central shape is a pentagon instead of a square; the diamond has not moved on from the fifth figure; the diamond is black instead of grey; **E** — the central shape is a pentagon instead of a square; the central shape is black instead of white; the diamond is white instead of grey.

**Question 70 — Answer: C**

**Rules:** the arrow rotates anticlockwise, alternately by 45° and 90°; the triangle moves anticlockwise around the frame, alternately 2 and 1 positions at a time; the triangle's shading alternates between grey and black.

**Why the other options are wrong:** **A** — the arrow has not rotated on from the fifth figure; the triangle has not moved on from the fifth figure; **B** — the arrow is rotated 45° too far anticlockwise; the triangle is grey instead of black; **D** — the arrow has not rotated on from the fifth figure; the triangle is white instead of black; **E** — the triangle is white instead of black.

**Question 71 — Answer: D**

**Rules:** the central shape cycles triangle → square → pentagon and then repeats; the central shape's shading alternates between white and black; the circle moves anticlockwise around the frame, alternately 3 and 1 positions at a time; the circle's shading cycles grey → black → white.

**Why the other options are wrong:** **A** — the central shape is grey instead of black; the circle is one position too far clockwise; the circle is grey instead of white; **B** — the central shape is a square instead of a pentagon; the circle has not moved on from the fifth figure; the circle is black instead of white; **C** — the central shape is a square instead of a pentagon; the central shape is white instead of black; the circle has not moved on from the fifth figure; the circle is black instead of white; **E** — the central shape is a square instead of a pentagon; the central shape is grey instead of black; the circle has not moved on from the fifth figure; the circle is black instead of white.

**Question 72 — Answer: A**

**Rules:** the arrow rotates clockwise, alternately by 90° and 135°; the diamond moves clockwise around the frame, alternately 3 and 1 positions at a time; the diamond's shading alternates between black and white.

**Why the other options are wrong:** **B** — the arrow is rotated 45° too far clockwise; the diamond is one position too far clockwise; the diamond is black instead of white; **C** — the arrow has not rotated on from the fifth figure; the diamond is one position too far clockwise; the diamond is grey instead of white; **D** — the arrow has not rotated on from the fifth figure; **E** — the arrow is rotated 45° too far clockwise; the diamond is black instead of white.

**Question 73 — Answer: B**

**Rules:** the arrow rotates anticlockwise, alternately by  $90^\circ$  and  $45^\circ$ ; the circle moves clockwise around the frame, alternately 1 and 2 positions at a time; the circle's shading alternates between white and grey; the square moves 2 positions clockwise around the frame each time.

**Why the other options are wrong:** **A** — the circle is one position too far clockwise; the circle is black instead of grey; **C** — the arrow has not rotated on from the fifth figure; the circle is one position too far clockwise; the square has not moved on from the fifth figure; **D** — the arrow has not rotated on from the fifth figure; the circle is one position too far clockwise; the circle is black instead of grey; the square has not moved on from the fifth figure; **E** — the circle is white instead of grey; the square has not moved on from the fifth figure.

**Question 74 — Answer: C**

**Rules:** the arrow rotates anticlockwise, alternately by  $90^\circ$  and  $135^\circ$ ; the square moves anticlockwise around the frame, alternately 3 and 1 positions at a time; the square's shading cycles white → grey → black; the triangle moves 2 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the arrow has not rotated on from the fifth figure; the square is one position too far clockwise; the square is white instead of black; **B** — the square has not moved on from the fifth figure; the square is white instead of black; **D** — the arrow is rotated  $45^\circ$  too far clockwise; the square has not moved on from the fifth figure; **E** — the arrow has not rotated on from the fifth figure; the square has not moved on from the fifth figure; the triangle is one position too far clockwise.

**Question 75 — Answer: E**

**Rules:** the arrow rotates clockwise, alternately by  $135^\circ$  and  $90^\circ$ ; the triangle moves anticlockwise around the frame, alternately 1 and 3 positions at a time; the triangle's shading cycles white → grey → black; the diamond moves 2 positions clockwise around the frame each time.

**Why the other options are wrong:** **A** — the arrow has not rotated on from the fifth figure; the triangle has not moved on from the fifth figure; the triangle is grey instead of black; the diamond is one position too far clockwise; **B** — the arrow is rotated  $45^\circ$  too far clockwise; the triangle is one position too far anticlockwise; the triangle is grey instead of black; **C** — the arrow is rotated  $45^\circ$  too far clockwise; the triangle is white instead of black; **D** — the triangle is grey instead of black; the diamond has not moved on from the fifth figure.



# Solutions — Situational Judgement

## Answer key

|          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|
| 1 C / A  | 2 A / D  | 3 D / B  | 4 B / C  | 5 A / C  | 6 D / A  |
| 7 B / D  | 8 C / B  | 9 B / A  | 10 D / C | 11 C / D | 12 A / B |
| 13 D / B | 14 A / D | 15 B / C | 16 C / A | 17 D / C | 18 A / D |
| 19 C / A | 20 B / D | 21 A / B | 22 C / B | 23 D / C | 24 B / A |

## Worked solutions

### Question 1 — Most effective: C · Least effective: A

**Ranking:** C > B > D > A (*Working with others*)

- A.** Least effective. Blaming a colleague in front of external stakeholders damages trust within the team and the image of the service, and does nothing to clear the backlog.
- B.** Fairly effective. It maintains the service to stakeholders and alerts the manager, but it goes round Aoife rather than working with her, so the underlying problem may continue.
- C.** Most effective. It shows good collaboration: it addresses the problem directly but supportively with the colleague, looks for a practical shared solution and involves the manager in agreeing the change.
- D.** Weak. Sticking rigidly to the rota ignores a shared responsibility and lets the service to stakeholders keep deteriorating, although it at least avoids conflict.

### Question 2 — Most effective: A · Least effective: D

**Ranking:** A > C > B > D (*Analysing & problem solving*)

- A.** Most effective. It tests the competing explanations against the evidence within the time available, which is exactly what sound analysis requires before advising on a design change.
- B.** Weak. It adopts an untested explanation and only hedges in a footnote, so the director may take a decision on a cause that has not been verified.
- C.** Fairly effective. It is balanced and transparent, but it defers analysis that could largely be done within the ten days, so the briefing gives the director less than it could.
- D.** Least effective. It recommends a costly policy change on the basis of an unverified assumption, ignoring a plausible alternative cause.

### Question 3 — Most effective: D · Least effective: B

**Ranking:** D > A > C > B (*Prioritising & organising*)

- A.** Fairly effective. The claims deadline is respected, but the most time-sensitive task is the room booking, and delaying it risks there being no suitable room for the workshop.
- B.** Least effective. It puts the one explicitly non-urgent request first and leaves both time-bound tasks waiting behind a long job, risking both Thursday's workshop and the Friday deadline.
- C.** Weak. Working in arrival order is systematic and protects the Friday deadline, but it ignores urgency: the room booking, received last, would be handled last and possibly too late.
- D.** Most effective. It deals first with the task where delay has the greatest consequences and plans the remaining tasks against their deadlines, keeping Marie informed.

**Question 4 — Most effective: B · Least effective: C**

**Ranking: B > D > A > C** (*Communicating*)

- A.** Weak. The content is accurate, but reading out approved text is not adapted to a non-specialist, multilingual audience and is unlikely to hold attention or aid understanding.
- B.** Most effective. It tailors the message to the audience, focuses on what matters to them and stays transparent by signposting the full evidence.
- C.** Least effective. Overloading a non-specialist audience with tables obscures the key messages; transparency is better ensured by making the full report available.
- D.** Fairly effective. It is transparent and gives participants a chance to raise their concerns, but few non-specialists will work through a 120-page report, so the key messages may not reach everyone.

**Question 5 — Most effective: A · Least effective: C**

**Ranking: A > D > B > C** (*Resilience*)

- A.** Most effective. It shows composure under pressure, handles the caller professionally with concrete information and puts the backlog into a sensible order.
- B.** Weak. It gives way to pressure rather than to objective priorities, which may cause other invoices to miss their legal deadline.
- C.** Least effective. The defensive, dismissive reply is likely to escalate the conflict and lead to a complaint, and it reflects poorly on the agency.
- D.** Fairly effective. It stays courteous and seeks support for an unmanageable load, but leaves the caller without any concrete information.

**Question 6 — Most effective: D · Least effective: A**

**Ranking: D > B > C > A** (*Leadership*)

- A.** Least effective. Avoiding the issue lets disengagement grow and signals that the behaviour is acceptable, which undermines the new team leader's authority.
- B.** Fairly effective. A structured format gives everyone a voice, but without feedback Luca may not change his behaviour outside the format.
- C.** Weak. Ideas are captured, but contributors are sidelined and the team dynamic that needs to change is left untouched.
- D.** Most effective. Direct, respectful feedback combined with a constructive role for Luca addresses the root cause while preserving his commitment, which is what good leadership requires.

**Question 7 — Most effective: B · Least effective: D**

**Ranking: B > A > C > D** (*Delivering quality & results*)

- A.** Fairly effective. The known errors are fixed and the deadline is met, but other errors may remain undetected and the source of the problem is not reported, so it may recur.
- B.** Most effective. It combines accuracy with timeliness: errors are fixed, the risk of further errors is checked proportionately, and the people concerned can address the cause.
- C.** Weak. Raising the issue is right, but a two-week delay to recheck everything is disproportionate when the errors can be corrected within the schedule.
- D.** Least effective. Knowingly publishing incorrect data damages the agency's credibility; taking ownership of the quality of one's output is central to this competency.

**Question 8 — Most effective: C · Least effective: B****Ranking: C > A > D > B** (*Learning & development*)

- A.** Fairly effective. It shows initiative in self-study, but working alone and without guidance is slower and may miss what matters most for the working group.
- B.** Least effective. Avoiding questions out of embarrassment blocks learning, and last-minute preparation leaves you poorly equipped to represent the unit.
- C.** Most effective. It takes ownership of development in a structured way, uses colleagues' expertise and aligns learning with the unit's needs.
- D.** Weak. Support at the first meetings is sensible, but it does not address the knowledge gap and relies on others to compensate for it.

**Question 9 — Most effective: B · Least effective: A****Ranking: B > C > D > A** (*Analysing & problem solving*)

- A.** Least effective. It creates a workaround that bypasses document management rules and risks data loss, while nobody investigates the cause.
- B.** Most effective. It looks for patterns across the evidence and gives specialists the analysis they need to find the root cause quickly.
- C.** Fairly effective. Escalation brings in the right experts, but without the pattern analysis they will start from scratch and the diagnosis will take longer.
- D.** Weak. It treats the symptom repeatedly and normalises a recurring fault instead of trying to solve it.

**Question 10 — Most effective: D · Least effective: C****Ranking: D > A > B > C** (*Working with others*)

- A.** Fairly effective. Clear deadlines and written conclusions help, but a one-size-fits-all message does not uncover why Ana is struggling and may sound confrontational.
- B.** Weak. Escalating before talking to the colleague is premature and may sour relations between the services.
- C.** Least effective. Excluding a partner service's input undermines the joint nature of the report and is likely to cause disagreement later at a higher level.
- D.** Most effective. It builds a working relationship by understanding the other service's constraints and agrees practical arrangements that keep everyone on board.

**Question 11 — Most effective: C · Least effective: D****Ranking: C > B > A > D** (*Communicating*)

- A.** Weak. It is a little more helpful than pointing to the guide, but he has already said the legal text is confusing, so the core problem remains.
- B.** Fairly effective. A call is personal and lets him ask questions, but he is left without a written record of the documents and deadline.
- C.** Most effective. It translates the rules into clear, actionable information, gives a written reference and keeps the door open for further questions.
- D.** Least effective. The reply is unhelpful and dismissive towards a new colleague who has explicitly asked for clarification, and it risks him missing the deadline.

**Question 12 — Most effective: A • Least effective: B****Ranking: A > C > D > B** (*Prioritising & organising*)

**A.** Most effective. It sets priorities by urgency and importance, and manages expectations early with everyone affected.

**B.** Least effective. Spreading effort evenly ignores urgency and makes it likely that the 48-hour briefing will be late or of poor quality.

**C.** Fairly effective. Involving the manager in a genuine conflict of priorities is reasonable, but the order here is fairly clear and you should come with a proposal rather than leave the whole decision to her.

**D.** Weak. The order is sensible and the colleague is warned, but no firm date or alternative reviewer is offered.

**Question 13 — Most effective: D • Least effective: B****Ranking: D > C > A > B** (*Delivering quality & results*)

**A.** Weak. Meeting the deadline matters, but applicants may receive feedback that does not reflect their proposal, exposing the unit to complaints.

**B.** Least effective. Cosmetic editing by someone who did not evaluate the proposals compromises the integrity of the evaluation and hides rather than fixes the problem.

**C.** Fairly effective. It takes quality seriously, but checking every report may be disproportionate, and it hands the whole task to the person who simply raised the alert; giving one evaluator access to all other evaluations may also breach the confidentiality rules of the evaluation.

**D.** Most effective. It targets the identified quality problem quickly through the proper people and balances accuracy with the published timetable.

**Question 14 — Most effective: A • Least effective: D****Ranking: A > C > B > D** (*Leadership*)

**A.** Most effective. It leads the change by giving it meaning, hearing objections, setting a clear expectation and providing peer support.

**B.** Weak. It may force compliance, but escalating without first trying to bring colleagues on board damages trust and does not build commitment to the change.

**C.** Fairly effective. Training addresses possible skills gaps, but on its own it may not tackle resistance or end the parallel paper system.

**D.** Least effective. Tolerating two systems prolongs confusion and duplication and fails to implement the director's decision.

**Question 15 — Most effective: B • Least effective: C****Ranking: B > D > A > C** (*Learning & development*)

**A.** Weak. Giving context is legitimate, but focusing on the wording of the appraisal rather than on improvement is defensive.

**B.** Most effective. It turns feedback into concrete learning, uses the people best placed to help and prevents the same errors recurring.

**C.** Least effective. Dismissing feedback means nothing changes and files will continue to be returned under the current rules.

**D.** Fairly effective. Accepting feedback openly is positive, but relying on future alerts is passive and does not address the existing gaps.



**Question 16 — Most effective: C · Least effective: A****Ranking: C > D > B > A** (*Resilience*)

- A.** Least effective. It feeds negativity in the team and lowers the quality of your own work, showing poor adaptability to legitimate changes in direction.
- B.** Weak. Some clarity is useful, but waiting passively loses valuable time from a tight deadline.
- C.** Most effective. It recognises the setback, recovers quickly, builds on what still has value and plans the new work constructively.
- D.** Fairly effective. It shows energy and adaptability, but discards useful earlier work and starts without agreeing the new direction with the manager.

**Question 17 — Most effective: D · Least effective: C****Ranking: D > B > A > C** (*Analysing & problem solving*)

- A.** Weak. It builds in a safeguard for the future but commits funds before checking a credible concern about the current programme's impact.
- B.** Fairly effective. It follows up the concern, but relies on the party with an interest in a positive result rather than on independent evidence.
- C.** Least effective. It draws a firm conclusion from a single anecdote and ignores the available evidence, which is the opposite of sound analysis.
- D.** Most effective. It seeks independent data to test both the report and the anecdote, and judges the programme on outcomes rather than outputs.

**Question 18 — Most effective: A · Least effective: D****Ranking: A > C > B > D** (*Communicating*)

- A.** Most effective. It communicates openly and consistently with the whole team, is honest about uncertainty and creates a channel for questions.
- B.** Weak. It is accurate, but impersonal and leaves people's concerns unaddressed, so rumours are likely to continue.
- C.** Fairly effective. It responds honestly to those who asked, but the rest of the team receives no information and may hear different versions.
- D.** Least effective. Making promises you cannot guarantee goes beyond what you are authorised to say and will destroy trust if they prove untrue.

**Question 19 — Most effective: C · Least effective: A****Ranking: C > D > B > A** (*Prioritising & organising*)

- A.** Least effective. Relying on suppliers to set the pace is reactive and risks missed deadlines and last-minute problems; travel reimbursements have no supplier to send reminders at all.
- B.** Weak. Delegating is right in principle, but giving a critical, complex booking to someone without experience creates a real risk of failure.
- C.** Most effective. It plans the work against deadlines and delegates tasks matched to the colleague's experience, while keeping critical items under control.
- D.** Fairly effective. The tasks are likely to be done, but it is not sustainable, wastes available help and leaves no margin if something goes wrong.

**Question 20 — Most effective: B · Least effective: D**

**Ranking: B > A > C > D** (*Resilience*)

**A.** Fairly effective. It shows perseverance and professionalism, but simply working harder does not reduce the overload.

**B.** Most effective. It stays composed, seeks support and finds a smart way to reduce pressure, while applying objective priorities.

**C.** Weak. Giving way to status rather than urgency is unfair to other callers and does little to manage the overall situation.

**D.** Least effective. Withdrawing from the main channel during a crisis leaves users without support and will make the backlog worse.

**Question 21 — Most effective: A · Least effective: B**

**Ranking: A > D > C > B** (*Working with others*)

**A.** Most effective. It treats colleagues fairly and respectfully, deals with the issue at your own level and builds trust through transparency.

**B.** Least effective. Favouritism damages team cohesion and fairness, and leaves a newer colleague carrying an unequal share of the work.

**C.** Weak. The intention is good, but it replaces one departure from the criteria with another rather than restoring fair allocation.

**D.** Fairly effective. Raising the issue is legitimate, but it passes to the manager a problem you can largely solve yourself by applying the agreed criteria.

**Question 22 — Most effective: C · Least effective: B**

**Ranking: C > A > D > B** (*Leadership*)

**A.** Fairly effective. It sets clear expectations and follows up, but it does not explore why a normally reliable colleague is struggling.

**B.** Least effective. Criticising an individual in front of the team is demotivating and humiliating, and is unlikely to solve the problem.

**C.** Most effective. It combines honest feedback with genuine care and agrees a way forward, which supports both the person and the team's results.

**D.** Weak. It may relieve short-term pressure, but it avoids the conversation, may be read as a loss of confidence and adds to colleagues' workload.

**Question 23 — Most effective: D · Least effective: C**

**Ranking: D > B > A > C** (*Delivering quality & results*)

**A.** Weak. It protects accuracy but misses the deadline without first trying to resolve the issue in the afternoon available.

**B.** Fairly effective. It is transparent and on time, but passes on data you could have checked yourself, so the report may contain avoidable errors.

**C.** Least effective. Replacing real data with an estimate without saying so produces misleading figures and is a serious breach of reliability.

**D.** Most effective. It takes ownership of the quality of the output, uses the available time to check the data and is transparent about what remains uncertain.

**Question 24 — Most effective: B • Least effective: A****Ranking: B > C > D > A** (*Learning & development*)

- A.** Least effective. Relying on general praise ignores clear feedback and misses the opportunity to improve.
- B.** Most effective. It involves the team in learning from experience and turns feedback into concrete improvements for the next event.
- C.** Fairly effective. Seeking specific feedback is useful, but it covers only one issue and does not involve the team in drawing lessons.
- D.** Weak. The feedback is kept, but nobody analyses it or decides on changes, so the same issues may recur.



# Solutions — Accuracy & Precision

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## Answer key

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 1 C  | 2 D  | 3 B  | 4 A  | 5 D  | 6 B  | 7 C  | 8 A  |
| 9 B  | 10 A | 11 C | 12 D | 13 A | 14 D | 15 A | 16 D |
| 17 B | 18 A | 19 D | 20 C | 21 A | 22 D | 23 B | 24 B |
| 25 A | 26 C | 27 D | 28 B | 29 B | 30 C |      |      |

## Worked solutions

### Question 1 — Answer: C

Compare the copy with row 8: the country code differs (BE → BG); all other fields match, so the answer is “Country code”. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

### Question 2 — Answer: D

Compare the copy with row 5: every field of the copy matches row 5 exactly, so “No error” is correct. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

### Question 3 — Answer: B

Compare the copy with row 8: the final two letters of the reference are reversed (CT → TC); all other fields match, so the answer is “Reference”. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

### Question 4 — Answer: A

Compare the copy with row 4: the amount differs (7,676.00 → 7,766.00); all other fields match, so the answer is “Amount (€)”. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

### Question 5 — Answer: D

Compare the copy with row 6: every field of the copy matches row 6 exactly, so “No error” is correct. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

### Question 6 — Answer: B

Compare the copy with row 2: the amount differs (75,238.15 → 75,328.15); all other fields match, so the answer is “Amount (€)”. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

### Question 7 — Answer: C

Compare the copy with row 5: two digits of the reference are transposed (TR-4931/FY → TR-4391/FY); all other fields match, so the answer is “Reference”. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 8 — Answer: A**

Compare the copy with row 4: two letters of the surname are transposed (Bērziņš → Brēziņš); all other fields match, so the answer is “Beneficiary”. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 9 — Answer: B**

Compare the copy with row 4: the country code differs (ES → EE); all other fields match, so the answer is “Country code”. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 10 — Answer: A**

Compare the copy with row 4: two letters of the surname are transposed (Weber → Wbeer); all other fields match, so the answer is “Beneficiary”. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 11 — Answer: C**

Compare the copy with row 2: the amount differs (10,821.00 → 10,812.00); all other fields match, so the answer is “Amount (€)”. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 12 — Answer: D**

Compare the copy with row 4: every field of the copy matches row 4 exactly, so “No error” is correct. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 13 — Answer: A**

A: matches row 3 exactly. B: compare with row 9: the country code differs (SI → SK). C: compare with row 6: one digit of the reference is wrong (TR-4550/UM → TR-4850/UM). D: compare with row 8: the final two letters of the reference are reversed (EY → YE). Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 14 — Answer: D**

A: compare with row 5: the first name is spelled differently (Katarina → Katharina). B: compare with row 10: two letters of the surname are transposed (Nowak → Noawk). C: compare with row 8: the country code differs (IT → IE). D: matches row 6 exactly. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 15 — Answer: A**

A: matches row 7 exactly. B: compare with row 10: one digit of the reference is wrong (CN-7050/VA → CN-7055/VA). C: compare with row 4: two digits of the reference are transposed (CN-8886/VM → CN-8868/VM). D: compare with row 2: the amount differs (51,619.90 → 51,169.90). Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 16 — Answer: D**

A: compare with row 1: the first name is spelled differently (Liis → Lis). B: compare with row 6: the amount differs (54,510.40 → 54,510.04). C: compare with row 4: the country code differs (PL → PT). D: matches row 7 exactly. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 17 — Answer: B**

A: compare with row 3: the amount differs (93,037.40 → 93,307.40). B: matches row 8 exactly. C: compare with row 4: the country code differs (ES → EE). D: compare with row 9: a double letter in the surname has become single (Andersson → Anderson). Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 18 — Answer: A**

A: matches row 7 exactly. B: compare with row 4: two letters of the surname are transposed (Georgiou → Georigou). C: compare with row 5: one digit of the reference is wrong (GR-8811/KV → GR-8911/KV). D: compare with row 9: an accent or diacritic is missing from the surname (Horváth → Horvath). Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 19 — Answer: D**

A: compare with row 8: the amount differs (60,696.25 → 60,696.52). B: compare with row 4: two letters of the surname are transposed (Ferrari → Ferrai). C: compare with row 3: the amount differs (8,161.40 → 1,861.40). D: matches row 7 exactly. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 20 — Answer: C**

A: compare with row 5: the country code differs (LV → LT). B: compare with row 1: two digits of the reference are transposed (FN-9072/FV → FN-9702/FV). C: matches row 10 exactly. D: compare with row 9: the amount differs (25,668.90 → 52,668.90). Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 21 — Answer: A**

A: matches row 1 exactly. B: compare with row 2: two digits of the reference are transposed (MS-7203/XJ → MS-7023/XJ). C: compare with row 3: the country code differs (LU → LT). D: compare with row 8: two letters of the surname are transposed (Petrović → Pertović). Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 22 — Answer: D**

A: compare with row 2: two letters of the surname are transposed (Lindqvist → Lindvqist). B: compare with row 7: two letters of the surname are transposed (Silva → Sivla). C: compare with row 5: the amount differs (81,012.40 → 80,112.40). D: matches row 3 exactly. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 23 — Answer: B**

copy 1 (row 9): no error; copy 2 (row 1): no error; copy 3 (row 2): a digit has been replaced by a look-alike letter (1 → I); copy 4 (row 3): the first name is spelled differently (Luca → Luka); copy 5 (row 4): one digit of the reference is wrong (MS-2958/CL → MS-3958/CL). Copies with an error: 3. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 24 — Answer: B**

copy 1 (row 1): two letters of the surname are transposed (García → Garíca); copy 2 (row 2): two letters of the surname are transposed (Petrović → Petroivć); copy 3 (row 7): no error; copy 4 (row 8): no error; copy 5 (row 3): no error. Copies with an error: 2. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 25 — Answer: A**

copy 1 (row 1): no error; copy 2 (row 6): no error; copy 3 (row 7): the country code differs (LV → LT); copy 4 (row 3): one digit of the reference is wrong (GR-5895/EH → GR-5395/EH); copy 5 (row 5): no error. Copies with an error: 2. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 26 — Answer: C**

copy 1 (row 10): the first name is spelled differently (Clara → Klara); copy 2 (row 4): no error; copy 3 (row 7): a digit has been replaced by a look-alike letter (8 → B); copy 4 (row 1): two letters of the surname are transposed (Georgiou → Geogriou); copy 5 (row 9): the amount differs (45,825.75 → 45,825.57). Copies with an error: 4. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 27 — Answer: D**

copy 1 (row 5): the final two letters of the reference are reversed (RC → CR); copy 2 (row 3): no error; copy 3 (row 8): the amount differs (95,305.25 → 95,305.52); copy 4 (row 10): no error; copy 5 (row 7): two letters of the surname are transposed (Lindqvist → Lindqivst). Copies with an error: 3. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 28 — Answer: B**

copy 1 (row 6): a digit has been replaced by a look-alike letter (0 → O); copy 2 (row 9): no error; copy 3 (row 5): no error; copy 4 (row 3): no error; copy 5 (row 2): no error. Copies with an error: 1. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 29 — Answer: B**

copy 1 (row 5): no error; copy 2 (row 3): no error; copy 3 (row 6): no error; copy 4 (row 10): the country code differs (PL → PT); copy 5 (row 2): no error. Copies with an error: 1. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.

**Question 30 — Answer: C**

copy 1 (row 3): the country code differs (LV → LT); copy 2 (row 9): no error; copy 3 (row 7): no error; copy 4 (row 1): two digits of the reference are transposed (FN-9671/TW → FN-6971/TW); copy 5 (row 2): no error. Copies with an error: 2. Method: compare one field at a time; read codes in chunks of two characters, check accents letter by letter, and check amounts digit by digit from the left.



# Solutions — Prioritising & Organising

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## Answer key

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 1 A  | 2 D  | 3 C  | 4 B  | 5 C  | 6 D  | 7 B  | 8 A  |
| 9 C  | 10 D | 11 A | 12 B | 13 D | 14 C | 15 B | 16 A |
| 17 B | 18 D | 19 A | 20 C | 21 D | 22 C | 23 B | 24 A |

## Worked solutions

### Question 1 — Answer: A

Check the slots in time order, applying every condition. Thursday 09:00: Bruno, Chiara, Eleni are all free and no earlier slot meets every condition — correct (Filip is busy, but attendance is optional); Wednesday 10:00: everyone required is free, but Wednesday is excluded; Tuesday 09:00: Bruno is busy; Thursday 15:00: possible, but not the earliest suitable slot.

### Question 2 — Answer: D

Check the slots in time order, applying every condition. Monday 09:00: everyone required is free, but Monday is excluded; Tuesday 10:00: Chiara is busy; Thursday 10:00: possible, but not the earliest suitable slot; Wednesday 09:00: Filip, Chiara, Bruno are all free and no earlier slot meets every condition — correct (Eleni is busy, but attendance is optional).

### Question 3 — Answer: C

Check the slots in time order, applying every condition. Monday 10:00: everyone required is free, but Monday is excluded; Tuesday 09:00: Filip is busy; Tuesday 15:00: Bruno, Dawid, Filip are all free and no earlier slot meets every condition — correct (Chiara is busy, but attendance is optional); Thursday 10:00: possible, but not the earliest suitable slot.

### Question 4 — Answer: B

Check the slots in time order, applying every condition. Wednesday 10:00: everyone required is free, but Wednesday is excluded; Thursday 11:00: Filip, Eleni, Chiara are all free and no earlier slot meets every condition — correct (Dawid is busy, but attendance is optional); Monday 16:00: everyone required is free, but it starts after 15:00; Monday 11:00: Filip is busy.

### Question 5 — Answer: C

Check the slots in time order, applying every condition. Monday 09:00: everyone required is free, but Monday is excluded; Tuesday 09:00: Filip is busy; Tuesday 11:00: Chiara, Bruno, Filip are all free and no earlier slot meets every condition — correct (Eleni is busy, but attendance is optional); Wednesday 10:00: possible, but not the earliest suitable slot.

### Question 6 — Answer: D

Check the slots in time order, applying every condition. Monday 11:00: everyone required is free, but Monday is excluded; Thursday 09:00: possible, but not the earliest suitable slot; Tuesday 09:00: Chiara is busy; Wednesday 10:00: Dawid, Filip, Chiara are all free and no earlier slot meets every condition — correct (Anna is busy, but attendance is optional).

**Question 7 — Answer: B**

Check the slots in time order, applying every condition. Thursday 15:00: possible, but not the earliest suitable slot; Tuesday 09:00: Anna, Eleni, Bruno are all free and no earlier slot meets every condition — correct (Dawid is busy, but attendance is optional); Monday 16:00: everyone required is free, but it starts after 15:00; Monday 14:00: Anna is busy.

**Question 8 — Answer: A**

Check the slots in time order, applying every condition. Thursday 09:00: Bruno, Anna, Chiara are all free and no earlier slot meets every condition — correct (Dawid is busy, but attendance is optional); Tuesday 10:00: everyone required is free, but Tuesday is excluded; Wednesday 09:00: Anna is busy; Thursday 10:00: possible, but not the earliest suitable slot.

**Question 9 — Answer: C**

Check the slots in time order, applying every condition. Monday 09:00: everyone required is free, but Monday is excluded; Thursday 10:00: possible, but not the earliest suitable slot; Wednesday 15:00: Bruno, Filip, Anna are all free and no earlier slot meets every condition — correct (Dawid is busy, but attendance is optional); Wednesday 09:00: Bruno is busy.

**Question 10 — Answer: D**

Earliest finish of each task = its duration + the latest finish among its prerequisites: Collect data 6; Draft analysis 12; Legal check 16; Translation 18; Layout 24; Final approval 29. The longest chain (the critical path) is Collect data → Draft analysis → Translation → Layout → Final approval, so the project needs 29 days. Typical errors: adding all durations (33) wrongly assumes that nothing runs in parallel; following a shorter branch underestimates the time; forgetting the final task, or adding a parallel task in series, also gives a wrong total.

**Question 11 — Answer: A**

Earliest finish of each task = its duration + the latest finish among its prerequisites: Book venue 4; Draft programme 1; Invite speakers 4; Open registration 8; Print materials 7; Final briefing 12. The longest chain (the critical path) is Draft programme → Invite speakers → Open registration → Final briefing, so the project needs 12 days. Typical errors: adding all durations (22) wrongly assumes that nothing runs in parallel; following a shorter branch underestimates the time; forgetting the final task, or adding a parallel task in series, also gives a wrong total.

**Question 12 — Answer: B**

Earliest finish of each task = its duration + the latest finish among its prerequisites: Gather requirements 3; Configure system 6; Security review 8; Migrate data 12; Train users 11; Go-live check 15. The longest chain (the critical path) is Gather requirements → Configure system → Migrate data → Go-live check, so the project needs 15 days. Typical errors: adding all durations (22) wrongly assumes that nothing runs in parallel; following a shorter branch underestimates the time; forgetting the final task, or adding a parallel task in series, also gives a wrong total.

**Question 13 — Answer: D**

Earliest finish of each task = its duration + the latest finish among its prerequisites: Publish vacancy 6; Screen applications 8; Prepare tests 4; Run tests 14; Hold interviews 18; Final decision 23. The longest chain (the critical path) is Publish vacancy → Screen applications → Run tests → Hold interviews → Final decision, so the project needs 23 days. Typical errors: adding all durations (27) wrongly assumes that nothing runs in parallel; following a shorter branch underestimates the time; forgetting the final task, or adding a parallel task in series, also gives a wrong total.

**Question 14 — Answer: C**

Earliest finish of each task = its duration + the latest finish among its prerequisites: Collect data 1; Draft analysis 3; Legal check 7; Translation 5; Layout 11; Final approval 17. The longest chain (the critical path) is Collect data → Draft analysis → Translation → Layout → Final approval, so the project needs 17 days. Typical errors: adding all durations (21) wrongly assumes that nothing runs in parallel; following a shorter branch underestimates the time; forgetting the final task, or adding a parallel task in series, also gives a wrong total.

**Question 15 — Answer: B**

Earliest finish of each task = its duration + the latest finish among its prerequisites: Book venue 2; Draft programme 3; Invite speakers 8; Open registration 6; Print materials 12; Final briefing 18. The longest chain (the critical path) is Draft programme → Invite speakers → Print materials → Final briefing, so the project needs 18 days. Typical errors: adding all durations (23) wrongly assumes that nothing runs in parallel; following a shorter branch underestimates the time; forgetting the final task, or adding a parallel task in series, also gives a wrong total.

**Question 16 — Answer: A**

Earliest finish of each task = its duration + the latest finish among its prerequisites: Gather requirements 1; Configure system 2; Security review 4; Migrate data 8; Train users 6; Go-live check 11. The longest chain (the critical path) is Gather requirements → Configure system → Migrate data → Go-live check, so the project needs 11 days. Typical errors: adding all durations (17) wrongly assumes that nothing runs in parallel; following a shorter branch underestimates the time; forgetting the final task, or adding a parallel task in series, also gives a wrong total.

**Question 17 — Answer: B**

Earliest finish of each task = its duration + the latest finish among its prerequisites: Publish vacancy 6; Screen applications 10; Prepare tests 8; Run tests 15; Hold interviews 19; Final decision 25. The longest chain (the critical path) is Publish vacancy → Screen applications → Run tests → Hold interviews → Final decision, so the project needs 25 days. Typical errors: adding all durations (27) wrongly assumes that nothing runs in parallel; following a shorter branch underestimates the time; forgetting the final task, or adding a parallel task in series, also gives a wrong total.

**Question 18 — Answer: D**

List, for each task, who is still allowed to take it, and start with the task or person that has the fewest possibilities; each forced choice removes that person from the other tasks. The only allocation that satisfies every constraint is: budget report — Chiara; event logistics — Bruno; website update — Filip; press summary — Dawid. So the budget report must go to Chiara.

**Question 19 — Answer: A**

List, for each task, who is still allowed to take it, and start with the task or person that has the fewest possibilities; each forced choice removes that person from the other tasks. The only allocation that satisfies every constraint is: budget report — Filip; event logistics — Bruno; website update — Chiara; press summary — Dawid. So the press summary must go to Dawid.

**Question 20 — Answer: C**

List, for each task, who is still allowed to take it, and start with the task or person that has the fewest possibilities; each forced choice removes that person from the other tasks. The only allocation that satisfies every constraint is: budget report — Eleni; event logistics — Chiara; website update — Dawid; press summary — Filip. So the press summary must go to Filip.

**Question 21 — Answer: D**

List, for each task, who is still allowed to take it, and start with the task or person that has the fewest possibilities; each forced choice removes that person from the other tasks. The only allocation that satisfies every constraint is: budget report — Bruno; event logistics — Anna; website update — Filip; press summary — Dawid. So the event logistics must go to Anna.

**Question 22 — Answer: C**

List, for each task, who is still allowed to take it, and start with the task or person that has the fewest possibilities; each forced choice removes that person from the other tasks. The only allocation that satisfies every constraint is: budget report — Chiara; event logistics — Anna; website update — Filip; press summary — Dawid. So the budget report must go to Chiara.

**Question 23 — Answer: B**

List, for each task, who is still allowed to take it, and start with the task or person that has the fewest possibilities; each forced choice removes that person from the other tasks. The only allocation that satisfies every constraint is: budget report — Bruno; event logistics — Filip; website update — Dawid; press summary — Anna. So the press summary must go to Anna.

**Question 24 — Answer: A**

List, for each task, who is still allowed to take it, and start with the task or person that has the fewest possibilities; each forced choice removes that person from the other tasks. The only allocation that satisfies every constraint is: budget report — Eleni; event logistics — Dawid; website update — Filip; press summary — Anna. So the budget report must go to Eleni.

# Solutions — Mock Exam 1

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## Answer key

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 1 A  | 2 B  | 3 D  | 4 B  | 5 A  | 6 D  | 7 C  | 8 B  |
| 9 D  | 10 A | 11 B | 12 B | 13 C | 14 C | 15 D | 16 A |
| 17 A | 18 C | 19 C | 20 A | 21 B | 22 A | 23 C | 24 D |
| 25 A | 26 C | 27 B | 28 D | 29 A | 30 C | 31 A | 32 D |
| 33 C | 34 B | 35 E | 36 C | 37 D | 38 B | 39 E | 40 A |

## Worked solutions

### Question 1 — Answer: A

- A.** Correct. Volunteers without clocks or daylight kept their rhythms, but these “drifted” into cycles slightly longer than 24 hours.
- B.** Incorrect – partial truth. Morning light advances the clock; bright evening light “tends to delay it”.
- C.** Incorrect – cause and effect. Researchers “continue to investigate” whether longer-term problems stem from the disruption itself or from other features of shift work.
- D.** Incorrect – extreme wording. The passage says circadian rhythms occur in many organisms, “from cyanobacteria to humans”, so they are not confined to mammals.

### Question 2 — Answer: B

- A.** Incorrect – contradiction. C4 plants pump carbon dioxide into inner cells “where Rubisco is located”, and photorespiration is only “largely” suppressed.
- B.** Correct. Closing the pores lowers the carbon dioxide inside the leaf, making wasteful photorespiration, which releases already-fixed carbon, more frequent.
- C.** Incorrect – extreme wording. In cool, moist climates, plants without the mechanism are “often equally or more productive”.
- D.** Incorrect – contradiction. Photorespiration becomes more frequent as temperatures rise.

### Question 3 — Answer: D

- A.** Incorrect – contradiction. Peatlands cover only a small share of the world’s land surface.
- B.** Incorrect – contradiction. Rewetting does not immediately restore the original condition, and vegetation may take many years to recover.
- C.** Incorrect – cause and effect. Waterlogged conditions slow decomposition, which is why plant material accumulates as peat.
- D.** Correct. Drained peat is exposed to air and begins to decompose, releasing carbon dioxide.

**Question 4 — Answer: B**

- A.** Incorrect – contradiction. The burden lies on the prosecution; it is not for the accused to prove innocence.
- B.** Correct. Because civil claims are decided on the lower “more likely than not” standard, the same facts may lead to acquittal in criminal proceedings yet to liability in civil proceedings.
- C.** Incorrect – contradiction. Civil disputes are usually decided on a lower standard.
- D.** Incorrect – extreme wording. Officials are expected to refrain from statements presenting a suspect as guilty, not from any comment at all.

**Question 5 — Answer: A**

- A.** Correct. Estimates move towards the initial figure even when participants have been told it was picked at random.
- B.** Incorrect – contradiction. The effect “has been observed among experts as well as among laypeople”.
- C.** Incorrect – contradiction. The strength of the effect “varies from one study to another”.
- D.** Incorrect – extreme wording. Insufficient adjustment is one of “several explanations” proposed; nothing has been established as the only cause.

**Question 6 — Answer: D**

- A.** Incorrect – supported. Surveys had identified women and older residents as “particularly deterred by mixed traffic”.
- B.** Incorrect – supported. The second phase has been “approved but not yet funded”.
- C.** Incorrect – supported. Officials acknowledge that part of the increase may reflect cyclists switching from parallel streets rather than new trips.
- D.** Correct – not supported. The survey “is intended to clarify” the question; no findings are reported, so a planned measure is presented as a result.

**Question 7 — Answer: C**

- A.** Incorrect – contradiction. Research suggests that sleep plays an active role “rather than simply” protecting new information from interference.
- B.** Incorrect – partial truth. Slow-wave sleep is linked to facts and events; REM sleep to emotional memories and certain skills.
- C.** Correct. Researchers caution that most experiments use simple tasks and short intervals, and that effects on complex learning over weeks or months are “less well understood”.
- D.** Incorrect – unsupported inference. The passage compares sleep with wakefulness after learning; it makes no prediction about examination results or relative amounts of sleep.

**Question 8 — Answer: B**

- A.** Incorrect – contradiction. English “continued to be spoken by the great majority of the population”.
- B.** Correct. Borrowing is not confined to words for new things; it may replace existing native words.
- C.** Incorrect – unsupported inference. The passage says nothing about how English is classified and notes that its most frequent words remain Germanic.
- D.** Incorrect – extreme wording. Linguists warn against treating such examples as a complete explanation; nothing is proved about every difference.

**Question 9 — Answer: D**

- A.** Incorrect – extreme wording. The passage says that when resources are fully employed, higher saving can finance investment and support growth.
- B.** Incorrect – scope shift. One family’s decision scarcely affects the wider economy; the harm arises only when most households act together.
- C.** Incorrect – extreme wording. Critics say the paradox applies chiefly when there is unused capacity, so they accept it in some conditions.
- D.** Correct. Falling incomes may leave total saving unchanged or even lower when households cut spending at the same time.

**Question 10 — Answer: A**

- A.** Correct. Such programmes have produced stronger reading skills, “including, after several years, in the second language itself”, through the transfer of underlying skills.
- B.** Incorrect – unsupported inference. The passage mentions only that some parents resist these programmes; it gives no evidence of general parental preference.
- C.** Incorrect – contradiction. Diverse areas require difficult decisions about which of many home languages to support.
- D.** Incorrect – partial truth. The benefit in the second language appears “after several years”, not from the first year.

**Question 11 — Answer: B**

- A.** Incorrect – extreme wording. Expectations are only one of three sources named, alongside natural fluctuation and the extra attention participants receive.
- B.** Correct. Assessors who know the allocation “may, without intending to, interpret ambiguous outcomes more favourably”; blinding guards against this.
- C.** Incorrect – contradiction. Placebo responses are largest for self-reported outcomes and smallest for objective measurements.
- D.** Incorrect – unsupported inference. The passage says only that blinding is usually impossible in such trials; it draws no conclusion about their reliability.

**Question 12 — Answer: B**

- A.** Incorrect – contradiction. Experienced staff may find their productivity unchanged or even improved.
- B.** Correct. Output indicators capture routine work but “tend to overlook” informal exchanges, so an organisation relying on them may miss the cost to junior staff’s learning.
- C.** Incorrect – scope shift. Only “some organisations” have designated office days, and others have formalised mentoring instead.
- D.** Incorrect – cause and effect. The passage mentions interruptions only as something experienced staff avoid when working remotely; it makes no claim about junior employees’ relative productivity.

**Question 13 — Answer: C**

- A.** Incorrect – partial truth. Reading scores were higher, but the mathematics difference was small and not statistically significant.
- B.** Incorrect – contradiction. Classroom observations found that many teachers continued to teach in much the same way.
- C.** Correct. The reading difference was larger for pupils from low-income families.
- D.** Incorrect – unsupported inference. The authority has not yet decided whether to extend smaller classes to all schools.

**Question 14 — Answer: C**

- A.** Incorrect – partial truth. Such a gift may be accepted, but only on behalf of the agency; it must be registered and handed over, so the official cannot keep it.
- B.** Incorrect – contradiction. Hospitality is excluded where the host is party to a procedure the official is handling, such as a tender, whatever the agency's interest.
- C.** Correct. A gift accepted because refusal would cause offence must be handed over and becomes the agency's property.
- D.** Incorrect – extreme wording. Consulting the adviser is expected only “in cases of doubt”, not before every acceptance of hospitality.

**Question 15 — Answer: D**

- A.** Incorrect – supported. Returns above a threshold are referred to inspectors, who decide whether to open an audit.
- B.** Incorrect – supported. The internal review found that these audits recovered more unpaid tax per inspector hour.
- C.** Incorrect – supported. The review warned that taxpayers in rarely audited groups may be under-represented in the data.
- D.** Correct – not supported. The authority has only been asked by an oversight body to explain how taxpayers can find this out; no such procedure is said to exist.

**Question 16 — Answer: A**

- A.** Correct. The council has funded only a feasibility assessment on one section; implementation depends on that assessment, so realignment itself has not been approved.
- B.** Incorrect – unsupported inference. The council will merely consider compensation, and only if the assessment recommends proceeding.
- C.** Incorrect – partial truth. The study said realignment would be cheaper to maintain over several decades, not cheaper to build.
- D.** Incorrect – unsupported inference. Realignment has not been approved, the habitat benefit depends on the land not being drained, and a marsh may take many years to become established.

**Question 17 — Answer: A**

- A.** Correct. Until 1986 known superconductors needed costly liquid helium, whereas some later materials can be cooled with far cheaper liquid nitrogen.
- B.** Incorrect – scope shift. The passage says only that some related materials found “soon afterwards” could be cooled with liquid nitrogen.
- C.** Incorrect – extreme wording. Cooling costs are named as an obstacle alongside brittleness.
- D.** Incorrect – contradiction. Claims of a room-temperature superconductor “have repeatedly failed to be confirmed”.

**Question 18 — Answer: C**

- A.** Incorrect – scope shift. Serfdom weakened in much of western Europe, to which the labour shortage “contributed”, while peasant obligations grew heavier in parts of eastern Europe.
- B.** Incorrect – contradiction. The laws' effectiveness varied, and in many places wages rose despite them.
- C.** Correct. Historians say the changed balance between land and labour contributed to the weakening of serfdom, but that the epidemic was not the sole cause.
- D.** Incorrect – partial truth. Estimates vary, and the range of a third to a half refers to losses in many regions, not an agreed figure for Europe as a whole.



**Question 19 — Answer: C**

- A.** Incorrect – cause and effect. The evaluators said the subsidy “may” have helped and warned that the result could partly reflect employers choosing the most employable candidates.
- B.** Incorrect – partial truth. The “roughly half” figure refers to deadweight (hires that would have happened anyway), not to displacement.
- C.** Correct. Deadweight means fewer additional jobs than subsidised hires, which is why the cost per additional job was considerably higher.
- D.** Incorrect – contradiction. The evaluators recommended more narrowly targeted schemes, not ending them.

**Question 20 — Answer: A**

- A.** Correct. The benefits “depended on standardisation” and multiplied only once international dimensions were agreed.
- B.** Incorrect – cause and effect. Economists “continue to debate” how much of the growth reflects containerisation rather than falling tariffs and other factors.
- C.** Incorrect – partial truth. Old ports did lose traffic, but the passage attributes this to a lack of space for cranes and storage yards, not to resistance by workers.
- D.** Incorrect – contradiction. Before containers a ship could spend as long in port as at sea; afterwards ships spent “a far greater proportion of their time travelling”.

**Question 21 — Answer: B**

- > Percentage change uses the earlier value as the base:  $(223 - 180) \div 180 \times 100$ .
- >  $= 43 \div 180 \times 100 \approx 23.9\%$ , i.e. an increase of about 23.9%.
- > Typical errors: dividing by the Year 3 value (wrong base), or adding the two year-on-year percentage changes (percentages do not add).

**Correct option: B** (23.9%)**Question 22 — Answer: A**

- > Spanish : Italian = 125 : 75. Both numbers divide by 25, giving 5:3.
- > Typical error: reversing the order of the ratio (3:5).

**Correct option: A** (5:3)**Question 23 — Answer: C**

- > Approved =  $2,300 \times 15\% = 345$ .
- > Rejected =  $2,300 - 345 = 1,955$ .

**Correct option: C** (1,955)**Question 24 — Answer: D**

- > Time = distance  $\div$  speed. Now:  $336 \div 110 \times 60 = 183.3$  minutes. New:  $336 \div 150 \times 60 = 134.4$  minutes.
- > Saving =  $183.3 - 134.4 = 48.9$ , i.e. about 49 minutes.
- > Typical error: assuming journey time falls by the same percentage as speed rises.

**Correct option: D** (49 minutes)**Question 25 — Answer: A**

- > Total salary bill =  $34 \times 5,400 + 43 \times 7,200 + 45 \times 8,400 + 15 \times 3,800 = \text{€}928,200$ .
- > Total staff = 137. Weighted average =  $\text{€}928,200 \div 137 \approx \text{€}6,775$ .
- > Typical error: the simple average of the four unit averages ( $\text{€}6,200$ ) ignores the different unit sizes.

**Correct option: A** ( $\text{€}6,775$ )

**Question 26 — Answer: C**

- › Spending per inhabitant = spending ÷ population (€ million ÷ million inhabitants = € per inhabitant):
- › Central region ≈ €202; Alpine region ≈ €204; Coastal region ≈ €231; Island region ≈ €115.
- › The highest figure is for the Coastal region. Trap: the Central region has the largest total spending, but not the highest spending per inhabitant.

**Correct option: C** (Coastal region)

**Question 27 — Answer: B**

- › For an average of 50 over 6 months, the total must be  $50 \times 6 = 300$ .
- › January–May total =  $43 + 39 + 59 + 22 + 37 = 200$ .
- › June must therefore be  $300 - 200 = 100$ .
- › Typical errors: answering with the target itself (50) or with the current average (about 40).

**Correct option: B** (100)

**Question 28 — Answer: D**

- › Relative change =  $(\text{new} - \text{old}) \div \text{old} \times 100 = (8.5 - 10.6) \div 10.6 \times 100 \approx -19.8\%$ , i.e. a fall of about 19.8%.
- › Typical errors: 2.1 is the change in percentage points, not a percentage change; another is dividing by the later value.

**Correct option: D** (19.8%)

**Question 29 — Answer: A**

- › Staff now =  $37\% \times \text{€}265 \text{ million} = \text{€}98.05 \text{ million}$ ; after +10%: €107.86 million.
- › Communication =  $21\% \times \text{€}265 \text{ million} = \text{€}55.65 \text{ million}$ .
- › Difference = €52.21 million ≈ €52.2 million. Typical error: adding 10 percentage points to the share instead of raising the amount by 10%.

**Correct option: A** (€52.2 million)

**Question 30 — Answer: C**

- › Compound growth:  $2,790 \times 1.15^3 = 2,790 \times 1.5209 \approx 4,243 \text{ tonnes}$ .
- › Typical error: simple growth,  $2,790 \times (1 + 3 \times 0.15) = 4,046 \text{ tonnes}$ , ignores compounding; others apply the rate for the wrong number of years.

**Correct option: C** (4,243 t)

**Question 31 — Answer: A**

**Rules:** the arrow rotates 45° anticlockwise each time; the number of dots decreases by 1 each time.

**Why the other options are wrong:** **B** — there are 3 dots instead of 1; **C** — the arrow is rotated 45° too far anticlockwise; there are 2 dots instead of 1; **D** — the arrow has not rotated on from the fifth figure; there are 3 dots instead of 1; **E** — the arrow has not rotated on from the fifth figure.

**Question 32 — Answer: D**

**Rules:** the triangle moves 1 position clockwise around the frame each time; the triangle's shading alternates between black and grey.

**Why the other options are wrong:** **A** — the triangle is one position too far clockwise; **B** — the triangle has not moved on from the fifth figure; the triangle is black instead of grey; **C** — the triangle has not moved on from the fifth figure; **E** — the triangle is one position too far clockwise; the triangle is white instead of grey.

**Question 33 — Answer: C**

**Rules:** the central shape cycles hexagon → pentagon → square → triangle and then repeats; the circle moves 1 position anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the circle has not moved on from the fifth figure; **B** — the circle is one position too far anticlockwise; **D** — the central shape is a triangle instead of a square; **E** — the central shape is a triangle instead of a square; the circle has not moved on from the fifth figure.

**Question 34 — Answer: B**

**Rules:** the shaded quarter of the square moves one quarter anticlockwise each time; the diamond moves 1 position anticlockwise around the frame each time; the diamond's shading alternates between white and black.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; the diamond is one position too far anticlockwise; the diamond is white instead of black; **C** — the wrong quarter of the square is shaded; the diamond has not moved on from the fifth figure; the diamond is grey instead of black; **D** — the wrong quarter of the square is shaded; the diamond is one position too far anticlockwise; the diamond is grey instead of black; **E** — the wrong quarter of the square is shaded; the diamond is one position too far anticlockwise.

**Question 35 — Answer: E**

**Rules:** the shaded quarter of the square moves one quarter clockwise each time; the triangle moves 2 positions clockwise around the frame each time; the diamond moves 1 position anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; the diamond is one position too far anticlockwise; **B** — the wrong quarter of the square is shaded; the diamond has not moved on from the fifth figure; **C** — the triangle is one position too far clockwise; the diamond has not moved on from the fifth figure; **D** — the wrong quarter of the square is shaded; the triangle is one position too far clockwise; the diamond is one position too far anticlockwise.

**Question 36 — Answer: C**

**Rules:** the central shape cycles triangle → square → pentagon and then repeats; the central shape's shading alternates between black and white; the circle moves 3 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the central shape is a pentagon instead of a triangle; the central shape is grey instead of white; **B** — the central shape is grey instead of white; the circle is one position too far clockwise; **D** — the central shape is a pentagon instead of a triangle; the central shape is grey instead of white; the circle is one position too far clockwise; **E** — the central shape is a pentagon instead of a triangle; the circle has not moved on from the fifth figure.

**Question 37 — Answer: D**

**Rules:** the shaded quarter of the square moves one quarter anticlockwise each time; the circle moves clockwise around the frame, alternately 1 and 2 positions at a time; the circle's shading alternates between white and grey.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; the circle has not moved on from the fifth figure; the circle is black instead of grey; **B** — the wrong quarter of the square is shaded; the circle has not moved on from the fifth figure; **C** — the wrong quarter of the square is shaded; the circle is white instead of grey; **E** — the wrong quarter of the square is shaded; the circle is one position too far clockwise; the circle is black instead of grey.

**Question 38 — Answer: B**

**Rules:** the arrow rotates 90° clockwise each time; the diamond moves 1 position clockwise around the frame each time; the diamond's shading alternates between white and black; the circle moves 3 positions clockwise around the frame each time.

**Why the other options are wrong:** **A** — the circle has not moved on from the fifth figure; **C** — the diamond is one position too far clockwise; **D** — the arrow is rotated 45° too far clockwise; the diamond has not moved on from the fifth figure; the diamond is grey instead of black; the circle is one position too far clockwise; **E** — the arrow is rotated 45° too far clockwise; the diamond is one position too far clockwise; the circle has not moved on from the fifth figure.

**Question 39 — Answer: E**

**Rules:** the central shape cycles triangle → square → pentagon and then repeats; the central shape's shading cycles black → white → grey; the circle moves 3 positions clockwise around the frame each time; the circle's shading cycles white → grey → black.

**Why the other options are wrong:** **A** — the circle is white instead of black; **B** — the central shape is a hexagon instead of a pentagon; the central shape is white instead of grey; the circle is one position too far clockwise; the circle is white instead of black; **C** — the central shape is a square instead of a pentagon; **D** — the central shape is a hexagon instead of a pentagon; the circle is one position too far clockwise; the circle is grey instead of black.

**Question 40 — Answer: A**

**Rules:** the arrow rotates 135° clockwise each time; the diamond moves 3 positions clockwise around the frame each time; the diamond's shading cycles black → white → grey; the circle moves 3 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **B** — the diamond has not moved on from the fifth figure; the circle is one position too far clockwise; **C** — the arrow is rotated 45° too far clockwise; **D** — the arrow has not rotated on from the fifth figure; the diamond is one position too far clockwise; the diamond is black instead of grey; the circle is one position too far clockwise; **E** — the arrow is rotated 45° too far clockwise; the diamond is white instead of grey; the circle has not moved on from the fifth figure.

# Solutions — Mock Exam 2

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## Answer key

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 1 A  | 2 B  | 3 C  | 4 A  | 5 A  | 6 D  | 7 D  | 8 B  |
| 9 C  | 10 A | 11 B | 12 D | 13 C | 14 D | 15 D | 16 A |
| 17 D | 18 D | 19 B | 20 D | 21 A | 22 B | 23 D | 24 B |
| 25 D | 26 A | 27 C | 28 D | 29 C | 30 B | 31 C | 32 E |
| 33 A | 34 B | 35 D | 36 A | 37 D | 38 B | 39 C | 40 E |

## Worked solutions

### Question 1 — Answer: A

- A. Correct. When the two managers disagree about priorities, employees “may receive conflicting instructions”.
- B. Incorrect – contradiction. Specialists can work on projects “without leaving their home department”.
- C. Incorrect – contradiction. Decisions “can slow down” as disagreements are escalated.
- D. Incorrect – contradiction. Matrix structures are most common in organisations running many complex projects at the same time.

### Question 2 — Answer: B

- A. Incorrect – contradiction. The network currently serves “around a fifth” of households.
- B. Correct. Conventional stations release much of their fuel’s energy as waste heat, which a combined heat and power plant captures and uses.
- C. Incorrect – contradiction. The council has only commissioned a feasibility study and has not yet taken a decision.
- D. Incorrect – contradiction. Opponents note that a network is most economical in densely built areas.

### Question 3 — Answer: C

- A. Incorrect – contradiction. This is the older view the passage says research has overturned; fluent bilinguals are often the most skilful switchers.
- B. Incorrect – contradiction. The passage says that in some communities code-switching remains stigmatised, particularly in formal settings such as schools and administration.
- C. Correct. The passage states that speakers rarely switch where the word orders of the two languages would clash.
- D. Incorrect – partial truth. Expressing emotion is only one of several functions listed, nothing makes it the chief reason, and the text speaks of one language seeming “better suited”, not of a lack of vocabulary.

### Question 4 — Answer: A

- A. Correct. The period runs from delivery of the goods rather than from the date of the order.
- B. Incorrect – extreme wording. The consumer may be asked to pay the direct cost of returning the item.
- C. Incorrect – scope shift. The right concerns distance purchases; its rationale is that such buyers, unlike shoppers in physical stores, cannot inspect goods beforehand.
- D. Incorrect – contradiction. Consumers may withdraw “without giving a reason”.

**Question 5 — Answer: A**

- A.** Correct. Grants cover up to half of eligible costs and producers meet the remainder, so their share is at least half.
- B.** Incorrect – unsupported inference. Officials say shorter chains do not automatically cut emissions, which is why emissions will be monitored; no expected reduction is stated.
- C.** Incorrect – extreme wording. Farms must sell at least a third of the relevant products directly or through no more than one intermediary.
- D.** Incorrect – contradiction. Extension across the region will be decided at the end of the pilot; eligibility is currently limited to producers within 80 kilometres of the capital.

**Question 6 — Answer: D**

- A.** Incorrect – contradiction. A plant “returns less electricity than it consumes”.
- B.** Incorrect – outside knowledge. The passage compares the two technologies but says nothing about batteries displacing pumped storage.
- C.** Incorrect – partial truth. Water is not enough: two reservoirs at substantially different heights and suitable geology are also required.
- D.** Correct. Despite cycle losses, the plant can be economically attractive because it buys power when prices are low and sells it when they are high.

**Question 7 — Answer: D**

- A.** Incorrect – cause and effect. The organisation warned that the association “did not prove” that the labels changed behaviour.
- B.** Incorrect – extreme wording. A larger pack is “not always” cheaper per unit, especially when a smaller pack is on promotion.
- C.** Incorrect – scope shift. The passage refers to “many” consumer-protection regimes, not to every country.
- D.** Correct. Unit prices were shown, but sometimes per 100 grams for one brand and per kilogram for another, which hindered comparison.

**Question 8 — Answer: B**

- A.** Incorrect – contradiction. A light-year is “a unit of distance, not of time”.
- B.** Correct. Observing more distant objects means “looking further into the past”, so young galaxies can be compared with older, nearer ones.
- C.** Incorrect – extreme wording. The passage says only that an observed object “may” have changed or ceased to exist; it makes no claim about most stars.
- D.** Incorrect – contradiction. The principle “applies equally to nearby objects”, such as the Sun, whose light takes about eight minutes to arrive.

**Question 9 — Answer: C**

- A.** Incorrect – contradiction. Calendar ageing “occurs even when a battery is not in use”.
- B.** Incorrect – contradiction. Cold reduces performance only “temporarily”, and much of it returns once the battery warms up.
- C.** Correct. Calendar ageing is accelerated both by high temperatures and by storage at a high state of charge.
- D.** Incorrect – scope shift. “Many” manufacturers “allow” drivers to set a limit; this is an option offered, not a requirement.

**Question 10 — Answer: A**

- A.** Correct. Missions are described as a variant of priority-setting that goes further by setting a concrete, measurable objective.
- B.** Incorrect – unsupported comparison. The passage reports arguments on both sides but no evidence on which model yields more useful applications.
- C.** Incorrect – contradiction. Missions mobilise participants “from several disciplines”.
- D.** Incorrect – contradiction. Critics fear that missions could crowd out fundamental research, the opposite of wanting them to replace it.

**Question 11 — Answer: B**

- A.** Incorrect – cause and effect. Railway companies adopted standard time first; legal recognition followed some years later.
- B.** Correct. Some public clocks had two minute hands, showing both local time and railway time.
- C.** Incorrect – cause and effect. The differences arose because local noon depends on the sun’s position, which changes from east to west.
- D.** Incorrect – extreme wording. Countries adopted the system at different speeds.

**Question 12 — Answer: D**

- A.** Incorrect – supported. A stock near carrying capacity grows slowly because individuals compete for food and space.
- B.** Incorrect – supported. Uncertain estimates of stock size are listed among the practical difficulties.
- C.** Incorrect – supported. Many scientists advise catches somewhat below the theoretical maximum, as a buffer against error.
- D.** Correct – not supported. Single-species models “overlook interactions within the ecosystem”, such as the target species being food for predators.

**Question 13 — Answer: C**

- A.** Incorrect – contradiction. The passage states that it is the bacteria, not the patients, that become resistant.
- B.** Incorrect – cause and effect. The passage describes variations already carried by some cells, which antibiotics then select for; it does not say the drugs create them.
- C.** Correct. Bacteria can acquire resistance genes from other bacteria, including those of different species.
- D.** Incorrect – extreme wording. Only “some researchers” argue that shorter courses may suit “certain cases”.

**Question 14 — Answer: D**

- A.** Incorrect – contradiction. Air-source units extract heat from outdoor air “even when the air is cold”.
- B.** Incorrect – unsupported inference. Ground-source models are more efficient on average but cost more to install; the passage draws no conclusion about overall cost-effectiveness.
- C.** Incorrect – contradiction. The coefficient falls as the temperature gap widens, so it tends to be lowest on the coldest days.
- D.** Correct. Lower water temperatures narrow the gap between heat source and heating system, and the coefficient declines as that gap widens; hence the engineers’ recommendation.

**Question 15 — Answer: D**

- A. Incorrect – supported. The participating departments “maintained their output”.
- B. Incorrect – supported. Sickness absence also fell in the departments that kept the five-day schedule, so other factors may have contributed.
- C. Incorrect – supported. The steering group noted that volunteering departments may have been those best suited to a shorter week.
- D. Correct – not supported. A second trial is only a steering-group recommendation and the board “has not yet decided” whether to extend the arrangement.

**Question 16 — Answer: A**

- A. Correct. Early arrivals “frequently paid for the passage of relatives or neighbours”.
- B. Incorrect – unsupported inference. The passage never links the origins of chain migration to steamships or says which came first.
- C. Incorrect – contradiction. Emigrants from one district often clustered in a few destinations rather than dispersing.
- D. Incorrect – unsupported inference. Only “some historians” hold this view and others dispute it, so there is no agreement.

**Question 17 — Answer: D**

- A. Incorrect – cause and effect. Early selection is only “associated with” greater inequality, and researchers warn that its effect is hard to isolate.
- B. Incorrect – contradiction. Some studies find modest gains for the highest-attaining pupils.
- C. Incorrect – extreme wording. Systems that postponed selection “have not always seen the expected reduction in inequality”.
- D. Correct. Gains for high attainers appear to be offset by losses among lower attainers, “so that average performance is not raised”.

**Question 18 — Answer: D**

- A. Incorrect – contradiction. Retroactive rules in taxation are “not automatically invalid” and may be accepted exceptionally.
- B. Incorrect – partial truth. The “objective requires it” exception concerns fields outside criminal law; in criminal law no heavier penalty than the one applicable at the time may be imposed.
- C. Incorrect – contradiction. The passage warns that a law that could never be amended would prevent necessary reform.
- D. Correct. Many systems allow “a more lenient penalty introduced later to benefit the offender”.

**Question 19 — Answer: B**

- A. Incorrect – extreme wording. Reversibility applies “as far as possible” and is treated as an aspiration, not an absolute requirement.
- B. Correct. New work must be distinguishable “on close inspection” but may be designed to blend in when seen from a normal distance.
- C. Incorrect – extreme wording. The ideas emerged “partly” in reaction to nineteenth-century restorations, and nothing is said about undoing that work.
- D. Incorrect – unsupported inference. Critics cite cleaning to show that perfect reversibility is rarely achievable; the passage never says practitioners avoid cleaning.



**Question 20 — Answer: D**

**A.** Incorrect – contradiction. An operator was on board throughout, and the report concludes that the technology is “not yet” ready for mixed traffic.

**B.** Incorrect – partial truth. The partnership recommends that a second phase be considered, but the authorities “have not committed” to it; a proposal is presented as a decision.

**C.** Incorrect – scope shift. Respondents were mostly regular commuters who had volunteered, so the results cannot be generalised to the public.

**D.** Correct. There were no injury-causing collisions, but a safety operator was present throughout and intervened “on average once every few kilometres”.

**Question 21 — Answer: A**

- › The North region’s total allocation is €1,430 million, of which Education received €270 million.
- ›  $270 \div 1,430 \times 100 \approx 18.9\%$ .
- › Typical errors: dividing by the Education column total (the region’s share of that sector) or by the remainder instead of the total.

**Correct option: A** (18.9%)

**Question 22 — Answer: B**

- › A change in percentage points is the simple difference between the two rates:  $7.2 - 9.5 = -2.3$  points, i.e. a fall of 2.3 percentage points.
- › Typical error: computing the relative (percentage) change instead.

**Correct option: B** (2.3 percentage points)

**Question 23 — Answer: D**

- › Dutch : French = 18 : 24. Both numbers divide by 6, giving 3:4.
- › Typical error: reversing the order of the ratio (4:3).

**Correct option: D** (3:4)

**Question 24 — Answer: B**

- › Spending per inhabitant = spending  $\div$  population (€ million  $\div$  million inhabitants = € per inhabitant):
- › Western region  $\approx$  €103; Northern region  $\approx$  €98; Eastern region  $\approx$  €163; Central region  $\approx$  €182.
- › The lowest figure is for the Northern region. Trap: the Western region has the smallest total spending, but not the lowest spending per inhabitant.

**Correct option: B** (Northern region)

**Question 25 — Answer: D**

- › Percentage change =  $(\text{Year 2} - \text{Year 1}) \div \text{Year 1} \times 100$ :
- › Hillcrest:  $(520 - 390) \div 390 \approx 33.3\%$ ; Brookhaven:  $(800 - 610) \div 610 \approx 31.1\%$ ; Kingsford:  $(960 - 730) \div 730 \approx 31.5\%$ ; Northport:  $(500 - 360) \div 360 \approx 38.9\%$ .
- › Largest: Northport. Trap: Kingsford has the largest increase in absolute terms (+230 thousand), but from a larger base.

**Correct option: D** (Northport)

**Question 26 — Answer: A**

- › Participants Q1 to Q2 inclusive:  $13 + 8 = 21$ .
- › Total cost =  $21 \times €150 = €3,150$ .
- › Typical errors: including the wrong quarters, using the whole year, or reading the wrong row of either table.

**Correct option: A** (€3,150)

**Question 27 — Answer: C**

- › Convert every price to euros by DIVIDING the local price by the number of units per €1:
- › NOK:  $6,957 \div 11.39 \approx €610.80$ ; ISK:  $75,400 \div 145.5 \approx €518.21$ ; CAD:  $1,000 \div 1.42 \approx €704.23$ ; CHF:  $558 \div 0.94 \approx €593.62$ .
- › Most expensive (CAD) – cheapest (ISK)  $\approx €704.23 - €518.21 = €186.01$ , i.e. about €186.
- › Typical errors: ranking the markets by local price (the currencies are not comparable) or comparing the wrong pair of markets.

**Correct option: C** (€186)

**Question 28 — Answer: D**

- › Travel now =  $26\% \times €290 \text{ million} = €75.4 \text{ million}$ ; after +15%: €86.71 million.
- › Staff =  $11\% \times €290 \text{ million} = €31.9 \text{ million}$ .
- › Difference = €54.81 million  $\approx €54.8 \text{ million}$ . Typical error: adding 15 percentage points to the share instead of raising the amount by 15%.

**Correct option: D** (€54.8 million)

**Question 29 — Answer: C**

- › This year = last year  $\times 1.25$ , so last year = this year  $\div 1.25$ .
- ›  $5,300.0 \div 1.25 = 4,240.0$  (€ thousand).
- › Typical error: reducing this year's figure by 25% applies the percentage to the wrong base.

**Correct option: C** (€4,240.0k)

**Question 30 — Answer: B**

- › Compound growth:  $4,490 \times 1.2^3 = 4,490 \times 1.7280 \approx 7,759 \text{ tonnes}$ .
- › Typical error: simple growth,  $4,490 \times (1 + 3 \times 0.2) = 7,184 \text{ tonnes}$ , ignores compounding; others apply the rate for the wrong number of years.

**Correct option: B** (7,759 t)

**Question 31 — Answer: C**

**Rules:** the shaded quarter of the square moves one quarter anticlockwise each time; the circle moves 1 position anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; **B** — the circle is one position too far anticlockwise; **D** — the wrong quarter of the square is shaded; the circle is one position too far anticlockwise; **E** — the wrong quarter of the square is shaded.

**Question 32 — Answer: E**

**Rules:** the number of dots decreases by 1 each time; the diamond moves 2 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **A** — there are 3 dots instead of 1; **B** — the diamond has not moved on from the fifth figure; **C** — there are 2 dots instead of 1; the diamond is one position too far clockwise; **D** — there are 2 dots instead of 1; the diamond has not moved on from the fifth figure.

**Question 33 — Answer: A**

**Rules:** the central shape cycles hexagon  $\rightarrow$  pentagon  $\rightarrow$  square and then repeats; the central shape's shading alternates between grey and white.

**Why the other options are wrong:** **B** — the central shape is grey instead of white; **C** — the central shape is a pentagon instead of a square; the central shape is grey instead of white; **D** — the central shape is a triangle instead of a square; the central shape is black instead of white; **E** — the central shape is a pentagon instead of a square.

**Question 34 — Answer: B**

**Rules:** the arrow rotates  $45^\circ$  clockwise each time; the triangle moves 3 positions clockwise around the frame each time; the triangle's shading cycles grey  $\rightarrow$  black  $\rightarrow$  white.

**Why the other options are wrong:** **A** — the arrow is rotated  $45^\circ$  too far clockwise; the triangle has not moved on from the fifth figure; the triangle is black instead of white; **C** — the arrow has not rotated on from the fifth figure; the triangle is one position too far clockwise; **D** — the arrow is rotated  $45^\circ$  too far clockwise; the triangle is one position too far clockwise; the triangle is black instead of white; **E** — the arrow is rotated  $45^\circ$  too far clockwise; the triangle is black instead of white.

**Question 35 — Answer: D**

**Rules:** the shaded quarter of the square moves one quarter anticlockwise each time; the diamond moves clockwise around the frame, alternately 1 and 2 positions at a time; the diamond's shading cycles grey  $\rightarrow$  black  $\rightarrow$  white.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; the diamond is one position too far clockwise; the diamond is black instead of white; **B** — the wrong quarter of the square is shaded; the diamond has not moved on from the fifth figure; **C** — the diamond is one position too far clockwise; the diamond is grey instead of white; **E** — the wrong quarter of the square is shaded; the diamond is grey instead of white.

**Question 36 — Answer: A**

**Rules:** the number of dots increases by 1 each time; the circle moves clockwise around the frame, alternately 1 and 3 positions at a time; the circle's shading cycles grey  $\rightarrow$  black  $\rightarrow$  white.

**Why the other options are wrong:** **B** — there are 5 dots instead of 6; the circle is grey instead of white; **C** — there are 7 dots instead of 6; the circle is black instead of white; **D** — there are 7 dots instead of 6; the circle has not moved on from the fifth figure; the circle is black instead of white; **E** — there are 5 dots instead of 6; the circle is black instead of white.

**Question 37 — Answer: D**

**Rules:** the shaded quarter of the square moves one quarter anticlockwise each time; the triangle moves anticlockwise around the frame, alternately 1 and 3 positions at a time; the triangle's shading alternates between black and grey.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; the triangle has not moved on from the fifth figure; **B** — the wrong quarter of the square is shaded; the triangle is one position too far anticlockwise; **C** — the wrong quarter of the square is shaded; the triangle has not moved on from the fifth figure; the triangle is white instead of grey; **E** — the wrong quarter of the square is shaded; the triangle is one position too far anticlockwise; the triangle is black instead of grey.

**Question 38 — Answer: B**

**Rules:** the central shape cycles hexagon  $\rightarrow$  pentagon  $\rightarrow$  square and then repeats; the central shape's shading cycles grey  $\rightarrow$  black  $\rightarrow$  white; the diamond moves anticlockwise around the frame, alternately 3 and 1 positions at a time.

**Why the other options are wrong:** **A** — the central shape is black instead of white; the diamond is one position too far clockwise; **C** — the central shape is a square instead of a hexagon; the diamond has not moved on from the fifth figure; **D** — the central shape is grey instead of white; the diamond has not moved on from the fifth figure; **E** — the central shape is a pentagon instead of a hexagon; the central shape is grey instead of white.

**Question 39 — Answer: C**

**Rules:** the arrow rotates 45° anticlockwise each time; the square moves anticlockwise by 1, then 2, then 3, then 4 positions (one more each time); the circle moves 3 positions clockwise around the frame each time.

**Why the other options are wrong:** **A** — the square is one position too far clockwise; the circle is one position too far clockwise; **B** — the arrow is rotated 45° too far anticlockwise; the square is one position too far clockwise; **D** — the circle is one position too far clockwise; **E** — the square has not moved on from the fifth figure; the circle is one position too far clockwise.

**Question 40 — Answer: E**

**Rules:** the arrow rotates clockwise, alternately by 90° and 135°; the square moves anticlockwise around the frame, alternately 3 and 1 positions at a time; the square's shading cycles white → grey → black.

**Why the other options are wrong:** **A** — the arrow has not rotated on from the fifth figure; the square has not moved on from the fifth figure; the square is grey instead of black; **B** — the square has not moved on from the fifth figure; **C** — the arrow has not rotated on from the fifth figure; the square is one position too far clockwise; **D** — the arrow is rotated 45° too far clockwise; the square is white instead of black.

# Solutions — Mock Exam 3

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## Answer key

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 1 C  | 2 A  | 3 A  | 4 B  | 5 B  | 6 C  | 7 D  | 8 B  |
| 9 B  | 10 A | 11 C | 12 C | 13 D | 14 A | 15 A | 16 D |
| 17 C | 18 D | 19 C | 20 A | 21 A | 22 B | 23 C | 24 D |
| 25 A | 26 B | 27 D | 28 C | 29 C | 30 A | 31 D | 32 B |
| 33 C | 34 E | 35 A | 36 E | 37 D | 38 A | 39 C | 40 B |

## Worked solutions

### Question 1 — Answer: C

- A. Incorrect – contradiction. Much canal building was “financed by private investors”.
- B. Incorrect – extreme wording. Some canals “continued to carry bulk cargoes for decades”.
- C. Correct. A horse towing a barge could move “many times as much” as it could pull in a cart, and this advantage drove canal building.
- D. Incorrect – contradiction. Locks added to journey times.

### Question 2 — Answer: A

- A. Correct. Because of airport access, security checks and early arrival, the train “can deliver passengers sooner despite its lower speed” on shorter routes.
- B. Incorrect – extreme wording. Rail “tends to win a large share” on such routes; “always” and “majority” go beyond the passage.
- C. Incorrect – extreme wording. The passage names other factors as well: ticket prices, service frequency and the ability to work during the journey.
- D. Incorrect – unsupported comparison. The passage says high-speed lines are very expensive but never compares their cost with that of airports.

### Question 3 — Answer: A

- A. Correct. Reverse osmosis “generally requires less energy than distillation” but considerably more than treating water from rivers or aquifers.
- B. Incorrect – extreme wording. The membrane holds back “most of the salt”, not all of it.
- C. Incorrect – contradiction. Brine is denser than seawater and “tends to sink and spread along the sea bed”.
- D. Incorrect – outside knowledge. The passage makes no claim about cost, and its point about high energy use suggests the opposite.

**Question 4 — Answer: B**

- A.** Incorrect – contradiction. Such food should not be eaten after the date “even if it looks and smells acceptable”.
- B.** Correct. Both kinds of date assume correct storage, and a product kept in a warm place may deteriorate before its date.
- C.** Incorrect – contradiction. “Best before” products generally remain safe after the date if stored correctly and undamaged.
- D.** Incorrect – extreme wording. Confusion is said to “contribute to” avoidable waste, not to be its main cause.

**Question 5 — Answer: B**

- A.** Incorrect – contradiction. Atmospheric nitrogen gas is “unusable by most organisms” until it has been fixed.
- B.** Correct. Run-off stimulates excessive algal growth; when the algae die and decompose, bacteria “consume the dissolved oxygen”.
- C.** Incorrect – contradiction. Nitrogen is fixed “chiefly” by bacteria; lightning contributes only “to a lesser extent”.
- D.** Incorrect – contradiction. Lightning helps to fix nitrogen; it is “other bacteria” that return nitrogen to the atmosphere, through denitrification.

**Question 6 — Answer: C**

- A.** Incorrect – contradiction. A display is commonly an invitation, so the seller “may accept or decline” the customer’s offer.
- B.** Incorrect – contradiction. A reply introducing different terms is a counter-offer that rejects the original offer; no contract is formed until it is itself accepted.
- C.** Correct. The passage states that in such cases the customer makes the offer by proposing to buy.
- D.** Incorrect – extreme wording. The passage refers to “many legal systems” and says such advertisements are “commonly” treated as invitations.

**Question 7 — Answer: D**

- A.** Incorrect – contradiction. Framing cannot be avoided because “every message must be phrased somehow”.
- B.** Incorrect – contradiction. Losses “weigh more heavily than gains of the same size”, not the reverse.
- C.** Incorrect – unsupported comparison. The passage never compares doctors with patients.
- D.** Correct. The survival-rate and mortality-rate descriptions carry identical information, yet one is chosen more readily in experiments.

**Question 8 — Answer: B**

- A.** Incorrect – contradiction. Few could rely on concert income alone, and many combined it with teaching, publishing or patronage.
- B.** Correct. The passage states that a paying public “also influenced the music itself”, favouring works that made a strong first impression.
- C.** Incorrect – contradiction. Public concerts became important “first in large commercial cities”.
- D.** Incorrect – cause and effect. The passage links larger orchestras to the use of large halls, not to composers’ demands.

**Question 9 — Answer: B**

- A.** Incorrect – contradiction. Once a garden city reached its intended size, growth was to occur through a new settlement nearby, not through expansion.
- B.** Correct. Land was to be held collectively so that rising land values would benefit the community rather than private landlords.
- C.** Incorrect – contradiction. Critics note that later garden suburbs abandoned the self-contained character of Howard’s scheme.
- D.** Incorrect – extreme wording. Howard was also troubled by the depopulation and limited opportunities of the countryside.

**Question 10 — Answer: A**

- A.** Correct – not supported. Outreach was only recommended, and the council has not decided whether to adopt it, so it cannot have broadened participation.
- B.** Incorrect – supported. Officials check proposals for legal and technical feasibility and cost before they go to a public vote.
- C.** Incorrect – supported. Any resident aged sixteen or over may submit a proposal, which includes sixteen- and seventeen-year-olds.
- D.** Incorrect – supported. The council keeps a formal right to reject a winning project but has never used it.

**Question 11 — Answer: C**

- A.** Incorrect – unsupported inference. The work ranged “from philosophy and theology to the crafts and trades”; the passage never says crafts were its main subject.
- B.** Incorrect – partial truth. D’Alembert co-edited the work only in its early years.
- C.** Correct. Contributors used cross-references to lead attentive readers from orthodox entries to subversive ones.
- D.** Incorrect – contradiction. Despite the withdrawal of permission, the remaining volumes were issued and the work was completed.

**Question 12 — Answer: C**

- A.** Incorrect – contradiction. The decisive feature was “not the shapes of the digits but the positional principle”.
- B.** Incorrect – extreme wording. Calculations were done on counting boards or the abacus; written arithmetic was cumbersome, not impossible.
- C.** Correct. With Roman numerals, calculations were usually made on a counting board or abacus and only the outcome was recorded.
- D.** Incorrect – outside knowledge. The passage mentions only a concern “sometimes given” for keeping the older notation, not any Europe-wide ban.

**Question 13 — Answer: D**

- A.** Incorrect – supported. The department expects electric vehicles to reduce noise “mainly at low speeds”.
- B.** Incorrect – supported. Porous asphalt must be cleaned or replaced more often than conventional surfaces.
- C.** Incorrect – supported. The effect of lower speed limits “depends on enforcement”.
- D.** Correct – not supported. Barriers help only buildings directly behind them and are “unsuitable for most city streets”; the superlative has no basis in the passage.

**Question 14 — Answer: A**

- A.** Correct – not supported. The passage states that “even before” expiry others can learn from the published description and build on it.
- B.** Incorrect – supported. The patent holder “can charge prices above those that would prevail under competition”.
- C.** Incorrect – supported. Patents are regarded as essential where products are easy to imitate but less so elsewhere, implying that ease of imitation is one relevant factor.
- D.** Incorrect – supported. Some researchers argue that overlapping patents in complex technologies can impede further innovation.

**Question 15 — Answer: A**

- A.** Correct. For Locke, persistent violation of rights entitles the people to resist and replace a government, so obedience is conditional on its conduct.
- B.** Incorrect – cause and effect. The passage notes only that Hobbes wrote during the civil war; it does not say that the war caused his view.
- C.** Incorrect – extreme wording. Locke justified resistance when a government “persistently” violated rights, not after any single violation.
- D.** Incorrect – contradiction. Hobbes argued that even a harsh ruler was preferable to a return to anarchy.

**Question 16 — Answer: D**

- A.** Incorrect – supported. The effort to prove the postulate from the others “was bound to fail”.
- B.** Incorrect – supported. The discovery was first seen as a curiosity, and its significance grew only when physicists later applied it.
- C.** Incorrect – supported. In such geometries the angles of a triangle “no longer add up to exactly two right angles”.
- D.** Correct – not supported. Euclidean geometry “remains an excellent approximation for everyday purposes”, so it is not shown to be unreliable in daily use.

**Question 17 — Answer: C**

- A.** Incorrect – contradiction. Gains were concentrated in irrigated areas, and rain-fed regions benefited far less.
- B.** Incorrect – extreme wording. Wealthier farmers were often able to adopt the package more quickly than smallholders; slower adoption is not inability.
- C.** Correct. Supporters argue that, without higher yields, far more forest and grassland would have been converted to farmland.
- D.** Incorrect – cause and effect. Intensive irrigation and agrochemical use have been “linked, in some districts” to these problems; the passage asserts neither causation nor universal effects.

**Question 18 — Answer: D**

- A.** Incorrect – contradiction. In the design described, all successful generators receive the price offered by the last, most expensive plant needed.
- B.** Incorrect – extreme wording. Renewables are usually at the front of the queue; the price is set by the last plant needed, often a gas-fired one.
- C.** Incorrect – outside knowledge. The passage concerns wholesale prices and says nothing about household bills; “guarantees” also overstates a tendency.
- D.** Correct. Every successful generator receives the price of the last plant needed, so in an hour when that plant burns gas, a wind farm is paid the gas-set price.



**Question 19 — Answer: C**

- A.** Incorrect – supported. Threads varied from maker to maker, so a bolt from one factory would rarely fit a nut from another.
- B.** Incorrect – supported. Engineers argued that agreed thread forms “would reduce costs and make parts interchangeable”.
- C.** Correct – not supported. Standard forms were taken up “first by individual firms and engineering societies” and only gradually by governments, so the option reverses the order.
- D.** Incorrect – supported. Different nations adopted different systems, so machinery with parts from several countries could still need several sets of tools.

**Question 20 — Answer: A**

- A.** Correct. Such a measure would fail the necessity stage, and failing any one stage makes a measure disproportionate.
- B.** Incorrect – unsupported inference. Suitability is only one stage; the measure must also be necessary, so advancing its aim does not guarantee that it will be upheld.
- C.** Incorrect – extreme wording. Only “some courts” add the balancing stage of proportionality in the strict sense.
- D.** Incorrect – contradiction. The intensity of review may vary with the importance of the right affected.

**Question 21 — Answer: A**

- > Unit A has 32 women and 16 men (48 staff).
- > New total =  $48 \times 1.25 = 60$ ; women =  $60 \times 32/48 = 40$ .
- > Shortcut: if the proportion is unchanged, the number of women also grows by 25%:  $32 \times 1.25 = 40$ .

**Correct option: A** (40)**Question 22 — Answer: B**

- > Percentage change uses the earlier value as the base:  $(285 - 326) \div 326 \times 100$ .
- >  $\approx -41 \div 326 \times 100 \approx -12.6\%$ , i.e. a decrease of about 12.6%.
- > Typical errors: dividing by the Year 3 value (wrong base), or adding the two year-on-year percentage changes (percentages do not add).

**Correct option: B** (12.6%)**Question 23 — Answer: C**

- > Approved =  $500 \times 18\% = 90$ .
- > Rejected =  $500 - 90 = 410$ .

**Correct option: C** (410)**Question 24 — Answer: D**

- > Total salary bill =  $20 \times 4,500 + 48 \times 4,200 + 15 \times 8,900 + 20 \times 9,000 = \text{€}605,100$ .
- > Total staff = 103. Weighted average =  $\text{€}605,100 \div 103 \approx \text{€}5,875$ .
- > Typical error: the simple average of the four unit averages (€6,650) ignores the different unit sizes.

**Correct option: D** (€5,875)**Question 25 — Answer: A**

- > With index numbers, the percentage change between two years is  $(\text{later index} - \text{earlier index}) \div \text{earlier index} \times 100$ .
- >  $(122.1 - 111.8) \div 111.8 \times 100 \approx 9.2\%$ .
- > Typical errors: subtracting the indices (10.3 is a change in index points, not per cent) or measuring from the base year (index – 100).

**Correct option: A** (9.2%)

**Question 26 — Answer: B**

- › Time = distance ÷ speed. Now:  $193 \div 120 \times 60 = 96.5$  minutes. New:  $193 \div 150 \times 60 = 77.2$  minutes.
- › Saving =  $96.5 - 77.2 = 19.3$ , i.e. about 19 minutes.
- › Typical error: assuming journey time falls by the same percentage as speed rises.

**Correct option: B** (19 minutes)

**Question 27 — Answer: D**

- › For an average of 50 over 6 months, the total must be  $50 \times 6 = 300$ .
- › January–May total =  $24 + 41 + 55 + 54 + 40 = 214$ .
- › June must therefore be  $300 - 214 = 86$ .
- › Typical errors: answering with the target itself (50) or with the current average (about 43).

**Correct option: D** (86)

**Question 28 — Answer: C**

- › Total salary bill =  $33 \times 5,300 + 19 \times 7,600 + 7 \times 9,200 + 16 \times 7,300 = €500,500$ .
- › Total staff = 75. Weighted average =  $€500,500 \div 75 \approx €6,673$ .
- › Typical error: the simple average of the four unit averages (€7,350) ignores the different unit sizes.

**Correct option: C** (€6,673)

**Question 29 — Answer: C**

- › Spending per inhabitant = spending ÷ population (€ million ÷ million inhabitants = € per inhabitant):
- › Coastal region  $\approx €122$ ; Central region  $\approx €157$ ; Western region  $\approx €193$ ; Eastern region  $\approx €95$ .
- › The highest figure is for the Western region. Trap: the Coastal region has the largest total spending, but not the highest spending per inhabitant.

**Correct option: C** (Western region)

**Question 30 — Answer: A**

- › Studies now =  $27\% \times €725 \text{ million} = €195.75 \text{ million}$ ; after +15%: €225.11 million.
- › Buildings =  $19\% \times €725 \text{ million} = €137.75 \text{ million}$ .
- › Difference =  $€87.36 \text{ million} \approx €87.4 \text{ million}$ . Typical error: adding 15 percentage points to the share instead of raising the amount by 15%.

**Correct option: A** (€87.4 million)

**Question 31 — Answer: D**

**Rules:** the shaded quarter of the square moves one quarter anticlockwise each time; the circle moves 2 positions clockwise around the frame each time.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; **B** — the circle has not moved on from the fifth figure; **C** — the wrong quarter of the square is shaded; **E** — the wrong quarter of the square is shaded; the circle is one position too far clockwise.

**Question 32 — Answer: B**

**Rules:** the number of dots increases by 1 each time; the square moves 2 positions clockwise around the frame each time.

**Why the other options are wrong:** **A** — there are 5 dots instead of 6; **C** — there are 5 dots instead of 6; the square is one position too far clockwise; **D** — there are 5 dots instead of 6; the square has not moved on from the fifth figure; **E** — the square is one position too far clockwise.

**Question 33 — Answer: C**

**Rules:** the shaded quarter of the square moves one quarter clockwise each time; the triangle moves 1 position clockwise around the frame each time; the triangle's shading alternates between black and white.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; the triangle is one position too far clockwise; the triangle is grey instead of white; **B** — the wrong quarter of the square is shaded; the triangle has not moved on from the fifth figure; the triangle is grey instead of white; **D** — the wrong quarter of the square is shaded; the triangle is black instead of white; **E** — the wrong quarter of the square is shaded; the triangle is one position too far clockwise.

**Question 34 — Answer: E**

**Rules:** the shaded quarter of the square moves one quarter anticlockwise each time; the circle moves 2 positions anticlockwise around the frame each time; the circle's shading alternates between white and grey.

**Why the other options are wrong:** **A** — the wrong quarter of the square is shaded; the circle is black instead of grey; **B** — the wrong quarter of the square is shaded; the circle is black instead of grey; **C** — the wrong quarter of the square is shaded; the circle has not moved on from the fifth figure; the circle is black instead of grey; **D** — the wrong quarter of the square is shaded; the circle is one position too far clockwise.

**Question 35 — Answer: A**

**Rules:** the central shape cycles triangle → square → pentagon → hexagon and then repeats; the circle moves 3 positions anticlockwise around the frame each time.

**Why the other options are wrong:** **B** — the central shape is a hexagon instead of a triangle; **C** — the circle has not moved on from the fifth figure; **D** — the circle is one position too far clockwise; **E** — the central shape is a square instead of a triangle; the circle has not moved on from the fifth figure.

**Question 36 — Answer: E**

**Rules:** the number of dots decreases by 1 each time; the triangle moves clockwise around the frame, alternately 1 and 2 positions at a time; the triangle's shading cycles black → white → grey.

**Why the other options are wrong:** **A** — there is 1 dot instead of 2; the triangle has not moved on from the fifth figure; the triangle is white instead of grey; **B** — there are 3 dots instead of 2; the triangle is one position too far clockwise; **C** — there is 1 dot instead of 2; the triangle is white instead of grey; **D** — there is 1 dot instead of 2; the triangle is one position too far clockwise; the triangle is white instead of grey.

**Question 37 — Answer: D**

**Rules:** the central shape cycles square → pentagon → hexagon and then repeats; the central shape's shading alternates between grey and white; the diamond moves 1 position anticlockwise around the frame each time; the diamond's shading cycles grey → black → white.

**Why the other options are wrong:** **A** — the central shape is a pentagon instead of a hexagon; the central shape is black instead of white; **B** — the central shape is a pentagon instead of a hexagon; the diamond is one position too far anticlockwise; the diamond is grey instead of white; **C** — the central shape is a square instead of a hexagon; the central shape is grey instead of white; the diamond is one position too far anticlockwise; the diamond is grey instead of white; **E** — the diamond has not moved on from the fifth figure; the diamond is black instead of white.

**Question 38 — Answer: A**

**Rules:** the arrow rotates 45° clockwise each time; the diamond moves clockwise around the frame, alternately 1 and 2 positions at a time; the square moves 1 position clockwise around the frame each time.

**Why the other options are wrong:** **B** — the arrow is rotated 45° too far clockwise; the square has not moved on from the fifth figure; **C** — the arrow is rotated 45° too far clockwise; the square is one position too far clockwise; **D** — the arrow has not rotated on from the fifth figure; the diamond has not moved on from the fifth figure; **E** — the square has not moved on from the fifth figure.

**Question 39 — Answer: C**

**Rules:** the arrow rotates anticlockwise, alternately by 90° and 45°; the triangle moves 1 position anticlockwise around the frame each time; the triangle's shading cycles black → white → grey; the circle moves 2 positions clockwise around the frame each time.

**Why the other options are wrong:** **A** — the arrow has not rotated on from the fifth figure; the triangle has not moved on from the fifth figure; **B** — the arrow is rotated 45° too far clockwise; the triangle is one position too far anticlockwise; the triangle is black instead of grey; **D** — the arrow is rotated 45° too far clockwise; the triangle is one position too far anticlockwise; the triangle is white instead of grey; the circle is one position too far clockwise; **E** — the arrow is rotated 45° too far clockwise; the triangle has not moved on from the fifth figure; the triangle is black instead of grey.

**Question 40 — Answer: B**

**Rules:** the arrow rotates anticlockwise, alternately by 45° and 90°; the diamond moves anticlockwise around the frame, alternately 3 and 1 positions at a time; the diamond's shading alternates between white and black.

**Why the other options are wrong:** **A** — the arrow is rotated 45° too far anticlockwise; the diamond is grey instead of black; **C** — the arrow is rotated 45° too far anticlockwise; the diamond is one position too far clockwise; **D** — the diamond is grey instead of black; **E** — the arrow has not rotated on from the fifth figure; the diamond is one position too far clockwise; the diamond is grey instead of black.

# Score Tracker

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Record every attempt. Seeing your scores rise is one of the best motivators there is.

| Set                       | Questions | Attempt 1 score | Date | Attempt 2 score | Date |
|---------------------------|-----------|-----------------|------|-----------------|------|
| Verbal Reasoning          | 65        |                 |      |                 |      |
| Numerical Reasoning       | 75        |                 |      |                 |      |
| Abstract Reasoning        | 75        |                 |      |                 |      |
| Situational Judgement     | 24        |                 |      |                 |      |
| Accuracy & Precision      | 30        |                 |      |                 |      |
| Prioritising & Organising | 24        |                 |      |                 |      |
| Mock Exam 1               | 40        |                 |      |                 |      |
| Mock Exam 2               | 40        |                 |      |                 |      |
| Mock Exam 3               | 40        |                 |      |                 |      |



# Error Log

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For every question you get wrong, write one line. Re-read this log before each mock exam and on the day before your test.

| Set / Q | What I did | What I should have done | Rule to remember |
|---------|------------|-------------------------|------------------|
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# Answer Sheets

Photocopy or tear out these sheets for the mock exams.

## Mock Exam 1 — answer sheet

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Time taken: \_\_\_\_\_ Score: \_\_\_\_ / 40

- 1

A

B

C

D
- 2

A

B

C

D
- 3

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B

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D
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- 5

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- 6

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- 21

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D
- 22

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B

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D
- 23

A

B

C

D
- 24

A

B

C

D
- 25

A

B

C

D
- 26

A

B

C

D
- 27

A

B

C

D
- 28

A

B

C

D
- 29

A

B

C

D
- 30

A

B

C

D
- 31

A

B

C

D

E
- 32

A

B

C

D

E
- 33

A

B

C

D

E
- 34

A

B

C

D

E
- 35

A

B

C

D

E
- 36

A

B

C

D

E
- 37

A

B

C

D

E
- 38

A

B

C

D

E
- 39

A

B

C

D

E
- 40

A

B

C

D

E

Questions 1–20: verbal · 21–30: numerical · 31–40: abstract



**Mock Exam 2 — answer sheet**

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Time taken: \_\_\_\_\_ Score: \_\_\_\_ / 40

- 1    (A) (B) (C) (D)  
2    (A) (B) (C) (D)  
3    (A) (B) (C) (D)  
4    (A) (B) (C) (D)  
5    (A) (B) (C) (D)  
6    (A) (B) (C) (D)  
7    (A) (B) (C) (D)  
8    (A) (B) (C) (D)  
9    (A) (B) (C) (D)  
10   (A) (B) (C) (D)  
11   (A) (B) (C) (D)  
12   (A) (B) (C) (D)  
13   (A) (B) (C) (D)  
14   (A) (B) (C) (D)  
15   (A) (B) (C) (D)  
16   (A) (B) (C) (D)  
17   (A) (B) (C) (D)  
18   (A) (B) (C) (D)  
19   (A) (B) (C) (D)  
20   (A) (B) (C) (D)

- 21   (A) (B) (C) (D)  
22   (A) (B) (C) (D)  
23   (A) (B) (C) (D)  
24   (A) (B) (C) (D)  
25   (A) (B) (C) (D)  
26   (A) (B) (C) (D)  
27   (A) (B) (C) (D)  
28   (A) (B) (C) (D)  
29   (A) (B) (C) (D)  
30   (A) (B) (C) (D)  
31   (A) (B) (C) (D) (E)  
32   (A) (B) (C) (D) (E)  
33   (A) (B) (C) (D) (E)  
34   (A) (B) (C) (D) (E)  
35   (A) (B) (C) (D) (E)  
36   (A) (B) (C) (D) (E)  
37   (A) (B) (C) (D) (E)  
38   (A) (B) (C) (D) (E)  
39   (A) (B) (C) (D) (E)  
40   (A) (B) (C) (D) (E)

*Questions 1–20: verbal · 21–30: numerical · 31–40: abstract*



**Mock Exam 3 — answer sheet**

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Time taken: \_\_\_\_\_ Score: \_\_\_\_ / 40

- 1    (A) (B) (C) (D)  
2    (A) (B) (C) (D)  
3    (A) (B) (C) (D)  
4    (A) (B) (C) (D)  
5    (A) (B) (C) (D)  
6    (A) (B) (C) (D)  
7    (A) (B) (C) (D)  
8    (A) (B) (C) (D)  
9    (A) (B) (C) (D)  
10   (A) (B) (C) (D)  
11   (A) (B) (C) (D)  
12   (A) (B) (C) (D)  
13   (A) (B) (C) (D)  
14   (A) (B) (C) (D)  
15   (A) (B) (C) (D)  
16   (A) (B) (C) (D)  
17   (A) (B) (C) (D)  
18   (A) (B) (C) (D)  
19   (A) (B) (C) (D)  
20   (A) (B) (C) (D)

- 21   (A) (B) (C) (D)  
22   (A) (B) (C) (D)  
23   (A) (B) (C) (D)  
24   (A) (B) (C) (D)  
25   (A) (B) (C) (D)  
26   (A) (B) (C) (D)  
27   (A) (B) (C) (D)  
28   (A) (B) (C) (D)  
29   (A) (B) (C) (D)  
30   (A) (B) (C) (D)  
31   (A) (B) (C) (D) (E)  
32   (A) (B) (C) (D) (E)  
33   (A) (B) (C) (D) (E)  
34   (A) (B) (C) (D) (E)  
35   (A) (B) (C) (D) (E)  
36   (A) (B) (C) (D) (E)  
37   (A) (B) (C) (D) (E)  
38   (A) (B) (C) (D) (E)  
39   (A) (B) (C) (D) (E)  
40   (A) (B) (C) (D) (E)

*Questions 1–20: verbal · 21–30: numerical · 31–40: abstract*



# A Final Word

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Reasoning tests reward method, calm and practice, and all three can be learned. If you have worked through this book, reviewed your errors and sat the mock exams under timed conditions, you have done exactly what high scorers do.

Good luck with your selection procedure, and with the career in European public service that it can open up.

If this workbook has helped you, a short review on the store where you bought it helps other candidates to find it. Thank you.